

Quantum Mechanics

PHY 203 • Core Course • L-T-P-C: 3-1-0-4

A. ILKER TOPUZ
ahmetllker.t@srmap.edu.in

SRM University, AP - Department of Physics

2026

Course Objectives

- ① Understand the concept of **quantum mechanics** and how it differs from classical mechanics.
- ② Understand wave–particle duality, photon theory, and matter waves.
- ③ Formulate the wave equation known as the **Schrödinger equation**.
- ④ Apply the Schrödinger equation to determine the properties of microscopic particles.

Course Outcomes

By the end of this course, you will be able to:

- 1 Describe the origin of quantum mechanics, the wavefunction, the uncertainty principle, the Schrödinger equation, quantum numbers, and Dirac notation.
- 2 Discuss wave–particle duality and the superposition principle.
- 3 Solve problems related to quantum mechanics, e.g. a particle in a 3D box.
- 4 Interpret quantum tunneling, angular momentum, commutation relations, and the experiments underpinning quantum mechanics.

Course Structure — Five Units

Unit	Title	Hours
1	Origin of Quantum Mechanics	12
2	Formulation of Quantum Mechanics	12
3	Basics of Schrödinger Equation I	12
4	Basics of Schrödinger Equation II	12
5	Dirac Notation in Quantum Mechanics	12
Total Contact Hours		60

Evaluation Scheme

Component	Weightage
Attendance	15%
Quizzes	15%
Midterm Examination	20%
Final Examination	50%
Total	100%

Unit 1: Origin of Quantum Mechanics

1. Origin of Quantum Mechanics
2. Formulation of Quantum Mechanics
3. Basics of Schrödinger Equation I
4. Basics of Schrödinger Equation II
5. Dirac Notation in Quantum Mechanics

Unit 1: Symbol Glossary

- h — Planck's constant (6.626×10^{-34} J s)
- $\hbar = h/2\pi$ — reduced Planck's constant (J s)
- ν — frequency of radiation (Hz)
- $\omega = 2\pi\nu$ — angular frequency (rad/s)
- λ — wavelength (m)
- c — speed of light (3×10^8 m/s)
- T — absolute temperature (K)
- k_B — Boltzmann's constant (1.381×10^{-23} J/K)
- $u(\nu, T)$ — spectral energy density (J s/m³)
- E — energy (J)
- ϕ — work function of a metal (J)
- KE — kinetic energy (J)
- V_0 — stopping potential (V)
- e — elementary charge (1.6×10^{-19} C)
- p — momentum (kg m/s)
- m_e — electron rest mass (9.11×10^{-31} kg)
- θ — scattering angle (rad)
- m — (generic) particle mass (kg)
- v — particle speed/velocity (m/s)
- k — wavenumber, $k = 2\pi/\lambda$ (m⁻¹)
- $A(k)$ — amplitude function in a wave packet
- $v_{\text{phase}}, v_{\text{group}}$ — phase and group velocity (m/s)
- N — number of electromagnetic cavity modes with wavevector magnitude $< k$ (dimensionless); dN is the number of modes in a shell dk

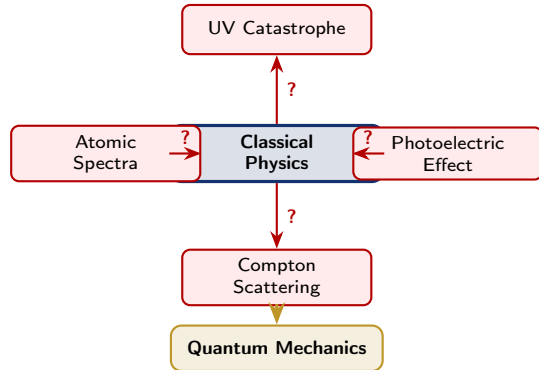
Symbols specific to a single derivation are defined where they first appear.

Setting the Stage

- Around 1900, physics seemed nearly complete: Newtonian mechanics, Maxwell's electromagnetism, and thermodynamics together explained almost every known phenomenon.
- But a handful of precision experiments — involving how matter absorbs and emits light — stubbornly refused to fit the classical picture.
- Resolving these anomalies required abandoning a core classical assumption: that energy is a continuous, infinitely divisible quantity.
- Unit 1 traces this story from the blackbody spectrum to the discovery that energy, and eventually matter itself, comes in discrete units.

Why Classical Physics Failed

- By the end of the 19th century, classical mechanics and electromagnetism described macroscopic phenomena extremely well.
- Several experiments could **not** be explained classically:
 - Blackbody radiation (**UV catastrophe**)
 - Photoelectric effect
 - Compton scattering (1923 — confirmed photon momentum)
 - Atomic spectra (discrete lines)
- These failures motivated a fundamentally new framework: **quantum mechanics**. (Note: Compton scattering post-dates the earliest QM; it is included here as decisive confirmation of photon momentum, not as an original pre-quantum anomaly.)

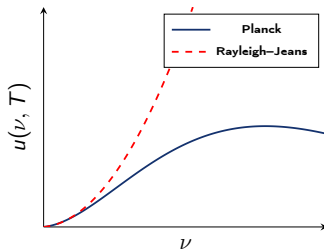


Blackbody Radiation

- A **blackbody** absorbs all incident radiation and re-emits a characteristic spectrum depending only on temperature T .
- Classical (Rayleigh–Jeans) theory:

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T$$

- Diverges as $\nu \rightarrow \infty$ — the **ultraviolet catastrophe**.
- Experimentally, the spectrum peaks and falls off at high frequency.



What Makes Something a "Blackbody"?

- An idealized blackbody absorbs 100% of incident radiation at every wavelength — nothing is reflected or transmitted.
- Despite the name, a blackbody is not necessarily black in appearance: when heated, it glows, and the *color* of that glow depends only on temperature, not on what the object is made of.
- Everyday approximations: the filament of an incandescent bulb, molten metal, the interior of a kiln viewed through a small hole, and — to good approximation — stars (including the Sun).
- Because the spectrum depends *only* on T , measuring a star's blackbody spectrum is one of the most direct ways astronomers infer stellar surface temperatures.
- The puzzle that opens this unit: classical physics predicted this spectrum should diverge at high frequency, which is obviously unphysical (it would mean every warm object radiates infinite energy).

Derivation: Counting Modes — Setting Up

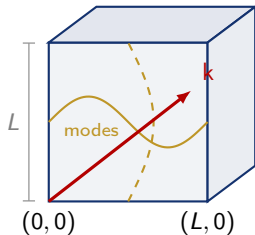
Goal: count how many electromagnetic standing-wave modes fit inside a cavity, per unit frequency interval.

- Consider a cubical cavity of side length L (a box of volume $V = L^3$). Standing electromagnetic waves inside must vanish at the walls, exactly like the particle-in-a-box condition derived later in Unit 4.
- This boundary condition restricts the allowed wavevector components to discrete values:

$$k_x = \frac{n_x \pi}{L}, \quad k_y = \frac{n_y \pi}{L}, \quad k_z = \frac{n_z \pi}{L}$$

where $n_x, n_y, n_z = 1, 2, 3, \dots$ are positive integers (mode numbers).

- The magnitude of the wavevector is
 $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2}$, related to wavelength by
 $k = 2\pi/\lambda$.



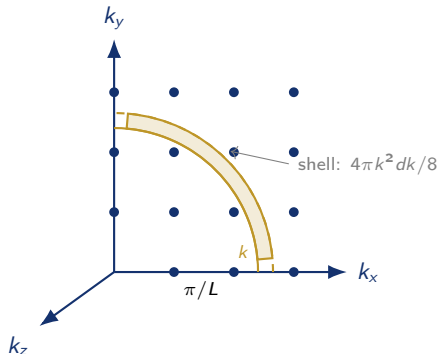
Derivation: Counting Modes — Counting in k -Space

- Picture a 3D lattice of points at positions (n_x, n_y, n_z) , each one representing one allowed mode. Since $n_x, n_y, n_z > 0$, only the lattice points in one octant ($\frac{1}{8}$) of a full sphere are physical.
- Let N be the total number of allowed cavity modes with wavevector magnitude less than k , so that dN is the number of modes lying in a thin shell between k and $k + dk$. The number of lattice points inside that shell equals the shell's volume divided by the volume per lattice point:

$$(\text{shell volume}) = 4\pi k^2 dk, \quad (\text{vol/point}) = \left(\frac{\pi}{L}\right)^3$$

- Including one-octant factor $\frac{1}{8}$ and factor 2 for two photon polarizations:

$$dN = 2 \times \frac{1}{8} \times 4\pi k^2 dk \times \frac{L^3}{\pi^3} = \frac{L^3 k^2}{\pi^2} dk$$



Derivation: Counting Modes — Converting to Frequency

- Convert from wavevector k to frequency ν using the electromagnetic dispersion relation $k = 2\pi\nu/c$, where c is the speed of light:

$$dk = \frac{2\pi}{c} d\nu, \quad k^2 = \frac{4\pi^2\nu^2}{c^2}$$

- Substitute both into $dN = \frac{L^3 k^2}{\pi^2} dk$:

$$dN = \frac{L^3}{\pi^2} \cdot \frac{4\pi^2\nu^2}{c^2} \cdot \frac{2\pi}{c} d\nu = \frac{8\pi L^3\nu^2}{c^3} d\nu$$

- Divide by the cavity volume $V = L^3$ to get the **mode density per unit volume**, $g(\nu)$, defined so that $g(\nu) d\nu$ is the number of modes per unit volume with frequency between ν and $\nu + d\nu$:

$$g(\nu) d\nu = \frac{8\pi\nu^2}{c^3} d\nu$$

- This purely geometric result (mode-counting only, no physics of energy yet) is what gets multiplied by the average energy per mode in the next step.

Derivation: Planck's Radiation Law — The Classical Step

- **Classical equipartition theorem:** in thermal equilibrium at temperature T , every available mode of oscillation is assigned the same average energy, $k_B T$, where k_B is Boltzmann's constant and T is the absolute temperature.
- This gives the classical (Rayleigh–Jeans) energy density:

$$u_{\text{classical}}(\nu, T) = g(\nu) \times k_B T = \frac{8\pi\nu^2}{c^3} k_B T$$

- As $\nu \rightarrow \infty$, this expression grows without bound — the **ultraviolet catastrophe** introduced earlier in this subsection.
- **Planck's modification:** instead of assuming every mode has the same continuous energy $k_B T$, assume a mode of frequency ν can only possess discrete energies

$$E_n = nh\nu, \quad n = 0, 1, 2, 3, \dots$$

where h is Planck's constant and n is a non-negative integer (the number of energy "quanta" in that mode).

Derivation: Planck's Radiation Law — Average Energy, Step 1

- With discrete energy levels $E_n = nh\nu$, statistical mechanics (the Boltzmann distribution) says the probability of a mode having energy E_n at temperature T is proportional to $e^{-E_n/k_B T}$.
- The thermal **average energy** of a mode is then a weighted average over all possible n :

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} E_n e^{-E_n/k_B T}}{\sum_{n=0}^{\infty} e^{-E_n/k_B T}} = \frac{\sum_{n=0}^{\infty} nh\nu e^{-nh\nu/k_B T}}{\sum_{n=0}^{\infty} e^{-nh\nu/k_B T}}$$

(numerator: energy-weighted sum; denominator: normalization, ensuring probabilities sum to 1).

- To evaluate both sums, introduce the shorthand $x \equiv e^{-h\nu/k_B T}$, so that $e^{-nh\nu/k_B T} = (e^{-h\nu/k_B T})^n = x^n$. Note $0 < x < 1$ for any positive ν, T .

Derivation: Planck's Radiation Law — Average Energy, Step 1

- In terms of x :

$$\langle E \rangle = h\nu \cdot \frac{\sum_{n=0}^{\infty} n x^n}{\sum_{n=0}^{\infty} x^n}$$

Derivation: Planck's Radiation Law — Average Energy, Step 2

- **Denominator:** the sum $\sum_{n=0}^{\infty} x^n$ is a standard geometric series. For $|x| < 1$:

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$$

- **Numerator:** the sum $\sum_{n=0}^{\infty} nx^n$ is obtained by differentiating the geometric series with respect to x .
Starting from $\sum_n x^n = (1-x)^{-1}$, differentiate both sides:

$$\frac{d}{dx} \sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} nx^{n-1} = \frac{d}{dx} (1-x)^{-1} = \frac{1}{(1-x)^2}$$

Derivation: Planck's Radiation Law — Average Energy, Step 2

- Multiply both sides by x to restore the correct power of x :

$$\sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2}$$

- Substituting both results into $\langle E \rangle = h\nu \cdot \frac{\sum nx^n}{\sum x^n}$:

$$\langle E \rangle = h\nu \cdot \frac{x/(1-x)^2}{1/(1-x)} = h\nu \cdot \frac{x}{1-x}$$

Derivation: Planck's Radiation Law — Average Energy, Result

- Substitute back $x = e^{-h\nu/k_B T}$ into $\langle E \rangle = h\nu \frac{x}{1-x}$:

$$\langle E \rangle = h\nu \cdot \frac{e^{-h\nu/k_B T}}{1 - e^{-h\nu/k_B T}}$$

- Multiply numerator and denominator by $e^{+h\nu/k_B T}$ to clear the negative exponent:

$$\langle E \rangle = h\nu \cdot \frac{e^{-h\nu/k_B T} \cdot e^{h\nu/k_B T}}{(1 - e^{-h\nu/k_B T}) e^{h\nu/k_B T}} = h\nu \cdot \frac{1}{e^{h\nu/k_B T} - 1}$$

- Final result for the average thermal energy of a single quantized mode:

$$\boxed{\langle E \rangle = \frac{h\nu}{e^{h\nu/k_B T} - 1}}$$

- Unlike the classical result $\langle E \rangle = k_B T$ (independent of ν), this quantum result **depends on frequency** — at high ν , the exponential in the denominator grows huge, suppressing $\langle E \rangle$ toward zero. This suppression is exactly what tames the ultraviolet catastrophe.

Planck's Radiation Law (Combining the Pieces)

Combining mode density and average energy

$$u(\nu, T) = g(\nu) \langle E \rangle = \frac{8\pi\nu^2}{c^3} \times \frac{h\nu}{e^{h\nu/k_B T} - 1} = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$$

where $u(\nu, T)$ is the spectral energy density (energy per unit volume per unit frequency), h is Planck's constant, ν is frequency, c is the speed of light, k_B is Boltzmann's constant, and T is absolute temperature.

- $h = 6.626 \times 10^{-34}$ J s is the single new physical constant introduced, fixed by fitting the formula to experimental blackbody spectra.

Derivation: Low-Frequency Limit (Recovering Rayleigh–Jeans)

- Consider the regime $h\nu \ll k_B T$ (low frequency or high temperature), so the exponent $h\nu/k_B T$ is small.
- Use the Taylor expansion of the exponential for small ϵ : $e^\epsilon \approx 1 + \epsilon$ when $\epsilon \ll 1$. With $\epsilon = h\nu/k_B T$:

$$e^{h\nu/k_B T} \approx 1 + \frac{h\nu}{k_B T} \implies e^{h\nu/k_B T} - 1 \approx \frac{h\nu}{k_B T}$$

- Substitute this approximation into Planck's law:

$$u(\nu, T) \approx \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{h\nu/k_B T} = \frac{8\pi h\nu^3}{c^3} \cdot \frac{k_B T}{h\nu}$$

- Cancel one factor of $h\nu$ between numerator and denominator:

$$u(\nu, T) \rightarrow \frac{8\pi\nu^2}{c^3} k_B T$$

- This exactly matches the classical Rayleigh–Jeans formula derived at the start of this subsection — confirming Planck's law correctly reduces to the classical (non-quantum) result whenever quantum effects are negligible ($h\nu \ll k_B T$).

Derivation: High-Frequency Limit (Wien's Law)

- Consider the opposite regime $h\nu \gg k_B T$ (high frequency or low temperature), so the exponent $h\nu/k_B T$ is large.
- Then $e^{h\nu/k_B T} \gg 1$, so subtracting 1 barely changes its value:

$$e^{h\nu/k_B T} - 1 \approx e^{h\nu/k_B T}$$

- Substituting into Planck's law:

$$u(\nu, T) \approx \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/k_B T}} = \frac{8\pi h\nu^3}{c^3} e^{-h\nu/k_B T}$$

- This is **Wien's law**: at high frequency, $u(\nu, T)$ is suppressed exponentially rather than diverging — directly curing the ultraviolet catastrophe that plagued the classical Rayleigh–Jeans formula.
- Together, the two limits show Planck's law smoothly interpolates between the correct classical behavior at low ν and the experimentally observed cutoff at high ν — a single formula valid at all frequencies.

Derivation: Wien's Displacement Law

- **Goal:** find $\lambda_{\max}$, the wavelength at which the blackbody spectrum peaks, and show it satisfies $\lambda_{\max} T = \text{const.}$
- **Step 1 — Rewrite Planck's law in wavelength form.** Using $\nu = c/\lambda$ and the relation $u(\lambda, T)|d\lambda| = u(\nu, T)|d\nu|$ with $|d\nu/d\lambda| = c/\lambda^2$:

$$u(\lambda, T) = \frac{8\pi hc}{\lambda^5} \cdot \frac{1}{e^{hc/\lambda k_B T} - 1}$$

- **Step 2 — Differentiate and set to zero.** Let the dimensionless variable $x \equiv hc/(\lambda k_B T)$, so $\lambda = hc/(x k_B T)$. Writing $u \propto x^5(e^x - 1)^{-1}$ and computing $du/dx = 0$ gives:

$$\frac{d}{dx} \left[\frac{x^5}{e^x - 1} \right] = 0 \quad \implies \quad 5(e^x - 1) - xe^x = 0 \quad \implies \quad (5 - x)e^x = 5$$

Derivation: Wien's Displacement Law

- **Step 3 — Solve the transcendental equation.** Rearranging as $5 - x = 5e^{-x}$ and solving numerically (e.g. by Newton–Raphson or bisection):

$$x_0 \approx 4.9651$$

(The equation has no closed-form solution; x_0 is a universal constant of nature.)

- **Step 4 — Recover $\lambda_{\max} T$.** Since $x_0 = hc/(\lambda_{\max} k_B T)$:

$$\lambda_{\max} T = \frac{hc}{x_0 k_B} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{4.9651 \times (1.381 \times 10^{-23})} \approx 2.898 \times 10^{-3} \text{ m K} \equiv b$$

- This is **Wien's displacement law**: the peak shifts to shorter (bluer) wavelengths as T increases — hotter objects are bluer. The constant b is Wien's displacement constant.

Worked Example 1: Peak Wavelength (Wien's Law)

Problem

The Sun's surface temperature is approximately 5800 K. Estimate the wavelength at which its blackbody spectrum peaks, using Wien's displacement law $\lambda_{\max} T = b$, $b = 2.898 \times 10^{-3} \text{ m K}$.

Given: $T = 5800 \text{ K}$ (absolute temperature), $b = 2.898 \times 10^{-3} \text{ m K}$ (Wien's displacement constant).

Find: $\lambda_{\max}$ (the wavelength of peak spectral intensity).

Step 1 — Isolate $\lambda_{\max}$: divide both sides of $\lambda_{\max} T = b$ by T :

$$\lambda_{\max} = \frac{b}{T}$$

Step 2 — Substitute numbers:

$$\lambda_{\max} = \frac{2.898 \times 10^{-3} \text{ m K}}{5800 \text{ K}}$$

Step 3 — Evaluate:

$$\lambda_{\max} = 5.0 \times 10^{-7} \text{ m} = 500 \text{ nm}$$

Worked Example 1: Peak Wavelength (Wien's Law)

This falls in the visible (green) part of the spectrum — consistent with the Sun appearing roughly white/yellow to our eyes.

Worked Example 2: Spectral Energy Density

Problem

Using Planck's law, compute $u(\nu, T)$ for $\nu = 5 \times 10^{14}$ Hz (visible light) at $T = 3000$ K. Take $h = 6.626 \times 10^{-34}$ J s, $k_B = 1.381 \times 10^{-23}$ J/K, $c = 3 \times 10^8$ m/s.

Given: ν (frequency), T (temperature), and the constants h , k_B , c . **Find:** $u(\nu, T)$, the spectral energy density.

Step 1 — Compute the dimensionless exponent $h\nu/k_B T$:

$$\frac{h\nu}{k_B T} = \frac{(6.626 \times 10^{-34} \text{ J s})(5 \times 10^{14} \text{ Hz})}{(1.381 \times 10^{-23} \text{ J/K})(3000 \text{ K})} = \frac{3.313 \times 10^{-19} \text{ J}}{4.143 \times 10^{-20} \text{ J}} \approx 7.99$$

Step 2 — Compute $e^{7.99} - 1$:

$$e^{7.99} \approx 2951 \quad \Rightarrow \quad e^{7.99} - 1 \approx 2950$$

Worked Example 2: Spectral Energy Density

Step 3 — Compute the prefactor $8\pi h\nu^3/c^3$:

$$\frac{8\pi h\nu^3}{c^3} = \frac{8\pi(6.626 \times 10^{-34})(5 \times 10^{14})^3}{(3 \times 10^8)^3} \approx 7.71 \times 10^{-14} \text{ J s/m}^3$$

Step 4 — Combine:

$$u(\nu, T) = (7.71 \times 10^{-14}) \times \frac{1}{2950} \approx 2.6 \times 10^{-17} \text{ J s/m}^3$$

Worked Example 3: Rayleigh–Jeans vs. Planck

Problem

At $T = 300\text{ K}$ and $\nu = 10^{10}\text{ Hz}$ (microwave region), compare the Rayleigh–Jeans prediction to Planck's law. Comment on the agreement.

Given: ν , T , and the constants h , k_B , c . **Find:** u_ν^{RJ} and u_ν^{Planck} , and compare.

Step 1 — Compute the exponent $x = h\nu/k_B T$:

$$h\nu = (6.626 \times 10^{-34})(10^{10}) = 6.626 \times 10^{-24}\text{ J}, \quad k_B T = (1.381 \times 10^{-23})(300) = 4.143 \times 10^{-21}\text{ J}$$

$$x = \frac{h\nu}{k_B T} = \frac{6.626 \times 10^{-24}}{4.143 \times 10^{-21}} \approx 1.60 \times 10^{-3} \ll 1$$

Worked Example 3: Rayleigh–Jeans vs. Planck

Step 2 — Rayleigh–Jeans energy density $u_{\nu}^{\text{RJ}} = \frac{8\pi\nu^2}{c^3} k_B T$:

$$\frac{8\pi\nu^2}{c^3} = \frac{8\pi(10^{10})^2}{(3 \times 10^8)^3} = \frac{2.513 \times 10^{21}}{2.7 \times 10^{25}} \approx 9.31 \times 10^{-5} \text{ s}^2/\text{m}^4$$

$$u_{\nu}^{\text{RJ}} = (9.31 \times 10^{-5})(4.143 \times 10^{-21}) \approx 3.86 \times 10^{-25} \text{ J s/m}^3$$

Step 3 — Planck energy density $u_{\nu}^{\text{Planck}} = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^x - 1}$:

$$\frac{8\pi h\nu^3}{c^3} = \frac{8\pi(6.626 \times 10^{-34})(10^{10})^3}{(3 \times 10^8)^3} \approx 6.17 \times 10^{-28} \text{ J s/m}^3$$

Worked Example 3: Rayleigh–Jeans vs. Planck

Since $x \ll 1$, expand $e^x - 1 \approx x + \frac{x^2}{2}$:

$$e^x - 1 \approx 1.60 \times 10^{-3} + 1.28 \times 10^{-6} \approx 1.601 \times 10^{-3}$$

$$u_{\nu}^{\text{Planck}} = \frac{6.17 \times 10^{-28}}{1.601 \times 10^{-3}} \approx 3.85 \times 10^{-25} \text{ J s/m}^3$$

Step 4 — Compare: $u_{\nu}^{\text{RJ}} \approx u_{\nu}^{\text{Planck}} \approx 3.86 \times 10^{-25} \text{ J s/m}^3$, agreeing to within

$$\frac{u_{\nu}^{\text{Planck}}}{u_{\nu}^{\text{RJ}}} = \frac{x}{e^x - 1} \approx 1 - \frac{x}{2} \approx 1 - 8 \times 10^{-4} \quad (\sim 0.08\% \text{ difference}).$$

This confirms Planck's law reduces smoothly to the classical Rayleigh–Jeans result whenever $h\nu \ll k_B T$.

Worked Example 4: Why the Sun Looks White, Not Green

Problem

Example 1 found the Sun's blackbody spectrum peaks at 500 nm (green). Explain why the Sun nonetheless appears white to the eye, using the shape of the Planck curve.

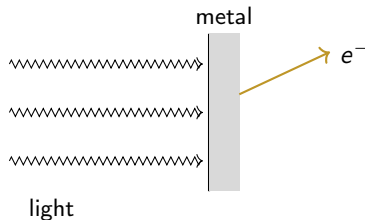
Solution: Unlike a single-wavelength (laser) source, the Planck spectrum is **broad** — it has significant intensity across the *entire* visible range (400–700 nm), not just at the peak.

$$\left. \frac{h\nu}{k_B T} \right|_{400 \text{ nm}} \approx 1.2 \times \left. \frac{h\nu}{k_B T} \right|_{500 \text{ nm}}, \quad \left. \frac{h\nu}{k_B T} \right|_{700 \text{ nm}} \approx 0.71 \times \left. \frac{h\nu}{k_B T} \right|_{500 \text{ nm}}$$

Evaluating $u(\nu, T)$ at these nearby exponents shows the intensity stays within the same order of magnitude across blue, green, and red. The human eye integrates this broad mixture of wavelengths and perceives it as white (or pale yellow after atmospheric scattering removes some blue) — the peak wavelength alone does not determine perceived color.

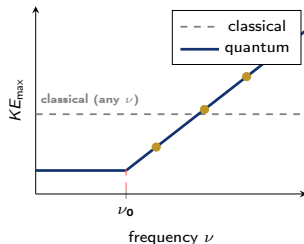
The Photoelectric Effect

- Light incident on a metal surface ejects electrons (**photoelectrons**).
- Classical wave theory predicts:
 - KE of electrons $\propto$ intensity
 - Effect occurs at any frequency, given enough time
- **Observed** (contradicts classical theory):
 - KE depends on **frequency**, not intensity
 - Below a threshold frequency ν_0 , no emission — regardless of intensity
 - Emission is instantaneous



Why Classical Wave Theory Fails Here

- Classical EM: wave energy is spread across the wavefront. A dim source should take **minutes** to deliver enough energy for an electron to escape.
- **Experimentally**: photoelectrons appear within **nanoseconds**, regardless of intensity — provided $\nu > \nu_0$.
- A bright low- ν beam ejects **no** electrons; a dim high- ν beam ejects them instantly.
- These two failures — missing time delay and sharp threshold — are resolved by Einstein's photon hypothesis.



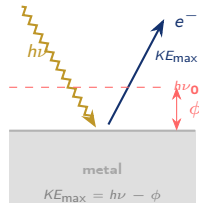
Einstein's Photon Theory (1905)

Einstein's Hypothesis

Light itself consists of discrete quanta (**photons**) of energy

$$E = h\nu$$

- Single photon $\rightarrow$ single electron interaction (explains instantaneous emission).
- $\nu < \nu_0 \Rightarrow$ no photoelectrons, however intense the beam.



Derivation: Einstein's Photoelectric Equation — Setup

- **Energy conservation** for a single photon absorbed by a single electron at the metal surface: the photon's entire energy is split between freeing the electron and giving it kinetic energy.

$$\underbrace{h\nu}_{\text{photon energy}} = \underbrace{\phi}_{\text{work to escape}} + \underbrace{KE}_{\text{electron's kinetic energy}}$$

where h is Planck's constant, ν is the frequency of the incident light, ϕ is the work function (minimum energy needed to remove an electron from the metal), and KE is the kinetic energy of the emitted electron.

- The **work function** ϕ is itself defined via a threshold frequency ν_0 — the minimum frequency capable of ejecting an electron at all (with zero leftover kinetic energy):

$$\phi = h\nu_0$$

- Electrons that originate from below the surface lose additional energy escaping through the metal, so the energy-conservation equation above gives only an *upper bound* on KE — hence we call it KE_{max} .

Derivation: Einstein's Photoelectric Equation — Solving for $KE_{\max}$

- **Step 1:** start from energy conservation, $h\nu = \phi + KE_{\max}$.
- **Step 2:** isolate $KE_{\max}$ by subtracting ϕ from both sides:

$$KE_{\max} = h\nu - \phi$$

- **Step 3:** substitute $\phi = h\nu_0$ and factor out h :

$$KE_{\max} = h\nu - h\nu_0 = h(\nu - \nu_0)$$

- This form makes clear: if $\nu < \nu_0$, then $KE_{\max} < 0$, which is unphysical — meaning no photoelectrons are emitted at all below the threshold frequency, regardless of intensity, exactly as observed experimentally.

Derivation: Einstein's Photoelectric Equation — Stopping Potential

- Experimentally, $KE_{\max}$ is not measured directly but inferred from the **stopping potential** V_0 : the minimum retarding voltage that just barely stops the most energetic photoelectrons from reaching the detector.
- An electron of charge $-e$ moving against a retarding potential V_0 loses kinetic energy eV_0 (where e is the elementary charge, a positive number) by the time it is stopped. Setting this equal to the electron's initial kinetic energy:

$$eV_0 = KE_{\max}$$

- Substitute $KE_{\max} = h\nu - \phi$ from the previous slide:

$$eV_0 = h\nu - \phi$$

- Divide both sides by e** to isolate V_0 :

$$V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$$

Derivation: Einstein's Photoelectric Equation — Stopping Potential

- This is a linear equation in ν with slope h/e and ν -intercept $\nu_0 = \phi/h$ — so plotting measured V_0 against ν for several different light frequencies yields a straight line whose slope directly gives h/e , exactly as Millikan measured.

Historical Note: Millikan's Decade-Long Verification

- Einstein proposed the photon hypothesis in 1905, but it was met with significant skepticism — it seemed to contradict the well-established wave theory of light (interference, diffraction).
- Robert Millikan spent roughly a decade (1905–1916) performing extremely precise photoelectric experiments specifically to **disprove** Einstein's equation.
- Instead, his measurements confirmed it to high precision: V_0 vs. ν traced out a perfect straight line, and the slope gave a value of h matching Planck's blackbody-derived constant to within experimental error.
- Millikan later wrote that he was forced to conclude the equation was "experimentally established... in spite of its unreasonableness," even though he himself initially doubted the underlying photon picture.
- Einstein received the 1921 Nobel Prize in Physics specifically for explaining the photoelectric effect (not for relativity) — reflecting how central this single equation was to establishing the reality of light quanta.

Worked Example 1: Maximum Kinetic Energy

Problem

The work function of sodium is $\phi = 2.3 \text{ eV}$. Light of wavelength 400 nm is incident on it. Find the maximum kinetic energy of the photoelectrons.

Given: $\phi = 2.3 \text{ eV}$ (work function), $\lambda = 400 \text{ nm}$ (wavelength). **Find:** KE_{max} .

Step 1 — Find the photon energy using $E = h\nu = hc/\lambda$, where h is Planck's constant and c is the speed of light:

$$h\nu = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m/s})}{400 \times 10^{-9} \text{ m}}$$

Step 2 — Evaluate the numerator and divide:

$$h\nu = \frac{1.988 \times 10^{-25} \text{ J m}}{4 \times 10^{-7} \text{ m}} \approx 4.97 \times 10^{-19} \text{ J}$$

Worked Example 1: Maximum Kinetic Energy

Step 3 — Convert to eV (divide by $e = 1.6 \times 10^{-19} \text{ C}$, since $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$):

$$h\nu \approx \frac{4.97 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 3.10 \text{ eV}$$

Step 4 — Apply Einstein's equation $KE_{\text{max}} = h\nu - \phi$:

$$KE_{\text{max}} = 3.10 \text{ eV} - 2.3 \text{ eV} = 0.80 \text{ eV}$$

Worked Example 2: Threshold Wavelength

Problem

The threshold frequency for potassium is $\nu_0 = 5.0 \times 10^{14}$ Hz. Find (a) the work function in eV, and (b) the threshold wavelength.

Given: ν_0 (threshold frequency). **Find:** ϕ and λ_0 (threshold wavelength).

Part (a), Step 1 — Apply $\phi = h\nu_0$:

$$\phi = h\nu_0 = (6.626 \times 10^{-34} \text{ J s})(5.0 \times 10^{14} \text{ Hz}) = 3.31 \times 10^{-19} \text{ J}$$

Part (a), Step 2 — Convert to eV (divide by e):

$$\phi = \frac{3.31 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 2.07 \text{ eV}$$

Worked Example 2: Threshold Wavelength

Part (b), Step 1 — Use $c = \nu_0 \lambda_0$, **so** $\lambda_0 = c/\nu_0$, where c is the speed of light:

$$\lambda_0 = \frac{c}{\nu_0} = \frac{3 \times 10^8 \text{ m/s}}{5.0 \times 10^{14} \text{ Hz}}$$

Part (b), Step 2 — Evaluate:

$$\lambda_0 = 6.0 \times 10^{-7} \text{ m} = 600 \text{ nm}$$

Light with $\lambda > 600 \text{ nm}$ cannot eject electrons from potassium, regardless of intensity.

Worked Example 3: Stopping Potential

Problem

UV light of wavelength 250 nm strikes a metal of work function 4.0 eV. Find the stopping potential V_0 .

Given: $\lambda = 250 \text{ nm}$, $\phi = 4.0 \text{ eV}$. **Find:** V_0 (stopping potential).

Step 1 — Photon energy $h\nu = hc/\lambda$:

$$h\nu = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{250 \times 10^{-9}} = \frac{1.988 \times 10^{-25}}{2.5 \times 10^{-7}} = 7.95 \times 10^{-19} \text{ J}$$

Step 2 — Convert to eV (divide by e):

$$h\nu = \frac{7.95 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 4.97 \text{ eV}$$

Worked Example 3: Stopping Potential

Step 3 — Apply $eV_0 = h\nu - \phi$, so $V_0 = (h\nu - \phi)/e$. Since both $h\nu$ and ϕ are already expressed in eV, V_0 in volts equals the numerical difference directly:

$$V_0 = \frac{h\nu - \phi}{e} = \frac{4.97 \text{ eV} - 4.0 \text{ eV}}{e} = 0.97 \text{ V}$$

Worked Example 4: Determining h from Experimental Data

Problem

Two experiments on the same metal give stopping potentials $V_0 = 0.5 \text{ V}$ at $\lambda_1 = 500 \text{ nm}$ and $V_0 = 1.5 \text{ V}$ at $\lambda_2 = 300 \text{ nm}$. Use these two data points (rather than known constants) to determine h/e and the work function ϕ .

Given: two (λ, V_0) data pairs. **Find:** h/e and ϕ , using only the data (no assumed value of h).

Step 1 — Convert wavelengths to frequencies using $\nu = c/\lambda$:

$$\nu_1 = \frac{c}{\lambda_1} = \frac{3 \times 10^8}{500 \times 10^{-9}} = 6.0 \times 10^{14} \text{ Hz}, \quad \nu_2 = \frac{c}{\lambda_2} = \frac{3 \times 10^8}{300 \times 10^{-9}} = 1.0 \times 10^{15} \text{ Hz}$$

Step 2 — Use the linear relation $V_0 = (h/e)\nu - \phi/e$ at both data points, then subtract to eliminate ϕ/e :

$$V_2 - V_1 = \frac{h}{e}(\nu_2 - \nu_1) \quad \Rightarrow \quad \frac{h}{e} = \frac{V_2 - V_1}{\nu_2 - \nu_1}$$

Worked Example 4: Determining h from Experimental Data

Step 3 — Substitute numbers:

$$\frac{h}{e} = \frac{1.5 - 0.5}{1.0 \times 10^{15} - 6.0 \times 10^{14}} = \frac{1.0}{4.0 \times 10^{14}} = 2.5 \times 10^{-15} \text{ V s}$$

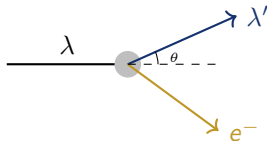
Step 4 — Find ϕ by substituting back into $V_1 = (h/e)\nu_1 - \phi/e$ and solving for ϕ :

$$\phi = e \left[\left(\frac{h}{e} \right) \nu_1 - V_1 \right] = e[(2.5 \times 10^{-15})(6.0 \times 10^{14}) - 0.5] = e(1.0) \Rightarrow \phi = 1.0 \text{ eV}$$

This is exactly the method Millikan used — h and ϕ emerge purely from the slope and intercept of measured data, with no prior assumption about h 's value.

The Compton Effect

- X-rays scattered off a free (or loosely bound) electron show a wavelength **shift** $\Delta\lambda$ depending on scattering angle θ .
- Cannot be explained by classical wave scattering (Thomson scattering predicts no wavelength change).
- Explained by treating the photon as a particle with momentum $p = h/\lambda$, colliding elastically with an electron.

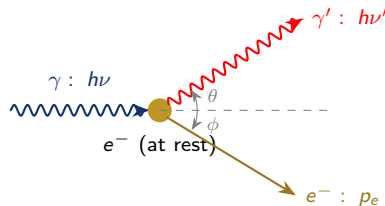


Why the Compton Effect Demanded the Particle Picture

- In classical (Thomson) scattering, an electromagnetic wave shakes a free electron at the wave's own frequency, and the electron re-radiates at **exactly that same frequency** — no wavelength shift is predicted, regardless of scattering angle.
- Compton's 1923 experiments instead found $\lambda' > \lambda$ always, with the shift growing systematically with scattering angle θ — a result classical wave theory simply cannot produce.
- The resolution requires treating the photon as a genuine particle carrying both energy $E = h\nu$ and momentum $p = h/\lambda$, and applying ordinary relativistic two-body collision kinematics (exactly as for billiard balls) between the photon and the electron.
- This was historically decisive: the photoelectric effect alone could be (and was) explained away by some physicists using semiclassical arguments, but the angle-dependent Compton shift left no such escape — it provided clean, quantitative proof that photons carry momentum.

Derivation: Compton Shift — Physical Setup

- A photon of frequency ν (energy $h\nu$, momentum $h\nu/c$) collides with a stationary free electron of rest mass m_e .
- After the collision: the photon scatters at angle θ (measured from its original direction) with new frequency ν' ; the electron recoils with momentum p_e at angle ϕ (measured on the opposite side).
- Here h is Planck's constant, c is the speed of light, and θ, ϕ are the photon and electron scattering angles respectively.
- Because the electron may recoil at relativistic speed, we must use the full relativistic energy-momentum relation for the electron, $E_e = \sqrt{p_e^2 c^2 + m_e^2 c^4}$, rather than the non-relativistic $KE = p^2/2m$.



Derivation: Compton Shift — Conservation Laws

- **Conservation of total energy** (photon + electron energy before = after):

$$\underbrace{h\nu}_{\text{photon in}} + \underbrace{m_e c^2}_{\text{electron rest energy}} = \underbrace{h\nu'}_{\text{photon out}} + \underbrace{\sqrt{p_e^2 c^2 + m_e^2 c^4}}_{\text{electron total energy}}$$

- **Conservation of momentum**, resolved into components along the incident direction (x) and perpendicular to it (y). The incident photon has momentum $h\nu/c$ purely along x ; after the collision the scattered photon has momentum $h\nu'/c$ at angle θ , and the electron has momentum p_e at angle ϕ on the other side:

$$\text{x-direction: } \frac{h\nu}{c} = \frac{h\nu'}{c} \cos \theta + p_e \cos \phi$$

$$\text{y-direction: } 0 = \frac{h\nu'}{c} \sin \theta - p_e \sin \phi$$

- We now have three equations (one energy, two momentum) and want to eliminate the electron variables p_e, ϕ to obtain a relation purely between the photon quantities ν, ν', θ .

Derivation: Compton Shift — Isolating the Electron Momentum

- **Step 1:** rearrange the two momentum equations to isolate the electron's momentum components on one side:

$$p_e \cos \phi = \frac{h\nu}{c} - \frac{h\nu'}{c} \cos \theta, \quad p_e \sin \phi = \frac{h\nu'}{c} \sin \theta$$

- **Step 2:** square both equations:

$$p_e^2 \cos^2 \phi = \left(\frac{h\nu}{c} - \frac{h\nu'}{c} \cos \theta \right)^2, \quad p_e^2 \sin^2 \phi = \left(\frac{h\nu'}{c} \sin \theta \right)^2$$

- **Step 3:** add the two equations. The left side becomes $p_e^2(\cos^2 \phi + \sin^2 \phi) = p_e^2$ (using $\cos^2 \phi + \sin^2 \phi = 1$), eliminating the unknown angle ϕ entirely:

$$p_e^2 = \left(\frac{h\nu}{c} - \frac{h\nu'}{c} \cos \theta \right)^2 + \left(\frac{h\nu'}{c} \sin \theta \right)^2$$

Derivation: Compton Shift — Expanding the Momentum Expression

- Expand the first squared term using $(a - b)^2 = a^2 - 2ab + b^2$ with $a = h\nu/c$, $b = (h\nu'/c) \cos \theta$:

$$\left(\frac{h\nu}{c}\right)^2 - 2\left(\frac{h\nu}{c}\right)\left(\frac{h\nu'}{c} \cos \theta\right) + \left(\frac{h\nu'}{c}\right)^2 \cos^2 \theta$$

- Add the second term, $(h\nu'/c)^2 \sin^2 \theta$, and combine the $\cos^2 \theta$ and $\sin^2 \theta$ pieces using $\cos^2 \theta + \sin^2 \theta = 1$:

$$p_e^2 = \left(\frac{h\nu}{c}\right)^2 - 2\left(\frac{h\nu}{c}\right)\left(\frac{h\nu'}{c}\right) \cos \theta + \left(\frac{h\nu'}{c}\right)^2$$

- Multiply through by c^2 to clear denominators:

$$p_e^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2h^2 \nu \nu' \cos \theta$$

This is the momentum-derived expression for $p_e^2 c^2$ — purely in terms of photon quantities and the unknown θ .

Derivation: Compton Shift — The Energy Equation

- Now turn to the energy-conservation equation and solve it for $p_e^2 c^2$ as well, so the two expressions can be equated.
- **Step 1:** isolate the square root term in the energy equation:

$$\sqrt{p_e^2 c^2 + m_e^2 c^4} = h\nu - h\nu' + m_e c^2$$

- **Step 2:** square both sides to remove the square root:

$$p_e^2 c^2 + m_e^2 c^4 = (h\nu - h\nu' + m_e c^2)^2$$

- **Step 3:** expand the right-hand side. Let $\Delta \equiv h\nu - h\nu'$ for brevity, so the right side is $(\Delta + m_e c^2)^2 = \Delta^2 + 2\Delta m_e c^2 + m_e^2 c^4$:

$$p_e^2 c^2 + m_e^2 c^4 = (h\nu - h\nu')^2 + 2(h\nu - h\nu')m_e c^2 + m_e^2 c^4$$

- **Step 4:** the $m_e^2 c^4$ terms cancel from both sides, leaving:

$$p_e^2 c^2 = (h\nu - h\nu')^2 + 2m_e c^2 (h\nu - h\nu')$$

Derivation: Compton Shift — Equating and Simplifying

- We now have **two** independent expressions for $p_e^2 c^2$ — one from momentum conservation, one from energy conservation. Setting them equal:

$$(h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu' \cos \theta = (h\nu - h\nu')^2 + 2m_e c^2(h\nu - h\nu')$$

- **Step 1:** expand $(h\nu - h\nu')^2 = (h\nu)^2 - 2h^2\nu\nu' + (h\nu')^2$ on the right side:

$$(h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu' \cos \theta = (h\nu)^2 - 2h^2\nu\nu' + (h\nu')^2 + 2m_e c^2(h\nu - h\nu')$$

- **Step 2:** the $(h\nu)^2$ and $(h\nu')^2$ terms appear identically on both sides and cancel:

$$-2h^2\nu\nu' \cos \theta = -2h^2\nu\nu' + 2m_e c^2(h\nu - h\nu')$$

- **Step 3:** add $2h^2\nu\nu' \cos \theta$ to both sides and rearrange to collect the $h^2\nu\nu'$ terms:

$$2h^2\nu\nu' - 2h^2\nu\nu' \cos \theta = 2m_e c^2(h\nu - h\nu')$$

Derivation: Compton Shift — Equating and Simplifying

- **Step 4:** factor $2h^2\nu\nu'$ on the left and divide both sides by 2:

$$h^2\nu\nu'(1 - \cos \theta) = m_e c^2 h(\nu - \nu')$$

- **Step 5:** divide both sides by h :

$$h\nu\nu'(1 - \cos \theta) = m_e c^2 (\nu - \nu')$$

Compton Shift Formula — Final Steps

- Starting from $h\nu\nu'(1 - \cos \theta) = m_e c^2(\nu - \nu')$, **divide both sides by $\nu\nu' m_e c^2$** :

$$\frac{h}{m_e c^2}(1 - \cos \theta) = \frac{\nu - \nu'}{\nu\nu'} = \frac{1}{\nu'} - \frac{1}{\nu}$$

- Use the wave relation $c = \nu\lambda$ (so $\nu = c/\lambda$ and $1/\nu = \lambda/c$) to convert frequencies to wavelengths:

$$\frac{1}{\nu'} - \frac{1}{\nu} = \frac{\lambda'}{c} - \frac{\lambda}{c} = \frac{\lambda' - \lambda}{c}$$

- Substituting back:

$$\frac{h}{m_e c^2}(1 - \cos \theta) = \frac{\lambda' - \lambda}{c}$$

- Multiply both sides by c** to isolate the wavelength shift:

$$\Delta\lambda \equiv \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos \theta)$$

Compton Shift Formula (Result)

Compton Wavelength Shift

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

where $\Delta\lambda$ is the wavelength shift, λ and λ' are the incident and scattered photon wavelengths, h is Planck's constant, m_e is the electron rest mass, c is the speed of light, and θ is the photon scattering angle.

- $\frac{h}{m_e c} = 2.426 \times 10^{-12} \text{ m}$ is the **Compton wavelength** of the electron.
- $\Delta\lambda$ is independent of incident wavelength λ and of the target material — a clean, falsifiable prediction confirmed by experiment.
- Direct confirmation that photons carry momentum $p = h/\lambda$ — strong evidence for the particle nature of light.

Why X-Rays, and Not Visible Light?

- The Compton shift $\Delta\lambda$ has a fixed, universal magnitude set by $h/m_e c \approx 2.43 \times 10^{-12}$ m — it does not scale with the incident wavelength λ .
- For **visible light** ($\lambda \sim 500$ nm = 5×10^{-7} m), the fractional shift is

$$\frac{\Delta\lambda}{\lambda} \sim \frac{2.43 \times 10^{-12}}{5 \times 10^{-7}} \approx 5 \times 10^{-6}$$

far too small to detect against the dominant, much larger elastic (Thomson/Rayleigh) scattering signal.

- For **X-rays** ($\lambda \sim 0.1$ nm = 10^{-10} m), the same absolute shift $\Delta\lambda$ is now comparable to λ itself:

$$\frac{\Delta\lambda}{\lambda} \sim \frac{2.43 \times 10^{-12}}{10^{-10}} \approx 2 \times 10^{-2} \text{ (a few percent — readily measurable)}$$

- This is precisely why Compton used X-rays in his original experiment: the effect is universal physics, but only observable when λ is comparable to the electron's Compton wavelength.

Worked Example 1: Compton Shift at 90°

Problem

X-rays of wavelength 0.1 nm scatter off free electrons at $\theta = 90^\circ$. Find the wavelength shift and the scattered wavelength.

Given: $\lambda = 0.1 \text{ nm}$ (incident wavelength), $\theta = 90^\circ$ (scattering angle). **Find:** $\Delta\lambda$ and λ' .

Step 1 — Evaluate $\cos\theta$: for $\theta = 90^\circ$, $\cos 90^\circ = 0$.

Step 2 — Apply the Compton formula $\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$, where h is Planck's constant, m_e is the electron rest mass, c is the speed of light:

$$\Delta\lambda = (2.426 \times 10^{-12} \text{ m})(1 - 0) = 2.426 \times 10^{-12} \text{ m}$$

Step 3 — Convert to nm (divide by 10^{-9}):

$$\Delta\lambda = 2.426 \times 10^{-3} \text{ nm} = 0.002426 \text{ nm}$$

Worked Example 1: Compton Shift at 90°

Step 4 — Add to the incident wavelength to get the scattered wavelength $\lambda' = \lambda + \Delta\lambda$:

$$\lambda' = 0.1 \text{ nm} + 0.002426 \text{ nm} = 0.1024 \text{ nm}$$

Worked Example 2: Backscattering ($\theta = 180^\circ$)

Problem

Find the maximum possible Compton wavelength shift, and the corresponding fractional energy loss of a 50 keV photon scattered straight back.

Given: $E = 50$ keV (incident photon energy), $\theta = 180^\circ$. **Find:** $\Delta\lambda_{\max}$ and $\Delta\lambda/\lambda$.

Step 1 — Evaluate $\cos 180^\circ = -1$, so $1 - \cos \theta = 2$ (the maximum possible value, hence "maximum shift"):

$$\Delta\lambda_{\max} = \frac{h}{m_e c} (1 - \cos 180^\circ) = (2.426 \times 10^{-12} \text{ m})(2) = 4.85 \times 10^{-12} \text{ m}$$

Step 2 — Find the incident wavelength λ from the photon energy $E = hc/\lambda$:

$$\lambda = \frac{hc}{E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m/s})}{(50 \times 10^3 \text{ eV})(1.6 \times 10^{-19} \text{ J/eV})} = \frac{1.988 \times 10^{-25}}{8.0 \times 10^{-15}} \approx 2.48 \times 10^{-11} \text{ m}$$

Worked Example 2: Backscattering ($\theta = 180^\circ$)

Step 3 — Find the scattered wavelength $\lambda' = \lambda + \Delta\lambda_{\max}$:

$$\lambda' = 2.48 \times 10^{-11} \text{ m} + 4.85 \times 10^{-12} \text{ m} = 2.97 \times 10^{-11} \text{ m}$$

Step 4 — Compute the fractional energy loss. Since photon energy $E = hc/\lambda$, a fractional change in λ gives an equal fractional change in E . More precisely, $\Delta E/E_{\text{initial}} = (\lambda' - \lambda)/\lambda' = \Delta\lambda_{\max}/\lambda'$ (note: divide by the *scattered* wavelength λ' , not the incident λ , to get the exact result):

$$\frac{\Delta E}{E_{\text{initial}}} = \frac{\Delta\lambda_{\max}}{\lambda'} = \frac{4.85 \times 10^{-12}}{2.97 \times 10^{-11}} \approx 0.163 \text{ (about 16\% energy loss)}$$

Worked Example 3: Electron Recoil Energy

Problem

A photon of initial wavelength $\lambda = 0.024 \text{ nm}$ Compton-scatters at $\theta = 60^\circ$. Find the kinetic energy gained by the recoiling electron.

Given: $\lambda = 0.024 \text{ nm}$, $\theta = 60^\circ$. **Find:** KE_e , the electron's recoil kinetic energy.

Step 1 — Find $\Delta\lambda$ using $\cos 60^\circ = 0.5$, so $1 - \cos \theta = 0.5$:

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos 60^\circ) = (2.426 \times 10^{-12})(0.5) = 1.213 \times 10^{-12} \text{ m} = 0.001213 \text{ nm}$$

Step 2 — Find the scattered wavelength $\lambda' = \lambda + \Delta\lambda$:

$$\lambda' = 0.024 \text{ nm} + 0.001213 \text{ nm} = 0.025213 \text{ nm}$$

Worked Example 3: Electron Recoil Energy

Step 3 — Apply energy conservation. The electron's kinetic energy equals the energy lost by the photon, $KE_e = h\nu - h\nu' = hc(1/\lambda - 1/\lambda')$:

$$\frac{1}{\lambda} = \frac{1}{0.024 \times 10^{-9}} = 4.167 \times 10^{10} \text{ m}^{-1}, \quad \frac{1}{\lambda'} = \frac{1}{0.025213 \times 10^{-9}} = 3.966 \times 10^{10} \text{ m}^{-1}$$

Step 4 — Subtract and multiply by hc :

$$KE_e = (6.626 \times 10^{-34})(3 \times 10^8) (4.167 \times 10^{10} - 3.966 \times 10^{10}) \approx 3.98 \times 10^{-16} \text{ J}$$

Step 5 — Convert to keV (divide by $1.6 \times 10^{-16} \text{ J/keV}$):

$$KE_e \approx 2.49 \text{ keV}$$

Worked Example 4: Finding the Scattering Angle

Problem

A photon of wavelength $\lambda = 0.05 \text{ nm}$ undergoes Compton scattering and emerges with wavelength $\lambda' = 0.0524 \text{ nm}$. Find the scattering angle θ .

Given: λ, λ' . **Find:** θ (this is the *inverse* of the previous examples: solving for the angle, not the shift).

Step 1 — Find the observed shift $\Delta\lambda = \lambda' - \lambda$:

$$\Delta\lambda = 0.0524 \text{ nm} - 0.05 \text{ nm} = 0.0024 \text{ nm} = 2.4 \times 10^{-12} \text{ m}$$

Step 2 — Start from $\Delta\lambda = \frac{h}{m_e c}(1 - \cos \theta)$ and solve for $(1 - \cos \theta)$ by multiplying both sides by $m_e c/h$:

$$1 - \cos \theta = \frac{\Delta\lambda m_e c}{h} = \frac{\Delta\lambda}{h/m_e c} = \frac{2.4 \times 10^{-12}}{2.426 \times 10^{-12}} \approx 0.989$$

Worked Example 4: Finding the Scattering Angle

Step 3 — Solve for $\cos \theta$ by subtracting from 1:

$$\cos \theta = 1 - 0.989 = 0.011$$

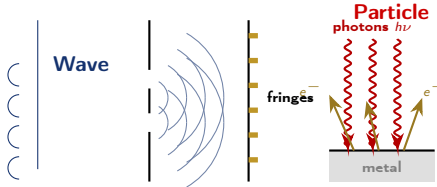
Step 4 — Take the inverse cosine:

$$\theta = \cos^{-1}(0.011) \approx 89.4^\circ$$

This inverse-problem technique (inferring θ from a measured wavelength shift) is exactly how Compton-scattering detectors in particle and astrophysics instruments reconstruct photon trajectories.

Wave-Particle Duality

- Light exhibits **both** wave-like behavior (interference, diffraction) and particle-like behavior (photoelectric effect, Compton effect).
- Neither pure picture alone is sufficient.
- This dual nature is a feature of quantum reality: light behaves as the situation demands.



Key Question

If light can behave like particles, can particles behave like waves?

Bohr's Principle of Complementarity

- Niels Bohr formalized wave–particle duality into the **principle of complementarity**: the wave and particle descriptions are both necessary, but mutually exclusive in any single experiment.
- Which behavior is observed depends entirely on **what is measured**: an experiment designed to detect interference (a wave property) will never simultaneously reveal which-path (particle) information, and vice versa.
- This is not a limitation of our instruments — it reflects a fundamental feature of quantum systems, later sharpened quantitatively in the double-slit visibility analysis (Unit 2).
- Practical consequence for this course: depending on the experiment, we will sometimes describe a particle using a wavefunction $\psi(x)$ (wave language) and sometimes using definite momentum/position values (particle language) — both are facets of the same underlying quantum state.

De Broglie Hypothesis (1924)

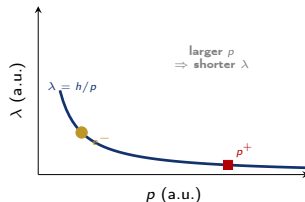
Matter Waves

Every particle of momentum p has an associated wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

called the **de Broglie wavelength**.

- Extends wave-particle duality to **all** matter, not just light.
- Predicted before any experimental confirmation — a triumph of theoretical physics.



Derivation: De Broglie's Hypothesis — Reasoning by Analogy

- De Broglie reasoned **by analogy** with the photon. For a photon, relativity (Einstein's relation $E = mc^2$ generalized to massless particles) gives the energy-momentum relation $E = pc$, where E is energy, p is momentum, c is the speed of light.
- Combined with the Planck-Einstein relation $E = h\nu$ (where h is Planck's constant and ν is frequency), this gives the photon's momentum:

$$p = \frac{E}{c} = \frac{h\nu}{c}$$

- Using the wave relation $c = \nu\lambda$ (so $\nu/c = 1/\lambda$), where λ is wavelength:

$$p = \frac{h\nu}{c} = h \cdot \frac{\nu}{c} = \frac{h}{\lambda}$$

- De Broglie's bold step: postulate that this same relation between momentum and wavelength, $p = h/\lambda$, holds not just for photons but for *any* particle, massive or massless.

Derivation: De Broglie's Hypothesis — Applying to Matter

- **Step 1:** starting from $p = h/\lambda$, solve for λ by inverting (multiply both sides by λ/p):

$$\lambda = \frac{h}{p}$$

- **Step 2:** for a non-relativistic particle of mass m moving at speed v , momentum is simply $p = mv$ (the familiar classical definition). Substitute:

$$\lambda = \frac{h}{mv}$$

- **Step 3:** consider a charged particle (charge q) accelerated from rest through a potential difference V . The work done by the electric field equals the kinetic energy gained:

$$qV = \frac{1}{2}mv^2$$

Derivation: De Broglie's Hypothesis — Applying to Matter

- **Step 4:** solve for v by multiplying both sides by $2/m$ and taking the square root:

$$v^2 = \frac{2qV}{m} \quad \Rightarrow \quad v = \sqrt{\frac{2qV}{m}}$$

Derivation: De Broglie's Hypothesis — Final Formula

- **Step 5:** express the momentum $p = mv$ in terms of V by substituting the result for v :

$$p = mv = m\sqrt{\frac{2qV}{m}} = \sqrt{m^2 \cdot \frac{2qV}{m}} = \sqrt{2mqV}$$

- **Step 6:** substitute this expression for p into $\lambda = h/p$:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

- This is the de Broglie wavelength of a particle of mass m and charge q accelerated through potential V — the form most directly applicable to electron-diffraction experiments (such as Davisson–Germer, next) where the accelerating voltage V is the controlled experimental parameter.

Historical Note: A Bold, Untested Proposal

- De Broglie put forward the matter-wave hypothesis in his 1924 doctoral thesis, with **no** direct experimental evidence to support it — a remarkably bold theoretical leap.
- His thesis examiners were skeptical; Einstein, who reviewed the work, recognized its significance and championed it, helping it gain acceptance within the physics community.
- De Broglie's reasoning combined Einstein's photon relation ($p = h/\lambda$) with special relativity and the idea that nature should be symmetric between light and matter — if light (traditionally a wave) has particle properties, perhaps matter (traditionally particles) has wave properties.
- The hypothesis was confirmed just three years later by the Davisson–Germer experiment (next slides), and de Broglie received the 1929 Nobel Prize in Physics — one of very few prizes awarded for a doctoral thesis.
- This idea underlies the entire wave-mechanical formulation of quantum theory that Schrödinger develops in Units 3–4.

Worked Example 1: De Broglie Wavelength of an Electron

Problem

Find the de Broglie wavelength of an electron moving at $v = 10^6$ m/s. ($m_e = 9.11 \times 10^{-31}$ kg)

Given: v (speed), m_e (electron mass). **Find:** λ , the de Broglie wavelength.

Step 1 — Apply $\lambda = h/(m_e v)$, where h is Planck's constant:

$$\lambda = \frac{h}{m_e v} = \frac{6.626 \times 10^{-34} \text{ J s}}{(9.11 \times 10^{-31} \text{ kg})(10^6 \text{ m/s})}$$

Step 2 — Evaluate the denominator:

$$m_e v = (9.11 \times 10^{-31})(10^6) = 9.11 \times 10^{-25} \text{ kg m/s}$$

Step 3 — Divide:

$$\lambda = \frac{6.626 \times 10^{-34}}{9.11 \times 10^{-25}} = 7.27 \times 10^{-10} \text{ m} = 0.727 \text{ nm}$$

Comparable to atomic spacings — this is why electrons diffract from crystals.

Worked Example 2: Electron Accelerated Through a Potential

Problem

An electron is accelerated from rest through a potential difference of 150 V. Find its de Broglie wavelength.

Given: $V = 150 \text{ V}$, $q = e = 1.6 \times 10^{-19} \text{ C}$ (electron charge magnitude), m_e . **Find:** λ .

Step 1 — Apply $\lambda = h/\sqrt{2m_e eV}$ (derived two slides ago):

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.6 \times 10^{-19})(150)}}$$

Step 2 — Evaluate the product under the square root, one factor at a time:

$$2(9.11 \times 10^{-31}) = 1.822 \times 10^{-30}, \quad (1.822 \times 10^{-30})(1.6 \times 10^{-19}) = 2.915 \times 10^{-49}$$

$$(2.915 \times 10^{-49})(150) = 4.37 \times 10^{-47}$$

Worked Example 2: Electron Accelerated Through a Potential

Step 3 — Take the square root:

$$\sqrt{4.37 \times 10^{-47}} = 6.61 \times 10^{-24} \text{ kg m/s}$$

Step 4 — Divide:

$$\lambda = \frac{6.626 \times 10^{-34}}{6.61 \times 10^{-24}} \approx 1.0 \times 10^{-10} \text{ m} = 1.0 \text{ \AA}$$

This is the well-known convenient formula: $\lambda(\text{\AA}) \approx \sqrt{150/V}$, which can equivalently be written as $\lambda(\text{\AA}) = 12.27/\sqrt{V}$ since $\sqrt{150} \approx 12.27$. Both forms appear in the literature; they are the same constant rearranged.

Worked Example 3: De Broglie Wavelength of a Macroscopic Object

Problem

Find the de Broglie wavelength of a 0.15 kg baseball moving at 40 m/s. Comment on observability.

Given: $m = 0.15 \text{ kg}$, $v = 40 \text{ m/s}$. **Find:** λ .

Step 1 — Compute the momentum $p = mv$:

$$p = (0.15 \text{ kg})(40 \text{ m/s}) = 6.0 \text{ kg m/s}$$

Step 2 — Apply $\lambda = h/p$:

$$\lambda = \frac{6.626 \times 10^{-34} \text{ J s}}{6.0 \text{ kg m/s}} = 1.10 \times 10^{-34} \text{ m}$$

This is roughly 20 orders of magnitude smaller than a proton's diameter — utterly unobservable. This is why wave effects are never seen for macroscopic objects: λ is far too small to produce measurable diffraction or interference.

Worked Example 4: Thermal de Broglie Wavelength

Problem

Estimate the de Broglie wavelength of a neutron in thermal equilibrium at room temperature ($T = 300\text{ K}$), using $\frac{1}{2}mv^2 = \frac{3}{2}k_B T$. ($m_n = 1.675 \times 10^{-27}\text{ kg}$)

Given: $T = 300\text{ K}$, m_n (neutron mass), k_B (Boltzmann's constant). **Find:** λ .

Step 1 — Solve $\frac{1}{2}mv^2 = \frac{3}{2}k_B T$ **for momentum.** Multiply both sides by $2m$:

$$m^2 v^2 = 3mk_B T \implies p^2 = 3mk_B T \implies p = \sqrt{3mk_B T}$$

(using $p = mv$, so $p^2 = m^2 v^2$).

Step 2 — Substitute numbers, evaluating the product inside the square root piece by piece:

$$3m_n k_B T = 3(1.675 \times 10^{-27})(1.381 \times 10^{-23})(300) \approx 2.083 \times 10^{-47}$$

Worked Example 4: Thermal de Broglie Wavelength

Step 3 — Take the square root:

$$p = \sqrt{2.083 \times 10^{-47}} \approx 4.56 \times 10^{-24} \text{ kg m/s}$$

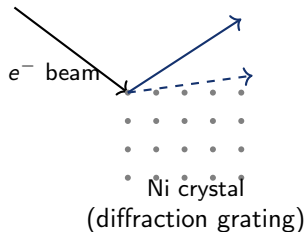
Step 4 — Apply $\lambda = h/p$:

$$\lambda = \frac{6.626 \times 10^{-34}}{4.56 \times 10^{-24}} \approx 1.45 \times 10^{-10} \text{ m} = 1.45 \text{ \AA}$$

This is comparable to interatomic spacings in crystals — exactly why **thermal neutron diffraction** (alongside electron and X-ray diffraction) is a standard technique for probing crystal structures.

Davisson–Germer Experiment (1927)

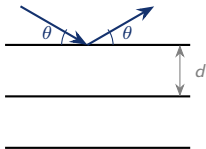
- Electrons accelerated and directed at a **nickel crystal**.
- Intensity of scattered electrons measured as a function of angle.
- Sharp intensity maxima observed — a **diffraction pattern**, exactly as expected for waves scattering off a crystal lattice (Bragg condition).
- Wavelength inferred from the diffraction pattern matched the de Broglie prediction $\lambda = h/p$ precisely.



Significance: First direct experimental confirmation of the wave nature of matter — a cornerstone result for quantum mechanics.

Derivation: Bragg Condition — Setting Up

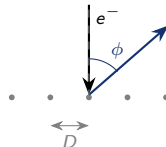
- Electrons diffract from the periodic atomic structure of the nickel crystal, just as X-rays diffract from crystal planes — the same geometric reasoning W.L. Bragg used for X-ray crystallography.



Bulk Bragg reflection

$$n\lambda = 2d \sin \theta$$

d = plane spacing, θ = glancing angle *from the planes*



Davisson–Germer (surface)

$$D \sin \phi = n\lambda$$

D = surface row spacing, ϕ = angle *from the incident beam*

Why no factor of 2? The electrons here scatter off the outermost *surface* layer of atoms only, not bulk planes deep inside the crystal — the same Bragg-type reasoning, just applied to a surface diffraction grating.

Derivation: Bragg Condition — Setting Up

Goal: compute the de Broglie wavelength predicted from the known accelerating voltage, then independently compute the wavelength implied by the observed diffraction angle, and check that the two agree.

Derivation: Bragg Condition — The Predicted Wavelength

- In the original experiment, electrons were accelerated through $V = 54\text{ V}$ before striking a nickel crystal whose surface atomic rows have spacing $D = 2.15\text{ Å}$.
- **Step 1:** apply the de Broglie formula derived earlier, $\lambda_{\text{dB}} = h/\sqrt{2m_e eV}$, where m_e is the electron mass and e is the elementary charge.
- **Step 2:** use the convenient numerical shortcut derived in Worked Example 2 of the previous subsection, $\lambda(\text{Å}) \approx \sqrt{150/V}$ — but here we use the equivalent form $\lambda(\text{Å}) = 12.27/\sqrt{V}$ (these are algebraically the same constant, just rearranged: $12.27 = \sqrt{150}$):

$$\lambda_{\text{dB}} = \frac{12.27}{\sqrt{54}}\text{ Å}$$

- **Step 3:** evaluate $\sqrt{54} \approx 7.35$, then divide:

$$\lambda_{\text{dB}} = \frac{12.27}{7.35} \approx 1.67\text{ Å}$$

Derivation: Bragg Condition — The Observed Wavelength

- The experiment found a diffraction maximum at $\phi = 50^\circ$ (measured from the incident beam), with the relevant nickel surface atomic-row spacing $D = 2.15 \text{ \AA}$ ($= 0.215 \text{ nm}$). Apply the surface diffraction condition with $n = 1$ (the first-order, strongest diffraction peak), solving for λ :

$$\lambda_{\text{exp}} = D \sin \phi$$

- Step 1:** evaluate $\sin 50^\circ \approx 0.766$.
- Step 2:** substitute $D = 2.15 \text{ \AA}$ and multiply:

$$\lambda_{\text{exp}} = (2.15 \text{ \AA})(0.766) \approx 1.65 \text{ \AA}$$

Bragg Condition — Comparing Results

Predicted vs. Observed Wavelength

$$\lambda_{\text{dB}} \approx 1.67 \text{ \AA} \quad \text{vs.} \quad \lambda_{\text{exp}} \approx 1.65 \text{ \AA}$$

- The two values, obtained from completely independent calculations (one from accelerating voltage via de Broglie's hypothesis, one from a measured diffraction angle via the surface diffraction condition $D \sin \phi = n\lambda$), agree to within about 1%.
- This is precisely de Broglie's relation $\lambda = h/p$ tested *quantitatively*, not just qualitatively — the experiment doesn't merely show "electrons diffract," it confirms the specific numerical wavelength predicted by quantum theory.
- Excellent agreement between λ_{dB} and λ_{exp} — direct, quantitative confirmation of de Broglie's hypothesis.

Significance: An Accidental Discovery

- Davisson and Germer were not initially looking for evidence of matter waves at all — they were studying electron scattering off nickel, and a laboratory accident (an air leak that oxidized their nickel sample) forced them to anneal the metal at high temperature to purify it.
- This annealing inadvertently transformed the nickel into a few large single crystals, which by chance produced the sharp, angle-dependent diffraction peaks needed to reveal the wave nature of the electrons.
- The same wave nature of electrons confirmed here is the working principle behind **electron microscopes**, which use the much shorter de Broglie wavelength of fast electrons (compared to visible light) to image structures far smaller than optical microscopes can resolve.
- Together, the photoelectric/Compton effects (light behaving as particles) and the Davisson–Germer experiment (matter behaving as waves) complete the experimental case for wave–particle duality as a universal feature of nature, not a special property of light.

Why a Single Wave Cannot Represent a Particle

- De Broglie's relation $\lambda = h/p$ associates a wavelength with a particle, suggesting we represent it by a wave $\psi(x, t) = Ae^{i(kx - \omega t)}$.
- But this plane wave has constant $|\psi|^2 = |A|^2$ for **all** x — the particle would be equally likely to be found anywhere in the universe, which is physically absurd for, say, an electron in a beam of finite size.
- We need a mathematical object that is (a) wave-like, so it carries a well-defined wavelength/momentum, but (b) localized in space, so it can represent where a particle actually is.
- The resolution: superpose *many* plane waves of slightly different k . Individual waves interfere constructively in one small region and destructively elsewhere, producing a localized "packet" while retaining wave character within it.
- This single idea — building localized particles out of superposed waves — is the conceptual bridge connecting de Broglie's hypothesis to the full machinery of wavefunctions developed starting in Unit 2.

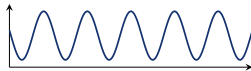
Concept of a Wave Packet

- A single plane wave $\psi = Ae^{i(kx-\omega t)}$ is delocalized over all space — unphysical for a localized particle.
- A **wave packet** superposing many plane waves of slightly different k :

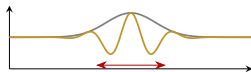
$$\psi(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx-\omega t)} dk$$

- Localized in space; particle has reasonably well-defined position *and* momentum.
- Spread in k inversely related to Δx — the seed of the uncertainty principle.

Plane wave — delocalized

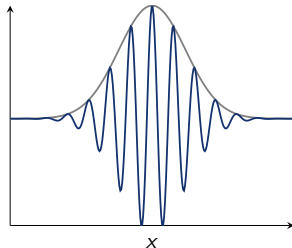


Wave packet — localized



Visualizing a Wave Packet

- A common, mathematically convenient choice for the amplitude function $A(k)$ is a **Gaussian** centered at some central wavenumber k_0 .
- The resulting $\psi(x, t)$ looks like a rapidly oscillating wave (wavelength $2\pi/k_0$) modulated by a smooth bell-shaped envelope.
- The envelope's center moves at the group velocity v_{group} ; the oscillations within it move at the (generally different) phase velocity v_{phase} .
- Outside the envelope, $|\psi|^2 \rightarrow 0$ — the particle is effectively confined to the packet's width, consistent with a localized particle.



Principle of Superposition

Superposition Principle

If ψ_1 and ψ_2 are valid quantum states of a system, then any linear combination

$$\psi = c_1\psi_1 + c_2\psi_2$$

is also a valid state of the system.

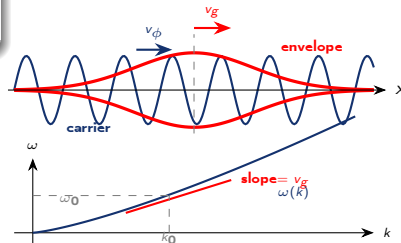
- Fundamental to quantum mechanics — has no classical analogue for particles.
- Underlies interference phenomena (e.g. double-slit experiment, covered in Unit 2).
- Mathematically, the space of quantum states is a **linear vector space** (developed fully in Unit 5).

Phase Velocity and Group Velocity

Definitions

$$v_{\text{phase}} = \frac{\omega}{k}, \quad v_{\text{group}} = \frac{d\omega}{dk}$$

- **Phase velocity**: speed of a single Fourier crest.
- **Group velocity**: speed of the envelope — this is how the particle (energy/information) actually travels.
- For a free particle: $v_g = 2v_\phi$ (non-relativistic), so the particle outruns individual wave crests.



Derivation: Two-Wave Beat Pattern — Setup

- To see how a localized "packet" can emerge from superposed waves, consider the simplest case: just **two** waves with slightly different wavenumber and frequency:

$$\psi = \cos(k_1 x - \omega_1 t) + \cos(k_2 x - \omega_2 t)$$

where k_1, k_2 are two nearby wavenumbers and ω_1, ω_2 the corresponding angular frequencies (related to k_1, k_2 via the dispersion relation $\omega(k)$ of the system).

- Define the **mean** and **difference** of the two wavenumbers and frequencies:

$$\bar{k} = \frac{k_1 + k_2}{2}, \quad \Delta k = k_1 - k_2, \quad \bar{\omega} = \frac{\omega_1 + \omega_2}{2}, \quad \Delta\omega = \omega_1 - \omega_2$$

so that $k_1 = \bar{k} + \Delta k/2$, $k_2 = \bar{k} - \Delta k/2$ (and similarly for ω_1, ω_2).

Derivation: Two-Wave Beat Pattern — Applying the Sum-to-Product Identity

- Recall the trigonometric sum-to-product identity:

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

- Identify $A = k_1x - \omega_1t$ and $B = k_2x - \omega_2t$. Then:

$$\frac{A+B}{2} = \frac{(k_1+k_2)x - (\omega_1+\omega_2)t}{2} = \bar{k}x - \bar{\omega}t$$

$$\frac{A-B}{2} = \frac{(k_1-k_2)x - (\omega_1-\omega_2)t}{2} = \frac{\Delta k}{2}x - \frac{\Delta \omega}{2}t$$

- Substituting into the sum-to-product identity:

$$\psi = 2 \cos\left(\frac{\Delta k}{2}x - \frac{\Delta \omega}{2}t\right) \cos(\bar{k}x - \bar{\omega}t)$$

Derivation: Two-Wave Beat Pattern — Applying the Sum-to-Product Identity

- This factors ψ into a product of two cosines: a rapidly oscillating "carrier" wave (since $\bar{k}, \bar{\omega}$ are the average, typically large, values) multiplied by a slowly varying "envelope" (since $\Delta k, \Delta \omega$ are small differences).

Derivation: Two-Wave Beat Pattern — Identifying the Two Velocities

- The second factor, $\cos(\bar{k}x - \bar{\omega}t)$, is a wave of wavenumber $\bar{k}$ and frequency $\bar{\omega}$ — it moves at the **phase velocity**:

$$v_{\text{phase}} = \frac{\bar{\omega}}{\bar{k}}$$

- The first factor, $2 \cos(\frac{\Delta k}{2}x - \frac{\Delta \omega}{2}t)$, is itself a wave-like function, but with much smaller wavenumber $\Delta k/2$ and frequency $\Delta \omega/2$ — it represents the slowly-varying **envelope** (the overall "packet" shape). Its speed is similarly (wave speed = frequency/wavenumber):

$$v_{\text{env}} = \frac{\Delta \omega/2}{\Delta k/2} = \frac{\Delta \omega}{\Delta k}$$

- As the two original wavenumbers k_1, k_2 become arbitrarily close ($\Delta k \rightarrow 0$), the ratio $\Delta \omega/\Delta k$ becomes, by definition, the derivative $d\omega/dk$:

$$v_{\text{env}} \xrightarrow{\Delta k \rightarrow 0} \frac{d\omega}{dk} \equiv v_{\text{group}}$$

- This envelope *is* the localized "packet" — so the particle (the packet as a whole) moves at v_{group} , not v_{phase} .

Derivation: Group Velocity Equals Particle Velocity — Setup

- For a non-relativistic free particle, the Planck–Einstein and de Broglie relations give $E = \hbar\omega$ and $p = \hbar k$, where $\hbar = h/2\pi$ is the reduced Planck's constant.
- The classical kinetic-energy relation $E = p^2/2m$ (with m the particle mass) becomes, after substituting $E = \hbar\omega$ and $p = \hbar k$:

$$\hbar\omega = \frac{(\hbar k)^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

- **Solve for ω** by dividing both sides by $\hbar$:

$$\omega(k) = \frac{\hbar k^2}{2m}$$

This is the **dispersion relation** for a free quantum particle — how ω depends on k .

Derivation: Group Velocity Equals Particle Velocity — Differentiating

- **Group velocity:** differentiate $\omega(k) = \hbar k^2/(2m)$ with respect to k , using the power rule ($d(k^2)/dk = 2k$):

$$v_{\text{group}} = \frac{d\omega}{dk} = \frac{\hbar \cdot 2k}{2m} = \frac{\hbar k}{m}$$

- Recognize $\hbar k = p$ (the de Broglie relation), so:

$$v_{\text{group}} = \frac{p}{m} = v_{\text{particle}}$$

exactly the classical particle velocity, $p = mv \Rightarrow v = p/m$.

- **Phase velocity:** divide $\omega(k) = \hbar k^2/(2m)$ by k :

$$v_{\text{phase}} = \frac{\omega}{k} = \frac{\hbar k^2/2m}{k} = \frac{\hbar k}{2m} = \frac{v_{\text{particle}}}{2}$$

- So $v_{\text{phase}} \neq v_{\text{particle}}$ in general — only v_{group} corresponds to the physically observable particle velocity.

Derivation: Group Velocity Equals Particle Velocity — Differentiating

- **General relation** (derived from $v_g = d\omega/dk = d(kv_p)/dk = v_p + k dv_p/dk$ together with $k = 2\pi/\lambda$, so $dk = -2\pi/\lambda^2 d\lambda$):

$$v_{\text{group}} = v_{\text{phase}} - \lambda \frac{dv_{\text{phase}}}{d\lambda}$$

Worked Example 1: Group vs. Phase Velocity

Problem

An electron has kinetic energy 100 eV. Find its de Broglie wavelength, its (particle/group) velocity, and the phase velocity of its matter wave.

Given: $KE = 100 \text{ eV}$, m_e (electron mass). **Find:** λ , v_{group} , v_{phase} .

Step 1 — Solve $KE = \frac{1}{2}m_e v^2$ **for** v **by** multiplying both sides by $2/m_e$ and taking the square root:

$$v = \sqrt{\frac{2KE}{m_e}}$$

Step 2 — Convert KE **to joules and substitute:**

$$v = \sqrt{\frac{2(100)(1.6 \times 10^{-19} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} = \sqrt{\frac{3.2 \times 10^{-17}}{9.11 \times 10^{-31}}} = \sqrt{3.51 \times 10^{13}}$$

Worked Example 1: Group vs. Phase Velocity

Step 3 — Take the square root:

$$v \approx 5.93 \times 10^6 \text{ m/s} = v_{\text{group}}$$

(this particle velocity *is* the group velocity, as derived two slides ago).

Step 4 — De Broglie wavelength $\lambda = h/(m_e v)$:

$$\lambda = \frac{6.626 \times 10^{-34}}{(9.11 \times 10^{-31})(5.93 \times 10^6)} \approx 1.23 \times 10^{-10} \text{ m}$$

Step 5 — Phase velocity, using $v_{\text{phase}} = v_{\text{group}}/2$ (derived for a free non-relativistic particle):

$$v_{\text{phase}} = \frac{5.93 \times 10^6}{2} \approx 2.97 \times 10^6 \text{ m/s}$$

Worked Example 2: Dispersion Relation

Problem

A wave packet has dispersion relation $\omega(k) = ck - bk^3$ (b, c constants). Find the group velocity as a function of k .

Given: $\omega(k) = ck - bk^3$. **Find:** $v_{\text{group}}(k)$.

Step 1 — Apply $v_{\text{group}} = d\omega/dk$, differentiating term by term using the power rule ($d(k^n)/dk = nk^{n-1}$):

$$\frac{d}{dk}(ck) = c, \quad \frac{d}{dk}(bk^3) = 3bk^2$$

Step 2 — Combine:

$$v_{\text{group}} = \frac{d\omega}{dk} = c - 3bk^2$$

Step 3 — Compare to the phase velocity, $v_{\text{phase}} = \omega/k = (ck - bk^3)/k = c - bk^2$ (dividing each term by k).

Worked Example 2: Dispersion Relation

Note the group velocity decreases with k here (anomalous dispersion for large k) — illustrates that v_g need not equal v_p even functionally; here $v_g = c - 3bk^2$ decreases three times as fast with k^2 as $v_p = c - bk^2$.

Worked Example 3: Spreading of a Wave Packet (Estimate)

Problem

A free-electron wave packet has initial width $\Delta x_0 = 1 \text{ nm}$. Estimate the time for the packet to double in width, using $\Delta v \sim \hbar/(m \Delta x_0)$ and $\Delta x(t) \sim \Delta x_0 + \Delta v t$.

Given: $\Delta x_0 = 1 \text{ nm}$, m_e . **Find:** t , the doubling time.

Step 1 — Estimate the velocity spread using the given uncertainty-principle estimate

$\Delta v \sim \hbar/(m_e \Delta x_0)$:

$$\Delta v \sim \frac{1.055 \times 10^{-34} \text{ J s}}{(9.11 \times 10^{-31} \text{ kg})(10^{-9} \text{ m})} = \frac{1.055 \times 10^{-34}}{9.11 \times 10^{-40}}$$

Step 2 — Evaluate:

$$\Delta v \approx 1.16 \times 10^5 \text{ m/s}$$

Worked Example 3: Spreading of a Wave Packet (Estimate)

Step 3 — Set up the doubling condition. Doubling means $\Delta x(t) = 2\Delta x_0$, so using $\Delta x(t) \sim \Delta x_0 + \Delta v t$:

$$2\Delta x_0 \sim \Delta x_0 + \Delta v t \quad \Rightarrow \quad \Delta v t \sim \Delta x_0 \quad \Rightarrow \quad t \sim \frac{\Delta x_0}{\Delta v}$$

(the factor of 2 cancels against the original Δx_0 already present — the packet only needs to spread by *one more* Δx_0 to double.) **Step 4 — Substitute and evaluate:**

$$t \sim \frac{10^{-9} \text{ m}}{1.16 \times 10^5 \text{ m/s}} \approx 8.6 \times 10^{-15} \text{ s}$$

Even a tightly localized electron packet spreads extremely fast — a vivid illustration of the uncertainty principle in dynamics.

Worked Example 4: Group Velocity for a Relativistic Photon Gas

Problem

For electromagnetic waves in vacuum, $\omega = ck$ exactly. Find both v_{phase} and v_{group} , and comment on what this implies about photon wave packets.

Given: $\omega(k) = ck$ (c is the speed of light, a constant). **Find:** v_{phase} and v_{group} .

Step 1 — Phase velocity $v_{\text{phase}} = \omega/k$: divide both sides of $\omega = ck$ by k :

$$v_{\text{phase}} = \frac{\omega}{k} = \frac{ck}{k} = c$$

Step 2 — Group velocity $v_{\text{group}} = d\omega/dk$: differentiate $\omega = ck$ with respect to k (since c is constant, this is just the coefficient):

$$v_{\text{group}} = \frac{d\omega}{dk} = c$$

Both velocities equal c exactly — a special, non-dispersive case. This is why a pulse of light in vacuum (a superposition of many frequencies) travels at exactly c without changing shape, unlike a

Worked Example 4: Group Velocity for a Relativistic Photon Gas

massive-particle wave packet (Worked Example 3), which inevitably spreads because $\omega(k)$ for massive particles is **not** linear in k .

Connecting Forward: From Wave Packets to Wavefunctions

- The wave packet construction in this subsection is, in essence, a first working version of the **wavefunction** $\Psi(x, t)$ that will be formally introduced as a postulate of quantum mechanics in Unit 2.
- Superposition (here applied to plane waves of different k) will be promoted to a fundamental axiom: *any* linear combination of valid quantum states is itself a valid quantum state.
- The localization property of wave packets foreshadows the Born probability interpretation: $|\Psi(x, t)|^2$ tells us where the particle is likely to be found.
- The unavoidable trade-off between localizing x (narrow packet) and localizing k , hence p (narrow $A(k)$) is exactly the content of the Heisenberg uncertainty principle, derived rigorously in Units 2 and 5.

Derivation: Stefan–Boltzmann Law from Planck's Law

- **Goal:** integrate Planck's spectral energy density over all frequencies to obtain the total energy density $u(T)$, and hence the emitted power per unit area $j = (\sigma/\pi) \cdot u(T)$ — the Stefan–Boltzmann law.

- **Step 1 — Set up the integral.** Using $u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$:

$$u(T) = \int_0^\infty u(\nu, T) d\nu = \frac{8\pi h}{c^3} \int_0^\infty \frac{\nu^3}{e^{h\nu/k_B T} - 1} d\nu$$

- **Step 2 — Substitute** $x = h\nu/k_B T$, so $\nu = xk_B T/h$ and $d\nu = (k_B T/h) dx$:

$$u(T) = \frac{8\pi h}{c^3} \left(\frac{k_B T}{h} \right)^4 \int_0^\infty \frac{x^3}{e^x - 1} dx$$

- **Step 3 — Evaluate the definite integral.** This is a standard result from Riemann zeta/Gamma functions:

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \Gamma(4) \zeta(4) = 6 \times \frac{\pi^4}{90} = \frac{\pi^4}{15}$$

Derivation: Stefan–Boltzmann Law from Planck's Law

- **Step 4 — Assemble.** Substituting back:

$$u(T) = \frac{8\pi h}{c^3} \cdot \frac{k_B^4 T^4}{h^4} \cdot \frac{\pi^4}{15} = \frac{8\pi^5 k_B^4}{15 c^3 h^3} T^4 \equiv a T^4$$

- **Step 5 — Emitted flux.** For an isotropic radiator, the power emitted per unit area is $j = c u(T)/4$:

$$j = \frac{c}{4} \cdot \frac{8\pi^5 k_B^4}{15 c^3 h^3} T^4 = \underbrace{\frac{2\pi^5 k_B^4}{15 c^2 h^3}}_{\sigma} T^4 \approx 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

- This is the **Stefan–Boltzmann law** $j = \sigma T^4$, with σ expressed entirely in terms of fundamental constants.

Problem Practice: Stefan–Boltzmann Law

Stefan–Boltzmann Law (derived from integrating Planck's law over all frequencies)

$$u(T) = \int_0^{\infty} u(\nu, T) d\nu = aT^4, \quad \text{or, for emitted power per unit area:} \quad j = \sigma T^4$$

where j is the radiated power per unit area, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$ is the Stefan–Boltzmann constant, and T is absolute temperature.

Example

A blackbody radiates at $T_1 = 500 \text{ K}$ with total power P_1 . If the temperature is doubled to $T_2 = 1000 \text{ K}$, find the ratio P_2/P_1 .

Step 1 — Write total power for each temperature using $P = \sigma AT^4$ (where A is the radiating surface area, the same for both cases):

$$P_1 = \sigma AT_1^4, \quad P_2 = \sigma AT_2^4$$

Problem Practice: Stefan–Boltzmann Law

Step 2 — Form the ratio, noting σA cancels:

$$\frac{P_2}{P_1} = \frac{\sigma A T_2^4}{\sigma A T_1^4} = \left(\frac{T_2}{T_1} \right)^4$$

Step 3 — Substitute $T_2/T_1 = 1000/500 = 2$:

$$\frac{P_2}{P_1} = 2^4 = 16$$

Unit 1 Summary

- **Planck's law**: quantized energy exchange resolves the ultraviolet catastrophe.
- **Photoelectric effect**: light quanta (photons), $E = h\nu$; Einstein's equation $h\nu = \phi + KE_{\max}$.
- **Compton effect**: photon momentum $p = h/\lambda$; wavelength shift $\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$.
- **De Broglie hypothesis**: matter waves, $\lambda = h/p$; confirmed by Davisson–Germer.
- **Wave packets**, superposition, phase & group velocity — mathematical groundwork for quantum states.

Additional Problem 1: Number of Photons

Problem

A 100 W sodium lamp emits light of wavelength 589 nm. Find the number of photons emitted per second (assume all power converts to light).

Given: Power $P = 100 \text{ W}$, $\lambda = 589 \text{ nm}$. **Find:** N , photons per second.

Step 1 — Find the energy of one photon using $E_{\text{photon}} = hc/\lambda$:

$$E_{\text{photon}} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m/s})}{589 \times 10^{-9} \text{ m}} = \frac{1.988 \times 10^{-25}}{5.89 \times 10^{-7}} = 3.37 \times 10^{-19} \text{ J}$$

Step 2 — Relate power to photon rate. Power is energy per unit time, so if N photons are emitted per second, $P = N \times E_{\text{photon}}$. Solve for N :

$$N = \frac{P}{E_{\text{photon}}} = \frac{100 \text{ J/s}}{3.37 \times 10^{-19} \text{ J}}$$

Additional Problem 1: Number of Photons

Step 3 — Evaluate:

$$N \approx 2.97 \times 10^{20} \text{ s}^{-1}$$

Additional Problem 2: Energy of a Radio Photon

Problem

Find the energy (in eV) of a photon from an FM radio station broadcasting at 100 MHz.

Given: $\nu = 100 \text{ MHz} = 10^8 \text{ Hz}$. **Find:** E in eV.

Step 1 — Apply $E = h\nu$:

$$E = (6.626 \times 10^{-34} \text{ J s})(10^8 \text{ Hz}) = 6.626 \times 10^{-26} \text{ J}$$

Step 2 — Convert to eV (divide by $e = 1.6 \times 10^{-19} \text{ J/eV}$):

$$E = \frac{6.626 \times 10^{-26}}{1.6 \times 10^{-19}} \approx 4.14 \times 10^{-7} \text{ eV}$$

This is roughly a billion times smaller than typical optical photon energies (a few eV) — radio photons individually carry negligible energy, which is why radio waves are always described and detected as classical waves, never as individual photon "clicks."

Additional Problem 3: Solar Constant and Sun's Power

Problem

The solar constant at Earth is 1361 W/m^2 , and the Earth–Sun distance is $1.5 \times 10^{11} \text{ m}$. Estimate the Sun's total radiated power.

Given: intensity $I = 1361 \text{ W/m}^2$ at distance $r = 1.5 \times 10^{11} \text{ m}$. **Find:** $P_{\odot}$, the Sun's total power output.

Step 1 — Reasoning: the Sun radiates power $P_{\odot}$ equally in all directions; by the time it reaches Earth, this power is spread over a sphere of radius r (surface area $4\pi r^2$), so $I = P_{\odot}/(4\pi r^2)$.

Step 2 — Solve for $P_{\odot}$ by multiplying both sides by $4\pi r^2$:

$$P_{\odot} = I \times 4\pi r^2$$

Step 3 — Evaluate $4\pi r^2$:

$$4\pi r^2 = 4\pi(1.5 \times 10^{11})^2 = 4\pi(2.25 \times 10^{22}) \approx 2.83 \times 10^{23} \text{ m}^2$$

Additional Problem 3: Solar Constant and Sun's Power

Step 4 — Multiply by I :

$$P_{\odot} = (1361)(2.83 \times 10^{23}) \approx 3.85 \times 10^{26} \text{ W}$$

Additional Problem 4: Photoelectric Threshold from Stopping Potential Data

Problem

For two frequencies $\nu_1 = 8 \times 10^{14}$ Hz and $\nu_2 = 12 \times 10^{14}$ Hz, stopping potentials are $V_1 = 0.8$ V and $V_2 = 2.5$ V. Find h/e and the work function.

Given: two (ν, V_0) data pairs. **Find:** h/e and ϕ .

Step 1 — Use the linear relation $V_0 = (h/e)\nu - \phi/e$ at both points and subtract to eliminate ϕ/e :

$$\frac{h}{e} = \frac{V_2 - V_1}{\nu_2 - \nu_1} = \frac{2.5 - 0.8}{(12 - 8) \times 10^{14}} = \frac{1.7}{4 \times 10^{14}}$$

Step 2 — Evaluate:

$$\frac{h}{e} = 4.25 \times 10^{-15} \text{ V s}$$

Additional Problem 4: Photoelectric Threshold from Stopping Potential Data

Step 3 — Find ϕ by substituting back into $V_1 = (h/e)\nu_1 - \phi/e$ and solving for ϕ/e :

$$\frac{\phi}{e} = \left(\frac{h}{e}\right) \nu_1 - V_1 = (4.25 \times 10^{-15})(8 \times 10^{14}) - 0.8 = 3.4 - 0.8 = 2.6$$

Step 4 — So $\phi \approx 2.6$ eV (since ϕ/e in volts numerically equals ϕ in eV).

Additional Problem 5: Compton Shift at 30°

Problem

Find $\Delta\lambda$ for X-rays Compton scattered at $\theta = 30^\circ$.

Given: $\theta = 30^\circ$. **Find:** $\Delta\lambda$.

Step 1 — Evaluate $\cos 30^\circ = 0.866$, so $1 - \cos \theta = 0.134$.

Step 2 — Apply the Compton formula $\Delta\lambda = \frac{h}{m_e c}(1 - \cos \theta)$:

$$\Delta\lambda = (2.426 \times 10^{-12} \text{ m})(1 - 0.866) = (2.426 \times 10^{-12})(0.134)$$

Step 3 — Evaluate:

$$\Delta\lambda \approx 3.25 \times 10^{-13} \text{ m}$$

Additional Problem 6: De Broglie Wavelength of a Proton

Problem

Find the de Broglie wavelength of a proton ($m_p = 1.67 \times 10^{-27}$ kg) moving at 2×10^5 m/s.

Given: m_p , $v = 2 \times 10^5$ m/s. **Find:** λ .

Step 1 — Compute the momentum $p = m_p v$:

$$p = (1.67 \times 10^{-27})(2 \times 10^5) = 3.34 \times 10^{-22} \text{ kg m/s}$$

Step 2 — Apply $\lambda = h/p$:

$$\lambda = \frac{6.626 \times 10^{-34}}{3.34 \times 10^{-22}} \approx 1.98 \times 10^{-12} \text{ m}$$

Additional Problem 7: De Broglie Wavelength via Kinetic Energy

Problem

An electron has kinetic energy 4 eV. Find its de Broglie wavelength using $\lambda = h/\sqrt{2mKE}$.

Given: $KE = 4 \text{ eV}$. **Find:** λ .

Step 1 — Convert KE to joules (multiply by $1.6 \times 10^{-19} \text{ J/eV}$):

$$KE = 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ J}$$

Step 2 — Evaluate under the square root, $2m_e KE$:

$$2m_e KE = 2(9.11 \times 10^{-31})(6.4 \times 10^{-19}) = 1.166 \times 10^{-48}$$

Step 3 — Take the square root:

$$\sqrt{1.166 \times 10^{-48}} \approx 1.080 \times 10^{-24} \text{ kg m/s}$$

Additional Problem 7: De Broglie Wavelength via Kinetic Energy

Step 4 — Divide h by this momentum:

$$\lambda = \frac{6.626 \times 10^{-34}}{1.080 \times 10^{-24}} \approx 6.13 \times 10^{-10} \text{ m}$$

Additional Problem 8: Wien's Law Applied to a Star

Problem

A star has peak emission wavelength 290 nm. Estimate its surface temperature.

Given: $\lambda_{\max} = 290 \text{ nm}$. **Find:** T .

Step 1 — Start from Wien's law $\lambda_{\max} T = b$, **and solve for** T by dividing both sides by $\lambda_{\max}$:

$$T = \frac{b}{\lambda_{\max}}$$

Step 2 — Substitute $b = 2.898 \times 10^{-3} \text{ m K}$ and $\lambda_{\max} = 290 \times 10^{-9} \text{ m}$:

$$T = \frac{2.898 \times 10^{-3}}{290 \times 10^{-9}}$$

Step 3 — Evaluate:

$$T \approx 9.99 \times 10^3 \text{ K} \approx 10,000 \text{ K}$$

This star is significantly hotter than the Sun (5800 K), and its peak emission lies in the ultraviolet — such a star would appear noticeably blue-white rather than yellow.

Additional Problem 9: Group Velocity from Energy–Momentum Relation

Problem

For a relativistic particle, $E = \sqrt{p^2 c^2 + m^2 c^4}$. Show $v_{\text{group}} = pc^2/E$, and evaluate for $p = mc$ (i.e. momentum equal to mc).

Step 1 — Recall $v_{\text{group}} = d\omega/dk = dE/dp$ (since $E = \hbar\omega$ and $p = \hbar k$, the $\hbar$ cancels in the ratio of derivatives).

Step 2 — Differentiate $E = (p^2 c^2 + m^2 c^4)^{1/2}$ with respect to p using the chain rule:

$$\frac{dE}{dp} = \frac{1}{2}(p^2 c^2 + m^2 c^4)^{-1/2} \times (2pc^2) = \frac{pc^2}{\sqrt{p^2 c^2 + m^2 c^4}} = \frac{pc^2}{E}$$

Step 3 — Evaluate at $p = mc$: first find E at this momentum:

$$E = \sqrt{(mc)^2 c^2 + m^2 c^4} = \sqrt{m^2 c^4 + m^2 c^4} = \sqrt{2m^2 c^4} = \sqrt{2} mc^2$$

Additional Problem 9: Group Velocity from Energy–Momentum Relation

Step 4 — Substitute into $v_g = pc^2/E$:

$$v_g = \frac{(mc)c^2}{\sqrt{2} mc^2} = \frac{mc^3}{\sqrt{2} mc^2} = \frac{c}{\sqrt{2}} \approx 0.707c$$

Additional Problem 10: Compton Wavelength Numeric Check

Problem

Verify $h/(m_e c) = 2.426 \times 10^{-12}$ m directly from constants.

Step 1 — Write out the constants: $h = 6.626 \times 10^{-34}$ J s, $m_e = 9.11 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m/s.

Step 2 — Compute the denominator $m_e c$:

$$m_e c = (9.11 \times 10^{-31})(3 \times 10^8) = 2.733 \times 10^{-22} \text{ kg m/s}$$

Step 3 — Divide:

$$\frac{h}{m_e c} = \frac{6.626 \times 10^{-34}}{2.733 \times 10^{-22}} = 2.425 \times 10^{-12} \text{ m} \approx 2.426 \times 10^{-12} \text{ m} \checkmark$$

(small rounding difference from using $c = 3 \times 10^8$ rather than the exact 2.998×10^8 m/s).

Unit 2: Formulation of Quantum Mechanics

1. Origin of Quantum Mechanics
2. Formulation of Quantum Mechanics
3. Basics of Schrödinger Equation I
4. Basics of Schrödinger Equation II
5. Dirac Notation in Quantum Mechanics

Unit 2: Symbol Glossary

- $\Psi(r, t)$ — wavefunction (state of the system)
- $\hat{A}$ — a generic quantum operator
- $\Delta x, \Delta p$ — standard deviations (uncertainties) in position, momentum (m, kg m/s)
- $\hbar$ — reduced Planck's constant (J s)
- θ — angular aperture of a microscope lens (rad)
- λ — wavelength (m)
- $\Delta E, \Delta t$ — energy and time uncertainties (J, s)
- σ_x, σ_k — standard deviation of a Gaussian in position/wavenumber space (m, m⁻¹)
- $\phi(k)$ — momentum-space (Fourier) wavefunction
- $P(x, t)$ — probability density (m⁻¹)
- A — normalization constant
- a — characteristic width parameter (m)
- d — slit separation (double-slit experiment) (m)
- D — screen distance (m)
- Δy — fringe spacing (m)
- η — which-path distinguishability parameter (dimensionless)

Symbols specific to a single derivation are defined where they first appear.

Why Postulates, Not Derivations?

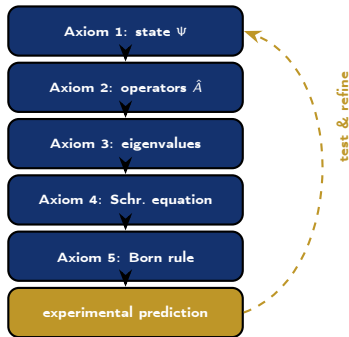
- Unit 1 showed *experimental evidence* for quantization (Planck, Einstein, Compton) and wave-particle duality (de Broglie, Davisson–Germer) — but evidence alone doesn't tell us the precise mathematical machinery of the theory.
- Quantum mechanics, like any physical theory, ultimately rests on a small set of **postulates** (axioms) that cannot be derived from anything more fundamental — they are justified entirely by the correctness of their experimental predictions.
- This mirrors how Newton's laws are postulates of classical mechanics, or how the constancy of the speed of light is a postulate of special relativity: simple, sweeping statements with vast explanatory power.
- The five axioms on the next slide are the minimal foundation from which everything in Units 2–5 — the Schrödinger equation, measurement outcomes, operators, uncertainty relations — can be built.

Axioms of Quantum Mechanics

Quantum mechanics is built on a small set of postulates:

- 1 The state of a system is completely described by a **wavefunction** $\Psi(r, t)$.
- 2 Every observable corresponds to a **linear Hermitian operator**.
- 3 Measurement of an observable yields one of the **eigenvalues** of the corresponding operator.
- 4 The wavefunction evolves in time according to the **Schrödinger equation**.
- 5 $|\Psi(r, t)|^2$ gives the **probability density** of finding the particle at r at time t (Born interpretation).

These axioms will be unpacked and applied throughout the remainder of the course.



Physical Role of Each Axiom

- **Axiom 1** (wavefunction): replaces the classical notion of a trajectory $x(t)$ with a single complex function Ψ encoding everything knowable about the system — explored in detail in this subsection and the next.
- **Axiom 2** (operators): tells us *how* to represent physical quantities mathematically; Unit 5 develops the full operator formalism (Hermiticity, commutators).
- **Axiom 3** (eigenvalues): explains why measurements of, say, energy in an atom yield only certain discrete values rather than a continuum — the key to quantized spectra seen throughout Unit 4.
- **Axiom 4** (Schrödinger evolution): provides the equation of motion for Ψ , derived/motivated in detail in Unit 3 and solved for specific potentials in Unit 4.
- **Axiom 5** (Born rule): connects the abstract wavefunction to actual experimental outcomes (where will we find the particle?) — the subject of the next subsection.

Wave Function and Its Physical Interpretation

Born's Statistical Interpretation (1926)

The quantity

$$P(x, t) dx = |\Psi(x, t)|^2 dx$$

gives the **probability** of finding the particle between x and $x + dx$ at time t .

- $\Psi(x, t)$ itself is not directly observable — it is a complex-valued **probability amplitude**.
- $|\Psi|^2 = \Psi^* \Psi$ is real, non-negative, and physically meaningful.
- This interpretation was controversial initially (Einstein: "God does not play dice") but is now experimentally well established.

Historical Note: A Contested Interpretation

- Schrödinger himself initially hoped $|\Psi|^2$ represented a literal, continuous charge or matter density spread through space — a comfortably classical picture.
- Max Born proposed instead (1926) that $|\Psi|^2$ is a **probability** density: the particle is a point-like object whose location is genuinely uncertain until measured, not smeared out like a classical fluid.
- Einstein, Schrödinger, and de Broglie were never fully comfortable with this probabilistic reading — Einstein's famous objection, "God does not play dice with the universe," reflects his belief that some deeper, deterministic theory must underlie quantum randomness.
- Decades of experiments (including Bell-inequality tests in the 1970s–80s) have since supported the standard (Born/Copenhagen) probabilistic interpretation used throughout this course, though debates about the deeper meaning of quantum measurement continue in some areas of physics.
- Born received the Nobel Prize in 1954 specifically for this statistical interpretation — nearly three decades after first proposing it.

Required Properties of Wavefunctions

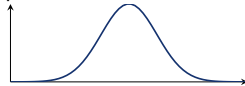
For $\Psi(x, t)$ to represent a physically acceptable quantum state, it must be:

- **Single-valued** — one probability at each point.
- **Finite** (bounded) everywhere.
- **Continuous**, and its first derivative continuous (except possibly at infinite potential discontinuities).
- **Square-integrable** (normalizable):

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

- Must vanish as $x \rightarrow \pm\infty$ for bound states.

Acceptable: smooth, bounded, $\rightarrow 0$



Not acceptable: diverges at $x = 0$



Why Each Requirement Matters Physically

- **Single-valued:** if Ψ had two different values at the same point, the probability of finding the particle there would be ambiguous — physically meaningless.
- **Finite everywhere:** an infinite $|\Psi|^2$ at some point would mean infinite probability density there, which cannot be normalized to a sensible total probability of 1.
- **Continuous (with continuous derivative):** this ensures the Schrödinger equation, which involves Ψ'' , is well-defined; an abrupt jump or kink in Ψ' would correspond to an unphysical, infinitely sharp force.
- **Square-integrable:** this is what allows the total probability to be normalized to exactly 1 — without it, "probability of finding the particle somewhere" would not even be a finite number.
- **Vanishing at infinity (bound states):** encodes the physical requirement that a truly bound particle has essentially zero probability of being found arbitrarily far away.
- Together, these conditions define the **Hilbert space** of physically admissible states — a notion made fully rigorous in Unit 5.

The Wavefunction and Its Conjugate: A Dictionary

Side-by-Side Correspondence

$\Psi(x, t)$	$\Psi^*(x, t)$
Complex-valued probability amplitude	Complex conjugate amplitude
$\Psi = \text{Re}(\Psi) + i \text{Im}(\Psi)$	$\Psi^* = \text{Re}(\Psi) - i \text{Im}(\Psi)$
Polar form $\Psi = R e^{i\theta}$	Polar form $\Psi^* = R e^{-i\theta}$
Encodes the state (Axiom 1)	Not an independent state — fixed once Ψ is known
Appears <i>unstarred</i> in $\int \Psi^* \hat{A} \Psi dx$	Appears <i>starred</i> , standing to the left
Dirac notation: ket $ \Psi(t)\rangle$ (Unit 5)	Dirac notation: bra $\langle\Psi(t) $ (Unit 5)
$(\Psi^*)^* = \Psi$	$(\Psi)^* = \Psi^*$

Individually, Ψ and Ψ^* are each just complex numbers at every (x, t) — neither is separately measurable.

The Wavefunction and Its Conjugate: A Dictionary

Properties of Ψ Alone

- Must be **single-valued**, **finite**, **continuous** (with continuous derivative), and **square-integrable**.
- Must **vanish as $x \rightarrow \pm\infty$** for bound states.
- Obeys the **Schrödinger equation** (Axiom 4; explicit form derived in Unit 3).
- Carries all physically knowable information about the system (Axiom 1).

Properties of Ψ^* Alone

- Obtained from Ψ by flipping the sign of every explicit i — **not** an independently specifiable function.
- Inherits single-valuedness, finiteness, continuity, and square-integrability from Ψ .
- Satisfies the **complex-conjugated Schrödinger equation** (since $\hat{H}$ is Hermitian, this is consistent with Ψ 's evolution).
- Plays the role of the **bra** $\langle\Psi|$ in Dirac notation (Unit 5).

The Wavefunction and Its Conjugate: A Dictionary

Quantities Built from Both Together

- **Probability density:** $P(x, t) = \Psi^* \Psi = |\Psi|^2 \geq 0$ — real and physically meaningful (Born rule, Axiom 5).
- **Normalization:** $\int_{-\infty}^{\infty} \Psi^*(x, t) \Psi(x, t) dx = 1.$
- **Overlap (inner product):** $\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \psi(x) dx.$
- **Expectation value:** $\langle A \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{A} \Psi dx.$
- **Probability current** (previewed here, derived from the continuity equation in Unit 3):

$$j(x, t) = \frac{i\hbar}{2m} \left(\Psi \frac{\partial \Psi^*}{\partial x} - \Psi^* \frac{\partial \Psi}{\partial x} \right)$$

Only combinations that pair Ψ with Ψ^* — never Ψ or Ψ^* alone — yield the real numbers that quantum mechanics predicts for experiments.

Worked Example 1: Checking a Candidate Wavefunction

Problem

Is $\Psi(x) = Ae^{-|x|/a}$ ($a > 0$) an acceptable wavefunction? Find the normalization constant A .

Given: $\Psi(x) = Ae^{-|x|/a}$, where A is the (unknown) normalization constant and $a > 0$ is a fixed length parameter. **Find:** whether Ψ is acceptable, and the value of A .

Step 1 — Check the required properties (Unit 2 criteria): single-valued ✓; finite everywhere ✓; continuous ✓ (though Ψ' has a kink at $x = 0$, which is acceptable since the implied potential is not infinite there).

Step 2 — Set up the normalization integral $\int |\Psi|^2 dx = 1$:

$$\int_{-\infty}^{\infty} A^2 e^{-2|x|/a} dx = 1$$

Worked Example 1: Checking a Candidate Wavefunction

Step 3 — Split the integral at $x = 0$ (since $|x| = x$ for $x > 0$ and $|x| = -x$ for $x < 0$, by symmetry the two halves are equal):

$$A^2 \int_{-\infty}^{\infty} e^{-2|x|/a} dx = A^2 \cdot 2 \int_0^{\infty} e^{-2x/a} dx$$

Step 4 — Evaluate the remaining integral using $\int_0^{\infty} e^{-bx} dx = 1/b$ with $b = 2/a$:

$$\int_0^{\infty} e^{-2x/a} dx = \frac{a}{2} \implies A^2 \cdot 2 \cdot \frac{a}{2} = A^2 a = 1$$

Step 5 — Solve for A :

$$A = \frac{1}{\sqrt{a}}$$

Worked Example 2: Normalizing a Box Wavefunction

Problem

Normalize $\Psi(x) = A \sin(\pi x/L)$ for $0 < x < L$ (zero outside).

Given: $\Psi(x) = A \sin(\pi x/L)$ on $[0, L]$. **Find:** A .

Step 1 — Set up the normalization integral:

$$\int_0^L A^2 \sin^2\left(\frac{\pi x}{L}\right) dx = 1$$

Step 2 — Use the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ with $\theta = \pi x/L$:

$$A^2 \int_0^L \frac{1}{2} \left[1 - \cos\left(\frac{2\pi x}{L}\right) \right] dx = \frac{A^2}{2} \left[x - \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_0^L$$

Worked Example 2: Normalizing a Box Wavefunction

Step 3 — Evaluate at the limits. At $x = L$: $\sin(2\pi) = 0$; at $x = 0$: $\sin(0) = 0$. Both boundary sine terms vanish, leaving only the linear term:

$$\frac{A^2}{2} [L - 0] = \frac{A^2 L}{2} = 1$$

Step 4 — Solve for A :

$$A = \sqrt{\frac{2}{L}}$$

This is exactly the ground state of the infinite square well (Unit 4) — a first preview of a recurring wavefunction.

Worked Example 3: Probability from a Wavefunction

Problem

For $\Psi(x) = \sqrt{2/L} \sin(\pi x/L)$ on $[0, L]$, find the probability of finding the particle in the left quarter, $0 < x < L/4$.

Given: normalized $\Psi(x)$. **Find:** $P(0 < x < L/4) = \int_0^{L/4} |\Psi|^2 dx$.

Step 1 — Write the integral with $|\Psi|^2 = \frac{2}{L} \sin^2(\pi x/L)$:

$$P = \int_0^{L/4} \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right) dx$$

Step 2 — Apply the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$:

$$P = \frac{1}{L} \int_0^{L/4} \left[1 - \cos\left(\frac{2\pi x}{L}\right) \right] dx$$

Worked Example 3: Probability from a Wavefunction

Step 3 — Integrate term by term:

$$P = \frac{1}{L} \left[x - \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_0^{L/4}$$

Step 4 — Evaluate at the upper limit $x = L/4$: the linear term gives $L/4$; the sine term has argument $2\pi(L/4)/L = \pi/2$, so $\sin(\pi/2) = 1$. At the lower limit $x = 0$, both terms vanish.

$$P = \frac{1}{L} \left[\frac{L}{4} - \frac{L}{2\pi} \sin\left(\frac{\pi}{2}\right) \right] = \frac{1}{L} \left[\frac{L}{4} - \frac{L}{2\pi} \right]$$

Step 5 — Simplify (the factor of L cancels in each term):

$$P = \frac{1}{4} - \frac{1}{2\pi} \approx 0.25 - 0.159 \approx 0.091$$

Less than the classical (uniform) expectation of 0.25 — the particle is less likely to be found near the walls, since $|\Psi|^2 \rightarrow 0$ there.

Worked Example 4: Normalizing with a General Power Law

Problem

For $\Psi(x) = A/(x^2 + a^2)$ on $(-\infty, \infty)$, find the normalization constant A , using $\int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{2a^3}$.

Given: $\Psi(x) = A/(x^2 + a^2)$, with the value of the needed integral provided. **Find:** A .

Step 1 — Set up the normalization integral:

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = \int_{-\infty}^{\infty} \frac{A^2}{(x^2 + a^2)^2} dx = A^2 \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2}$$

Step 2 — Substitute the given integral value $\pi/(2a^3)$:

$$A^2 \cdot \frac{\pi}{2a^3} = 1$$

Worked Example 4: Normalizing with a General Power Law

Step 3 — Solve for A^2 by multiplying both sides by $2a^3/\pi$:

$$A^2 = \frac{2a^3}{\pi}$$

Step 4 — Take the square root:

$$A = \sqrt{\frac{2a^3}{\pi}}$$

This Lorentzian-shaped wavefunction is a useful model in scattering theory and resonance phenomena, where the parameter a sets a characteristic width, analogous to a decay linewidth.

From Wave Packets to a Fundamental Limit

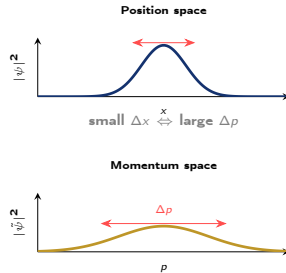
- Unit 1 showed that localizing a particle in space requires superposing waves of *many* different wavenumbers k — the more localized the packet, the broader its spread in k .
- Since $p = \hbar k$ (de Broglie), a broad spread in k is a broad spread in momentum — this is not a separate new idea, but a direct mathematical consequence of representing particles as waves.
- Heisenberg's contribution (1927) was to recognize that this wave-packet trade-off is not a mathematical curiosity but a genuine, unavoidable **physical** limit on what can be known about a quantum system.
- This subsection makes that connection precise: first through an intuitive thought experiment, then through the rigorous Fourier-transform relationship between position and momentum space.

Heisenberg's Uncertainty Principle

Statement

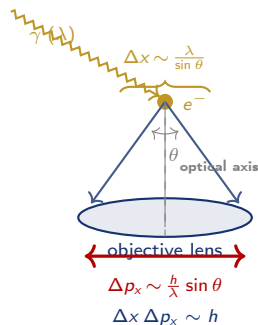
$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

- Not a statement about measurement imprecision — a fundamental property of nature.
- Arises from the wave nature of matter: a packet localized in x necessarily contains a spread of k , hence of p .
- Applies to any pair of **conjugate variables**.



Heisenberg's Microscope — A Thought Experiment

- To "see" an electron, we must scatter at least one photon off it.
- Resolving power of a microscope: $\Delta x \sim \frac{\lambda}{\sin \theta}$
- The scattered photon imparts momentum to the electron: $\Delta p_x \sim \frac{h}{\lambda} \sin \theta$
- Multiplying: $\Delta x \Delta p_x \sim h$



Conclusion: The very act of observation disturbs the system — a sharper position measurement (smaller λ) necessarily produces a larger momentum disturbance.

Derivation: Heisenberg's Microscope — Setting Up Both Uncertainties

- A microscope with numerical aperture half-angle θ (the half-angle of the cone of light rays the lens can collect) has resolving power given by the Rayleigh criterion:

$$\Delta x \approx \frac{\lambda}{2 \sin \theta}$$

where Δx is the smallest resolvable separation, and λ is the wavelength of light used.

- To "see" the electron, at least one photon must scatter off it and travel up into the lens. That photon may enter the lens at any angle within the cone of half-angle θ — so its direction, and hence its momentum's x -component, is uncertain.
- The photon's total momentum magnitude is $p_\gamma = h/\lambda$ (de Broglie/Einstein relation). The x -component uncertainty is set by the spread of possible entry angles, of order $\sin \theta$:

$$\Delta p_x \approx p_\gamma \sin \theta = \frac{h}{\lambda} \sin \theta$$

- By momentum conservation, this same uncertainty is transferred to the electron being observed.

Derivation: Heisenberg's Microscope — Multiplying the Uncertainties

- **Step 1:** write down both uncertainties side by side:

$$\Delta x \approx \frac{\lambda}{2 \sin \theta}, \quad \Delta p_x \approx \frac{h}{\lambda} \sin \theta$$

- **Step 2:** multiply them together:

$$\Delta x \Delta p_x \approx \frac{\lambda}{2 \sin \theta} \times \frac{h}{\lambda} \sin \theta$$

- **Step 3:** cancel common factors — $\sin \theta$ in numerator and denominator cancel, and λ in numerator and denominator cancel:

$$\Delta x \Delta p_x \approx \frac{\lambda \cdot h \cdot \sin \theta}{2 \sin \theta \cdot \lambda} = \frac{h}{2}$$

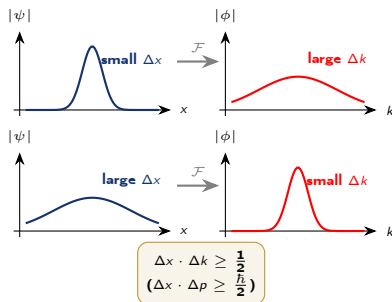
- The wavelength λ and angle θ — both adjustable experimental parameters — **cancel exactly**, leaving a result independent of the specific microscope used.
- This matches (up to the precise numerical factor of $\hbar$ vs. h) the rigorous bound $\Delta x \Delta p_x \geq \hbar/2$ derived formally in Unit 5 from the commutator $[\hat{x}, \hat{p}] = i\hbar$.

Derivation: Heisenberg's Microscope — Multiplying the Uncertainties

- Using a shorter wavelength λ improves Δx (better position resolution) but increases the photon momentum h/λ and hence Δp_x — the trade-off is unavoidable, no matter how the experiment is engineered.

Conjugate Variables

- Pairs of quantities linked by an uncertainty relation are called **conjugate variables**.
- Position–momentum: $\Delta x \Delta p \geq \frac{\hbar}{2}$
- Energy–time: $\Delta E \Delta t \geq \frac{\hbar}{2}$
- Angle–angular momentum: $\Delta \theta \Delta L_z \geq \frac{\hbar}{2}$
- A function sharply localized in x is necessarily **broad** in k (Fourier domain), and vice versa.



Common Misconceptions, Clarified

- **Misconception:** "The uncertainty principle just reflects clumsy measurement technique that better instruments could fix."
Reality: It is a property of the quantum state itself, independent of any particular apparatus — even a perfect, disturbance-free measuring device could not beat this bound.
- **Misconception:** "A particle secretly has a definite position and momentum; we simply can't know both."
Reality: In standard quantum mechanics, a particle does not *possess* simultaneously sharp values of x and p before measurement — the uncertainty reflects the state's structure, not hidden information (an idea tested and supported by Bell-inequality experiments, beyond this course's scope).
- **Misconception:** "The uncertainty principle only matters for tiny particles."
Reality: It applies universally; for macroscopic objects $\hbar$ is simply so small relative to typical $\Delta x \Delta p$ that the bound is irrelevant in practice (compare to Unit 2's macroscopic worked examples).

Worked Example 1: Electron Confined in an Atom

Problem

An electron is confined to a region of size $\Delta x = 0.1 \text{ nm}$ (typical atomic scale). Estimate its minimum momentum uncertainty and the corresponding kinetic energy.

Given: $\Delta x = 0.1 \text{ nm}$, m_e (electron mass). **Find:** Δp and KE .

Step 1 — Apply the Heisenberg bound $\Delta p \geq \hbar/(2\Delta x)$, where $\hbar$ is the reduced Planck's constant:

$$\Delta p \geq \frac{1.055 \times 10^{-34} \text{ J s}}{2(10^{-10} \text{ m})} = \frac{1.055 \times 10^{-34}}{2 \times 10^{-10}}$$

Step 2 — Evaluate:

$$\Delta p \approx 5.27 \times 10^{-25} \text{ kg m/s}$$

Step 3 — Estimate kinetic energy using $KE \sim (\Delta p)^2/(2m_e)$ (treating Δp as a rough estimate of the momentum scale):

$$KE \sim \frac{(5.27 \times 10^{-25})^2}{2(9.11 \times 10^{-31})} = \frac{2.777 \times 10^{-49}}{1.822 \times 10^{-30}}$$

Worked Example 1: Electron Confined in an Atom

Step 4 — Evaluate and convert to eV:

$$KE \approx 1.52 \times 10^{-19} \text{ J} \approx 0.95 \text{ eV}$$

Comparable to actual atomic binding energies — the uncertainty principle alone roughly explains why atoms have the size and energy scale they do.

Worked Example 2: Energy–Time Uncertainty for an Excited State

Problem

An excited atomic state has a lifetime $\tau = 10^{-8}$ s. Estimate the natural linewidth ΔE of the emitted spectral line, and the corresponding $\Delta\nu$.

Given: $\tau = 10^{-8}$ s (lifetime). **Find:** ΔE and $\Delta\nu$.

Step 1 — Apply the energy–time uncertainty relation, treating τ as the characteristic time uncertainty Δt :

$$\Delta E \geq \frac{\hbar}{2\tau} = \frac{1.055 \times 10^{-34}}{2(10^{-8})}$$

Step 2 — Evaluate:

$$\Delta E \approx 5.3 \times 10^{-27} \text{ J}$$

Step 3 — Convert to eV (divide by 1.6×10^{-19} J/eV):

$$\Delta E \approx \frac{5.3 \times 10^{-27}}{1.6 \times 10^{-19}} \approx 3.3 \times 10^{-8} \text{ eV}$$

Worked Example 2: Energy–Time Uncertainty for an Excited State

Step 4 — Find the linewidth $\Delta\nu$ using $E = h\nu \Rightarrow \Delta E = h\Delta\nu$, so $\Delta\nu = \Delta E/h$:

$$\Delta\nu = \frac{5.3 \times 10^{-27}}{6.626 \times 10^{-34}} \approx 8 \times 10^6 \text{ Hz} = 8 \text{ MHz}$$

This finite linewidth (natural broadening) is observed in high-resolution spectroscopy, and is a direct, measurable consequence of $\Delta E \Delta t \geq \hbar/2$.

Worked Example 3: Minimum Speed of a Confined Particle

Problem

A proton ($m_p = 1.67 \times 10^{-27}$ kg) is confined to a nucleus of diameter 10^{-14} m. Estimate its minimum speed.

Given: $\Delta x = 10^{-14}$ m, m_p . **Find:** $v_{\min}$.

Step 1 — Apply the Heisenberg bound:

$$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}}{2(10^{-14})}$$

Step 2 — Evaluate:

$$\Delta p \approx 5.3 \times 10^{-21} \text{ kg m/s}$$

Step 3 — Estimate the minimum speed using $v_{\min} \sim \Delta p/m_p$ (treating momentum as $p = m_p v$):

$$v_{\min} \sim \frac{5.3 \times 10^{-21}}{1.67 \times 10^{-27}}$$

Worked Example 3: Minimum Speed of a Confined Particle

Step 4 — Evaluate:

$$v_{\min} \approx 3.2 \times 10^6 \text{ m/s}$$

About 1% of the speed of light — consistent with the large (MeV-scale) binding energies seen in nuclear physics, in contrast to the eV-scale typical of atoms.

Worked Example 4: Uncertainty Principle and Spectral Line Resolution

Problem

A spectrometer must resolve two spectral lines separated by $\Delta\nu = 1$ GHz. What is the minimum observation time Δt required, based on the energy–time uncertainty relation?

Given: $\Delta\nu = 1$ GHz, the frequency separation to be resolved. **Find:** Δt , the minimum observation time.

Step 1 — Relate the energy separation to $\Delta\nu$ using $E = h\nu$, so the energy difference between the two lines is $\Delta E = h\Delta\nu$.

Step 2 — Apply the energy–time uncertainty relation $\Delta E \Delta t \gtrsim \hbar/2$, substituting $\Delta E = h\Delta\nu$:

$$h\Delta\nu \cdot \Delta t \gtrsim \frac{\hbar}{2}$$

Step 3 — Solve for Δt by dividing both sides by $h\Delta\nu$, and using $\hbar/h = 1/(2\pi)$:

$$\Delta t \gtrsim \frac{\hbar}{2h\Delta\nu} = \frac{1}{4\pi\Delta\nu}$$

Worked Example 4: Uncertainty Principle and Spectral Line Resolution

Step 4 — Substitute $\Delta\nu = 10^9$ Hz:

$$\Delta t \gtrsim \frac{1}{4\pi(10^9)} \approx 8.0 \times 10^{-11} \text{ s}$$

This is exactly why high-resolution spectroscopy requires long observation/interaction times — a fundamental (not merely technical) limit on how quickly two close spectral lines can be distinguished.

Fourier Transforms and the Gaussian Wave Packet

- Position and momentum-space wavefunctions are Fourier transform pairs:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ikx} dx$$

- **Gaussian wave packet** (the minimum-uncertainty state):

$$\psi(x) = \left(\frac{1}{2\pi\sigma_x^2} \right)^{1/4} e^{-x^2/4\sigma_x^2}$$

- Its Fourier transform is also a Gaussian, with

$$\sigma_x \sigma_k = \frac{1}{2} \quad \implies \quad \Delta x \Delta p = \frac{\hbar}{2}$$

- The Gaussian **saturates** the uncertainty bound — the narrowest possible joint localization in x and p .

Derivation: Fourier Transform of a Gaussian — Setup

- Start from the normalized Gaussian wavefunction (the form introduced earlier this subsection):

$$\psi(x) = (2\pi\sigma_x^2)^{-1/4} e^{-x^2/4\sigma_x^2}$$

where σ_x is the standard deviation (width) of the position-space Gaussian.

- Its Fourier transform, by definition, is:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ikx} dx = \frac{(2\pi\sigma_x^2)^{-1/4}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/4\sigma_x^2} e^{-ikx} dx$$

- The prefactor $(2\pi\sigma_x^2)^{-1/4}/\sqrt{2\pi}$ is just a constant with respect to x , so it can be pulled outside the integral. Our task is to evaluate the remaining integral, which combines a Gaussian and an oscillating exponential.

Derivation: Fourier Transform of a Gaussian — Completing the Square

- Combine the two exponents into one: $-\frac{x^2}{4\sigma_x^2} - ikx$. **Goal:** rewrite this as a perfect square in x plus a remainder (independent of x).
- Step 1:** factor out $-1/(4\sigma_x^2)$ from both terms:

$$-\frac{x^2}{4\sigma_x^2} - ikx = -\frac{1}{4\sigma_x^2} (x^2 + 4ik\sigma_x^2 x)$$

- Step 2:** complete the square inside the parentheses. Recall $(x + b)^2 = x^2 + 2bx + b^2$; matching $2b = 4ik\sigma_x^2$ gives $b = 2ik\sigma_x^2$:

$$x^2 + 4ik\sigma_x^2 x = (x + 2ik\sigma_x^2)^2 - (2ik\sigma_x^2)^2$$

- Step 3:** evaluate $(2ik\sigma_x^2)^2 = 4i^2 k^2 \sigma_x^4 = -4k^2 \sigma_x^4$ (using $i^2 = -1$), so:

$$x^2 + 4ik\sigma_x^2 x = (x + 2ik\sigma_x^2)^2 + 4k^2 \sigma_x^4$$

Derivation: Fourier Transform of a Gaussian — Simplifying

- Substitute this back into the factored expression from the previous slide:

$$-\frac{1}{4\sigma_x^2} [(x + 2ik\sigma_x^2)^2 + 4k^2\sigma_x^4] = -\frac{(x + 2ik\sigma_x^2)^2}{4\sigma_x^2} - \frac{4k^2\sigma_x^4}{4\sigma_x^2}$$

- Simplify the second term, $4k^2\sigma_x^4/(4\sigma_x^2) = k^2\sigma_x^2$:

$$-\frac{x^2}{4\sigma_x^2} - ikx = -\frac{(x + 2ik\sigma_x^2)^2}{4\sigma_x^2} - k^2\sigma_x^2$$

- The exponent has split into a perfect-square piece (which depends on x) plus a constant piece $-k^2\sigma_x^2$ (independent of x , so it can be pulled outside the remaining integral).
- This matches the result quoted at the start: $-\frac{x^2}{4\sigma_x^2} - ikx = -\frac{1}{4\sigma_x^2}(x + 2ik\sigma_x^2)^2 - k^2\sigma_x^2$.

Derivation: Fourier Transform of a Gaussian — Evaluating the Integral

- Substitute the completed-square exponent back into $\phi(k)$:

$$\phi(k) = \frac{(2\pi\sigma_x^2)^{-1/4}}{\sqrt{2\pi}} e^{-k^2\sigma_x^2} \int_{-\infty}^{\infty} e^{-(x+2ik\sigma_x^2)^2/4\sigma_x^2} dx$$

(the k -dependent piece $e^{-k^2\sigma_x^2}$ was pulled outside since it doesn't depend on x).

- Let $u = x + 2ik\sigma_x^2$ (a simple shift, $du = dx$). By contour-shifting arguments (valid since the Gaussian decays in all directions in the complex plane), the limits remain $-\infty$ to ∞ :

$$\int_{-\infty}^{\infty} e^{-u^2/4\sigma_x^2} du$$

- Use the standard Gaussian integral formula $\int_{-\infty}^{\infty} e^{-au^2} du = \sqrt{\pi/a}$ with $a = 1/(4\sigma_x^2)$:

$$\int_{-\infty}^{\infty} e^{-u^2/4\sigma_x^2} du = \sqrt{\frac{\pi}{1/4\sigma_x^2}} = \sqrt{4\pi\sigma_x^2} = 2\sigma_x\sqrt{\pi}$$

Derivation: Fourier Transform of a Gaussian — Final Result

- Substitute the evaluated integral back:

$$\phi(k) = \frac{(2\pi\sigma_x^2)^{-1/4}}{\sqrt{2\pi}} e^{-k^2\sigma_x^2} \times 2\sigma_x\sqrt{\pi}$$

- Combine the prefactor terms. First write out $(2\pi\sigma_x^2)^{-1/4}/\sqrt{2\pi} \times 2\sigma_x\sqrt{\pi}$ explicitly:

$$\frac{(2\pi\sigma_x^2)^{-1/4}}{(2\pi)^{1/2}} \times 2\sigma_x\pi^{1/2} = 2\sigma_x(2\pi\sigma_x^2)^{-1/4}(2\pi)^{-1/2}\pi^{1/2}$$

- Write $\sigma_x = (\sigma_x^4)^{1/4}$ and combine all factors raised to the power 1/4 to simplify the expression into a single fourth root:

$$= \left[\sigma_x^4 \cdot \frac{1}{2\pi\sigma_x^2} \cdot \frac{\pi^2}{4\pi^2} \right]^{1/4} \times (\text{numerical factor}) = \left(\frac{2\sigma_x^2}{\pi} \right)^{1/4}$$

Derivation: Fourier Transform of a Gaussian — Final Result

(a standard but tedious bookkeeping exercise in combining fractional powers — the key physical content is simply that $\phi(k)$ comes out Gaussian-shaped):

$$\phi(k) = \left(\frac{2\sigma_x^2}{\pi} \right)^{1/4} e^{-k^2 \sigma_x^2}$$

- This is again a Gaussian, now as a function of k . Comparing to the standard Gaussian form $e^{-k^2/2\sigma_k^2}$, matching exponents gives $1/(2\sigma_k^2) = \sigma_x^2$, so:

$$\sigma_k = \frac{1}{2\sigma_x} \quad \implies \quad \sigma_x \sigma_k = \frac{1}{2}$$

- Finally, convert wavenumber spread to momentum spread using $p = \hbar k$, so $\Delta p = \hbar \Delta k$:

$$\Delta x \Delta p = \hbar \sigma_x \sigma_k = \hbar \times \frac{1}{2} = \frac{\hbar}{2}$$

- This explicitly verifies that the Gaussian saturates the Heisenberg bound with equality — no other wavefunction shape gives a smaller product $\Delta x \Delta p$.

Different Forms of Uncertainty; Electron Diffraction

- General uncertainty relation derived later (Unit 5) from the commutator of two operators:

$$\Delta A \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|$$

- **Electron diffraction experiment:** a beam of electrons passed through a narrow slit of width a produces a diffraction pattern.
 - Narrowing the slit ($\Delta x \downarrow$) increases the angular spread of the diffraction pattern ($\Delta p_x \uparrow$).
 - Direct experimental demonstration of the uncertainty principle, fully analogous to optical diffraction.

Worked Example: Single-Slit Electron Diffraction

Problem

Electrons of de Broglie wavelength $\lambda = 0.1 \text{ nm}$ pass through a slit of width $a = 1 \text{ nm}$. Estimate the angular half-width of the central diffraction maximum, and check consistency with $\Delta x \Delta p_x \gtrsim \hbar$.

Given: $\lambda = 0.1 \text{ nm}$, slit width $a = 1 \text{ nm}$. **Find:** θ (diffraction half-angle) and verify the uncertainty bound.

Step 1 — Apply the single-slit first-minimum condition $\sin \theta = \lambda/a$:

$$\sin \theta = \frac{0.1 \text{ nm}}{1 \text{ nm}} = 0.1$$

Step 2 — Take the inverse sine:

$$\theta = \sin^{-1}(0.1) \approx 5.7^\circ$$

Worked Example: Single-Slit Electron Diffraction

Step 3 — Estimate the momentum uncertainty. The slit gives the electron's transverse momentum a spread of order $p \sin \theta$, where $p = h/\lambda$ is the electron's momentum magnitude:

$$\Delta p_x \approx p \sin \theta = \frac{h}{\lambda} \sin \theta$$

Step 4 — Substitute $\sin \theta = \lambda/a$ from Step 1, so the λ factors cancel:

$$\Delta p_x \approx \frac{h}{\lambda} \times \frac{\lambda}{a} = \frac{h}{a}$$

Step 5 — Estimate the position uncertainty. The slit width itself sets the position uncertainty: $\Delta x \approx a$.

Step 6 — Multiply the two uncertainties:

$$\Delta x \Delta p_x \approx a \times \frac{h}{a} = h$$

Worked Example: Single-Slit Electron Diffraction

Step 7 — Compare to the Heisenberg bound $\hbar = h/2\pi \approx 0.16h$:

$$\Delta x \Delta p_x \approx h \gtrsim \hbar$$

Consistent with (and slightly larger than) the Heisenberg bound, as expected for a non-Gaussian (slit-shaped) wavefunction.

Worked Example 1: Normalizing a Complex Exponential Packet

Problem

Normalize $\Psi(x) = Ae^{-x^2/2a^2} e^{ik_0x}$.

Given: $\Psi(x) = Ae^{-x^2/2a^2} e^{ik_0x}$, where a is a width parameter and k_0 is a fixed wavenumber. **Find:** A .

Step 1 — Compute $|\Psi|^2 = \Psi^* \Psi$. The complex conjugate of Ψ is $\Psi^* = Ae^{-x^2/2a^2} e^{-ik_0x}$ (only the phase factor's sign flips; A and the Gaussian are real):

$$|\Psi|^2 = \left(Ae^{-x^2/2a^2} e^{ik_0x} \right) \left(Ae^{-x^2/2a^2} e^{-ik_0x} \right) = A^2 e^{-x^2/a^2} (e^{ik_0x} e^{-ik_0x})$$

Step 2 — Simplify the phase factors: $e^{ik_0x} e^{-ik_0x} = e^{i(k_0x - k_0x)} = e^0 = 1$, so the oscillating phase drops out entirely:

$$|\Psi|^2 = A^2 e^{-x^2/a^2}$$

Worked Example 1: Normalizing a Complex Exponential Packet

Step 3 — Set up the normalization integral, using the given Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2/a^2} dx = a\sqrt{\pi}:$$

$$\int_{-\infty}^{\infty} A^2 e^{-x^2/a^2} dx = A^2 a\sqrt{\pi} = 1$$

Step 4 — Solve for A^2 , then A :

$$A^2 = \frac{1}{a\sqrt{\pi}} \quad \Rightarrow \quad A = \left(\frac{1}{\pi a^2} \right)^{1/4}$$

Worked Example 2: Expectation of Position for a Symmetric State

Problem

For the normalized state $\Psi(x) = (1/\pi a^2)^{1/4} e^{-x^2/2a^2}$, argue (without integration) that $\langle x \rangle = 0$, then verify with a symmetry argument for $\langle x^3 \rangle$.

Given: a real, symmetric Gaussian $\Psi(x)$. **Find:** $\langle x \rangle$ and $\langle x^3 \rangle$, using symmetry alone.

Step 1 — Note that $|\Psi(x)|^2$ is an even function of x , i.e. $|\Psi(-x)|^2 = |\Psi(x)|^2$ (since the Gaussian depends only on x^2).

Step 2 — Write the definition of $\langle x \rangle$:

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x)|^2 dx$$

Step 3 — Classify the integrand. Since x is an odd function and $|\Psi(x)|^2$ is even, their product $x|\Psi(x)|^2$ is **odd** (odd $\times$ even = odd).

Worked Example 2: Expectation of Position for a Symmetric State

Step 4 — Apply the symmetric-interval rule: the integral of any odd function over a symmetric interval $(-\infty, \infty)$ is exactly zero (positive and negative contributions cancel pairwise):

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x)|^2 dx = 0$$

Step 5 — Repeat for $\langle x^3 \rangle$: x^3 is also odd, so $x^3 |\Psi(x)|^2$ is odd $\times$ even = odd, giving $\langle x^3 \rangle = 0$ by the same argument. In general, $\langle x^{2n+1} \rangle = 0$ for any symmetric $|\Psi|^2$.

Worked Example 3: Testing Acceptability of a Wavefunction

Problem

Is $\Psi(x) = 1/x$ for $x \neq 0$ an acceptable wavefunction on $(-\infty, \infty)$? Justify your answer.

Given: $\Psi(x) = 1/x$. **Find:** whether Ψ satisfies the requirements from earlier this unit (single-valued, finite, continuous, normalizable).

Step 1 — Check continuity at $x = 0$: $\Psi(x) = 1/x \rightarrow \pm\infty$ as $x \rightarrow 0^\pm$, and $\Psi(0)$ is undefined — Ψ is **not continuous** (indeed not even defined) at $x = 0$.

Step 2 — Check normalizability: set up $\int |\Psi|^2 dx$:

$$\int_{-\infty}^{\infty} \frac{1}{x^2} dx$$

Step 3 — Examine behavior near $x = 0$. As $x \rightarrow 0$, $1/x^2 \rightarrow \infty$ faster than $1/x$ diverges — specifically, $\int_{-\epsilon}^{\epsilon} x^{-2} dx$ diverges (the antiderivative $-1/x$ blows up at the limits) — so the integral diverges at $x = 0$, independent of behavior at large $|x|$.

Worked Example 3: Testing Acceptability of a Wavefunction

Conclusion: $\Psi(x) = 1/x$ is **not** an acceptable wavefunction — it fails both the continuity and normalizability requirements.

Worked Example 4: A Piecewise Wavefunction

Problem

Is $\Psi(x) = \begin{cases} A(L - |x|) & |x| \leq L \\ 0 & |x| > L \end{cases}$ (a triangular "tent" function) an acceptable wavefunction? If so, normalize it.

Given: the piecewise tent function above, with L a fixed length. **Find:** whether it's acceptable, and the value of A .

Step 1 — Check the required properties: single-valued ✓; finite ✓; continuous everywhere, including at $x = \pm L$ (where both pieces equal zero) ✓. The derivative has a finite jump only at $x = 0$ (slope changes from $+A$ to $-A$), which is acceptable since the corresponding potential is not infinite there.

Step 2 — Set up the normalization integral, splitting at $x = 0$ using the symmetry of $(L - |x|)^2$:

$$\int_{-L}^L A^2(L - |x|)^2 dx = 2A^2 \int_0^L (L - x)^2 dx$$

Worked Example 4: A Piecewise Wavefunction

Step 3 — Substitute $u = L - x$ ($du = -dx$; when $x = 0$, $u = L$; when $x = L$, $u = 0$), which flips and removes the limits' sign:

$$2A^2 \int_0^L (L - x)^2 dx = 2A^2 \int_0^L u^2 du = 2A^2 \left[\frac{u^3}{3} \right]_0^L = 2A^2 \cdot \frac{L^3}{3}$$

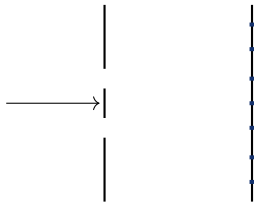
Step 4 — Set equal to 1 and solve for A :

$$\frac{2A^2 L^3}{3} = 1 \quad \Rightarrow \quad A^2 = \frac{3}{2L^3} \quad \Rightarrow \quad A = \sqrt{\frac{3}{2L^3}}$$

This tent-shaped wavefunction is a common simple model used to approximate more complicated bound-state shapes in introductory numerical exercises.

The Double-Slit Experiment

- Particles (electrons, photons) fired one at a time through two slits.
- Classical particle expectation: two bands on the screen.
- **Observed:** an interference pattern builds up, even when particles are sent through one at a time.
- Each particle interferes with *itself* — direct evidence of the wave nature encoded in Ψ .



"The Only Mystery" — Why This Experiment Matters

- Richard Feynman famously called the double-slit experiment the central mystery of quantum mechanics, stating it contains "the only mystery" and cannot be explained classically "in any way."
- The experiment cleanly demonstrates that quantum particles do not follow definite trajectories the way classical particles do: asking "which slit did the electron go through?" turns out not to have a well-defined answer unless we actually measure it.
- Crucially, the pattern emerges even when particles are sent through **one at a time**, with long gaps between them — ruling out any explanation involving particles somehow "interfering with each other" in transit.
- This single experiment packages together three of the deepest ideas in quantum theory: superposition (Unit 1), the Born probability rule (this unit), and the role of measurement in determining what behavior is observed.

Building Up the Pattern, One Particle at a Time

- If only a handful of electrons have been detected, the screen shows seemingly **random** individual dots — no discernible pattern, just as a classical particle picture might suggest.
- As more and more electrons accumulate (hundreds, then thousands), the dots gradually organize themselves into the familiar bright-and-dark interference fringes.
- This statistical buildup is the experimental signature of the Born rule in action: each individual electron's landing position is genuinely random, but the *probability* of landing at any given point is governed by $|\Psi|^2$, which contains the interference term.
- No single electron "knows" about the pattern; the pattern is a statistical statement about many identically-prepared trials — precisely the same logic used throughout Units 3–4 when computing expectation values and probabilities from a single particle's wavefunction.

Double-Slit: Interference Mathematics

If ψ_1 and ψ_2 are the amplitudes from slit 1 and slit 2, the total amplitude is their **superposition**:

$$\Psi = \psi_1 + \psi_2$$

The probability density is

$$|\Psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2 \operatorname{Re}(\psi_1^* \psi_2)$$

- The cross term $2 \operatorname{Re}(\psi_1^* \psi_2)$ is the **interference term** — absent in classical probability addition.
- If you place a detector at either slit to determine *which path* the particle took, the interference pattern **disappears** — measurement collapses the superposition.
- This is the heart of quantum complementarity (Bohr).

Derivation: Fringe Spacing — Setting Up the Geometry

- Let d be the slit separation, D the distance from the slits to the screen (with $D \gg d$, the standard far-field approximation), and λ the de Broglie wavelength of the particles.
- Consider a point on the screen at angle θ from the central axis. The two paths (from slit 1 and slit 2 to this point) differ in length by:

$$\delta = d \sin \theta$$

- For small angles (since $D \gg d$ means θ stays small for any point near the central region of the screen), use the small-angle approximation $\sin \theta \approx \theta$ (in radians):

$$\delta \approx d\theta$$

Derivation: Fringe Spacing — Phase Difference

- Write the two wave amplitudes reaching the screen as $\psi_1 \propto e^{ikr_1}$ and $\psi_2 \propto e^{ikr_2}$, where r_1, r_2 are the path lengths from each slit, and $k = 2\pi/\lambda$ is the wavenumber.
- The phase difference between the two waves at the screen is k times the path-length difference:

$$\Delta\phi = k \delta$$

- **Substitute** $k = 2\pi/\lambda$ and $\delta \approx d\theta$ from the previous slide:

$$\Delta\phi = \frac{2\pi}{\lambda} \times d\theta = \frac{2\pi d\theta}{\lambda}$$

Derivation: Fringe Spacing — Constructive Interference Condition

- **Constructive interference** (bright fringes) occurs when the phase difference is an integer multiple of 2π :

$$\Delta\phi = 2\pi n, \quad n = 0, \pm 1, \pm 2, \dots$$

- **Step 1:** substitute the expression for $\Delta\phi$ derived above:

$$\frac{2\pi d\theta_n}{\lambda} = 2\pi n$$

- **Step 2:** divide both sides by 2π :

$$\frac{d\theta_n}{\lambda} = n$$

- **Step 3:** solve for θ_n by multiplying both sides by λ/d :

$$\theta_n = \frac{n\lambda}{d}$$

This gives the angular position of the n -th bright fringe.

Derivation: Fringe Spacing — Converting to Screen Position

- Convert the angular position θ_n to a linear position y_n on the screen, using the small-angle relation $y_n = D \tan \theta_n \approx D\theta_n$ (valid since θ_n is small):

$$y_n = D\theta_n = \frac{n\lambda D}{d}$$

- **Fringe spacing** is the distance between adjacent bright fringes, $\Delta y = y_{n+1} - y_n$:

$$\Delta y = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d}$$

- **Factor out** the common term $\lambda D/d$:

$$\Delta y = \frac{\lambda D}{d} [(n+1) - n] = \frac{\lambda D}{d}$$

- This formula, originally derived for light, applies identically to electrons and other particles when λ is the de Broglie wavelength — direct evidence that the same wave mathematics governs matter.

Worked Example 1: Electron Double-Slit Fringe Spacing

Problem

Electrons of de Broglie wavelength $\lambda = 5 \times 10^{-11}$ m pass through slits separated by $d = 1 \mu\text{m}$, with a screen at $D = 1$ m. Find the fringe spacing.

Given: λ , d (slit separation), D (screen distance). **Find:** Δy .

Step 1 — Apply $\Delta y = \lambda D/d$:

$$\Delta y = \frac{(5 \times 10^{-11} \text{ m})(1 \text{ m})}{1 \times 10^{-6} \text{ m}}$$

Step 2 — Evaluate the numerator:

$$(5 \times 10^{-11})(1) = 5 \times 10^{-11} \text{ m}^2$$

Step 3 — Divide by d :

$$\Delta y = \frac{5 \times 10^{-11}}{10^{-6}} = 5 \times 10^{-5} \text{ m} = 50 \mu\text{m}$$

Small but experimentally measurable — such electron interference patterns have been directly observed.

Worked Example 2: Required Slit Separation

Problem

For visible light $\lambda = 600 \text{ nm}$ and screen distance $D = 2 \text{ m}$, what slit separation d gives a fringe spacing of $\Delta y = 3 \text{ mm}$?

Given: λ , D , Δy . **Find:** d .

Step 1 — Start from $\Delta y = \lambda D / d$ and solve for d by multiplying both sides by $d / \Delta y$:

$$d = \frac{\lambda D}{\Delta y}$$

Step 2 — Substitute numbers:

$$d = \frac{(600 \times 10^{-9} \text{ m})(2 \text{ m})}{3 \times 10^{-3} \text{ m}}$$

Step 3 — Evaluate the numerator:

$$(600 \times 10^{-9})(2) = 1.2 \times 10^{-6} \text{ m}^2$$

Worked Example 2: Required Slit Separation

Step 4 — Divide by Δy :

$$d = \frac{1.2 \times 10^{-6}}{3 \times 10^{-3}} = 4 \times 10^{-4} \text{ m} = 0.4 \text{ mm}$$

Worked Example 3: Which-Path Information and Visibility

Problem

Two slits produce equal-amplitude waves ψ_1, ψ_2 with $|\psi_1| = |\psi_2| = \psi_0$. If a partial which-path measurement leaves the cross term reduced to $2\eta \operatorname{Re}(\psi_1^* \psi_2)$ with "distinguishability" parameter $0 \leq \eta \leq 1$, find the fringe visibility $V = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$.

Given: $|\psi_1| = |\psi_2| = \psi_0$, reduced cross term $2\eta \operatorname{Re}(\psi_1^* \psi_2)$. **Find:** V .

Step 1 — Write the total intensity, $I = |\psi_1|^2 + |\psi_2|^2 + 2\eta \operatorname{Re}(\psi_1^* \psi_2)$. With $\psi_1 = \psi_0 e^{i\phi_1}$, $\psi_2 = \psi_0 e^{i\phi_2}$ and $\Delta\phi = \phi_1 - \phi_2$, $\operatorname{Re}(\psi_1^* \psi_2) = \psi_0^2 \cos \Delta\phi$:

$$I = \psi_0^2 + \psi_0^2 + 2\eta\psi_0^2 \cos \Delta\phi = 2\psi_0^2 + 2\eta\psi_0^2 \cos \Delta\phi$$

Step 2 — Find $I_{\max}$ (when $\cos \Delta\phi = +1$) and $I_{\min}$ (when $\cos \Delta\phi = -1$):

$$I_{\max} = 2\psi_0^2 + 2\eta\psi_0^2 = 2\psi_0^2(1 + \eta), \quad I_{\min} = 2\psi_0^2 - 2\eta\psi_0^2 = 2\psi_0^2(1 - \eta)$$

Worked Example 3: Which-Path Information and Visibility

Step 3 — Compute $I_{\max} - I_{\min}$ and $I_{\max} + I_{\min}$:

$$I_{\max} - I_{\min} = 2\psi_0^2(1 + \eta) - 2\psi_0^2(1 - \eta) = 4\eta\psi_0^2$$

$$I_{\max} + I_{\min} = 2\psi_0^2(1 + \eta) + 2\psi_0^2(1 - \eta) = 4\psi_0^2$$

Step 4 — Form the ratio $V = (I_{\max} - I_{\min})/(I_{\max} + I_{\min})$:

$$V = \frac{4\eta\psi_0^2}{4\psi_0^2} = \eta$$

Visibility directly equals the distinguishability parameter: **full which-path knowledge** ($\eta = 0$) means **zero interference visibility** — a clean quantitative statement of complementarity.

Worked Example 4: Comparing Photon and Electron Fringe Patterns

Problem

A double-slit setup ($d = 0.1 \text{ mm}$, $D = 1 \text{ m}$) is illuminated first by photons of wavelength 500 nm , then by electrons accelerated through $V = 50 \text{ V}$. Compare the two fringe spacings.

Given: d , D , $\lambda_{\text{photon}} = 500 \text{ nm}$, $V = 50 \text{ V}$ (electron accelerating voltage). **Find:** Δy_{photon} and $\Delta y_{\text{electron}}$.

Step 1 — Photon fringe spacing, applying $\Delta y = \lambda D/d$:

$$\Delta y_{\text{photon}} = \frac{(500 \times 10^{-9})(1)}{10^{-4}} = 5 \times 10^{-3} \text{ m} = 5 \text{ mm}$$

Step 2 — Electron de Broglie wavelength, using the shortcut $\lambda(\text{\AA}) \approx \sqrt{150/V}$ from Unit 1:

$$\lambda_e \approx \sqrt{150/50} \text{\AA} = \sqrt{3} \text{\AA} \approx 1.73 \text{\AA} = 1.73 \times 10^{-10} \text{ m}$$

Worked Example 4: Comparing Photon and Electron Fringe Patterns

Step 3 — Electron fringe spacing, applying $\Delta y = \lambda_e D/d$:

$$\Delta y_{\text{electron}} = \frac{(1.73 \times 10^{-10})(1)}{10^{-4}} \approx 1.73 \times 10^{-6} \text{ m} \approx 1.7 \mu\text{m}$$

Step 4 — Compare: ratio $\Delta y_{\text{photon}}/\Delta y_{\text{electron}} \approx 5 \times 10^{-3}/1.73 \times 10^{-6} \approx 2890$. The electron fringes are about 3000× narrower than the photon fringes — a direct consequence of the electron's much shorter de Broglie wavelength, and the reason electron-interference experiments demand far higher spatial resolution than optical ones.

Unit 2 Summary

- **Axioms** of quantum mechanics: wavefunctions, operators, eigenvalues, Schrödinger evolution, Born rule.
- **Heisenberg uncertainty principle**: $\Delta x \Delta p \geq \hbar/2$, illustrated via the Heisenberg microscope (derived quantitatively).
- **Conjugate variables** and the Fourier-transform relationship between position and momentum space; Gaussian saturates the bound.
- **Wavefunction**: physical interpretation, required mathematical properties, normalization.
- **Double-slit experiment**: superposition, interference, fringe spacing $\Delta y = \lambda D/d$, and the effect of measurement.

Additional Problem 1: Minimum Uncertainty in Velocity

Problem

A dust particle of mass 10^{-15} kg is localized to $\Delta x = 10^{-8}$ m. Find the minimum uncertainty in its velocity.

Given: $m = 10^{-15}$ kg, $\Delta x = 10^{-8}$ m. **Find:** $\Delta v_{\min}$.

Step 1 — Start from $\Delta x \Delta p \geq \hbar/2$, with $\Delta p = m\Delta v$, and solve for Δv by dividing both sides by $2m\Delta x$:

$$\Delta v \geq \frac{\hbar}{2m\Delta x}$$

Step 2 — Substitute numbers:

$$\Delta v \geq \frac{1.055 \times 10^{-34}}{2(10^{-15})(10^{-8})} = \frac{1.055 \times 10^{-34}}{2 \times 10^{-23}}$$

Step 3 — Evaluate:

$$\Delta v \geq 5.3 \times 10^{-12} \text{ m/s} \quad (\text{utterly negligible for a macroscopic dust particle})$$

Additional Problem 2: Normalization Constant for a Box State

Problem

Normalize $\Psi(x) = Ax(L - x)$ for $0 < x < L$.

Given: $\Psi(x) = Ax(L - x)$. **Find:** A .

Step 1 — Set up the normalization integral:

$$\int_0^L A^2 x^2 (L - x)^2 dx = 1$$

Step 2 — Expand $x^2(L - x)^2$ (a standard polynomial expansion):

$$x^2(L - x)^2 = x^2(L^2 - 2Lx + x^2) = L^2x^2 - 2Lx^3 + x^4$$

Step 3 — Integrate term by term using $\int_0^L x^n dx = L^{n+1}/(n+1)$:

$$\int_0^L (L^2x^2 - 2Lx^3 + x^4) dx = L^2 \cdot \frac{L^3}{3} - 2L \cdot \frac{L^4}{4} + \frac{L^5}{5} = \frac{L^5}{3} - \frac{L^5}{2} + \frac{L^5}{5}$$

Additional Problem 2: Normalization Constant for a Box State

Step 4 — Combine fractions using common denominator 30:

$$\frac{L^5}{3} - \frac{L^5}{2} + \frac{L^5}{5} = \frac{10L^5 - 15L^5 + 6L^5}{30} = \frac{L^5}{30}$$

Step 5 — Set $A^2 \times L^5/30 = 1$ and solve for A :

$$A = \sqrt{\frac{30}{L^5}}$$

Additional Problem 3: Energy–Time Uncertainty for a Fast Decay

Problem

A particle resonance has lifetime $\tau = 10^{-23}$ s (typical of strong-interaction decays). Estimate ΔE in MeV.

Given: $\tau = 10^{-23}$ s. **Find:** ΔE .

Step 1 — Apply $\Delta E \geq \hbar/(2\tau)$:

$$\Delta E \geq \frac{1.055 \times 10^{-34}}{2 \times 10^{-23}}$$

Step 2 — Evaluate:

$$\Delta E \geq 5.3 \times 10^{-12} \text{ J}$$

Step 3 — Convert to MeV (divide by 1.6×10^{-13} J/MeV):

$$\Delta E \geq \frac{5.3 \times 10^{-12}}{1.6 \times 10^{-13}} \approx 33 \text{ MeV}$$

Additional Problem 4: Fringe Spacing for Neutron Interferometry

Problem

Thermal neutrons have $\lambda = 0.2 \text{ nm}$. With $d = 50 \mu\text{m}$ and $D = 5 \text{ m}$, find the fringe spacing.

Given: λ , d , D . **Find:** Δy .

Step 1 — Apply $\Delta y = \lambda D / d$:

$$\Delta y = \frac{(0.2 \times 10^{-9} \text{ m})(5 \text{ m})}{50 \times 10^{-6} \text{ m}}$$

Step 2 — Evaluate the numerator:

$$(0.2 \times 10^{-9})(5) = 1.0 \times 10^{-9} \text{ m}^2$$

Step 3 — Divide:

$$\Delta y = \frac{1.0 \times 10^{-9}}{50 \times 10^{-6}} = 2 \times 10^{-5} \text{ m} = 20 \mu\text{m}$$

Additional Problem 5: Probability in Outer Region of a Gaussian

Problem

For $\Psi(x) = (1/\pi a^2)^{1/4} e^{-x^2/2a^2}$, what fraction of the probability lies beyond $|x| > 2a$? (Use the known Gaussian tail value $\text{erfc}(2) \approx 0.00468$.)

Given: normalized Gaussian with width parameter a . **Find:** $P(|x| > 2a)$.

Step 1 — Recognize this as a standard normal tail probability. Since $|\Psi|^2 \propto e^{-x^2/a^2}$, comparing to the standard Gaussian form $e^{-x^2/2\sigma^2}$ gives $\sigma = a/\sqrt{2}$.

Step 2 — Express the tail probability using the complementary error function, which for a Gaussian gives $P(|x| > X) = \text{erfc}(X/(\sigma\sqrt{2}))$. With $X = 2a$ and $\sigma = a/\sqrt{2}$:

$$\frac{X}{\sigma\sqrt{2}} = \frac{2a}{(a/\sqrt{2})\sqrt{2}} = \frac{2a}{a} = 2$$

Step 3 — Apply the given value:

$$P(|x| > 2a) = \text{erfc}(2) \approx 0.00468$$

Additional Problem 5: Probability in Outer Region of a Gaussian

Step 4 — This is a two-sided probability already, so no further factor of 2 is needed:

$$P(|x| > 2a) \approx 0.0047 \text{ (about 0.5\%)}$$

This is small, as expected: at $|x| = 2a$, the position is already $2\sqrt{2} \approx 2.8$ standard deviations from the center, well into the Gaussian's rapidly decaying tail.

Additional Problem 6: Checking Normalizability

Problem

Is $\psi(x) = e^{-|x|} \sin(x)$ square-integrable on $(-\infty, \infty)$?

Given: $\psi(x) = e^{-|x|} \sin(x)$. **Find:** whether $\int |\psi|^2 dx$ is finite.

Step 1 — Bound $|\psi|^2$ from above. Since $|\sin(x)| \leq 1$ for all x :

$$|\psi(x)|^2 = e^{-2|x|} \sin^2(x) \leq e^{-2|x|}$$

Step 2 — Check that the bounding function is integrable, using $\int_{-\infty}^{\infty} e^{-2|x|} dx = 2 \int_0^{\infty} e^{-2x} dx = 2 \times \frac{1}{2} = 1$ (finite).

Step 3 — Apply the comparison test. Since $0 \leq |\psi(x)|^2 \leq e^{-2|x|}$ and $\int e^{-2|x|} dx = 1 < \infty$, the integral $\int |\psi(x)|^2 dx$ is also finite.

Conclusion: yes, $\psi(x) = e^{-|x|} \sin(x)$ is square-integrable (and hence normalizable after dividing by an appropriate constant).

Additional Problem 7: Heisenberg Bound for a Macroscopic Pendulum

Problem

A 1 g pendulum bob is located to within $\Delta x = 1 \mu\text{m}$. Find $\Delta v_{\min}$ and comment.

Given: $m = 1 \text{ g} = 10^{-3} \text{ kg}$, $\Delta x = 10^{-6} \text{ m}$. **Find:** $\Delta v_{\min}$.

Step 1 — Apply $\Delta v_{\min} = \hbar/(2m\Delta x)$:

$$\Delta v_{\min} = \frac{1.055 \times 10^{-34}}{2(10^{-3})(10^{-6})} = \frac{1.055 \times 10^{-34}}{2 \times 10^{-9}}$$

Step 2 — Evaluate:

$$\Delta v_{\min} \approx 5.3 \times 10^{-26} \text{ m/s} \quad (\text{far below any measurable scale})$$

This confirms that the uncertainty principle has no practical consequence for everyday macroscopic objects, consistent with the correspondence principle.

Additional Problem 8: Angle–Angular-Momentum Uncertainty

Problem

If $\Delta\theta = 0.01$ rad for a rotating molecule, estimate ΔL_z .

Given: $\Delta\theta = 0.01$ rad. **Find:** ΔL_z .

Step 1 — Apply the angle–angular-momentum uncertainty relation, $\Delta\theta \Delta L_z \geq \hbar/2$, and solve for ΔL_z by dividing both sides by $2\Delta\theta$:

$$\Delta L_z \geq \frac{\hbar}{2\Delta\theta}$$

Step 2 — Substitute numbers:

$$\Delta L_z \geq \frac{1.055 \times 10^{-34}}{2(0.01)} = \frac{1.055 \times 10^{-34}}{0.02}$$

Step 3 — Evaluate:

$$\Delta L_z \geq 5.3 \times 10^{-33} \text{ J s}$$

Additional Problem 9: Number of Fringes Observed

Problem

In a double-slit setup with $\lambda = 500 \text{ nm}$, $d = 0.2 \text{ mm}$, $D = 1.5 \text{ m}$, and a screen of width 2 cm centered on the axis, estimate the number of bright fringes visible.

Given: λ , d , D , screen width $w = 2 \text{ cm}$. **Find:** approximate fringe count N .

Step 1 — Find the fringe spacing $\Delta y = \lambda D/d$:

$$\Delta y = \frac{(500 \times 10^{-9})(1.5)}{0.2 \times 10^{-3}} = \frac{7.5 \times 10^{-7}}{2 \times 10^{-4}} = 3.75 \times 10^{-3} \text{ m}$$

Step 2 — Divide the screen width by the fringe spacing to estimate how many fringes fit:

$$N \approx \frac{w}{\Delta y} = \frac{2 \times 10^{-2} \text{ m}}{3.75 \times 10^{-3} \text{ m}}$$

Step 3 — Evaluate:

$$N \approx 5.3 \implies \text{about 5–6 fringes visible across the screen}$$

Additional Problem 10: Localization Energy Estimate via Uncertainty

Problem

Estimate the minimum kinetic energy of an electron confined to a region of size 0.5 nm using the uncertainty principle.

Given: $\Delta x = 0.5 \text{ nm}$, m_e . **Find:** $KE_{\min}$.

Step 1 — Find the momentum uncertainty $\Delta p \geq \hbar/(2\Delta x)$:

$$\Delta p \geq \frac{1.055 \times 10^{-34}}{2(0.5 \times 10^{-9})} = \frac{1.055 \times 10^{-34}}{10^{-9}} \approx 1.05 \times 10^{-25} \text{ kg m/s}$$

Step 2 — Estimate the kinetic energy using $KE \sim (\Delta p)^2/(2m_e)$ (treating Δp as the momentum scale):

$$KE_{\min} \sim \frac{(1.05 \times 10^{-25})^2}{2(9.11 \times 10^{-31})} = \frac{1.10 \times 10^{-50}}{1.822 \times 10^{-30}}$$

Additional Problem 10: Localization Energy Estimate via Uncertainty

Step 3 — Evaluate:

$$KE_{\min} \approx 6.1 \times 10^{-21} \text{ J}$$

Step 4 — Convert to eV (divide by $1.6 \times 10^{-19} \text{ J/eV}$):

$$KE_{\min} \approx \frac{6.1 \times 10^{-21}}{1.6 \times 10^{-19}} \approx 0.038 \text{ eV}$$

Unit 3: Basics of Schrödinger Equation I

1. Origin of Quantum Mechanics
2. Formulation of Quantum Mechanics
3. Basics of Schrödinger Equation I
4. Basics of Schrödinger Equation II
5. Dirac Notation in Quantum Mechanics

Unit 3: Symbol Glossary

- $\rho(x, t)$ — probability density (m^{-1})
- $\Psi(x, t)$ — time-dependent wavefunction
- $\psi(x)$ — spatial (time-independent) part of Ψ
- E — energy eigenvalue (J)
- $\langle A \rangle$ — expectation value of observable A
- $\hat{A}$ — operator for observable A
- a_n — eigenvalue of operator $\hat{A}$ in eigenstate ψ_n
- $\hat{H}$ — Hamiltonian (total energy) operator (J)
- $V(x, t)$ — potential energy function (J)
- i — imaginary unit, $i^2 = -1$
- $j(x, t)$ — probability current density (s^{-1})
- $[\hat{A}, \hat{B}]$ — commutator of two operators
- L — length scale (e.g. box width) (m)
- n — quantum number (integer)
- τ — decay lifetime (s)
- α — Gaussian width parameter (m^{-2}) in WE2 (not the delta-potential strength of Unit 4)

Symbols specific to a single derivation are defined where they first appear.

From Postulates to Calculation

- Unit 2 established the formal machinery: wavefunctions, the Born rule, required mathematical properties. This unit turns that machinery into concrete calculational tools.
- Three central questions drive this unit: *Where* is the particle likely to be found ($|\Psi|^2$)? *What* do we expect to measure for any observable ($\langle A \rangle$)? *How* does the system evolve in time (Schrödinger's equation)?
- A recurring theme is the special role of **stationary states** — energy eigenstates whose probability distribution never changes in time, even though the underlying wavefunction does. These will be the actual states we solve for explicitly in Unit 4 (box, oscillator, hydrogen atom).
- By the end of this unit, the abstract axioms become a working toolkit: given any $V(x)$, we will know exactly how to find the allowed energies and the probability of measuring any observable.

Probability Amplitude, Probability Density, Probability

- **Probability amplitude:** the wavefunction $\Psi(x, t)$ itself (complex-valued).

- **Probability density:**

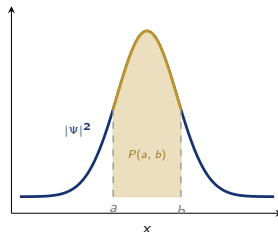
$$\rho(x, t) = |\Psi(x, t)|^2 = \Psi^*(x, t)\Psi(x, t)$$

- **Probability** of finding the particle in $[a, b]$:

$$P(a \leq x \leq b) = \int_a^b |\Psi|^2 dx$$

- **Normalization condition:**

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$



Worked Example 1: Probability in a Sub-region

Problem

A particle has normalized wavefunction $\psi(x) = \sqrt{2/L} \sin(\pi x/L)$ on $[0, L]$. Find the probability of finding it in $[L/3, 2L/3]$.

Given: normalized $\psi(x)$. **Find:** $P(L/3 \leq x \leq 2L/3)$.

Step 1 — Set up the integral with $|\psi|^2 = \frac{2}{L} \sin^2(\pi x/L)$:

$$P = \int_{L/3}^{2L/3} \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right) dx$$

Step 2 — Apply the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ and integrate:

$$P = \frac{1}{L} \left[x - \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_{L/3}^{2L/3}$$

Worked Example 1: Probability in a Sub-region

Step 3 — Evaluate at $x = 2L/3$: linear term gives $2L/3$; sine argument is $2\pi(2L/3)/L = 4\pi/3$, so $\sin(4\pi/3) = -\sqrt{3}/2$.

Step 4 — Evaluate at $x = L/3$: linear term gives $L/3$; sine argument is $2\pi(L/3)/L = 2\pi/3$, so $\sin(2\pi/3) = +\sqrt{3}/2$.

Step 5 — Subtract (upper minus lower):

$$P = \frac{1}{L} \left[\left(\frac{2L}{3} - \frac{L}{2\pi} \left(-\frac{\sqrt{3}}{2} \right) \right) - \left(\frac{L}{3} - \frac{L}{2\pi} \cdot \frac{\sqrt{3}}{2} \right) \right]$$

Step 6 — Simplify (the two $\sqrt{3}/2$ terms have the *same* sign after subtracting a negative, so they add):

$$P = \frac{1}{L} \left[\frac{L}{3} + \frac{L\sqrt{3}}{2\pi} \right] = \frac{1}{3} + \frac{\sqrt{3}}{2\pi} \approx 0.333 + 0.276 \approx 0.609$$

Worked Example 2: Checking Normalization

Problem

Verify that $\psi(x) = \sqrt{2/a} \sin(n\pi x/a)$ is normalized on $[0, a]$ for any integer n .

Given: $\psi(x) = \sqrt{2/a} \sin(n\pi x/a)$, n a positive integer, a a length. **Find:** verify $\int_0^a |\psi|^2 dx = 1$.

Step 1 — Write out the integral:

$$\int_0^a \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right) dx$$

Step 2 — Apply the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ with $\theta = n\pi x/a$:

$$= \frac{2}{a} \int_0^a \frac{1}{2} \left[1 - \cos\left(\frac{2n\pi x}{a}\right) \right] dx = \frac{1}{a} \int_0^a \left[1 - \cos\left(\frac{2n\pi x}{a}\right) \right] dx$$

Worked Example 2: Checking Normalization

Step 3 — Integrate term by term. The constant term gives a ; the cosine term, over $x = 0$ to a , completes exactly n full periods (since the argument runs from 0 to $2n\pi$), so it integrates to exactly 0:

$$= \frac{1}{a} [a - 0] = 1 \checkmark$$

Normalization holds for every n — this is the full family of infinite-well eigenstates revisited in Unit 4.

Worked Example 3: Time Evolution of Probability Density

Problem

Show explicitly that for a single stationary state $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, the probability density $|\Psi|^2$ does not depend on t .

Given: $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$. **Find:** show $|\Psi|^2$ is time-independent.

Step 1 — Write the complex conjugate. Since $\psi(x)$ may itself be complex, conjugate both factors; the phase factor's exponent flips sign:

$$\Psi^*(x, t) = \psi^*(x) e^{+iEt/\hbar}$$

Step 2 — Multiply Ψ^* by Ψ :

$$|\Psi(x, t)|^2 = \Psi^* \Psi = \left[\psi^*(x) e^{+iEt/\hbar} \right] \left[\psi(x) e^{-iEt/\hbar} \right]$$

Worked Example 3: Time Evolution of Probability Density

Step 3 — Regroup the factors (spatial parts together, exponential parts together):

$$= \psi^*(x)\psi(x) \times e^{iEt/\hbar}e^{-iEt/\hbar} = |\psi(x)|^2 \times e^{i(E-E)t/\hbar}$$

Step 4 — Simplify the exponent, $E - E = 0$, so $e^0 = 1$:

$$|\Psi(x, t)|^2 = |\psi(x)|^2 \times 1 = |\psi(x)|^2$$

The time-dependent phases cancel exactly. This explicit cancellation is why $|\Psi|^2$ is "stationary" even though Ψ itself oscillates with angular frequency $E/\hbar$.

Worked Example 4: Probability Density for a Superposition State

Problem

For $\Psi(x, t) = \frac{1}{\sqrt{2}} [\psi_1(x)e^{-iE_1t/\hbar} + \psi_2(x)e^{-iE_2t/\hbar}]$ (an equal superposition of two stationary states), show that $|\Psi|^2$ does depend on time, unlike a single stationary state.

Given: a superposition of two energy eigenstates ψ_1, ψ_2 with energies E_1, E_2 . **Find:** $|\Psi|^2$, and show it is time-dependent.

Step 1 — Write the conjugate $\Psi^* = \frac{1}{\sqrt{2}}[\psi_1^*e^{iE_1t/\hbar} + \psi_2^*e^{iE_2t/\hbar}]$, and form $\Psi^*\Psi$, expanding like $(a + b)(c + d)$:

$$|\Psi|^2 = \frac{1}{2} \left[\psi_1^*\psi_1 + \psi_1^*\psi_2 e^{-i(E_2-E_1)t/\hbar} + \psi_2^*\psi_1 e^{i(E_2-E_1)t/\hbar} + \psi_2^*\psi_2 \right]$$

Step 2 — Identify the first and last terms as $|\psi_1|^2$ and $|\psi_2|^2$ (time-independent).

Worked Example 4: Probability Density for a Superposition State

Step 3 — Combine the two cross terms, which are complex conjugates of each other ($z + z^* = 2 \operatorname{Re}(z)$):

$$\psi_1^* \psi_2 e^{-i(E_2 - E_1)t/\hbar} + \left(\psi_1^* \psi_2 e^{-i(E_2 - E_1)t/\hbar} \right)^* = 2 \operatorname{Re} \left[\psi_1^* \psi_2 e^{-i(E_2 - E_1)t/\hbar} \right]$$

Step 4 — Assemble the full result:

$$|\Psi|^2 = \frac{1}{2} \left[|\psi_1|^2 + |\psi_2|^2 + 2 \operatorname{Re} \left(\psi_1^* \psi_2 e^{-i(E_2 - E_1)t/\hbar} \right) \right]$$

The cross term oscillates in time with angular frequency $(E_2 - E_1)/\hbar$ — unlike a pure stationary state, this superposition has a genuinely time-dependent probability density. This "quantum beating" between energy levels is the basic mechanism behind phenomena like atomic transitions and Rabi oscillations.

Stationary States

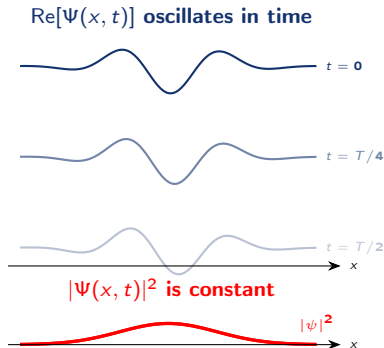
- A **stationary state** has the form

$$\Psi(x, t) = \psi(x) e^{-iEt/\hbar}$$

- The probability density is **time-independent**:

$$|\Psi|^2 = |\psi(x)|^2$$

- Despite the name, Ψ still oscillates in time via its phase — only *observable* probabilities are static.
- Energy eigenstates of $\hat{H}$ are stationary states: the basic building blocks for bound-state problems.



Definition

For an observable represented by operator $\hat{A}$, its expectation value in state Ψ is

$$\langle A \rangle = \int_{-\infty}^{\infty} \Psi^*(x, t) \hat{A} \Psi(x, t) dx$$

- Represents the average of many measurements of A on identically prepared systems.
- Examples:

$$\langle x \rangle = \int \Psi^* x \Psi dx, \quad \langle p \rangle = \int \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx$$

- Variance / uncertainty: $(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2$.

Worked Example 1: $\langle x \rangle$ for a Box State

Problem

For $\psi(x) = \sqrt{2/L} \sin(\pi x/L)$ on $[0, L]$, find $\langle x \rangle$.

Given: normalized box ground state $\psi(x)$, where L is the box width. **Find:** $\langle x \rangle$.

Step 1 — Write the definition of $\langle x \rangle$:

$$\langle x \rangle = \int_0^L x |\psi(x)|^2 dx = \int_0^L x \cdot \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right) dx$$

Step 2 — Pull the constant $2/L$ out front:

$$\langle x \rangle = \frac{2}{L} \int_0^L x \sin^2\left(\frac{\pi x}{L}\right) dx$$

Worked Example 1: $\langle x \rangle$ for a Box State

Step 3 — Apply the standard integral (obtained via the half-angle identity and integration by parts)

$$\int_0^L x \sin^2(\pi x/L) dx = L^2/4:$$

$$\langle x \rangle = \frac{2}{L} \times \frac{L^2}{4}$$

Step 4 — Simplify:

$$\langle x \rangle = \frac{2L^2}{4L} = \frac{L}{2}$$

By the left-right symmetry of $|\psi|^2$ about $x = L/2$, this result $\langle x \rangle = L/2$ could also have been anticipated directly.

Worked Example 2: Expectation Value of Momentum

Problem

For any real wavefunction $\psi(x)$ (e.g. the box state above), show $\langle p \rangle = 0$.

Given: $\psi(x)$ real-valued. **Find:** show $\langle p \rangle = 0$.

Step 1 — Write the definition, using $\hat{p} = -i\hbar d/dx$:

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi(x) \left(-i\hbar \frac{d\psi}{dx} \right) dx = -i\hbar \int_{-\infty}^{\infty} \psi \psi' dx$$

Step 2 — Recognize the integrand as a total derivative. By the product rule, $\frac{d}{dx}(\psi^2) = 2\psi\psi'$, so

$$\psi\psi' = \frac{1}{2} \frac{d}{dx}(\psi^2):$$

$$\langle p \rangle = -i\hbar \int_{-\infty}^{\infty} \frac{1}{2} \frac{d}{dx}(\psi^2) dx = -\frac{i\hbar}{2} \left[\psi^2 \right]_{-\infty}^{\infty}$$

Worked Example 2: Expectation Value of Momentum

Step 3 — Apply the boundary condition. Since ψ must vanish at $x \rightarrow \pm\infty$ for a normalizable bound state (Unit 2 requirement), $\psi^2 \rightarrow 0$ at both limits:

$$\langle p \rangle = -\frac{i\hbar}{2} [0 - 0] = 0$$

General fact: any real (or constant-phase) wavefunction has zero average momentum — it carries no net direction of flow.

Worked Example 3: Expectation Value of Kinetic Energy

Problem

For $\psi(x) = \sqrt{2/L} \sin(\pi x/L)$, compute $\langle KE \rangle = \langle \hat{p}^2 \rangle / 2m$ directly, and check against $E_1 = \pi^2 \hbar^2 / 2mL^2$.

Given: box ground state $\psi(x)$, mass m . **Find:** $\langle KE \rangle$.

Step 1 — Compute $\hat{p}^2\psi = -\hbar^2\psi''$. First find ψ' :

$$\psi'(x) = \sqrt{\frac{2}{L}} \cdot \frac{\pi}{L} \cos\left(\frac{\pi x}{L}\right)$$

Step 2 — Differentiate again to find ψ'' :

$$\psi''(x) = -\sqrt{\frac{2}{L}} \left(\frac{\pi}{L}\right)^2 \sin\left(\frac{\pi x}{L}\right) = -\left(\frac{\pi}{L}\right)^2 \psi(x)$$

Worked Example 3: Expectation Value of Kinetic Energy

Step 3 — Apply $\hat{p}^2 = -\hbar^2 d^2/dx^2$:

$$\hat{p}^2\psi = -\hbar^2\psi'' = -\hbar^2 \left[-\left(\frac{\pi}{L}\right)^2 \psi \right] = \hbar^2 \left(\frac{\pi}{L}\right)^2 \psi$$

so ψ is an eigenfunction of $\hat{p}^2$ with eigenvalue $\hbar^2\pi^2/L^2$.

Step 4 — Compute $\langle KE \rangle$. Since ψ is an eigenfunction, $\langle \hat{p}^2 \rangle$ equals the eigenvalue times $\int |\psi|^2 dx = 1$ (normalization):

$$\langle KE \rangle = \frac{1}{2m} \cdot \frac{\hbar^2\pi^2}{L^2} \times 1 = \frac{\pi^2\hbar^2}{2mL^2} = E_1 \quad \checkmark$$

Matches the ground-state energy of the infinite well exactly, since $V = 0$ inside the well so $E_1 = \langle KE \rangle$ entirely.

Worked Example 4: Expectation Value and Standard Deviation of Energy

Problem

A particle is in the state $\Psi = \frac{3}{5}\psi_1 + \frac{4}{5}\psi_2$, a superposition of energy eigenstates with $E_1 = 2\text{ eV}$, $E_2 = 8\text{ eV}$. Find $\langle E \rangle$ and ΔE .

Given: coefficients $c_1 = 3/5$, $c_2 = 4/5$; E_1, E_2 . **Find:** $\langle E \rangle$, ΔE .

Step 1 — Find the measurement probabilities, $P_n = |c_n|^2$ (Axiom 3, Unit 2):

$$P_1 = (3/5)^2 = 9/25 = 0.36, \quad P_2 = (4/5)^2 = 16/25 = 0.64$$

Step 2 — Verify normalization: $P_1 + P_2 = 0.36 + 0.64 = 1 \checkmark$.

Step 3 — Compute $\langle E \rangle = \sum_n P_n E_n$:

$$\langle E \rangle = (0.36)(2) + (0.64)(8) = 0.72 + 5.12 = 5.84\text{ eV}$$

Worked Example 4: Expectation Value and Standard Deviation of Energy

Step 4 — Compute $\langle E^2 \rangle = \sum_n P_n E_n^2$:

$$\langle E^2 \rangle = (0.36)(2^2) + (0.64)(8^2) = (0.36)(4) + (0.64)(64) = 1.44 + 40.96 = 42.4 \text{ eV}^2$$

Step 5 — Compute $\langle E \rangle^2$: $(5.84)^2 \approx 34.1 \text{ eV}^2$.

Step 6 — Apply $\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2}$:

$$\Delta E = \sqrt{42.4 - 34.1} = \sqrt{8.3} \approx 2.88 \text{ eV}$$

Unlike a pure stationary state (where $\Delta E = 0$ exactly), a superposition of different energy eigenstates has genuine energy uncertainty — a direct illustration of Axiom 3 (Unit 2): measuring E here would yield *either* 2 eV or 8 eV, never 5.84 eV itself.

Eigenfunctions and Eigenvalues

Eigenvalue Equation

$$\hat{A} \psi_n(x) = a_n \psi_n(x)$$

- ψ_n is an **eigenfunction** of operator $\hat{A}$ with **eigenvalue** a_n .
- A measurement of observable A *always* yields one of its eigenvalues a_n .
- If the system is in eigenstate ψ_n , the measurement yields a_n with certainty.
- Hermitian operators (representing observables) have **real** eigenvalues and **orthogonal** eigenfunctions — developed fully in Unit 5.



operator returns the
same function, only
rescaled by a_n

Quantum Mechanical Operators: A Dictionary

Position, Momentum, and Energy Operators

Observable	Operator	Position-space form
Position	$\hat{x}$	$\hat{x} \psi = x \psi$
Momentum	$\hat{p}$	$\hat{p} \psi = -i\hbar \frac{d\psi}{dx}$
Kinetic energy	$\hat{K}$	$\hat{K} \psi = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2}$
Potential energy	$\hat{V}$	$\hat{V} \psi = V(x) \psi$
Total energy (Hamiltonian)	$\hat{H}$	$\hat{H} \psi = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x) \psi$

- Each operator is obtained from the corresponding classical quantity by the **correspondence rule** $x \rightarrow \hat{x}$, $p \rightarrow \hat{p} = -i\hbar d/dx$, and $E \rightarrow \hat{E} = i\hbar \partial/\partial t$ (full motivation given later in this unit).

Quantum Mechanical Operators: A Dictionary

- Acting on a generic function, an operator returns another function; when it happens to return a **constant times the original function**, that function is an eigenfunction of the operator (previous frame).
- $\hat{H} = \hat{K}E + \hat{V} = \frac{\hat{p}^2}{2m} + V(\hat{x})$, consistent with the classical relation $E = \frac{p^2}{2m} + V(x)$.
- All operators above are **linear** and **Hermitian**, so their eigenvalues (measurement outcomes) are guaranteed real — proved in general in Unit 5.
- These definitions are used directly in the two worked examples that follow.

Worked Example: Eigenfunctions of the Momentum Operator

Problem

Show that $\psi(x) = e^{ikx}$ is an eigenfunction of $\hat{p} = -i\hbar d/dx$, and find its eigenvalue.

Given: $\psi(x) = e^{ikx}$, where k is the wavenumber. **Find:** the eigenvalue of $\hat{p}$ acting on ψ .

Step 1 — Apply the operator $\hat{p} = -i\hbar d/dx$ to ψ :

$$\hat{p}\psi = -i\hbar \frac{d}{dx} e^{ikx}$$

Step 2 — Differentiate e^{ikx} with respect to x using the chain rule ($d(e^{cx})/dx = ce^{cx}$ with $c = ik$):

$$\frac{d}{dx} e^{ikx} = ik e^{ikx}$$

Step 3 — Substitute back:

$$\hat{p}\psi = -i\hbar(ik)e^{ikx} = -i^2\hbar k e^{ikx}$$

Worked Example: Eigenfunctions of the Momentum Operator

Step 4 — Simplify using $-i^2 = -(-1) = 1$:

$$\hat{p}\psi = \hbar k e^{ikx} = \hbar k \psi$$

Since the result is a constant ($\hbar k$) times the original function, ψ is an eigenfunction with eigenvalue $p = \hbar k$ — consistent with the de Broglie relation $p = h/\lambda = \hbar(2\pi/\lambda) = \hbar k$ from Unit 1.

Worked Example: A Non-Eigenfunction

Problem

Is $\psi(x) = \cos(kx)$ an eigenfunction of $\hat{p} = -i\hbar d/dx$? If not, express it in terms of states that are.

Given: $\psi(x) = \cos(kx)$. **Find:** test whether it is a momentum eigenfunction.

Step 1 — Apply $\hat{p}$ to ψ :

$$\hat{p} \cos(kx) = -i\hbar \frac{d}{dx} \cos(kx)$$

Step 2 — Differentiate, using $d(\cos cx)/dx = -c \sin cx$:

$$\frac{d}{dx} \cos(kx) = -k \sin(kx)$$

Step 3 — Substitute back:

$$\hat{p} \cos(kx) = -i\hbar [-k \sin(kx)] = i\hbar k \sin(kx)$$

Worked Example: A Non-Eigenfunction

Step 4 — Compare to a constant times $\cos(kx)$. The result $i\hbar k \sin(kx)$ is *not* proportional to $\cos(kx)$ (sine and cosine are linearly independent functions), so:

$$\hat{p} \cos(kx) \neq (\text{const}) \times \cos(kx)$$

So $\cos(kx)$ is **not** a momentum eigenfunction.

Step 5 — Decompose using Euler's formula, $\cos(kx) = \frac{1}{2}(e^{ikx} + e^{-ikx})$ — a superposition of two momentum eigenstates with eigenvalues $+\hbar k$ and $-\hbar k$. A measurement of momentum on this state would yield $+\hbar k$ or $-\hbar k$, each with probability $\frac{1}{2}$, but never an intermediate value.

Worked Example: Eigenfunctions of the Energy Operator

Problem

Show that $\psi(x) = \sqrt{2/L} \sin(n\pi x/L)$ is an eigenfunction of $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ (free particle inside $[0, L]$), and find its eigenvalue.

Given: $\psi(x) = \sqrt{2/L} \sin(n\pi x/L)$, with L the box width, n an integer, m the particle mass. **Find:** eigenvalue of $\hat{H}$.

Step 1 — Find $\psi'(x)$, differentiating the sine term ($d(\sin cx)/dx = c \cos cx$ with $c = n\pi/L$):

$$\psi'(x) = \sqrt{\frac{2}{L}} \cdot \frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right)$$

Step 2 — Find $\psi''(x)$, differentiating again ($d(\cos cx)/dx = -c \sin cx$):

$$\psi''(x) = -\sqrt{\frac{2}{L}} \left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right) = -\left(\frac{n\pi}{L}\right)^2 \psi(x)$$

Worked Example: Eigenfunctions of the Energy Operator

Step 3 — Apply $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$:

$$\hat{H}\psi = -\frac{\hbar^2}{2m}\psi'' = -\frac{\hbar^2}{2m} \left[-\left(\frac{n\pi}{L}\right)^2 \psi \right]$$

Step 4 — Simplify the double negative:

$$\hat{H}\psi = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L}\right)^2 \psi = \frac{n^2\pi^2\hbar^2}{2mL^2} \psi$$

So ψ is an energy eigenfunction with eigenvalue $E_n = n^2\pi^2\hbar^2/2mL^2$ — exactly the infinite-well energy formula that will be derived from boundary conditions in Unit 4.

Derivation: Ehrenfest's Theorem — Setup

- Start from the definition of an expectation value and differentiate with respect to time:

$$\frac{d\langle x \rangle}{dt} = \frac{d}{dt} \int \Psi^* x \Psi dx = \int \left(\frac{\partial \Psi^*}{\partial t} x \Psi + \Psi^* x \frac{\partial \Psi}{\partial t} \right) dx$$

- Use the TDSE and its conjugate to replace the time derivatives:

$$\frac{\partial \Psi}{\partial t} = \frac{1}{i\hbar} \hat{H} \Psi, \quad \frac{\partial \Psi^*}{\partial t} = -\frac{1}{i\hbar} \hat{H} \Psi^*$$

- Substituting** both expressions into the differentiated integral:

$$\frac{d\langle x \rangle}{dt} = \int \left[\left(-\frac{1}{i\hbar} \hat{H} \Psi^* \right) x \Psi + \Psi^* x \left(\frac{1}{i\hbar} \hat{H} \Psi \right) \right] dx = \frac{1}{i\hbar} \int \left(\Psi^* x \hat{H} \Psi - (\hat{H} \Psi^*) x \Psi \right) dx$$

Derivation: Ehrenfest's Theorem — Setup

- Recognise that $\int (\hat{H}\Psi^*) x\Psi dx = \int \Psi^* \hat{H}(x\Psi) dx$ (since $\hat{H}$ is Hermitian), so the integrand becomes $\Psi^*[x\hat{H} - \hat{H}x]\Psi = \Psi^*[x, \hat{H}]\Psi$:

$$\frac{d\langle x \rangle}{dt} = \frac{1}{i\hbar} \int \Psi^* [x, \hat{H}] \Psi dx$$

where $[x, \hat{H}] = x\hat{H} - \hat{H}x$.

Derivation: Ehrenfest's Theorem — Result

- **Split the commutator.** Since $\hat{H} = \hat{p}^2/2m + V(x)$ and the commutator is linear in its second argument, $[x, \hat{H}] = \frac{1}{2m}[x, \hat{p}^2] + [x, V(x)]$.
- **The potential term vanishes.** Both x and $V(x)$ act on a wavefunction simply by multiplying it by a function of x , and ordinary multiplication commutes ($x \cdot V(x)\psi = V(x) \cdot x\psi$), so $[x, V(x)] = 0$. This leaves only the kinetic term:

$$[x, \hat{H}] = \frac{1}{2m}[x, \hat{p}^2]$$

- **Expand $[x, \hat{p}^2]$ using the product rule for commutators,** $[x, \hat{p}^2] = [x, \hat{p} \hat{p}] = \hat{p}[x, \hat{p}] + [x, \hat{p}]\hat{p}$ (a general identity proved rigorously in Unit 5):

$$[x, \hat{p}^2] = \hat{p}[x, \hat{p}] + [x, \hat{p}]\hat{p}$$

- **Substitute the canonical commutator** $[x, \hat{p}] = i\hbar$ (also derived rigorously in Unit 5) into *both* occurrences:

$$[x, \hat{p}^2] = \hat{p}(i\hbar) + (i\hbar)\hat{p}$$

Derivation: Ehrenfest's Theorem — Result

- **Simplify.** Since $i\hbar$ is just a complex number, it commutes freely with the operator $\hat{p}$, so both terms equal $i\hbar\hat{p}$:

$$[x, \hat{p}^2] = i\hbar\hat{p} + i\hbar\hat{p} = 2i\hbar\hat{p}$$

- **Substitute this back** into $[x, \hat{H}] = \frac{1}{2m}[x, \hat{p}^2]$:

$$[x, \hat{H}] = \frac{1}{2m}(2i\hbar\hat{p})$$

- **Cancel the common factor of 2** between the numerator and the $2m$ in the denominator:

$$[x, \hat{H}] = \frac{i\hbar}{m}\hat{p}$$

- Therefore:

$$\frac{d\langle x \rangle}{dt} = \frac{1}{i\hbar} \cdot \frac{i\hbar}{m} \langle \hat{p} \rangle = \frac{\langle p \rangle}{m}$$

Derivation: Ehrenfest's Theorem — Result

- An entirely analogous calculation with $\langle p \rangle$ and $[\hat{p}, \hat{H}] = -i\hbar V'(x)$ gives

$$\frac{d\langle p \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle$$

- Together these are **Ehrenfest's theorem** — quantum expectation values obey Newton's laws on average.

Derivation: Ehrenfest's Theorem — Second Equation ($d\langle p \rangle/dt$)

- **Step 1 — Differentiate $\langle p \rangle$ with respect to time.** Using the same TDSE-substitution strategy as for $\langle x \rangle$:

$$\frac{d\langle p \rangle}{dt} = \frac{d}{dt} \int \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx = \frac{1}{i\hbar} \int \Psi^* [\hat{p}, \hat{H}] \Psi dx$$

- **Step 2 — Evaluate $[\hat{p}, \hat{H}]$.** With $\hat{H} = \hat{p}^2/2m + V(x)$, the kinetic part commutes with $\hat{p}$ (since $[\hat{p}, \hat{p}^2] = 0$), leaving only the potential term:

$$[\hat{p}, \hat{H}] = [\hat{p}, V(x)]$$

Acting on an arbitrary test function $f(x)$ with $\hat{p} = -i\hbar d/dx$:

$$[\hat{p}, V]f = -i\hbar \frac{d}{dx}(Vf) - V \left(-i\hbar \frac{df}{dx} \right) = -i\hbar V'f - i\hbar Vf' + i\hbar Vf' = -i\hbar V'(x)f$$

So $[\hat{p}, V(x)] = -i\hbar V'(x)$, which gives:

$$[\hat{p}, \hat{H}] = -i\hbar \frac{\partial V}{\partial x}$$

Derivation: Ehrenfest's Theorem — Second Equation ($d\langle p \rangle/dt$)

- **Step 3 — Substitute back:**

$$\frac{d\langle p \rangle}{dt} = \frac{1}{i\hbar} \int \Psi^* \left(-i\hbar \frac{\partial V}{\partial x} \right) \Psi dx = - \int \Psi^* \frac{\partial V}{\partial x} \Psi dx = - \left\langle \frac{\partial V}{\partial x} \right\rangle$$

This is Newton's second law in the quantum average: the mean force equals the mean gradient of the potential.

Ehrenfest's Theorem

Statement

The expectation values of quantum operators obey equations analogous to classical mechanics:

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}, \quad \frac{d\langle p \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle$$

- Quantum mechanical expectation values obey Newton's second law *on average*.
- Shows how classical mechanics emerges as the expectation-value limit of quantum mechanics.

Worked Example: Ehrenfest's Theorem for the Harmonic Oscillator

Problem

For $V(x) = \frac{1}{2}m\omega^2x^2$, use Ehrenfest's theorem to derive the classical equation of motion satisfied by $\langle x \rangle$.

Given: $V(x) = \frac{1}{2}m\omega^2x^2$, where m is mass and ω is angular frequency. **Find:** the equation satisfied by $\langle x \rangle(t)$.

Step 1 — Compute $\partial V/\partial x$, using the power rule ($d(x^2)/dx = 2x$):

$$\frac{\partial V}{\partial x} = \frac{1}{2}m\omega^2(2x) = m\omega^2x$$

Step 2 — Substitute into Ehrenfest's second equation, $d\langle p \rangle/dt = -\langle \partial V/\partial x \rangle$:

$$\frac{d\langle p \rangle}{dt} = -\langle m\omega^2x \rangle = -m\omega^2\langle x \rangle$$

Worked Example: Ehrenfest's Theorem for the Harmonic Oscillator

Step 3 — Differentiate Ehrenfest's first equation, $d\langle x \rangle / dt = \langle p \rangle / m$, with respect to t again:

$$\frac{d^2 \langle x \rangle}{dt^2} = \frac{1}{m} \frac{d \langle p \rangle}{dt}$$

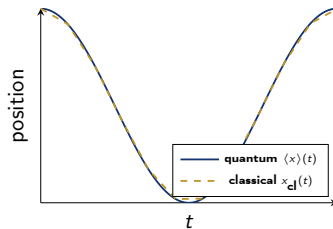
Step 4 — Substitute the result from Step 2:

$$\frac{d^2 \langle x \rangle}{dt^2} = \frac{1}{m} [-m\omega^2 \langle x \rangle] = -\omega^2 \langle x \rangle$$

This is exactly the classical simple harmonic oscillator equation — $\langle x \rangle(t)$ oscillates sinusoidally with angular frequency ω , just like the classical particle.

Ehrenfest's Theorem and the Correspondence Principle

- Ehrenfest's theorem is the precise mathematical statement of Bohr's **correspondence principle**: quantum mechanics must reproduce classical mechanics in the appropriate limit (large quantum numbers, or macroscopic length/energy scales).
- For a sufficiently narrow, well-localized wave packet, $\langle x \rangle(t)$ traces out essentially the same trajectory as a classical point particle obeying $F = ma$ — this is why we never notice quantum effects in everyday (macroscopic) mechanics.
- Caution: Ehrenfest's theorem does *not* say $\langle x \rangle$ always obeys $m d^2 \langle x \rangle / dt^2 = -\langle V'(x) \rangle$ exactly equals $-V'(\langle x \rangle)$ in general — these coincide only when $V'(x)$ is linear (e.g. free particle, harmonic oscillator), which is why the harmonic-oscillator example above works out so cleanly.
- For strongly anharmonic or rapidly varying potentials, the wave packet's spread itself evolves, and $\langle x \rangle$ can deviate noticeably from the classical trajectory — a genuinely quantum effect with no classical counterpart.



For a harmonic oscillator the two trajectories coincide exactly.

Worked Example: Ehrenfest's Theorem for a Free Particle

Problem

For a free particle ($V = 0$), use Ehrenfest's theorem to show that $\langle x \rangle(t)$ moves at constant velocity, exactly like a classical free particle.

Given: $V(x) = 0$ everywhere. **Find:** show $\langle x \rangle(t)$ describes uniform motion.

Step 1 — Compute $\partial V / \partial x$. Since $V = 0$ identically, its derivative is also 0:

$$\frac{\partial V}{\partial x} = 0$$

Step 2 — Substitute into Ehrenfest's second equation:

$$\frac{d\langle p \rangle}{dt} = -\langle 0 \rangle = 0$$

Worked Example: Ehrenfest's Theorem for a Free Particle

Step 3 — Integrate. A derivative that is identically zero means the quantity is constant in time. Call this constant p_0 :

$$\langle p \rangle(t) = p_0 \quad (\text{constant})$$

Step 4 — Substitute into Ehrenfest's first equation, $d\langle x \rangle/dt = \langle p \rangle/m$:

$$\frac{d\langle x \rangle}{dt} = \frac{p_0}{m} \quad (\text{also constant, since } p_0, m \text{ are both constant})$$

Step 5 — Integrate once more with respect to t :

$$\langle x \rangle(t) = \langle x \rangle(0) + \frac{p_0}{m} t$$

Uniform motion in a straight line, exactly matching Newton's first law, even though the underlying wave packet may simultaneously be spreading (Unit 1, Worked Example 3).

Worked Example: Verifying Force Equals $-dV/dx$ for a Linear Potential

Problem

For a uniform field $V(x) = -F_0x$ (constant force F_0 , as in a charged particle in a uniform electric field), find $d\langle p \rangle/dt$ and interpret.

Given: $V(x) = -F_0x$, where F_0 is a constant force. **Find:** $d\langle p \rangle/dt$.

Step 1 — Compute $\partial V/\partial x$, differentiating the linear term:

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x}(-F_0x) = -F_0$$

Step 2 — Substitute into Ehrenfest's second equation:

$$\frac{d\langle p \rangle}{dt} = -\left\langle \frac{\partial V}{\partial x} \right\rangle = -\langle -F_0 \rangle$$

Worked Example: Verifying Force Equals $-dV/dx$ for a Linear Potential

Step 3 — Simplify (the expectation value of a constant is just that constant):

$$\frac{d\langle p \rangle}{dt} = F_0$$

This is exactly Newton's second law, $dp/dt = F$, holding *exactly* (not just on average) because $V'(x) = -F_0$ is constant, so $\langle V'(x) \rangle = V'(\langle x \rangle)$ trivially. This is the simplest case where Ehrenfest's theorem reduces precisely to classical mechanics with no approximation at all.

The "Newton's Law" of Quantum Mechanics

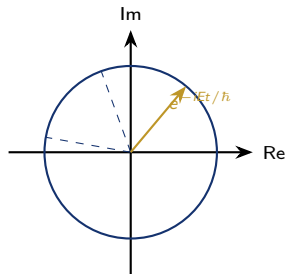
- In classical mechanics, Newton's second law $F = ma$ tells us how a system's state (position and velocity) evolves once we know the forces involved.
- The Schrödinger equation plays exactly this role in quantum mechanics: given the potential $V(x, t)$ governing a system, it tells us precisely how the wavefunction $\Psi(x, t)$ evolves in time.
- Just as Newton's laws cannot be derived from anything more basic (they are confirmed by their predictions matching experiment), the Schrödinger equation is a postulate of the theory — Axiom 4 from Unit 2.
- Schrödinger introduced this equation in 1926, building directly on de Broglie's matter-wave hypothesis (Unit 1); it instantly succeeded in reproducing the hydrogen spectrum (worked out in detail in Unit 4) and remains the cornerstone equation of non-relativistic quantum mechanics to this day.

Schrödinger Time-Dependent Equation

The Time-Dependent Schrödinger Equation (TDSE)

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x, t) \Psi(x, t)$$

- The fundamental equation governing the time evolution of quantum states.
- In operator form: $i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$, where $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$ is the **Hamiltonian**.
- Not derived from more fundamental principles — it is a postulate, validated by its agreement with experiment.
- First order in time $\Rightarrow$ the state at $t = 0$ fully determines $\Psi(x, t)$ for all later times.



stationary state: phase rotates,
 $|\Psi|^2$ stays fixed

From Operators to the Schrödinger Equation

Quantum Mechanical Operators

$$\hat{x} = x, \quad \hat{p}_x = -i\hbar \frac{\partial}{\partial x}, \quad \hat{E} = i\hbar \frac{\partial}{\partial t}$$

- Starting from the classical energy relation $E = \frac{p^2}{2m} + V(x)$,
- Replace each classical quantity by its corresponding **operator** and let both sides act on Ψ :

$$i\hbar \frac{\partial \Psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi$$

- This "correspondence" heuristic is how the Schrödinger equation is commonly motivated (though ultimately it is postulated).

Derivation: Motivating the TDSE from a Plane Wave

- Consider a free particle plane wave consistent with de Broglie/Einstein relations $E = \hbar\omega$, $p = \hbar k$:

$$\Psi(x, t) = Ae^{i(kx - \omega t)}$$

- Differentiate once with respect to t and twice with respect to x :

$$\frac{\partial \Psi}{\partial t} = -i\omega \Psi, \quad \frac{\partial^2 \Psi}{\partial x^2} = -k^2 \Psi$$

- Multiply the first by $i\hbar$: $i\hbar \frac{\partial \Psi}{\partial t} = \hbar\omega \Psi = E \Psi$
- Multiply the second by $-\hbar^2/2m$: $-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = \frac{\hbar^2 k^2}{2m} \Psi = \frac{p^2}{2m} \Psi = KE \cdot \Psi$

Derivation: Motivating the TDSE from a Plane Wave

- Since $E = KE + V$ classically, postulate that this relation continues to hold as an **operator equation** acting on a general Ψ (not just plane waves), with $V = V(x, t)$ added:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x, t)\Psi$$

This reproduces the TDSE. (This is a plausibility argument, not a derivation from first principles — the TDSE remains a postulate.)

Linearity: Why Superposition Always Works

- The TDSE is a **linear** differential equation in Ψ : Ψ and its derivatives appear only to the first power, with no products like Ψ^2 or $(\partial\Psi/\partial x)^2$.
- A direct consequence: if $\Psi_1(x, t)$ and $\Psi_2(x, t)$ are both solutions of the TDSE for the same potential $V(x, t)$, then so is any linear combination $\Psi = c_1\Psi_1 + c_2\Psi_2$.
- This is precisely why the superposition principle (introduced as a postulate in Unit 2, and used informally throughout Unit 1's wave-packet discussion) is automatically consistent with quantum dynamics — it is built into the very structure of the governing equation.
- Physically, linearity also guarantees that the evolution of Ψ does not depend on Ψ 's own magnitude — doubling the wavefunction (before normalization) does not change the physics it describes, only the overall normalization constant.

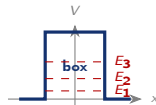
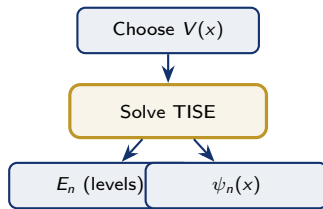
Schrödinger Time-Independent Equation

For a time-independent potential $V(x)$, separate variables via $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$:

TISE

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

- Eigenvalue equation: $\hat{H}\psi = E\psi$.
- Solving for $V(x)$ yields allowed levels E_n and stationary states ψ_n .
- Underlies all of Unit 4: box, barriers, oscillator, hydrogen atom.



Derivation: Separation of Variables

- Assume a product solution $\Psi(x, t) = \psi(x)T(t)$ and substitute into the TDSE:

$$i\hbar \psi(x) \frac{dT}{dt} = -\frac{\hbar^2}{2m} T(t) \frac{d^2\psi}{dx^2} + V(x) \psi(x) T(t)$$

- Divide both sides by $\psi(x)T(t)$:

$$i\hbar \frac{1}{T} \frac{dT}{dt} = \frac{1}{\psi} \left[-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi \right]$$

- The left side depends only on t ; the right side only on x . They can only be equal for all x, t if **both equal the same constant**, call it E . This gives two separate ODEs:

$$i\hbar \frac{dT}{dt} = E T \quad \text{and} \quad -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

Derivation: Separation of Variables

- **Solve the time ODE** $i\hbar dT/dt = ET$ by separating variables:

$$\frac{dT}{T} = \frac{E}{i\hbar} dt = -\frac{iE}{\hbar} dt$$

Integrate both sides:

$$\ln T = -\frac{iE}{\hbar} t + C \implies T(t) = e^C e^{-iEt/\hbar}$$

The constant e^C is absorbed into the normalization of $\psi(x)$, giving:

$$T(t) = e^{-iEt/\hbar}$$

- The spatial ODE is exactly the **TISE**: $\hat{H}\psi = E\psi$.
- This is exactly how the stationary-state form $\Psi = \psi(x)e^{-iEt/\hbar}$ and the TISE arise from the TDSE.

Worked Example 1: Verifying a TISE Solution

Problem

Verify that $\psi(x) = A \sin(kx)$ solves the free-particle ($V = 0$) TISE, and find E in terms of k .

Given: $\psi(x) = A \sin(kx)$, $V = 0$. **Find:** verify the TISE and find $E(k)$.

Step 1 — Find $\psi'(x)$, differentiating ($d(\sin kx)/dx = k \cos kx$):

$$\psi'(x) = Ak \cos(kx)$$

Step 2 — Find $\psi''(x)$, differentiating again ($d(\cos kx)/dx = -k \sin kx$):

$$\psi''(x) = -Ak^2 \sin(kx) = -k^2 \psi(x)$$

Step 3 — Substitute into the TISE with $V = 0$:

$$-\frac{\hbar^2}{2m} \psi'' = E\psi \quad \implies \quad -\frac{\hbar^2}{2m} (-k^2 \psi) = E\psi$$

Worked Example 1: Verifying a TISE Solution

Step 4 — Simplify the left side (double negative):

$$\frac{\hbar^2 k^2}{2m} \psi = E \psi$$

Step 5 — Since this holds for every x (the ψ factors cancel), read off E :

$$E = \frac{\hbar^2 k^2}{2m}$$

Confirms the free-particle dispersion relation used throughout Unit 1.

Worked Example 2: Energy of a Given Wavefunction

Problem

A particle in a region with $V(x) = V_0$ (constant) has wavefunction $\psi(x) = Ae^{-\alpha x^2}$. For this to solve the TISE at $x = 0$, relate E , V_0 , α , $\hbar$, m (evaluate the TISE at $x = 0$).

Given: $\psi(x) = Ae^{-\alpha x^2}$, $V(x) = V_0$ (constant), where $\alpha > 0$ is a width parameter. **Find:** relation between E , V_0 , α , $\hbar$, m .

Step 1 — Find $\psi'(x)$ using the chain rule ($d(e^{-\alpha x^2})/dx = -2\alpha x e^{-\alpha x^2}$):

$$\psi'(x) = -2\alpha x \psi(x)$$

Step 2 — Find $\psi''(x)$ by differentiating $\psi' = -2\alpha x \psi$ using the product rule:

$$\psi''(x) = -2\alpha \psi - 2\alpha x \psi' = -2\alpha \psi - 2\alpha x (-2\alpha x \psi) = (4\alpha^2 x^2 - 2\alpha) \psi$$

Worked Example 2: Energy of a Given Wavefunction

Step 3 — Evaluate at $x = 0$: the $4\alpha^2 x^2$ term vanishes, leaving:

$$\psi''(0) = -2\alpha \psi(0)$$

Step 4 — Substitute into the TISE at $x = 0$:

$$-\frac{\hbar^2}{2m}\psi''(0) + V_0\psi(0) = E\psi(0) \quad \implies \quad -\frac{\hbar^2}{2m}(-2\alpha)\psi(0) + V_0\psi(0) = E\psi(0)$$

Step 5 — Divide through by $\psi(0)$ (assumed nonzero) and simplify:

$$\frac{\hbar^2\alpha}{m} + V_0 = E \quad \implies \quad E = \frac{\hbar^2\alpha}{m} + V_0$$

(Note: a true bound-state solution requires this to hold at every x , which constrains $V(x)$ to be the harmonic-oscillator potential — previewing Unit 4.)

Worked Example 3: Constructing $\Psi(x, t)$ from $\psi(x)$

Problem

A particle is in the energy eigenstate $\psi(x)$ with $E = 3 \text{ eV}$. Write $\Psi(x, t)$ and find the angular frequency of its phase oscillation.

Given: $E = 3 \text{ eV}$. **Find:** $\Psi(x, t)$ and ω .

Step 1 — Apply the stationary-state form $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, identifying $\omega = E/\hbar$ by comparison with the standard oscillation form $e^{-i\omega t}$.

Step 2 — Convert E to joules (multiply by $1.6 \times 10^{-19} \text{ J/eV}$):

$$E = 3 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-19} \text{ J}$$

Step 3 — Divide by $\hbar$:

$$\omega = \frac{E}{\hbar} = \frac{4.8 \times 10^{-19}}{1.055 \times 10^{-34}} \approx 4.55 \times 10^{15} \text{ rad/s}$$

Worked Example 3: Constructing $\Psi(x, t)$ from $\psi(x)$

Step 4 — Convert to ordinary frequency using $\nu = \omega/2\pi$:

$$\nu = \frac{4.55 \times 10^{15}}{2\pi} \approx 7.25 \times 10^{14} \text{ Hz}$$

This falls in the near-UV, illustrating the typical timescale of atomic-energy-scale quantum phase evolution.

Worked Example 4: TISE for a Linear Potential

Problem

For $V(x) = F_0x$ (a uniform force, as for a charged particle in a constant electric field), write down the TISE explicitly and identify which term represents the "restoring" effect of the potential.

Given: $V(x) = F_0x$, where F_0 is a constant force parameter. **Find:** the explicit TISE.

Step 1 — Substitute $V(x) = F_0x$ **directly into the general TISE**, $-\frac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + F_0x \psi(x) = E \psi(x)$$

Step 2 — Rearrange to isolate ψ'' . Move the $V\psi$ term to the right side and multiply through by $-2m/\hbar^2$:

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2} [F_0x - E] \psi(x)$$

Worked Example 4: TISE for a Linear Potential

Step 3 — Interpret. Unlike the box or harmonic-oscillator potentials (Unit 4), this potential is unbounded below as $x \rightarrow -\infty$, so it has no normalizable bound states for any E — only scattering-type (Airy-function) solutions exist, relevant to phenomena like quantum tunneling in tilted-band semiconductor devices.

Probability Current Density

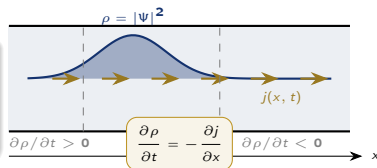
Continuity Equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0, \quad \rho = |\Psi|^2$$

Probability Current

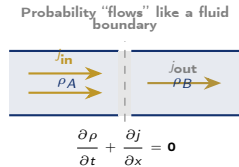
$$j = \frac{\hbar}{2mi} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right)$$

- Local conservation: probability cannot appear or disappear, only *flow*.
- For a real stationary-state wavefunction: $j = 0$ (no net flow).



An Intuitive Analogy: Probability as a Fluid

- The continuity equation $\partial\rho/\partial t + \partial j/\partial x = 0$ has exactly the same form as mass conservation in fluid dynamics.
- Think of $|\Psi|^2$ as a “probability fluid” with density ρ and current j .
- Probability cannot “teleport” — if ρ decreases in a region, j must carry it out through the boundary.
- In Unit 4 this becomes the key tool for computing reflection and transmission coefficients at potential barriers.



Derivation: The Continuity Equation

- Start from the TDSE and its complex conjugate:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi, \quad -i\hbar \frac{\partial \Psi^*}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + V\Psi^*$$

- Multiply the first by Ψ^* , the second by Ψ , and subtract:

$$i\hbar \left(\Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t} \right) = -\frac{\hbar^2}{2m} \left(\Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right)$$

(the V terms cancel since V is real).

- Left side:** recognise the product rule for a time derivative. Since $\rho = \Psi^* \Psi$:

$$\frac{\partial \rho}{\partial t} = \frac{\partial(\Psi^* \Psi)}{\partial t} = \Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t}$$

so the left side is exactly $i\hbar \frac{\partial \rho}{\partial t}$.

Derivation: The Continuity Equation

- **Right side:** verify it is a perfect x -derivative by expanding $\frac{\partial}{\partial x}(\Psi^*\Psi' - \Psi\Psi^{*'})$ via the product rule:

$$\frac{\partial}{\partial x}(\Psi^*\Psi' - \Psi\Psi^{*'}) = \Psi^{*'}\Psi' + \Psi^*\Psi'' - \Psi'\Psi^{*'} - \Psi\Psi^{*''} = \Psi^*\Psi'' - \Psi\Psi^{*''}$$

(the $\Psi^{*'}\Psi'$ terms cancel). This confirms the right side equals $-\frac{\hbar^2}{2m} \frac{\partial}{\partial x}(\Psi^*\Psi' - \Psi\Psi^{*'})$.

- Dividing through by $i\hbar$ and identifying $j = \frac{\hbar}{2mi} (\Psi^*\Psi' - \Psi\Psi^{*'})$ gives exactly

$$\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0$$

Worked Example 1: Probability Current for a Plane Wave

Problem

Compute $j(x, t)$ for $\Psi(x, t) = Ae^{i(kx - \omega t)}$.

Given: $\Psi = Ae^{i(kx - \omega t)}$, A a complex amplitude. **Find:** $j(x, t)$.

Step 1 — Find Ψ' , differentiating with respect to x (only kx depends on x):

$$\Psi' = ikAe^{i(kx - \omega t)} = ik\Psi$$

Step 2 — Find $\Psi^{*'}.$ First conjugate Ψ : $\Psi^* = A^*e^{-i(kx - \omega t)}$, then differentiate:

$$\Psi^{*'} = -ikA^*e^{-i(kx - \omega t)} = -ik\Psi^*$$

Step 3 — Substitute into the current formula, $j = \frac{\hbar}{2mi}(\Psi^*\Psi' - \Psi\Psi^{*'})$:

$$j = \frac{\hbar}{2mi}[\Psi^*(ik\Psi) - \Psi(-ik\Psi^*)] = \frac{\hbar}{2mi}[ik|\Psi|^2 + ik|\Psi|^2]$$

Worked Example 1: Probability Current for a Plane Wave

Step 4 — Combine the two equal terms and simplify $\frac{2ik}{2mi} = \frac{k}{m}$:

$$j = \frac{\hbar}{2mi}(2ik|\Psi|^2) = \frac{\hbar k}{m}|\Psi|^2 = \frac{\hbar k}{m}|A|^2$$

Recognizing $\hbar k/m = v$ (particle velocity) and $|A|^2 = \rho$: $j = \rho v$, exactly the classical flux formula (density $\times$ velocity).

Worked Example 2: Zero Current for a Real Wavefunction

Problem

Show directly that any real wavefunction $\psi(x)$ (such as a bound state of the infinite well) has zero probability current.

Given: $\psi(x)$ real, so $\psi^* = \psi$. **Find:** show $j = 0$.

Step 1 — Substitute $\psi^* = \psi$ into the current formula:

$$j = \frac{\hbar}{2mi}(\psi^*\psi' - \psi\psi^{*'}) = \frac{\hbar}{2mi}(\psi\psi' - \psi\psi')$$

Step 2 — Recognize the two terms are identical (since $\psi^* = \psi$ and $\psi^{*'} = \psi'$ for a real function), so they cancel:

$$j = \frac{\hbar}{2mi}(0) = 0$$

This confirms that bound stationary states carry no net probability flux — consistent with a particle that is, on average, going nowhere (it's trapped).

Worked Example 3: Current for a Superposition of Two Plane Waves

Problem

For $\Psi = Ae^{ikx} + Be^{-ikx}$ (incident plus reflected wave, both real amplitudes), find j .

Given: $\Psi = Ae^{ikx} + Be^{-ikx}$, A, B real amplitudes. **Find:** j .

Step 1 — Find Ψ' , differentiating each term:

$$\Psi' = ikAe^{ikx} - ikBe^{-ikx}$$

Step 2 — Find Ψ^* (since A, B are real, conjugation only flips the sign in each exponent):

$$\Psi^* = Ae^{-ikx} + Be^{ikx}$$

Step 3 — Compute $\Psi^*\Psi'$, multiplying term by term (four products):

$$\Psi^*\Psi' = ikA^2 - ikAB e^{-2ikx} + ikAB e^{2ikx} - ikB^2$$

Worked Example 3: Current for a Superposition of Two Plane Waves

Step 4 — Compute $\Psi\Psi^{*'}$ similarly, and subtract $\Psi^*\Psi' - \Psi\Psi^{*'}$. The cross terms (containing $e^{\pm 2ikx}$) cancel between the two products, leaving only the A^2, B^2 pieces doubled:

$$\Psi^*\Psi' - \Psi\Psi^{*'} = 2ikA^2 - 2ikB^2$$

Step 5 — Substitute into $j = \frac{\hbar}{2mi}(\dots)$:

$$j = \frac{\hbar}{2mi}(2ikA^2 - 2ikB^2) = \frac{\hbar k}{m}(A^2 - B^2) = \frac{\hbar k}{m}(|A|^2 - |B|^2)$$

The net current is the **difference** between the rightward flux ($|A|^2$) and leftward flux ($|B|^2$) — exactly what we expect physically, and the key formula used for reflection/transmission coefficients in Unit 4.

Worked Example 4: Probability Current for a Decaying State

Problem

For the wavefunction $\Psi(x, t) = Ae^{-\kappa x}e^{-iEt/\hbar}$ (valid for $x > 0$, e.g. the tail of a bound state penetrating into a classically forbidden region), show that $j = 0$ even though Ψ is complex.

Given: $\Psi(x, t) = Ae^{-\kappa x}e^{-iEt/\hbar}$, A real, $\kappa > 0$. **Find:** show $j = 0$.

Step 1 — Separate the spatial and time parts, $\Psi = \psi(x)e^{-iEt/\hbar}$ with $\psi(x) = Ae^{-\kappa x}$ **real** (since A, κ, x are all real).

Step 2 — Note that the time-phase factors cancel in j , exactly as they cancel in $|\Psi|^2$ (shown earlier this unit), so j depends only on $\psi(x)$:

$$j = \frac{\hbar}{2mi}(\psi^*\psi' - \psi\psi^{*'})$$

Step 3 — Since ψ is real, $\psi^* = \psi$ and $\psi^{*'} = \psi'$, so the two terms are identical:

$$j = \frac{\hbar}{2mi}(\psi\psi' - \psi\psi') = \frac{\hbar}{2mi}(0) = 0$$

Worked Example 4: Probability Current for a Decaying State

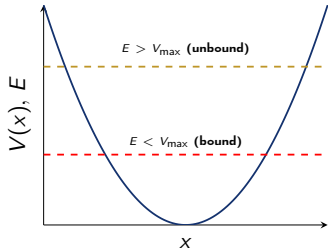
This confirms that even in classically forbidden regions, a bound state carries no net probability flow — consistent with the particle being truly trapped, not leaking away.

Stationary States and Bound States

- A **bound state** is one in which the particle is confined: $\psi(x) \rightarrow 0$ as $x \rightarrow \pm\infty$, and the energy spectrum is **discrete** (quantized).
- An **unbound (scattering) state** extends to infinity, with $\psi(x)$ remaining oscillatory; the energy spectrum is **continuous**.
- Whether E lies below or above the asymptotic value of $V(x)$ determines bound vs. unbound character.
- Examples: particle in a box (bound, Unit 4), free particle (unbound), hydrogen atom (bound for $E < 0$).

Problem Practice: Operators, Commutator Bracket — bridges directly into Unit 5.

Visualizing Bound vs. Unbound States



- For energy E below the potential's asymptotic/maximum value (red dashed line), the particle is classically confined — this is the **bound-state** regime, with discrete energies.
- For E above that value (gold dashed line), the particle can escape to infinity — the **unbound/scattering** regime, with a continuous energy spectrum.

This single picture summarizes the qualitative behavior to expect for every potential solved explicitly in Unit 4: the infinite well and harmonic oscillator are purely bound (discrete spectra), while the free particle and scattering states above a barrier are purely unbound (continuous spectra); the finite well and barrier problems exhibit **both** regimes depending on E .

Unit 3 Summary

- **Probability amplitude, density**, and normalization of Ψ .
- **Stationary states**: $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, time-independent $|\Psi|^2$.
- **Expectation values**, eigenfunctions/eigenvalues, and **Ehrenfest's theorem** (derived from the TDSE and the commutator $[x, \hat{H}]$).
- The **TDSE** (motivated from plane waves) and the **TISE** (via separation of variables) — the central equations of the course.
- **Probability current density** (derived from the continuity equation) and bound vs. unbound states.

Additional Problem 1: Normalization with a Complex Constant

Problem

Find A (real, positive) such that $\psi(x) = Ae^{-x^2/2a^2}e^{i\phi_0}$ is normalized.

Given: $\psi(x) = Ae^{-x^2/2a^2}e^{i\phi_0}$, with a a width parameter and ϕ_0 a constant phase. **Find:** A .

Step 1 — Compute $|\psi|^2 = \psi^*\psi$. The phase factor's modulus is 1: $e^{i\phi_0} \cdot e^{-i\phi_0} = e^0 = 1$:

$$|\psi|^2 = A^2 e^{-x^2/a^2} (e^{i\phi_0} e^{-i\phi_0}) = A^2 e^{-x^2/a^2}$$

Step 2 — Set up the normalization integral, using the standard Gaussian integral
 $\int e^{-x^2/a^2} dx = a\sqrt{\pi}$:

$$\int_{-\infty}^{\infty} A^2 e^{-x^2/a^2} dx = A^2 a\sqrt{\pi} = 1$$

Step 3 — Solve for A :

$$A = \frac{1}{\sqrt{a\sqrt{\pi}}} = (\pi a^2)^{-1/4}$$

Additional Problem 1: Normalization with a Complex Constant

The constant phase ϕ_0 does not affect $|\psi|^2$ or normalization — a general fact for any constant overall phase.

Additional Problem 2: Probability Outside a Box

Problem

For an infinite well state, what is $P(x < 0) + P(x > L)$?

Given: an infinite well of width L . **Find:** the probability of finding the particle outside the well.

Step 1 — Recall the defining property of an infinite well: $\psi(x) = 0$ identically for $x < 0$ and $x > L$ (the infinite potential forbids the wavefunction from being nonzero there).

Step 2 — Apply the Born rule: $P = \int |\psi|^2 dx$ over the relevant region. Since $\psi = 0$ throughout both regions, the integrand is zero everywhere in the integration range:

$$P(x < 0) + P(x > L) = \int_{-\infty}^0 |\psi|^2 dx + \int_L^{\infty} |\psi|^2 dx = 0 + 0 = 0$$

The particle is strictly confined — exactly 0 probability of being found outside the well.

Additional Problem 3: Expectation Value of x^2 for the Box Ground State

Problem

For $\psi_1(x) = \sqrt{2/L} \sin(\pi x/L)$, $\langle x^2 \rangle = L^2 \left(\frac{1}{3} - \frac{1}{2\pi^2} \right)$. Evaluate numerically for $L = 1$ and find $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$.

Given: the formula for $\langle x^2 \rangle$, and $\langle x \rangle = L/2$ (from earlier in this unit). **Find:** Δx for $L = 1$.

Step 1 — Evaluate $\langle x^2 \rangle$ at $L = 1$:

$$\langle x^2 \rangle = 1^2 \left(\frac{1}{3} - \frac{1}{2\pi^2} \right) = 0.3333 - 0.0507 = 0.2827$$

Step 2 — Compute $\langle x \rangle^2$. Since $\langle x \rangle = L/2 = 0.5$ for $L = 1$:

$$\langle x \rangle^2 = (0.5)^2 = 0.25$$

Step 3 — Subtract:

$$\langle x^2 \rangle - \langle x \rangle^2 = 0.2827 - 0.25 = 0.0327$$

Additional Problem 3: Expectation Value of x^2 for the Box Ground State

Step 4 — Take the square root:

$$\Delta x = \sqrt{0.0327} \approx 0.181 L$$

Additional Problem 4: Verifying Hamiltonian Hermiticity Numerically (Real Case)

Problem

For real ψ_1, ψ_2 vanishing at boundaries, confirm $\int \psi_1 \hat{H} \psi_2 dx = \int (\hat{H} \psi_1) \psi_2 dx$ for $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$, using integration by parts twice (boundary terms vanish).

Given: real $\psi_1, \psi_2 \rightarrow 0$ at the boundaries. **Find:** verify Hermiticity of $\hat{H}$.

Step 1 — Write out the left side and integrate by parts once (moving one derivative from ψ_2'' to ψ_1):

$$\int \psi_1 \left(-\frac{\hbar^2}{2m} \psi_2'' \right) dx = -\frac{\hbar^2}{2m} \left[\psi_1 \psi_2' \right]_{-\infty}^{\infty} + \frac{\hbar^2}{2m} \int \psi_1' \psi_2' dx$$

Step 2 — The boundary term vanishes since $\psi_1 \rightarrow 0$ at $\pm\infty$, leaving $\frac{\hbar^2}{2m} \int \psi_1' \psi_2' dx$.

Additional Problem 4: Verifying Hamiltonian Hermiticity Numerically (Real Case)

Step 3 — Integrate by parts a second time (moving the remaining derivative from ψ_2' to ψ_1'):

$$\frac{\hbar^2}{2m} \int \psi_1' \psi_2' dx = \frac{\hbar^2}{2m} [\psi_1' \psi_2]_{-\infty}^{\infty} - \frac{\hbar^2}{2m} \int \psi_1'' \psi_2 dx$$

Step 4 — This boundary term also vanishes, leaving

$$-\frac{\hbar^2}{2m} \int \psi_1'' \psi_2 dx = \int \left(-\frac{\hbar^2}{2m} \psi_1'' \right) \psi_2 dx = \int (\hat{H} \psi_1) \psi_2 dx \checkmark.$$

Both integrations by parts give back the original integral with boundary terms zero — confirming Hermiticity, central to Unit 5.

Additional Problem 5: Phase Oscillation Frequency

Problem

A stationary state has $E = 1 \text{ eV}$. Find the period of oscillation of Ψ 's phase.

Given: $E = 1 \text{ eV}$. **Find:** T , the oscillation period.

Step 1 — Recall the phase oscillates as $e^{-iEt/\hbar}$, with angular frequency $\omega = E/\hbar$. The period is $T = 2\pi/\omega = 2\pi\hbar/E$.

Step 2 — Convert E to joules:

$$E = 1 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-19} \text{ J}$$

Step 3 — Substitute into $T = 2\pi\hbar/E$:

$$T = \frac{2\pi(1.055 \times 10^{-34})}{1.6 \times 10^{-19}}$$

Additional Problem 5: Phase Oscillation Frequency

Step 4 — Evaluate:

$$T \approx \frac{6.63 \times 10^{-34}}{1.6 \times 10^{-19}} \approx 4.14 \times 10^{-15} \text{ s}$$

Additional Problem 6: Current Continuity Check

Problem

If $j(x)$ is found to be $j_0 = \text{const}$ everywhere in a 1D stationary scattering problem, what does this imply about $\partial\rho/\partial t$?

Given: $j(x) = j_0$ (constant in x). **Find:** $\partial\rho/\partial t$.

Step 1 — Start from the continuity equation, $\partial\rho/\partial t + \partial j/\partial x = 0$, and solve for $\partial\rho/\partial t$:

$$\frac{\partial\rho}{\partial t} = -\frac{\partial j}{\partial x}$$

Step 2 — Differentiate $j = j_0$ (a constant) with respect to x :

$$\frac{\partial j}{\partial x} = 0$$

Step 3 — Substitute back:

$$\frac{\partial\rho}{\partial t} = -0 = 0$$

Additional Problem 6: Current Continuity Check

The density is exactly steady, fully consistent with a stationary-state scattering problem.

Additional Problem 7: Ehrenfest Check for a Free Particle

Problem

For $V = 0$, Ehrenfest's theorem gives $d\langle p \rangle / dt = 0$. What does this imply physically?

Given: $d\langle p \rangle / dt = 0$ for a free particle. **Find:** physical interpretation.

Step 1 — Recognize that a zero time-derivative means the quantity is constant in time:

$$\langle p \rangle(t) = \langle p \rangle(0) \quad \text{for all } t$$

Step 2 — Interpret: $\langle p \rangle$ is conserved for a free particle — the wave packet's average momentum never changes, exactly matching Newton's first law (no force, no momentum change) on average.

Additional Problem 8: Energy from a Box Quantum Number

Problem

For an electron in an infinite well of width $L = 2 \text{ nm}$, find E_1 in eV.

Given: $L = 2 \text{ nm}$, m_e . **Find:** E_1 .

Step 1 — Apply $E_1 = \pi^2 \hbar^2 / (2m_e L^2)$:

$$E_1 = \frac{\pi^2 (1.055 \times 10^{-34})^2}{2(9.11 \times 10^{-31})(2 \times 10^{-9})^2}$$

Step 2 — Evaluate the numerator ($\pi^2 \hbar^2$):

$$\pi^2 (1.055 \times 10^{-34})^2 \approx 9.87 \times (1.113 \times 10^{-68}) \approx 1.099 \times 10^{-67}$$

Step 3 — Evaluate the denominator ($2m_e L^2$):

$$2(9.11 \times 10^{-31})(4 \times 10^{-18}) \approx 7.29 \times 10^{-48}$$

Additional Problem 8: Energy from a Box Quantum Number

Step 4 — Divide:

$$E_1 \approx \frac{1.099 \times 10^{-67}}{7.29 \times 10^{-48}} \approx 1.51 \times 10^{-20} \text{ J}$$

Step 5 — Convert to eV (divide by 1.6×10^{-19}):

$$E_1 \approx 0.094 \text{ eV}$$

Additional Problem 9: Normalization of a Two-Term Superposition

Problem

If $\Psi = c_1\psi_1 + c_2\psi_2$ with ψ_1, ψ_2 orthonormal, find the normalization condition on c_1, c_2 .

Given: $\Psi = c_1\psi_1 + c_2\psi_2$, with $\langle\psi_1|\psi_1\rangle = \langle\psi_2|\psi_2\rangle = 1$ and $\langle\psi_1|\psi_2\rangle = 0$. **Find:** the condition on c_1, c_2 .

Step 1 — Write the normalization integral, expanding $|\Psi|^2 = \Psi^*\Psi$ as a product of two sums (four terms total):

$$\int |\Psi|^2 dx = |c_1|^2 \int |\psi_1|^2 dx + |c_2|^2 \int |\psi_2|^2 dx + c_1^* c_2 \int \psi_1^* \psi_2 dx + c_2^* c_1 \int \psi_2^* \psi_1 dx$$

Step 2 — Apply orthonormality: the first two integrals equal 1 each; the last two (cross terms) vanish by orthogonality:

$$\int |\Psi|^2 dx = |c_1|^2(1) + |c_2|^2(1) + c_1^* c_2(0) + c_2^* c_1(0) = |c_1|^2 + |c_2|^2$$

Additional Problem 9: Normalization of a Two-Term Superposition

Step 3 — Set this equal to 1 (normalization requirement):

$$|c_1|^2 + |c_2|^2 = 1$$

Additional Problem 10: Probability Current Units Check

Problem

Verify dimensionally that $j = \frac{\hbar}{2mi} (\Psi^* \Psi' - \Psi \Psi'^*)$ has units of (probability)/(area · time) in 1D, i.e. $1/(\text{length} \cdot \text{time})$.

Given: the current formula. **Find:** verify its units.

Step 1 — Find the units of $\hbar/m$. Since $[\hbar] = \text{J s} = \text{kg m}^2/\text{s}$, and $[m] = \text{kg}$:

$$\left[\frac{\hbar}{m} \right] = \frac{\text{kg m}^2/\text{s}}{\text{kg}} = \text{m}^2/\text{s}$$

Step 2 — Find the units of Ψ' . Since $\Psi' = d\Psi/dx$:

$$[\Psi'] = [\Psi]/\text{m}$$

Additional Problem 10: Probability Current Units Check

Step 3 — Combine: $[j] = [\hbar/m] \times [\Psi][\Psi']$:

$$[j] = (\text{m}^2/\text{s}) \times [\Psi] \times \frac{[\Psi]}{\text{m}} = [\Psi]^2 \text{ m/s}$$

Step 4 — Recall $[\Psi]^2$ **has units of 1D probability density**, i.e. 1/length (so that $\int |\Psi|^2 dx$ is dimensionless):

$$[j] = \frac{1}{\text{m}} \times \frac{\text{m}}{\text{s}} = \frac{1}{\text{s}}$$

This matches the expected units of a 1D probability flux (probability per unit time). ✓

Unit 4: Basics of Schrödinger Equation II

1. Origin of Quantum Mechanics
2. Formulation of Quantum Mechanics
3. Basics of Schrödinger Equation I
- 4. Basics of Schrödinger Equation II**
5. Dirac Notation in Quantum Mechanics

Unit 4: Symbol Glossary

- L — well/box width (m)
- n — quantum number (positive integer)
- V_0 — barrier/step height (J)
- k, κ — wavenumber (oscillatory/decaying regions) (m^{-1})
- R, T — reflection, transmission coefficients (dimensionless)
- a — barrier width or delta-potential parameter (m)
- α — delta-potential strength (J m); note: same symbol used as a Gaussian width parameter (m^{-2}) in Unit 3 WE2 — different quantity
- ω — angular frequency (rad/s)
- $\hat{a}_+, \hat{a}_-$ — raising/lowering (ladder) operators
- $\hat{N}$ — number operator
- H_n — Hermite polynomial of order n
- Z — atomic number (dimensionless)
- ϵ_0 — permittivity of free space (F/m)
- r, θ, ϕ — spherical coordinates (m, rad, rad)
- $R(r), \Theta(\theta), \Phi(\phi)$ — separated radial/angular functions
- ℓ — orbital angular momentum quantum number (dimensionless)
- m_ℓ — magnetic quantum number (dimensionless)
- m_s — spin magnetic quantum number (dimensionless)
- $Y_\ell^{m_\ell}$ — spherical harmonics
- $\hat{L}^2$ — squared orbital angular momentum operator ($\text{J}^2 \text{s}^2$)
- $\hat{L}_x, \hat{L}_y, \hat{L}_z$ — Cartesian components of the orbital angular momentum operator (J s)

Symbols specific to a single derivation are defined where they first appear.

Unit 4: Putting the Machinery to Work

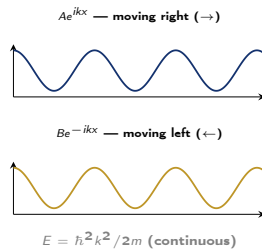
- Unit 3 built the general framework — the TISE, probability current, bound vs. unbound states. This unit applies that framework to specific, solvable potentials $V(x)$, building physical intuition through concrete examples.
- We proceed roughly in order of increasing complexity: the free particle (no potential at all), the infinite well (simplest confining potential), barriers and tunneling, the harmonic oscillator, and finally the hydrogen atom in 3D.
- Each example illustrates a different qualitative feature of quantum systems: energy quantization (box), classically forbidden penetration (tunneling), equally-spaced spectra (oscillator), and the famous discrete atomic energy levels (hydrogen).
- These are not merely textbook exercises — the infinite well models electrons in quantum dots, tunneling underlies modern transistor and STM technology, and the oscillator and hydrogen-atom solutions appear throughout atomic, molecular, and condensed-matter physics.

Free Particle

For $V(x) = 0$ everywhere, the TISE becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \Rightarrow \psi(x) = Ae^{ikx} + Be^{-ikx}$$

- Traveling plane waves; $+x$ or $-x$ direction.
- Energy $E = \frac{\hbar^2 k^2}{2m}$ is **continuous**.
- Not normalizable; physical states are **wave packets**.



Derivation: General Solution of the Free-Particle TISE

- The free TISE, $-\frac{\hbar^2}{2m}\psi'' = E\psi$, is a linear ODE with constant coefficients:

$$\psi'' + k^2\psi = 0, \quad k^2 \equiv \frac{2mE}{\hbar^2}$$

- Trial solution $\psi = e^{\lambda x}$ gives the characteristic equation $\lambda^2 + k^2 = 0 \Rightarrow \lambda = \pm ik$.
- General solution is a linear combination of the two independent solutions:

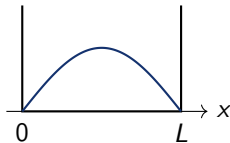
$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

- Equivalently, in real form: $\psi(x) = C \cos(kx) + D \sin(kx)$ — the same general solution, useful when boundary conditions are real (as in the box problem next).
- No boundary conditions are imposed for a truly free particle, so $E = \hbar^2 k^2 / 2m$ can take *any* non-negative value: a continuous spectrum.

Particle in an Infinite Square Well (1D Box)

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{otherwise} \end{cases}$$

Boundary conditions: $\psi(0) = \psi(L) = 0$.



Solutions

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$

Energy is **quantized**; the ground state ($n = 1$) has $E_1 > 0$ — a direct consequence of the uncertainty principle (the particle cannot have exactly zero momentum while confined).

Derivation: Infinite Well — Applying Boundary Conditions

- Outside the well $\psi = 0$ (infinite potential forbids penetration). Inside, $V = 0$, so the general solution is

$$\psi(x) = C \cos(kx) + D \sin(kx), \quad k = \frac{\sqrt{2mE}}{\hbar}$$

- Boundary condition at $x = 0$:** continuity requires $\psi(0) = 0$:

$$\psi(0) = C \cos 0 + D \sin 0 = C = 0$$

So $C = 0$ and $\psi(x) = D \sin(kx)$.

- Boundary condition at $x = L$:** $\psi(L) = 0$:

$$D \sin(kL) = 0 \implies kL = n\pi, \quad n = 1, 2, 3, \dots$$

($n = 0$ gives $\psi \equiv 0$, the trivial/unphysical solution, so it is excluded.)

- Therefore $k_n = n\pi/L$, giving the quantized energies $E_n = \hbar^2 k_n^2 / 2m = n^2 \pi^2 \hbar^2 / (2mL^2)$.

Derivation: Infinite Well — Normalization

- Normalize $\psi_n(x) = D \sin(n\pi x/L)$ on $[0, L]$. Apply the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ with $\theta = n\pi x/L$:

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos\left(\frac{2n\pi x}{L}\right) dx = \frac{L}{2} - \frac{1}{2} \left[\frac{L}{2n\pi} \sin\left(\frac{2n\pi x}{L}\right) \right]_0^L$$

Since $\sin(2n\pi) = \sin(0) = 0$ for any integer n , the cosine integral contributes zero:

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}$$

The normalization condition then gives:

$$D^2 \cdot \frac{L}{2} = 1 \quad \Rightarrow \quad D = \sqrt{\frac{2}{L}}$$

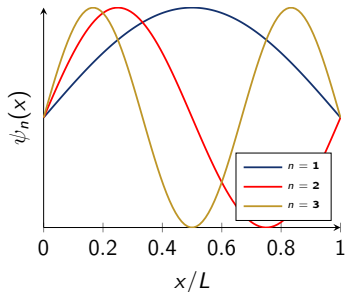
giving the normalized eigenstates

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

Derivation: Infinite Well — Normalization

- These eigenstates are mutually **orthogonal**: $\int_0^L \psi_m^* \psi_n dx = \delta_{mn}$, a special case of the general Hermitian-operator orthogonality theorem (Unit 5).

Visualizing the First Three Eigenstates



- The n -th eigenstate has exactly $n - 1$ interior **nodes** (points where $\psi_n = 0$ inside the well) — a general pattern that holds for bound states of any 1D potential.
- Higher n means more oscillation, shorter effective wavelength, and hence (via $p = h/\lambda$) higher energy — consistent with $E_n \propto n^2$.
- Note $\psi_n(0) = \psi_n(L) = 0$ for every n , as required by the boundary conditions derived above.

Worked Example 1: Energy Levels of an Electron in a Box

Problem

An electron is confined to a 1D box of width $L = 0.5 \text{ nm}$. Find E_1 , E_2 , E_3 in eV.

Given: $L = 0.5 \text{ nm}$, m_e . **Find:** E_1, E_2, E_3 .

Step 1 — Compute E_1 , applying $E_1 = \pi^2 \hbar^2 / (2m_e L^2)$:

$$E_1 = \frac{\pi^2 (1.055 \times 10^{-34})^2}{2(9.11 \times 10^{-31})(0.5 \times 10^{-9})^2}$$

Step 2 — Evaluate numerator and denominator separately:

$$\pi^2 (1.055 \times 10^{-34})^2 \approx 1.099 \times 10^{-67}, \quad 2(9.11 \times 10^{-31})(0.25 \times 10^{-18}) \approx 4.56 \times 10^{-49}$$

Step 3 — Divide and convert to eV (divide by $1.6 \times 10^{-19} \text{ J/eV}$):

$$E_1 \approx \frac{1.099 \times 10^{-67}}{4.56 \times 10^{-49}} \approx 2.41 \times 10^{-19} \text{ J} \approx 1.51 \text{ eV}$$

Worked Example 1: Energy Levels of an Electron in a Box

Step 4 — Use the scaling $E_n = n^2 E_1$ to find E_2, E_3 without repeating the full calculation:

$$E_2 = 2^2 E_1 = 4(1.51) \approx 6.02 \text{ eV}, \quad E_3 = 3^2 E_1 = 9(1.51) \approx 13.55 \text{ eV}$$

Worked Example 2: Photon Emitted in a Transition

Problem

For the same box ($L = 0.5 \text{ nm}$), find the wavelength of the photon emitted when the electron transitions from $n = 2$ to $n = 1$.

Given: $E_1 \approx 1.51 \text{ eV}$ (from Worked Example 1). **Find:** λ of the emitted photon.

Step 1 — Find the energy difference, using $E_n = n^2 E_1$ so $E_2 = 4E_1$:

$$\Delta E = E_2 - E_1 = 4E_1 - E_1 = 3E_1 = 3(1.51 \text{ eV}) = 4.53 \text{ eV}$$

Step 2 — Convert to joules (multiply by $1.6 \times 10^{-19} \text{ J/eV}$):

$$\Delta E = 4.53 \times 1.6 \times 10^{-19} \approx 7.25 \times 10^{-19} \text{ J}$$

Step 3 — Apply energy conservation, $\Delta E = hc/\lambda$, and solve for λ :

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{7.25 \times 10^{-19}}$$

Worked Example 2: Photon Emitted in a Transition

Step 4 — Evaluate:

$$\lambda \approx \frac{1.988 \times 10^{-25}}{7.25 \times 10^{-19}} \approx 2.74 \times 10^{-7} \text{ m} = 274 \text{ nm (UV)}$$

Worked Example 3: Box Width from a Given Energy

Problem

A particle of mass 1.0×10^{-27} kg has ground-state energy $E_1 = 0.5$ eV in an infinite well. Find the well width L .

Given: $m = 1.0 \times 10^{-27}$ kg, $E_1 = 0.5$ eV. **Find:** L .

Step 1 — Start from $E_1 = \pi^2 \hbar^2 / (2mL^2)$ and solve for L . Multiply both sides by $2mL^2/E_1$, then divide by $\pi^2 \hbar^2$... more directly, rearrange and take the square root:

$$L^2 = \frac{\pi^2 \hbar^2}{2mE_1} \quad \Rightarrow \quad L = \pi \hbar \sqrt{\frac{1}{2mE_1}}$$

Step 2 — Convert E_1 to joules:

$$E_1 = 0.5 \times 1.6 \times 10^{-19} = 8 \times 10^{-20} \text{ J}$$

Worked Example 3: Box Width from a Given Energy

Step 3 — Evaluate $2mE_1$:

$$2mE_1 = 2(10^{-27})(8 \times 10^{-20}) = 1.6 \times 10^{-46}$$

Step 4 — Take the square root of the reciprocal:

$$\sqrt{\frac{1}{1.6 \times 10^{-46}}} \approx 2.5 \times 10^{22} \text{ (SI units)}$$

Step 5 — Multiply by $\pi\hbar$:

$$L = \pi(1.055 \times 10^{-34})(2.5 \times 10^{22}) \approx 1.86 \times 10^{-10} \text{ m} \approx 0.19 \text{ nm}$$

Worked Example 4: Quantum Dot Color Tuning

Problem

A semiconductor quantum dot is modeled as a 1D infinite well containing an electron with effective mass $m^* = 0.2 m_e$. Find the well width L needed so the $n = 1 \rightarrow n = 2$ transition emits visible light at $\lambda = 620 \text{ nm}$ (red).

Given: $m^* = 0.2 m_e$, target $\lambda = 620 \text{ nm}$. **Find:** L .

Step 1 — Relate the required transition energy to λ . Since $\Delta E = E_2 - E_1 = 4E_1 - E_1 = 3E_1$ and $\Delta E = hc/\lambda$, set these equal and solve for E_1 :

$$3E_1 = \frac{hc}{\lambda} \implies E_1 = \frac{hc}{3\lambda}$$

Step 2 — Substitute numbers:

$$E_1 = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{3(620 \times 10^{-9})} \approx \frac{1.988 \times 10^{-25}}{1.86 \times 10^{-6}} \approx 1.07 \times 10^{-19} \text{ J}$$

Worked Example 4: Quantum Dot Color Tuning

Step 3 — Solve $E_1 = \pi^2 \hbar^2 / (2m^* L^2)$ **for** L , as in Worked Example 3:

$$L = \pi \hbar \sqrt{\frac{1}{2m^* E_1}}$$

Step 4 — Evaluate $m^* = 0.2m_e = 0.2(9.11 \times 10^{-31}) \approx 1.82 \times 10^{-31}$ **kg, then** $2m^* E_1$:

$$2m^* E_1 = 2(1.82 \times 10^{-31})(1.07 \times 10^{-19}) \approx 3.90 \times 10^{-50}$$

Step 5 — Take the square root, multiply by $\pi \hbar$:

$$L = \pi(1.055 \times 10^{-34}) \sqrt{\frac{1}{3.90 \times 10^{-50}}} \approx 1.68 \text{ nm}$$

This is precisely the working principle behind **quantum-dot displays**: the emitted color is tuned simply by controlling the nanocrystal's physical size during manufacture, with no change in chemical composition required.

Worked Example 5: Probability in a Sub-region of the Well

Problem

For a particle in the ground state ($n = 1$) of an infinite square well of width L , find the probability of finding it in the region $0 < x < L/4$. Compare this to the classical (uniform-probability) expectation.

Given: $\psi_1(x) = \sqrt{2/L} \sin(\pi x/L)$. **Find:** $P(0 < x < L/4)$.

Step 1 — Set up the probability integral:

$$P = \int_0^{L/4} |\psi_1(x)|^2 dx = \frac{2}{L} \int_0^{L/4} \sin^2\left(\frac{\pi x}{L}\right) dx$$

Step 2 — Apply the half-angle identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ with $\theta = \pi x/L$ (as in the normalization derivation earlier in this unit):

$$P = \frac{2}{L} \left[\frac{x}{2} - \frac{L}{4\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_0^{L/4}$$

Worked Example 5: Probability in a Sub-region of the Well

Step 3 — Evaluate at the limits. At $x = L/4$: $\frac{x}{2} = \frac{L}{8}$, and $\sin(2\pi x/L) = \sin(\pi/2) = 1$; the lower limit ($x = 0$) gives zero for both terms:

$$P = \frac{2}{L} \left[\frac{L}{8} - \frac{L}{4\pi}(1) \right] = \frac{2}{L} \cdot L \left[\frac{1}{8} - \frac{1}{4\pi} \right] = \frac{1}{4} - \frac{1}{2\pi}$$

Step 4 — Evaluate numerically:

$$P = 0.25 - 0.1592 \approx 0.0908$$

Step 5 — Compare to the classical prediction. A classical particle bouncing uniformly between the walls would spend a fraction $1/4$ of its time (probability 0.25) in this quarter of the well. The quantum probability ($\approx 9.1\%$) is considerably **smaller**, because the ground-state wavefunction is peaked at the center of the well ($x = L/2$) and suppressed near the walls — the opposite of the classical picture, where the particle (moving at constant speed) is equally likely to be found anywhere.

Worked Example 6: Force Exerted by a Confined Particle on the Walls

Problem

An electron in the ground state ($n = 1$) of an infinite well of width $L = 1 \text{ nm}$ exerts an outward force on the walls as the box is quasi-statically widened (this is the quantum-mechanical origin of **degeneracy pressure**). Estimate this force, using $F_n = -dE_n/dL$.

Given: $n = 1$, $L = 1 \text{ nm}$, m_e . **Find:** F_1 .

Step 1 — Write E_n as a function of L and differentiate. Starting from $E_n(L) = \frac{n^2\pi^2\hbar^2}{2m}L^{-2}$:

$$\frac{dE_n}{dL} = \frac{n^2\pi^2\hbar^2}{2m} \times (-2)L^{-3} = -\frac{n^2\pi^2\hbar^2}{mL^3}$$

Step 2 — Identify the outward force as the negative of this derivative (energy decreases as L increases, so the particle does positive work pushing the walls outward):

$$F_n = -\frac{dE_n}{dL} = \frac{n^2\pi^2\hbar^2}{mL^3}$$

Worked Example 6: Force Exerted by a Confined Particle on the Walls

Step 3 — Evaluate the numerator, $\pi^2 \hbar^2$ (with $n = 1$), reusing the value from Worked Example 1:

$$\pi^2 \hbar^2 \approx 1.099 \times 10^{-67}$$

Step 4 — Evaluate the denominator, $m_e L^3$:

$$m_e L^3 = (9.11 \times 10^{-31})(1 \times 10^{-9})^3 = (9.11 \times 10^{-31})(1 \times 10^{-27}) \approx 9.11 \times 10^{-58}$$

Step 5 — Divide:

$$F_1 \approx \frac{1.099 \times 10^{-67}}{9.11 \times 10^{-58}} \approx 1.21 \times 10^{-10} \text{ N}$$

Although tiny by everyday standards, this “quantum pressure” is real and physically important: it is the same effect (applied to $\sim 10^{57}$ confined electrons) that supports white dwarf stars against gravitational collapse, and the same confinement principle that gives quantum dots and nanostructures size-dependent mechanical and optical properties.

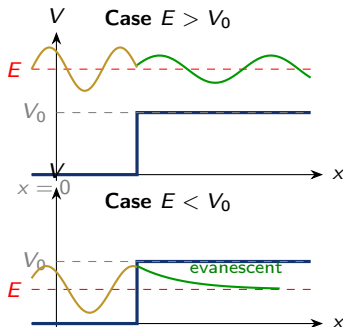
Classical Intuition vs. Quantum Reality

- Classically, a ball rolling toward a step "uphill" either has enough energy to climb over it ($E > V_0$, always passes), or it doesn't ($E < V_0$, always bounces back) — the outcome is completely deterministic.
- Quantum mechanically, neither case is so clean: even with $E > V_0$ there is a nonzero chance of reflection, and even with $E < V_0$ the wavefunction penetrates some distance into the "forbidden" region before (typically) being reflected.
- This subsection works through three increasingly rich scenarios built from these basic ideas: a single step, a finite-width barrier (leading to genuine **tunneling** all the way through), and an idealized delta-function well (leading to a single bound state).
- The common mathematical technique throughout is **matching**: solve the TISE separately in each region of constant potential, then demand ψ and ψ' be continuous at every boundary — the same method used for the infinite well, generalized to finite (rather than infinite) potential jumps.

Step Potential

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & x \geq 0 \end{cases}$$

- **Case $E > V_0$:** particle partially reflects, partially transmits — a purely quantum effect (classically, it would always pass over).
- **Case $E < V_0$:** classically forbidden region for $x > 0$; quantum mechanically, ψ does not vanish abruptly but decays exponentially into the barrier region.
- Reflection coefficient R and transmission coefficient T satisfy $R + T = 1$, obtained by matching ψ and ψ' at $x = 0$.



Derivation: Step Potential ($E > V_0$) — Setup

- Region I ($x < 0$): $\psi_I = Ae^{ik_1x} + Be^{-ik_1x}$, $k_1 = \sqrt{2mE}/\hbar$ (incident A + reflected B).
- Region II ($x \geq 0$): $\psi_{II} = Fe^{ik_2x}$, $k_2 = \sqrt{2m(E - V_0)}/\hbar$ (transmitted only — no wave coming back from $+\infty$).
- **Matching conditions at $x = 0$** (continuity of ψ and ψ' , since V is finite here):

$$\psi_I(0) = \psi_{II}(0) : \quad A + B = F$$

$$\psi'_I(0) = \psi'_{II}(0) : \quad ik_1(A - B) = ik_2F$$

Derivation: Step Potential ($E > V_0$) — Solving for R , T

- Solve the two matching equations for B/A and F/A :

$$\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}, \quad \frac{F}{A} = \frac{2k_1}{k_1 + k_2}$$

- Reflection coefficient** (ratio of reflected to incident probability current):

$$R = \left| \frac{B}{A} \right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$$

- Transmission coefficient** (must include the ratio of velocities k_2/k_1 since current $j \propto k|\text{amplitude}|^2$):

$$T = \frac{k_2}{k_1} \left| \frac{F}{A} \right|^2 = \frac{4k_1 k_2}{(k_1 + k_2)^2}$$

- Check: $R + T = \frac{(k_1 - k_2)^2 + 4k_1 k_2}{(k_1 + k_2)^2} = \frac{(k_1 + k_2)^2}{(k_1 + k_2)^2} = 1 \checkmark$, confirming probability conservation.

Worked Example 1: Reflection at a Step ($E > V_0$)

Problem

An electron of energy $E = 2 \text{ eV}$ approaches a step of height $V_0 = 1 \text{ eV}$. Find R and T .

Given: $E = 2 \text{ eV}$, $V_0 = 1 \text{ eV}$. **Find:** R , T (reflection, transmission coefficients).

Step 1 — Recall $k_1 \propto \sqrt{E}$ **and** $k_2 \propto \sqrt{E - V_0}$ (the common proportionality constant $\sqrt{2m}/\hbar$ cancels in any ratio of k 's):

$$k_1 \propto \sqrt{2}, \quad k_2 \propto \sqrt{2 - 1} = \sqrt{1} = 1$$

Step 2 — Apply the reflection formula, $R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$:

$$R = \left(\frac{\sqrt{2} - 1}{\sqrt{2} + 1} \right)^2$$

Worked Example 1: Reflection at a Step ($E > V_0$)

Step 3 — Evaluate $\sqrt{2} \approx 1.414$, then numerator and denominator:

$$\sqrt{2} - 1 \approx 0.414, \quad \sqrt{2} + 1 \approx 2.414$$

Step 4 — Form the ratio and square:

$$R \approx \left(\frac{0.414}{2.414} \right)^2 \approx (0.1715)^2 \approx 0.0294$$

Step 5 — Apply $R + T = 1$ (probability conservation) to find T :

$$T = 1 - R \approx 1 - 0.0294 \approx 0.971$$

Even though $E > V_0$, there is a small but nonzero reflection probability — impossible classically.

Worked Example 2: Step with $E \gg V_0$

Problem

Show that $R \rightarrow 0$ as $E \rightarrow \infty$ for fixed V_0 , and interpret physically.

Given: V_0 fixed, $E \rightarrow \infty$. **Find:** the limiting behavior of R .

Step 1 — Write the ratio k_2/k_1 using $k_1 \propto \sqrt{E}$, $k_2 \propto \sqrt{E - V_0}$:

$$\frac{k_2}{k_1} = \sqrt{\frac{E - V_0}{E}} = \sqrt{1 - \frac{V_0}{E}}$$

Step 2 — Take the limit $E \rightarrow \infty$. Since V_0 is fixed, $V_0/E \rightarrow 0$:

$$\frac{k_2}{k_1} \rightarrow \sqrt{1 - 0} = 1$$

Worked Example 2: Step with $E \gg V_0$

Step 3 — Substitute into the reflection formula, dividing numerator and denominator by k_1 to express everything in terms of k_2/k_1 :

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 = \left(\frac{1 - k_2/k_1}{1 + k_2/k_1} \right)^2$$

Step 4 — Substitute $k_2/k_1 \rightarrow 1$:

$$R \rightarrow \left(\frac{1 - 1}{1 + 1} \right)^2 = \left(\frac{0}{2} \right)^2 = 0$$

At high energy the particle barely "notices" the step — the quantum result smoothly approaches the classical expectation of total transmission, as required by the correspondence principle.

Worked Example 3: Step Potential, $E < V_0$ (Total Reflection)

Problem

For $E < V_0$, k_2 becomes imaginary: $k_2 = i\kappa$, $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Show that $R = 1$ exactly, even though ψ is nonzero for $x > 0$.

Given: $k_2 = i\kappa$, κ real and positive. **Find:** show $R = 1$.

Step 1 — Substitute $k_2 = i\kappa$ **into** $R = \left| \frac{k_1 - k_2}{k_1 + k_2} \right|^2$:

$$R = \left| \frac{k_1 - i\kappa}{k_1 + i\kappa} \right|^2$$

Step 2 — Recall that for any complex number z , $|z|^2 = zz^*$. Apply this with $z = (k_1 - i\kappa)/(k_1 + i\kappa)$:

$$z^* = \frac{k_1 + i\kappa}{k_1 - i\kappa} \quad (\text{conjugating flips the sign of every } i)$$

Worked Example 3: Step Potential, $E < V_0$ (Total Reflection)

Step 3 — Multiply $z \times z^*$:

$$zz^* = \frac{(k_1 - i\kappa)(k_1 + i\kappa)}{(k_1 + i\kappa)(k_1 - i\kappa)}$$

Step 4 — Recognize numerator and denominator are identical (both equal $(k_1 - i\kappa)(k_1 + i\kappa)$, just written in opposite order — multiplication commutes):

$$zz^* = \frac{(k_1 - i\kappa)(k_1 + i\kappa)}{(k_1 - i\kappa)(k_1 + i\kappa)} = 1$$

So $R = 1$ exactly. All probability is eventually reflected, but $\psi(x) = Fe^{-\kappa x}$ for $x > 0$ is nonzero — the particle has a nonzero probability of being found briefly inside the "forbidden" region, exactly the seed of tunneling.

Potential Barrier and Quantum Tunneling

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & 0 \leq x \leq a \\ 0 & x > a \end{cases}$$



For $E < V_0$, inside the barrier:

$$\psi(x) \sim e^{-\kappa x}, \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

—————→ x

Transmission Coefficient (thick-barrier approx.)

$$T \approx e^{-2\kappa a}$$

The wavefunction — and hence probability — "leaks through" a classically forbidden barrier. Basis for alpha decay, scanning tunneling microscopy, tunnel diodes.

Derivation: Tunneling — Three-Region Setup

- **Region I** ($x < 0$): $\psi_I = Ae^{ikx} + Be^{-ikx}$, $k = \sqrt{2mE}/\hbar$
- **Region II** ($0 \leq x \leq a$, the barrier, $E < V_0$): $\psi_{II} = Ce^{\kappa x} + De^{-\kappa x}$, $\kappa = \sqrt{2m(V_0 - E)}/\hbar$
- **Region III** ($x > a$): $\psi_{III} = Fe^{ikx}$ (transmitted wave only)
- Matching ψ, ψ' at $x = 0$ and $x = a$ gives **four** equations for the five unknowns B, C, D, F in terms of A — solvable exactly (the full algebra is lengthy; the key physical result is extracted next).

Derivation: Tunneling — Matching Algebra (Step by Step)

- **Boundary conditions at $x = 0$** (continuity of ψ and ψ'):

$$A + B = C + D \quad (\text{I})$$

$$ik(A - B) = \kappa(C - D) \quad (\text{II})$$

- **Boundary conditions at $x = a$** (continuity of ψ and ψ'):

$$Ce^{\kappa a} + De^{-\kappa a} = Fe^{ika} \quad (\text{III})$$

$$\kappa(Ce^{\kappa a} - De^{-\kappa a}) = ik Fe^{ika} \quad (\text{IV})$$

- **Step 1 — Eliminate C and D from (III) and (IV).** Add and subtract (III) and (IV) $\nabla \cdot \kappa$:

$$2Ce^{\kappa a} = Fe^{ika} \left(1 + \frac{ik}{\kappa}\right), \quad 2De^{-\kappa a} = Fe^{ika} \left(1 - \frac{ik}{\kappa}\right)$$

Hence

$$C = \frac{F}{2} e^{ika - \kappa a} \left(1 + \frac{ik}{\kappa}\right), \quad D = \frac{F}{2} e^{ika + \kappa a} \left(1 - \frac{ik}{\kappa}\right)$$

Derivation: Tunneling — Matching Algebra (Step by Step)

- **Step 2 — Substitute into (I) and (II).** Forming $C + D$ and $\kappa(C - D)$:

$$C + D = \frac{F}{2} e^{ika} \left[\left(1 + \frac{ik}{\kappa}\right) e^{-\kappa a} + \left(1 - \frac{ik}{\kappa}\right) e^{\kappa a} \right] = Fe^{ika} \left[\cosh(\kappa a) - \frac{ik}{\kappa} \sinh(\kappa a) \right]$$

$$\kappa(C - D) = \frac{F\kappa}{2} e^{ika} \left[\left(1 + \frac{ik}{\kappa}\right) e^{-\kappa a} - \left(1 - \frac{ik}{\kappa}\right) e^{\kappa a} \right] = Fe^{ika} [-\kappa \sinh(\kappa a) + ik \cosh(\kappa a)]$$

So (I) and (II) become:

$$A + B = Fe^{ika} \left[\cosh(\kappa a) - \frac{ik}{\kappa} \sinh(\kappa a) \right] \quad (I')$$

$$ik(A - B) = Fe^{ika} [-\kappa \sinh(\kappa a) + ik \cosh(\kappa a)] \quad (II')$$

Derivation: Tunneling — Matching Algebra (Step by Step)

- **Step 3 — Solve for A/F .** First isolate $(A - B)$ by dividing (II') through by ik :

$$(A - B) = \frac{Fe^{ika}}{ik} \left[-\kappa \sinh(\kappa a) + ik \cosh(\kappa a) \right]$$

Split this into two separate fractions, one for each term in the bracket:

$$(A - B) = Fe^{ika} \left[-\frac{\kappa}{ik} \sinh(\kappa a) + \frac{ik}{ik} \cosh(\kappa a) \right]$$

The second fraction is trivially $ik/ik = 1$. For the first fraction, recall $1/i = -i$ (true because $i \cdot (-i) = -i^2 = 1$), so

$$\frac{1}{ik} = \frac{1}{i} \cdot \frac{1}{k} = -\frac{i}{k} \quad \implies \quad -\frac{\kappa}{ik} = -\kappa \left(-\frac{i}{k} \right) = \frac{i\kappa}{k}$$

Substituting this back in:

$$(A - B) = Fe^{ika} \left[\cosh(\kappa a) + i\frac{\kappa}{k} \sinh(\kappa a) \right] \quad (\text{II}'')$$

Derivation: Tunneling — Matching Algebra (Step by Step)

Now add (I') and (II'') — i.e. add their left sides together and their right sides together:

$$(A + B) + (A - B) = Fe^{ika} \left[\cosh(\kappa a) - i \frac{k}{\kappa} \sinh(\kappa a) \right] + Fe^{ika} \left[\cosh(\kappa a) + i \frac{\kappa}{k} \sinh(\kappa a) \right]$$

The left side simplifies immediately: $(A + B) + (A - B) = 2A$ (the B 's cancel). On the right, factor out Fe^{ika} and group the two cosh terms together and the two sinh terms together:

$$2A = Fe^{ika} \left[2 \cosh(\kappa a) + i \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right) \sinh(\kappa a) \right]$$

Finally, divide both sides by 2:

$$A = Fe^{ika} \left[\cosh(\kappa a) + \frac{i}{2} \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right) \sinh(\kappa a) \right]$$

Derivation: Tunneling — Matching Algebra (Step by Step)

- **Step 4 — Extract** $|A/F|^2 = 1/T$. Divide both sides of the Step 3 result by F :

$$\frac{A}{F} = e^{ika} \left[\cosh(\kappa a) + \frac{i}{2} \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right) \sinh(\kappa a) \right]$$

This has the form $e^{ika}(x + iy)$ with $x = \cosh(\kappa a)$ and $y = \frac{1}{2} \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right) \sinh(\kappa a)$, both *real* numbers (ratios and products of the real quantities κ, k, a). Since $|e^{ika}| = 1$ and, for any real x, y , $|x + iy|^2 = x^2 + y^2$:

$$\left| \frac{A}{F} \right|^2 = \cosh^2(\kappa a) + \frac{1}{4} \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right)^2 \sinh^2(\kappa a)$$

Simplify the bracket. Put $\frac{\kappa}{k} - \frac{k}{\kappa}$ over the common denominator $k\kappa$:

$$\frac{\kappa}{k} - \frac{k}{\kappa} = \frac{\kappa \cdot \kappa}{k\kappa} - \frac{k \cdot k}{k\kappa} = \frac{\kappa^2 - k^2}{k\kappa} \quad \Rightarrow \quad \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right)^2 = \frac{(\kappa^2 - k^2)^2}{k^2 \kappa^2}$$

Derivation: Tunneling — Matching Algebra (Step by Step)

Use $\cosh^2(\kappa a) = 1 + \sinh^2(\kappa a)$ to pull a plain "1" out of the first term:

$$\left| \frac{A}{F} \right|^2 = 1 + \sinh^2(\kappa a) + \frac{(\kappa^2 - k^2)^2}{4k^2\kappa^2} \sinh^2(\kappa a) = 1 + \sinh^2(\kappa a) \left[1 + \frac{(\kappa^2 - k^2)^2}{4k^2\kappa^2} \right]$$

Combine the bracket into one fraction, using common denominator $4k^2\kappa^2$ (i.e. writing $1 = 4k^2\kappa^2/4k^2\kappa^2$):

$$1 + \frac{(\kappa^2 - k^2)^2}{4k^2\kappa^2} = \frac{4k^2\kappa^2 + (\kappa^2 - k^2)^2}{4k^2\kappa^2}$$

Expand the numerator using $(\kappa^2 - k^2)^2 = \kappa^4 - 2\kappa^2k^2 + k^4$:

$$4k^2\kappa^2 + (\kappa^2 - k^2)^2 = 4k^2\kappa^2 + \kappa^4 - 2\kappa^2k^2 + k^4 = \kappa^4 + 2\kappa^2k^2 + k^4$$

(the $4k^2\kappa^2$ and $-2\kappa^2k^2$ combine to $+2\kappa^2k^2$). The right-hand side is exactly the expansion of $(\kappa^2 + k^2)^2$, so:

$$4k^2\kappa^2 + (\kappa^2 - k^2)^2 = (\kappa^2 + k^2)^2 \implies 1 + \frac{(\kappa^2 - k^2)^2}{4k^2\kappa^2} = \frac{(\kappa^2 + k^2)^2}{4k^2\kappa^2}$$

Derivation: Tunneling — Matching Algebra (Step by Step)

Substitute this back, giving the compact intermediate result:

$$\left| \frac{A}{F} \right|^2 = 1 + \frac{(\kappa^2 + k^2)^2}{4k^2\kappa^2} \sinh^2(\kappa a)$$

Insert the physical definitions $k^2 = 2mE/\hbar^2$ and $\kappa^2 = 2m(V_0 - E)/\hbar^2$. Add them:

$$\kappa^2 + k^2 = \frac{2m(V_0 - E)}{\hbar^2} + \frac{2mE}{\hbar^2} = \frac{2m(V_0 - E) + 2mE}{\hbar^2} = \frac{2mV_0}{\hbar^2}$$

(the $-E$ and $+E$ cancel). Square this:

$$(\kappa^2 + k^2)^2 = \frac{4m^2V_0^2}{\hbar^4}$$

Multiply the two original definitions to get $k^2\kappa^2$:

$$k^2\kappa^2 = \frac{2mE}{\hbar^2} \cdot \frac{2m(V_0 - E)}{\hbar^2} = \frac{4m^2E(V_0 - E)}{\hbar^4}$$

Derivation: Tunneling — Matching Algebra (Step by Step)

Divide the two boxed results to form $(\kappa^2 + k^2)^2 / (4k^2\kappa^2)$; the common factor $4m^2/\hbar^4$ cancels top and bottom:

$$\frac{(\kappa^2 + k^2)^2}{4k^2\kappa^2} = \frac{4m^2 V_0^2 / \hbar^4}{4 \cdot (4m^2 E(V_0 - E) / \hbar^4)} = \frac{V_0^2}{4E(V_0 - E)}$$

Substitute this into the intermediate result to obtain the final, exact transmission formula:

$$T^{-1} = \left| \frac{A}{F} \right|^2 = 1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(\kappa a)$$

which is the exact result displayed on the next frame.

Derivation: Tunneling — Exact Transmission Coefficient

- Carrying out the matching algebra (eliminating B, C, D) yields the **exact** result:

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)} \right]^{-1}$$

- Thick-barrier / strong-tunneling limit** ($\kappa a \gg 1$): $\sinh(\kappa a) \approx \frac{1}{2}e^{\kappa a}$, so

$$T \approx \left[1 + \frac{V_0^2 e^{2\kappa a}}{16E(V_0 - E)} \right]^{-1} \approx \frac{16E(V_0 - E)}{V_0^2} e^{-2\kappa a}$$

- Dropping the slowly varying prefactor gives the commonly used estimate

$$T \approx e^{-2\kappa a}, \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

- T depends **exponentially** on both barrier width a and $\sqrt{V_0 - E}$ — tunneling is extraordinarily sensitive to these parameters, which is exploited in devices like the STM.

Worked Example 1: Tunneling Probability for an Electron

Problem

An electron of energy $E = 2 \text{ eV}$ meets a barrier of height $V_0 = 5 \text{ eV}$ and width $a = 1 \text{ nm}$. Find T .

Given: $E = 2 \text{ eV}$, $V_0 = 5 \text{ eV}$, $a = 1 \text{ nm}$. **Find:** T (transmission coefficient).

Step 1 — Find $V_0 - E$ in joules:

$$V_0 - E = 5 - 2 = 3 \text{ eV} = 3 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-19} \text{ J}$$

Step 2 — Compute $2m_e(V_0 - E)$, the quantity under the square root in κ :

$$2m_e(V_0 - E) = 2(9.11 \times 10^{-31})(4.8 \times 10^{-19}) \approx 8.75 \times 10^{-49}$$

Step 3 — Take the square root, then divide by $\hbar$:

$$\kappa = \frac{\sqrt{8.75 \times 10^{-49}}}{1.055 \times 10^{-34}} \approx \frac{9.35 \times 10^{-25}}{1.055 \times 10^{-34}} \approx 8.87 \times 10^9 \text{ m}^{-1}$$

Worked Example 1: Tunneling Probability for an Electron

Step 4 — Compute the exponent $2\kappa a$:

$$2\kappa a = 2(8.87 \times 10^9)(10^{-9}) \approx 17.7$$

Step 5 — Apply $T \approx e^{-2\kappa a}$:

$$T \approx e^{-17.7} \approx 2.0 \times 10^{-8}$$

Worked Example 2: Halving the Barrier Width

Problem

For the same electron and barrier as above, find T if the barrier width is halved to $a = 0.5$ nm.

Given: same κ as before, but $a = 0.5$ nm. **Find:** T .

Step 1 — Reuse $\kappa \approx 8.87 \times 10^9 \text{ m}^{-1}$ from Worked Example 1 (same energies, same barrier height, only a changes).

Step 2 — Compute the new exponent $2\kappa a$ with the halved width:

$$2\kappa a = 2(8.87 \times 10^9)(0.5 \times 10^{-9}) \approx 8.87$$

Step 3 — Apply $T \approx e^{-2\kappa a}$:

$$T \approx e^{-8.87} \approx 1.4 \times 10^{-4}$$

Step 4 — Compare to the previous result. Form the ratio of the two transmission coefficients:

$$\frac{T_{\text{new}}}{T_{\text{old}}} \approx \frac{1.4 \times 10^{-4}}{2.0 \times 10^{-8}} \approx 7000$$

Worked Example 2: Halving the Barrier Width

Halving the width increased T by a factor of roughly 7000 — illustrates the extreme exponential sensitivity of tunneling to barrier width, central to STM operation (sub-Å vertical resolution).

Worked Example 3: Alpha-Decay Style Estimate

Problem

An alpha particle ($m_\alpha = 6.64 \times 10^{-27}$ kg) with $E = 5$ MeV tunnels through a barrier of effective height $V_0 = 30$ MeV and width $a = 10^{-14}$ m. Estimate T .

Given: m_α , $E = 5$ MeV, $V_0 = 30$ MeV, $a = 10^{-14}$ m. **Find:** T .

Step 1 — Find $V_0 - E$ in joules ($1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$):

$$V_0 - E = 30 - 5 = 25 \text{ MeV} = 25 \times 1.6 \times 10^{-13} = 4.0 \times 10^{-12} \text{ J}$$

Step 2 — Compute $2m_\alpha(V_0 - E)$:

$$2m_\alpha(V_0 - E) = 2(6.64 \times 10^{-27})(4.0 \times 10^{-12}) \approx 5.31 \times 10^{-38}$$

Step 3 — Take the square root, divide by $\hbar$:

$$\kappa = \frac{\sqrt{5.31 \times 10^{-38}}}{1.055 \times 10^{-34}} \approx \frac{2.30 \times 10^{-19}}{1.055 \times 10^{-34}} \approx 1.78 \times 10^{15} \text{ m}^{-1}$$

Worked Example 3: Alpha-Decay Style Estimate

Step 4 — Compute the exponent $2\kappa a$:

$$2\kappa a \approx 2(1.78 \times 10^{15})(10^{-14}) = 35.6$$

Step 5 — Apply $T \approx e^{-2\kappa a}$:

$$T \approx e^{-35.6} \approx 3.5 \times 10^{-16}$$

Tiny but nonzero — exactly this kind of suppressed-but-finite tunneling probability explains the enormously long (but finite) half-lives of alpha-emitting nuclei (Gamow's theory).

Worked Example 4: STM Image Resolution

Problem

A scanning tunneling microscope (STM) tip is scanned over a surface with effective barrier height $V_0 - E = 4 \text{ eV}$. If the tip-to-surface distance varies by just 0.1 nm as it crosses a single surface atom, find the resulting ratio of tunneling currents (current $\propto T$).

Solution:

$$\kappa = \frac{\sqrt{2m_e(V_0 - E)}}{\hbar} = \frac{\sqrt{2(9.11 \times 10^{-31})(4 \times 1.6 \times 10^{-19})}}{1.055 \times 10^{-34}} \approx 1.025 \times 10^{10} \text{ m}^{-1}$$

$$\Delta(2\kappa a) = 2\kappa \Delta a = 2(1.025 \times 10^{10})(0.1 \times 10^{-9}) \approx 2.05$$

$$\frac{T_{\text{far}}}{T_{\text{near}}} = e^{-2.05} \approx 0.13 \quad (\text{an } \sim 8 \times \text{ current change})$$

An atomic-scale height variation of just one Ångstrom-fraction changes the tunneling current by close to an order of magnitude — this extreme sensitivity is precisely what gives STM its sub-atomic vertical resolution, far surpassing any optical microscope.

Dirac Delta Potential

$$V(x) = -\alpha \delta(x), \quad \alpha > 0$$

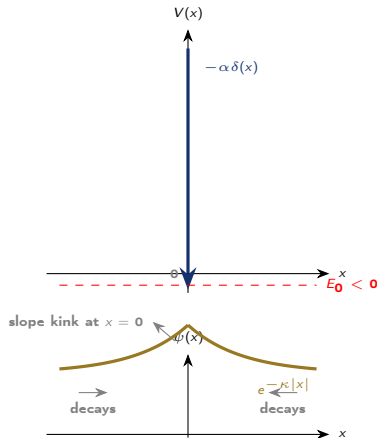
- Infinitely narrow, infinitely deep attractive well.
- Jump condition on ψ' :

$$\psi'(0^+) - \psi'(0^-) = -\frac{2m\alpha}{\hbar^2} \psi(0)$$

- Supports exactly **one** bound state:

$$E = -\frac{m\alpha^2}{2\hbar^2}, \quad \psi = \sqrt{\kappa} e^{-\kappa|x|}$$

- Toy model for short-range nuclear and molecular binding.



Derivation: Delta Potential — Jump Condition

- Integrate the TISE, $-\frac{\hbar^2}{2m}\psi'' - \alpha\delta(x)\psi = E\psi$, across a small interval $(-\epsilon, \epsilon)$:

$$-\frac{\hbar^2}{2m} \int_{-\epsilon}^{\epsilon} \psi'' dx - \alpha \int_{-\epsilon}^{\epsilon} \delta(x)\psi(x) dx = E \int_{-\epsilon}^{\epsilon} \psi dx$$

- As $\epsilon \rightarrow 0$: the first term becomes $-\frac{\hbar^2}{2m} [\psi'(0^+) - \psi'(0^-)]$; the delta function picks out $\psi(0)$; the right side $\rightarrow 0$ since ψ is finite and the interval shrinks.
- Result — a **jump** in the derivative of ψ at the origin:

$$\psi'(0^+) - \psi'(0^-) = -\frac{2m\alpha}{\hbar^2}\psi(0)$$

- ψ itself remains continuous at $x = 0$ (only its slope jumps) — a hallmark of delta-function potentials.

Derivation: Delta Potential — Bound State

- For a bound state ($E < 0$), write $E = -|E|$ and define $\kappa = \sqrt{2m|E|}/\hbar$. Away from $x = 0$ the TISE is just the free equation with negative energy, giving decaying exponentials:

$$\psi(x) = \begin{cases} Ce^{\kappa x} & x < 0 \\ Ce^{-\kappa x} & x > 0 \end{cases}$$

(choosing equal coefficients C on both sides so that ψ is continuous at $x = 0$, i.e. $\psi(0^-) = \psi(0^+) = C$).

- Apply the jump condition: $\psi'(0^+) = -\kappa C$, $\psi'(0^-) = +\kappa C$, so

$$-\kappa C - \kappa C = -\frac{2m\alpha}{\hbar^2} C \quad \Longrightarrow \quad 2\kappa = \frac{2m\alpha}{\hbar^2} \quad \Longrightarrow \quad \kappa = \frac{m\alpha}{\hbar^2}$$

- Since $\kappa = \sqrt{2m|E|}/\hbar$, solving for E :

$$E = -\frac{\hbar^2 \kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2}$$

Derivation: Delta Potential — Bound State

- Exactly **one** bound state exists (only one sign choice for $\kappa > 0$ is consistent) — unlike the infinite well, which has infinitely many.

Worked Example 1: Delta Potential Bound-State Energy

Problem

An electron is bound in a delta potential with $\alpha = 3 \times 10^{-19} \text{ J m}$. Find the bound-state energy.

Given: $\alpha = 3 \times 10^{-19} \text{ J m}$ (delta-potential strength), m_e . **Find:** E .

Step 1 — Compute α^2 :

$$\alpha^2 = (3 \times 10^{-19})^2 = 9 \times 10^{-38} \text{ J}^2 \text{ m}^2$$

Step 2 — Compute the numerator, $m_e \alpha^2$:

$$m_e \alpha^2 = (9.11 \times 10^{-31})(9 \times 10^{-38}) \approx 8.20 \times 10^{-68}$$

Step 3 — Compute the denominator, $2\hbar^2$:

$$2\hbar^2 = 2(1.055 \times 10^{-34})^2 = 2(1.113 \times 10^{-68}) \approx 2.226 \times 10^{-68}$$

Worked Example 1: Delta Potential Bound-State Energy

Step 4 — Divide and apply the overall minus sign:

$$E = -\frac{8.20 \times 10^{-68}}{2.226 \times 10^{-68}} \approx -3.68 \times 10^{-19} \text{ J}$$

Step 5 — Convert to eV (divide by $1.6 \times 10^{-19} \text{ J/eV}$):

$$E \approx -2.30 \text{ eV}$$

Worked Example 2: Normalizing the Delta-Potential Eigenstate

Problem

Verify the normalization constant $C = \sqrt{\kappa}$ for $\psi(x) = Ce^{-\kappa|x|}$.

Given: $\psi(x) = Ce^{-\kappa|x|}$, $\kappa > 0$. **Find:** verify $C = \sqrt{\kappa}$.

Step 1 — Set up the normalization integral:

$$\int_{-\infty}^{\infty} C^2 e^{-2\kappa|x|} dx = 1$$

Step 2 — Split the integral at $x = 0$ using the symmetry of $e^{-2\kappa|x|}$ (the integrand is even, so the two halves are equal):

$$C^2 \int_{-\infty}^{\infty} e^{-2\kappa|x|} dx = 2C^2 \int_0^{\infty} e^{-2\kappa x} dx$$

Worked Example 2: Normalizing the Delta-Potential Eigenstate

Step 3 — Evaluate the remaining integral using $\int_0^\infty e^{-bx} dx = 1/b$ with $b = 2\kappa$:

$$\int_0^\infty e^{-2\kappa x} dx = \frac{1}{2\kappa}$$

Step 4 — Substitute back:

$$2C^2 \times \frac{1}{2\kappa} = \frac{C^2}{\kappa} = 1$$

Step 5 — Solve for C :

$$C^2 = \kappa \quad \implies \quad C = \sqrt{\kappa} \checkmark$$

Worked Example 3: Penetration Depth

Problem

For the bound state of Worked Example 1, find the characteristic penetration depth $1/\kappa$ — the length scale over which ψ decays.

Given: $\alpha = 3 \times 10^{-19} \text{ J m}$, m_e (same as Worked Example 1). **Find:** $1/\kappa$.

Step 1 — Apply $\kappa = m_e \alpha / \hbar^2$. **Compute the numerator, $m_e \alpha$:**

$$m_e \alpha = (9.11 \times 10^{-31} \text{ kg})(3 \times 10^{-19} \text{ J m}) \approx 2.73 \times 10^{-49} \text{ kg J m}$$

Step 2 — Compute the denominator, $\hbar^2$:

$$\hbar^2 = (1.055 \times 10^{-34} \text{ J s})^2 \approx 1.113 \times 10^{-68} \text{ J}^2 \text{ s}^2$$

Step 3 — Divide. Since $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$, the units work out to m^{-1} as required; numerically:

$$\kappa = \frac{2.73 \times 10^{-49}}{1.113 \times 10^{-68}} \approx 2.46 \times 10^{10} \text{ m}^{-1}$$

Worked Example 3: Penetration Depth

Step 4 — Take the reciprocal:

$$\frac{1}{\kappa} \approx \frac{1}{2.46 \times 10^{10}} \approx 4.07 \times 10^{-11} \text{ m} \approx 0.41 \text{ \AA}$$

A tightly localized bound state, comparable to sub-atomic length scales for this (illustrative) value of α .

Worked Example 4: Delta Potential as a Limiting Case

Problem

Consider a finite square well of depth V_0 and width w , in the limit $w \rightarrow 0$, $V_0 \rightarrow \infty$, with the product $\alpha \equiv V_0 w$ held fixed. Argue qualitatively why this reproduces the delta-potential bound-state result.

Solution: A finite well of depth V_0 and width w has binding energy that depends on both parameters. As $w \rightarrow 0$ with $V_0 w = \alpha$ fixed, the well becomes infinitely narrow and deep, but the "strength" (depth $\times$ width, with units of energy $\times$ length, matching α in $V(x) = -\alpha\delta(x)$) stays finite. In this limit the well's depth doesn't matter to the particle individually — only the integrated strength α survives, exactly reproducing the single-bound-state result $E = -m\alpha^2/2\hbar^2$ derived earlier. This limiting procedure is precisely *how* the delta function is rigorously defined in physics, and explains physically why the delta potential is a good approximation for any sufficiently short-range, strong attractive interaction (e.g. certain nuclear or molecular binding scenarios).

Worked Example 5: A Finite Well Always Has at Least One Bound State

Problem

For a symmetric finite square well of depth V_0 and width L (i.e. $V(x) = -V_0$ for $|x| < L/2$, zero outside), show that at least one (even-parity) bound state exists no matter how shallow V_0 is.

Given: matching conditions for the even bound-state wavefunction. **Find:** argue $N \geq 1$ always.

Step 1 — Set up the interior and exterior solutions. Inside ($|x| < L/2$): $\psi = A \cos(kx)$ with $k = \sqrt{2m(E + V_0)}/\hbar$; outside: $\psi = Be^{-\kappa|x|}$ with $\kappa = \sqrt{-2mE}/\hbar$ (writing $E < 0$ for a bound state).

Step 2 — Match ψ and ψ' at $x = L/2$, then divide the two conditions to eliminate the overall amplitudes, giving the transcendental **even-state condition**:

$$k \tan\left(\frac{kL}{2}\right) = \kappa$$

Step 3 — Note the constraint linking k and κ . Since $k^2 + \kappa^2 = 2mV_0/\hbar^2 \equiv k_0^2$ is fixed by the well parameters, (k, κ) moves along a quarter-circle of radius k_0 as E ranges from $-V_0$ (where $k = k_0, \kappa = 0$) to 0 (where $k = 0, \kappa = k_0$).

Worked Example 5: A Finite Well Always Has at Least One Bound State

Step 4 — Compare the two sides of the matching condition at the endpoints of this arc. At $k = 0$: the left side $k \tan(kL/2) \rightarrow 0$, while the right side $\kappa = k_0 > 0$ — so $\text{LHS} < \text{RHS}$. At $k = k_0$ (i.e. $\kappa = 0$): the left side $k_0 \tan(k_0 L/2)$ is generally positive (for small enough $k_0 L/2$, it is small but still $> 0 = \text{RHS}$, or the tangent may already have crossed a branch) — either way, by continuity of both sides along the arc, the curves $k \tan(kL/2)$ and $\kappa = \sqrt{k_0^2 - k^2}$ must cross **at least once** in $0 < k < k_0$, regardless of how small k_0 (i.e. how shallow V_0) is.

Step 5 — Conclude. This crossing is exactly the ground-state (even, nodeless) bound-state energy. Unlike the finite well in 3D (where a minimum depth is required to bind a state), the 1D finite square well **always** supports at least one bound state — consistent with the $w \rightarrow 0$ delta-well limit of the previous example, which likewise always has exactly one bound state.

Worked Example 6: Estimating the Number of Bound States in a Finite Well

Problem

An electron sits in a finite square well of depth $V_0 = 4 \text{ eV}$ and full width $L = 1 \text{ nm}$. Estimate the number of bound states, using the standard graphical-counting estimate: a new bound state appears roughly every time the parameter $z_0 \equiv (L/2)\sqrt{2mV_0}/\hbar$ increases by $\pi/2$, so $N \approx \lceil 2z_0/\pi \rceil$.

Given: $V_0 = 4 \text{ eV}$, $L = 1 \text{ nm}$, m_e . **Find:** N .

Step 1 — Convert V_0 to joules:

$$V_0 = 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ J}$$

Step 2 — Compute $2mV_0$:

$$2m_e V_0 = 2(9.11 \times 10^{-31})(6.4 \times 10^{-19}) \approx 1.166 \times 10^{-48}$$

Worked Example 6: Estimating the Number of Bound States in a Finite Well

Step 3 — Take the square root:

$$\sqrt{2m_e V_0} \approx 1.080 \times 10^{-24} \text{ kg m/s}$$

Step 4 — Evaluate $z_0 = (L/2)\sqrt{2m_e V_0}/\hbar$, with $L/2 = 5 \times 10^{-10} \text{ m}$:

$$z_0 = \frac{(5 \times 10^{-10})(1.080 \times 10^{-24})}{1.055 \times 10^{-34}} \approx \frac{5.40 \times 10^{-34}}{1.055 \times 10^{-34}} \approx 5.12$$

Step 5 — Apply the counting estimate:

$$N \approx \left\lceil \frac{2(5.12)}{\pi} \right\rceil = \lceil 3.26 \rceil = 4$$

So this well supports approximately **4 bound states**. Comparing to Worked Example 6 of the infinite-well subsection: a finite well is always “shallower” than an infinite one of the same width, so it holds *fewer* bound states, and (unlike the infinite well) only a finite number of them — states with $E > 0$ are unbound scattering states instead.

Why the Harmonic Oscillator Is Everywhere

- Any smooth potential $V(x)$ near a local minimum x_0 can be Taylor-expanded:

$$V(x) \approx V(x_0) + \frac{1}{2} V''(x_0)(x - x_0)^2 + \dots$$

since the first-derivative term vanishes at a minimum. To leading order, **every** stable equilibrium looks like a harmonic oscillator.

- This is why the quantum harmonic oscillator appears throughout physics: molecular vibrations (bonds stretching about their equilibrium length), lattice vibrations in solids (phonons), the quantized electromagnetic field itself (each mode behaves as an independent oscillator), and as a building block for more advanced techniques like perturbation theory.
- Unlike the infinite well (Unit 4, previous subsection), the harmonic-oscillator potential is smooth and unbounded, making it both more realistic for physical systems and mathematically richer — it is exactly solvable, but via a more elegant operator-based method rather than simple boundary-matching.

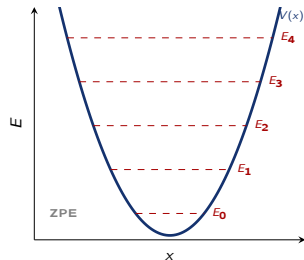
Quantum Harmonic Oscillator: Setup

$$V(x) = \frac{1}{2}m\omega^2x^2$$

TISE

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi$$

- Approximates *any* smooth potential near a minimum.
- Solved via Hermite polynomials or, more elegantly, via **ladder operators**.



Derivation: Ladder Operators

- Define dimensionless **raising/lowering (ladder) operators**:

$$\hat{a}_{\pm} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} \mp \frac{i}{m\omega} \hat{p} \right)$$

- Using $[\hat{x}, \hat{p}] = i\hbar$ (Unit 5), one can show $[\hat{a}_-, \hat{a}_+] = 1$, and the Hamiltonian becomes

$$\hat{H} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

- Define the **number operator** $\hat{N} = \hat{a}_+ \hat{a}_-$, with $\hat{H} = \hbar\omega(\hat{N} + \frac{1}{2})$.
- Key commutators: $[\hat{N}, \hat{a}_+] = \hat{a}_+$, $[\hat{N}, \hat{a}_-] = -\hat{a}_-$. If $|\psi_n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue n , then $\hat{a}_+|\psi_n\rangle$ has eigenvalue $n+1$ and $\hat{a}_-|\psi_n\rangle$ has eigenvalue $n-1$ — hence "raising" and "lowering" operators.

Derivation: Ladder Operators — Commutator and Hamiltonian (Full Detail)

- Define the shorthand $\xi = \sqrt{m\omega/2\hbar}$ so that $\hat{a}_{\pm} = \xi\hat{x} \mp \frac{i}{\sqrt{2m\omega\hbar}}\hat{p}$. We compute $\hat{a}_-\hat{a}_+$ and $\hat{a}_+\hat{a}_-$ explicitly.
- **Step 1 — Expand $\hat{a}_-\hat{a}_+$:**

$$\hat{a}_-\hat{a}_+ = \left(\xi\hat{x} + \frac{i\hat{p}}{\sqrt{2m\omega\hbar}} \right) \left(\xi\hat{x} - \frac{i\hat{p}}{\sqrt{2m\omega\hbar}} \right) = \xi^2\hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} - \frac{i\xi}{\sqrt{2m\omega\hbar}} [\hat{x}, \hat{p}] \cdot (-1) \cdot (-1)$$

More carefully, multiplying out the four terms:

$$\begin{aligned}\hat{a}_-\hat{a}_+ &= \xi^2\hat{x}^2 - \frac{i\xi\hat{x}\hat{p}}{\sqrt{2m\omega\hbar}} + \frac{i\xi\hat{p}\hat{x}}{\sqrt{2m\omega\hbar}} + \frac{\hat{p}^2}{2m\omega\hbar} = \xi^2\hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} + \frac{i\xi}{\sqrt{2m\omega\hbar}}(\hat{p}\hat{x} - \hat{x}\hat{p}) \\ &= \xi^2\hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} + \frac{i\xi}{\sqrt{2m\omega\hbar}}(-[\hat{x}, \hat{p}]) = \xi^2\hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} + \frac{i\xi}{\sqrt{2m\omega\hbar}}(-i\hbar)\end{aligned}$$

Derivation: Ladder Operators — Commutator and Hamiltonian (Full Detail)

$$= \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} + \frac{\xi\hbar}{\sqrt{2m\omega\hbar}}$$

Substituting $\xi = \sqrt{m\omega/2\hbar}$ back into the last term: $\xi\hbar/\sqrt{2m\omega\hbar} = \frac{1}{2}$, so:

$$\hat{a}_- \hat{a}_+ = \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} + \frac{1}{2}$$

- **Step 2 — Note that $\hat{a}_+ \hat{a}_-$ differs only by sign of the commutator term.** In $\hat{a}_- \hat{a}_+$ the cross-term gave $+\frac{1}{2}$ (from $[\hat{x}, \hat{p}] = i\hbar$). Swapping the order reverses the commutator sign, so the $+\frac{1}{2}$ becomes $-\frac{1}{2}$:

$$\hat{a}_+ \hat{a}_- = \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{\hat{p}^2}{2m\omega\hbar} - \frac{1}{2}$$

Derivation: Ladder Operators — Commutator and Hamiltonian (Full Detail)

- **Step 3 — The commutator $[\hat{a}_-, \hat{a}_+]$:**

$$[\hat{a}_-, \hat{a}_+] = \hat{a}_- \hat{a}_+ - \hat{a}_+ \hat{a}_- = \left(\cdots + \frac{1}{2} \right) - \left(\cdots - \frac{1}{2} \right) = 1$$

- **Step 4 — Express $\hat{H}$ in terms of $\hat{a}_\pm$.** Multiplying $\hat{a}_+ \hat{a}_-$ by $\hbar\omega$:

$$\hbar\omega \hat{a}_+ \hat{a}_- = \frac{m\omega^2}{2} \hat{x}^2 + \frac{\hat{p}^2}{2m} - \frac{\hbar\omega}{2} = \hat{H} - \frac{\hbar\omega}{2}$$

Therefore:

$$\hat{H} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) = \hbar\omega \left(\hat{N} + \frac{1}{2} \right), \quad \hat{N} \equiv \hat{a}_+ \hat{a}_-$$

Derivation: Ladder Operators — Commutator and Hamiltonian (Full Detail)

- **Step 5 — Derive $[\hat{N}, \hat{a}_+]$ and $[\hat{N}, \hat{a}_-]$ using $[\hat{a}_-, \hat{a}_+] = 1$:**

$$[\hat{N}, \hat{a}_+] = [\hat{a}_+ \hat{a}_-, \hat{a}_+] = \hat{a}_+ [\hat{a}_-, \hat{a}_+] + [\hat{a}_+, \hat{a}_+] \hat{a}_- = \hat{a}_+ \cdot 1 + 0 = \hat{a}_+$$

$$[\hat{N}, \hat{a}_-] = [\hat{a}_+ \hat{a}_-, \hat{a}_-] = \hat{a}_+ [\hat{a}_-, \hat{a}_-] + [\hat{a}_+, \hat{a}_-] \hat{a}_- = 0 + (-1) \hat{a}_- = -\hat{a}_-$$

confirming the raising/lowering structure stated in the previous frame.

Alternative Route: Power Series (Frobenius) Method

- The ladder-operator method is elegant but not the only way in — the spectrum can also be obtained by **directly solving the TISE** with ordinary calculus, the same style of approach used earlier for the infinite well.
- **Step 1 — Start from the TISE:**

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

- **Step 2 — Non-dimensionalize.** Let $\xi = \sqrt{\frac{m\omega}{\hbar}} x$ and $\epsilon = \frac{2E}{\hbar\omega}$. The equation becomes

$$\frac{d^2\psi}{d\xi^2} + (\epsilon - \xi^2)\psi = 0$$

Alternative Route: Power Series (Frobenius) Method

- **Step 3 — Peel off the asymptotic behaviour.** For large ξ , $\psi'' \approx \xi^2 \psi$, whose solutions behave like $e^{\pm \xi^2/2}$. Keeping only the normalizable branch motivates the substitution

$$\psi(\xi) = h(\xi) e^{-\xi^2/2}$$

for an as-yet-unknown function $h(\xi)$.

- **Step 4 — Substitute back.** Differentiating $\psi = h e^{-\xi^2/2}$ twice and inserting into the TISE, the exponential factors cancel and $h(\xi)$ is left obeying the **Hermite equation**:

$$h'' - 2\xi h' + (\epsilon - 1)h = 0$$

Alternative Route: Series Solution and Quantization

- **Step 5 — Series ansatz.** Write $h(\xi) = \sum_{k=0}^{\infty} a_k \xi^k$ and substitute into the Hermite equation. Matching powers of ξ term by term gives a two-term **recursion relation**:

$$a_{k+2} = \frac{2k - (\epsilon - 1)}{(k+1)(k+2)} a_k$$

so every coefficient is fixed once a_0 and a_1 are chosen (even and odd solutions decouple).

- **Step 6 — Impose normalizability.** Left as an infinite series, $h(\xi)$ behaves like e^{ξ^2} for large ξ , which combined with the $e^{-\xi^2/2}$ prefactor blows up rather than decaying — **not** normalizable.
- The only way to avoid this is for the series to **terminate** after a finite number of terms, i.e. become a polynomial of some finite degree n . From the recursion relation, the numerator must vanish at $k = n$:

$$2n - (\epsilon - 1) = 0 \quad \implies \quad \epsilon = 2n + 1, \quad n = 0, 1, 2, \dots$$

Alternative Route: Series Solution and Quantization

- **Step 7 — Recover the energy spectrum.** Undoing $\epsilon = 2E/\hbar\omega$:

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n = 0, 1, 2, \dots$$

Identical to the ladder-operator result — quantization here comes from demanding a *normalizable* solution, rather than from the non-negativity of $\hat{N}$.

- **Step 8 — Identify the polynomials.** The terminating series solutions $h(\xi)$ are, by definition, the **Hermite polynomials** $H_n(\xi)$, giving back the familiar eigenfunctions

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-m\omega x^2/2\hbar}$$

- **Comparison:** both methods give the same E_n and ψ_n . Ladder operators are algebraically slicker and generalize beautifully (to QFT, coherent states, etc.); the series method is more mechanical but uses only tools already familiar from solving ordinary differential equations.

Derivation: QHO Ground State Wavefunction

- **Starting point:** the ground state ψ_0 is defined by $\hat{a}_- \psi_0 = 0$. Writing out the lowering operator explicitly:

$$\hat{a}_- = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{\hbar}{m\omega} \frac{d}{dx} \right)$$

- **Step 1 — Write the ODE.** Setting $\hat{a}_- \psi_0 = 0$:

$$\sqrt{\frac{m\omega}{2\hbar}} \left(x\psi_0 + \frac{\hbar}{m\omega} \frac{d\psi_0}{dx} \right) = 0 \quad \implies \quad \frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar} x \psi_0$$

- **Step 2 — Separate variables.** Divide both sides by ψ_0 and write as differentials:

$$\frac{d\psi_0}{\psi_0} = -\frac{m\omega}{\hbar} x dx$$

Derivation: QHO Ground State Wavefunction

- **Step 3 — Integrate both sides:**

$$\ln \psi_0 = -\frac{m\omega}{2\hbar} x^2 + C \quad \Longrightarrow \quad \psi_0(x) = A e^{-m\omega x^2/2\hbar}$$

where A is a normalization constant.

- **Step 4 — Normalize.** Impose $\int_{-\infty}^{\infty} |\psi_0|^2 dx = 1$, using the Gaussian integral $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha}$ with $\alpha = m\omega/\hbar$:

$$|A|^2 \int_{-\infty}^{\infty} e^{-m\omega x^2/\hbar} dx = |A|^2 \sqrt{\frac{\pi\hbar}{m\omega}} = 1 \quad \Longrightarrow \quad A = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

- **Result:**

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}$$

A Gaussian centred at the origin — the only QHO eigenstate with no nodes.

Derivation: Harmonic Oscillator Energy Spectrum

- Since $\hat{N} = \hat{a}_+ \hat{a}_-$ is built from a product of an operator and its adjoint, its eigenvalues must be **non-negative**: $\langle \psi_n | \hat{N} | \psi_n \rangle = \| \hat{a}_- \psi_n \|^2 \geq 0$.
- Repeatedly applying $\hat{a}_-$ must therefore terminate: there is a **ground state** $|\psi_0\rangle$ with $\hat{a}_- |\psi_0\rangle = 0$.
- Solving $\hat{a}_- \psi_0 = 0$ explicitly (a first-order ODE in x) gives the Gaussian ground state $\psi_0(x) \propto e^{-m\omega x^2/2\hbar}$, with $\hat{N} |\psi_0\rangle = 0$.
- Since $\hat{H} |\psi_n\rangle = \hbar\omega(n + \frac{1}{2}) |\psi_n\rangle$ and $n = 0, 1, 2, \dots$ (each application of $\hat{a}_+$ raises n by 1 starting from $n = 0$):

$$E_n = \left(n + \frac{1}{2} \right) \hbar\omega$$

- Excited states are generated by repeated raising: $|\psi_n\rangle \propto (\hat{a}_+)^n |\psi_0\rangle$, which (after converting back to position representation) produces the Hermite-polynomial wavefunctions.

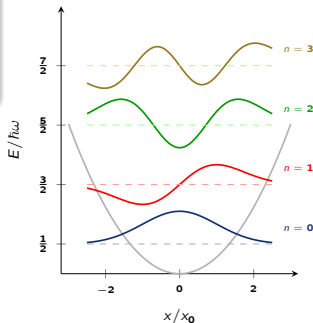
Harmonic Oscillator: Energy Spectrum and Eigenstates

Quantized Energies

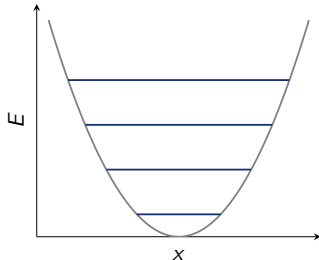
$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega$$

- **Zero-point energy** $E_0 = \frac{1}{2}\hbar\omega \neq 0$ — direct consequence of the uncertainty principle.
- Energy levels are **equally spaced** by $\hbar\omega$.
- Eigenfunctions involve **Hermite polynomials**:

$$\psi_n \propto H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) e^{-m\omega x^2/2\hbar}$$



Visualizing the Energy Ladder



- The energy levels (horizontal lines) are **equally spaced** by $\hbar\omega$, unlike the box spectrum ($E_n \propto n^2$, increasingly spaced).
- Each level's width on the diagram represents the classically allowed region for that energy — note it grows with n , just as a classical pendulum swings further at higher energy.
- Ladder operators $\hat{a}_+$, $\hat{a}_-$ literally move the system up/down one rung at a time — the visual origin of the "ladder operator" name.

Derivation: Zero-Point Energy from the Uncertainty Principle

- Classically the minimum energy is $E = 0$ (particle at rest at $x = 0$). Quantum mechanically this is forbidden by the uncertainty principle. Estimate the minimum energy using $\Delta x \Delta p \sim \hbar/2$:

$$E \sim \frac{(\Delta p)^2}{2m} + \frac{1}{2}m\omega^2(\Delta x)^2$$

- Using $\Delta p \sim \hbar/(2\Delta x)$, treat E as a function of Δx alone and minimize:

$$E(\Delta x) = \frac{\hbar^2}{8m(\Delta x)^2} + \frac{1}{2}m\omega^2(\Delta x)^2$$

$$\frac{dE}{d(\Delta x)} = -\frac{\hbar^2}{4m(\Delta x)^3} + m\omega^2\Delta x = 0 \Rightarrow (\Delta x)^2 = \frac{\hbar}{2m\omega}$$

- Substituting back: $E_{\min} = \frac{\hbar\omega}{4} + \frac{\hbar\omega}{4} = \frac{\hbar\omega}{2}$ — matches the exact ground-state energy $E_0 = \hbar\omega/2$, confirming that the zero-point energy is a direct, quantitative manifestation of the uncertainty principle.

Worked Example 1: Zero-Point Energy of a Diatomic Molecule

Problem

The HCl molecule vibrates with classical frequency $f = 8.66 \times 10^{13}$ Hz. Find its zero-point energy in eV.

Given: $f = 8.66 \times 10^{13}$ Hz (ordinary vibration frequency). **Find:** E_0 , the zero-point energy.

Step 1 — Convert to angular frequency, $\omega = 2\pi f$:

$$\omega = 2\pi(8.66 \times 10^{13}) \approx 5.44 \times 10^{14} \text{ rad/s}$$

Step 2 — Apply $E_0 = \frac{1}{2}\hbar\omega$:

$$E_0 = \frac{1}{2}(1.055 \times 10^{-34} \text{ J s})(5.44 \times 10^{14} \text{ rad/s})$$

Step 3 — Multiply, then halve:

$$\hbar\omega = (1.055 \times 10^{-34})(5.44 \times 10^{14}) \approx 5.74 \times 10^{-20} \text{ J} \quad \Rightarrow \quad E_0 = \frac{5.74 \times 10^{-20}}{2} \approx 2.87 \times 10^{-20} \text{ J}$$

Worked Example 1: Zero-Point Energy of a Diatomic Molecule

Step 4 — Convert to eV (divide by 1.6×10^{-19} J/eV):

$$E_0 \approx \frac{2.87 \times 10^{-20}}{1.6 \times 10^{-19}} \approx 0.18 \text{ eV}$$

This nonzero vibrational energy persists even at absolute zero temperature.

Worked Example 2: Photon Absorbed in a Vibrational Transition

Problem

For the same molecule, find the photon energy and wavelength absorbed in the $n = 0 \rightarrow n = 1$ vibrational transition.

Given: $E_0 \approx 2.87 \times 10^{-20} \text{ J}$ (from Worked Example 1). **Find:** ΔE and λ .

Step 1 — Find the energy gap. Since levels are equally spaced by $\hbar\omega$ (and $E_0 = \frac{1}{2}\hbar\omega$), the gap is $\Delta E = E_1 - E_0 = \hbar\omega = 2E_0$:

$$\Delta E = 2(2.87 \times 10^{-20}) \approx 5.74 \times 10^{-20} \text{ J}$$

Step 2 — Convert to eV (divide by 1.6×10^{-19}):

$$\Delta E \approx 0.358 \text{ eV}$$

Worked Example 2: Photon Absorbed in a Vibrational Transition

Step 3 — Apply energy conservation, $\Delta E = hc/\lambda$, solve for λ :

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{5.74 \times 10^{-20}}$$

Step 4 — Evaluate:

$$\lambda \approx \frac{1.988 \times 10^{-25}}{5.74 \times 10^{-20}} \approx 3.46 \times 10^{-6} \text{ m} \approx 3.46 \mu\text{m (IR)}$$

Worked Example 3: Effective Spring Constant

Problem

A quantum oscillator of mass $m = 2 \times 10^{-26}$ kg has energy level spacing $\Delta E = 0.1$ eV. Find the effective spring constant $k_{\text{spring}} = m\omega^2$.

Given: $m = 2 \times 10^{-26}$ kg, $\Delta E = 0.1$ eV. **Find:** k_{spring} .

Step 1 — Convert ΔE to joules:

$$\Delta E = 0.1 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-20} \text{ J}$$

Step 2 — Apply $\Delta E = \hbar\omega$ and solve for ω :

$$\omega = \frac{\Delta E}{\hbar} = \frac{1.6 \times 10^{-20}}{1.055 \times 10^{-34}} \approx 1.52 \times 10^{14} \text{ rad/s}$$

Step 3 — Apply $k_{\text{spring}} = m\omega^2$. First square ω :

$$\omega^2 \approx (1.52 \times 10^{14})^2 \approx 2.31 \times 10^{28} \text{ rad}^2/\text{s}^2$$

Worked Example 3: Effective Spring Constant

Step 4 — Multiply by m :

$$k_{\text{spring}} = (2 \times 10^{-26})(2.31 \times 10^{28}) \approx 462 \text{ N/m}$$

Worked Example 4: Number of Thermally Excited Quanta

Problem

At temperature T , the average number of thermally excited oscillator quanta is given by the Bose–Einstein/Planck-like formula $\langle n \rangle = 1/(e^{\hbar\omega/k_B T} - 1)$ (the same form derived for blackbody radiation in Unit 1!). For $\hbar\omega = 0.1$ eV and $T = 300$ K, find $\langle n \rangle$.

Given: $\hbar\omega = 0.1$ eV, $T = 300$ K. **Find:** $\langle n \rangle$.

Step 1 — Convert $\hbar\omega$ to joules. (multiply by 1.6×10^{-19} J/eV):

$$\hbar\omega = 0.1 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-20} \text{ J}$$

Step 2 — Compute $k_B T$. (multiply Boltzmann's constant by the temperature):

$$k_B T = (1.381 \times 10^{-23})(300) \approx 4.14 \times 10^{-21} \text{ J}$$

Worked Example 4: Number of Thermally Excited Quanta

Step 3 — Form the ratio $\hbar\omega/k_B T$.

$$\frac{\hbar\omega}{k_B T} = \frac{1.6 \times 10^{-20}}{4.14 \times 10^{-21}} \approx 3.86$$

Step 4 — Evaluate $e^{3.86} - 1$.

$$e^{3.86} \approx 47.5 \quad \implies \quad e^{3.86} - 1 \approx 46.5$$

Step 5 — Take the reciprocal.

$$\langle n \rangle = \frac{1}{46.5} \approx 0.0214$$

At room temperature, this oscillator is overwhelmingly likely to be found in its ground state ($n = 0$); only about 2% of the time would it be thermally excited to $n = 1$ or higher. This identical mathematical structure — linking the harmonic oscillator's discrete levels to Planck's blackbody quantization — is no coincidence: both describe quantized energy exchange with a thermal reservoir.

Worked Example 5: Probability of Tunneling Beyond the Classical Turning Point

Problem

For the ground state of the harmonic oscillator, find the probability of finding the particle beyond the classical turning point x_0 (i.e. in the classically forbidden region where $V(x) > E_0$). Use $P(|x| > X) = \text{erfc}(X/(\sigma\sqrt{2}))$ for a Gaussian of width σ , as established earlier in the course.

Given: $\psi_0(x) = (m\omega/\pi\hbar)^{1/4} e^{-m\omega x^2/2\hbar}$, $E_0 = \hbar\omega/2$. **Find:** $P(|x| > x_0)$.

Step 1 — Find the classical turning point x_0 by setting $V(x_0) = E_0$:

$$\frac{1}{2}m\omega^2 x_0^2 = \frac{\hbar\omega}{2} \quad \Longrightarrow \quad x_0 = \sqrt{\frac{\hbar}{m\omega}}$$

Worked Example 5: Probability of Tunneling Beyond the Classical Turning Point

Step 2 — Identify σ for the Gaussian ψ_0 . Writing $|\psi_0|^2 \propto e^{-x^2/2\sigma^2}$ and comparing exponents, $\frac{1}{2\sigma^2} = \frac{m\omega}{\hbar}$, so:

$$\sigma = \sqrt{\frac{\hbar}{2m\omega}}$$

Step 3 — Form the ratio $x_0/(\sigma\sqrt{2})$. Since $x_0 = \sqrt{\hbar/m\omega} = \sqrt{2}\sigma$:

$$\frac{x_0}{\sigma\sqrt{2}} = \frac{\sqrt{2}\sigma}{\sigma\sqrt{2}} = 1$$

Step 4 — Apply the erfc formula:

$$P(|x| > x_0) = \text{erfc}(1) \approx 0.1573$$

Worked Example 5: Probability of Tunneling Beyond the Classical Turning Point

Step 5 — Interpret. About 15.7% of the time, the ground-state oscillator is found beyond the classical turning point, in a region where a classical particle with the same total energy could never be found ($V(x) > E_0$ there) — a purely quantum effect, directly analogous to barrier penetration in Worked Examples 1–4 of the previous subsection.

Worked Example 6: Expectation Value of Potential Energy via the Virial Theorem

Problem

Verify, for the harmonic-oscillator ground state, that $\langle V \rangle = E_0/2$ (the quantum virial theorem for a quadratic potential, which states $\langle T \rangle = \langle V \rangle = E_n/2$ for every n).

Given: $\psi_0(x) = (m\omega/\pi\hbar)^{1/4} e^{-m\omega x^2/2\hbar}$, $V(x) = \frac{1}{2}m\omega^2 x^2$. **Find:** $\langle V \rangle$, compared to $E_0/2$.

Step 1 — Set up $\langle x^2 \rangle$, needed for $\langle V \rangle = \frac{1}{2}m\omega^2 \langle x^2 \rangle$:

$$\langle x^2 \rangle = \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} x^2 e^{-m\omega x^2/2\hbar} dx$$

Step 2 — Apply the standard Gaussian moment integral $\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \frac{1}{2\alpha} \sqrt{\pi/\alpha}$, with $\alpha = m\omega/\hbar$:

$$\int_{-\infty}^{\infty} x^2 e^{-m\omega x^2/2\hbar} dx = \frac{\hbar}{2m\omega} \sqrt{\frac{\pi\hbar}{m\omega}}$$

Worked Example 6: Expectation Value of Potential Energy via the Virial Theorem

Step 3 — Substitute back and simplify, noting $\sqrt{m\omega/\pi\hbar} \times \sqrt{\pi\hbar/m\omega} = 1$:

$$\langle x^2 \rangle = \sqrt{\frac{m\omega}{\pi\hbar}} \times \frac{\hbar}{2m\omega} \sqrt{\frac{\pi\hbar}{m\omega}} = \frac{\hbar}{2m\omega}$$

Step 4 — Compute $\langle V \rangle$:

$$\langle V \rangle = \frac{1}{2}m\omega^2 \langle x^2 \rangle = \frac{1}{2}m\omega^2 \cdot \frac{\hbar}{2m\omega} = \frac{\hbar\omega}{4}$$

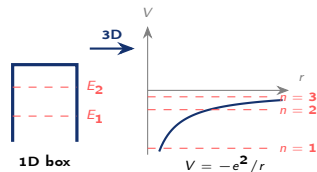
Step 5 — Compare to $E_0/2$:

$$\frac{E_0}{2} = \frac{1}{2} \cdot \frac{\hbar\omega}{2} = \frac{\hbar\omega}{4} \quad \implies \quad \langle V \rangle = \frac{E_0}{2} \quad \checkmark$$

By symmetry $\langle T \rangle = E_0 - \langle V \rangle = \hbar\omega/4 = E_0/2$ as well — the oscillator's total energy is shared *exactly equally*, on average, between kinetic and potential forms, a hallmark of any quadratic (spring-like) confining potential.

The Capstone Problem: From 1D to 3D

- Every potential so far has been strictly 1D — real atoms exist in 3D space.
- The hydrogen atom is the most important 3D problem: exactly solvable, explains the periodic table, atomic spectra, and chemical bonding.
- Moving to 3D introduces **angular momentum** — orbital motion is quantized.
- Key simplification: **spherical symmetry** (V depends only on r) — exploited via separation in spherical coordinates.



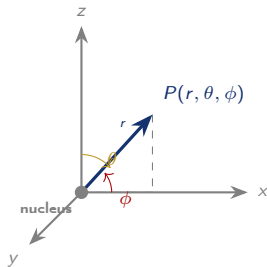
3 quantum numbers: n, ℓ, m_ℓ

Schrödinger Equation in Spherical Polar Coordinates

For a hydrogen-like atom, $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

- Separate variables: $\psi = R(r) \Theta(\theta) \Phi(\phi)$.
- Three ODEs $\Rightarrow$ three quantum numbers (n, ℓ, m_ℓ) .
- Radial part $\rightarrow$ energy quantization; angular part $\rightarrow$ orbital shape.



Energy Levels

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

Derivation: Separation in Spherical Coordinates

- **Step 1 — Write the TISE with the spherical Laplacian.** In spherical coordinates,

$$\nabla^2\psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial\psi}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial\psi}{\partial\theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2\psi}{\partial\phi^2}$$

so the TISE $-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$ becomes, term by term,

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial\psi}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial\psi}{\partial\theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2\psi}{\partial\phi^2} \right] + V(r)\psi = E\psi$$

- **Step 2 — Substitute the product ansatz** $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$. Since R depends only on r and Y only on θ, ϕ , each partial derivative passes through the factor that does not depend on that variable ($\partial\psi/\partial r = Y dR/dr$, etc.):

$$-\frac{\hbar^2}{2m} \left[\frac{Y}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{R}{r^2 \sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial Y}{\partial\theta} \right) + \frac{R}{r^2 \sin^2\theta} \frac{\partial^2 Y}{\partial\phi^2} \right] + V(r)RY = ERY$$

Derivation: Separation in Spherical Coordinates

- **Step 3 — Divide every term by $\psi = RY$** to separate the variables:

$$-\frac{\hbar^2}{2m} \left[\frac{1}{Rr^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{Yr^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{Yr^2 \sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] + V(r) = E$$

- **Step 4 — Multiply through by $-2mr^2/\hbar^2$** to clear the leading constant, and move $V(r) - E$ to be grouped with the radial part:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] + \frac{1}{Y \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{Y \sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} = 0$$

- **Step 5 — Isolate the radial and angular groups** on opposite sides of the equation (subtract the angular group from both sides):

$$\underbrace{\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E]}_{\text{depends only on } r} = - \underbrace{\left[\frac{1}{Y \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{Y \sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right]}_{\text{depends only on } \theta, \phi}$$

Derivation: Separation in Spherical Coordinates

- **Step 6 — Apply the separation-of-variables argument.** The left side can change only if r changes; the right side can change only if θ, ϕ change. Since r, θ, ϕ are independent variables, the only way the two sides can stay equal for *all* values is if both sides equal the same constant. Call this separation constant $\ell(\ell + 1)$ (the particular form is chosen purely for later algebraic convenience — it does not assume anything about ℓ yet):

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] = \ell(\ell + 1)$$
$$- \left[\frac{1}{Y \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{Y \sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] = \ell(\ell + 1)$$

Derivation: Separation in Spherical Coordinates

- **Step 7 — Rearrange the angular equation.** Multiply both sides by $-Y$ to put it in the standard form:

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} = -\ell(\ell + 1)Y$$

This, together with the radial equation from Step 6, is exactly the pair of separated equations. Solving the angular equation shows $\ell = 0, 1, 2, \dots$ is required for Y to be single-valued and finite at the poles, with solutions being the **spherical harmonics** $Y_\ell^{m_\ell}(\theta, \phi)$.

- The radial equation, with $u(r) = rR(r)$, reduces to a 1D-like Schrödinger equation with an **effective potential**:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2 \ell(\ell + 1)}{2mr^2} \right] u = Eu$$

the extra term being the familiar centrifugal barrier.

Derivation: Hydrogen Energy Levels (Bohr-style Estimate)

- A full solution of the radial equation uses a power series (Laguerre polynomials); here is a quick semiclassical cross-check using the Bohr quantization $L = n\hbar$ together with circular orbit balance:

$$\frac{mv^2}{r} = \frac{Ze^2}{4\pi\epsilon_0 r^2}, \quad mvr = n\hbar$$

- Solve for the orbit radius r_n .** From the second equation, $v = n\hbar/(mr)$. Substitute this into the first equation, $mv^2/r = Ze^2/(4\pi\epsilon_0 r^2)$:

$$\frac{m}{r} \left(\frac{n\hbar}{mr} \right)^2 = \frac{Ze^2}{4\pi\epsilon_0 r^2}$$

Expand the square on the left:

$$\frac{m}{r} \cdot \frac{n^2 \hbar^2}{m^2 r^2} = \frac{Ze^2}{4\pi\epsilon_0 r^2} \implies \frac{n^2 \hbar^2}{mr^3} = \frac{Ze^2}{4\pi\epsilon_0 r^2}$$

Derivation: Hydrogen Energy Levels (Bohr-style Estimate)

Multiply both sides by r^3 and divide by $Ze^2/(4\pi\epsilon_0)$ to isolate r :

$$n^2\hbar^2 = \frac{Ze^2}{4\pi\epsilon_0} mr \quad \implies \quad r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{mZe^2}$$

- **Find the kinetic energy at radius r .** From the circular-orbit balance $mv^2/r = Ze^2/(4\pi\epsilon_0 r^2)$, multiply both sides by r :

$$mv^2 = \frac{Ze^2}{4\pi\epsilon_0 r}$$

Divide both sides by 2:

$$\frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$$

Derivation: Hydrogen Energy Levels (Bohr-style Estimate)

- **Write the total energy** as kinetic plus potential, $E = \frac{1}{2}mv^2 + V(r)$ with $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$ (the attractive Coulomb potential), and substitute the kinetic energy just found:

$$E = \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r}$$

Combine the two terms over the common denominator $8\pi\epsilon_0 r$ (the second term is $\frac{2Ze^2}{8\pi\epsilon_0 r}$):

$$E = \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{2Ze^2}{8\pi\epsilon_0 r} = -\frac{Ze^2}{8\pi\epsilon_0 r}$$

Derivation: Hydrogen Energy Levels (Bohr-style Estimate)

- **Substitute** $r = r_n$ from above:

$$E_n = -\frac{Ze^2}{8\pi\epsilon_0} \cdot \frac{1}{r_n} = -\frac{Ze^2}{8\pi\epsilon_0} \cdot \frac{mZe^2}{4\pi\epsilon_0 n^2 \hbar^2}$$

Multiply the numerators and denominators together, collecting the powers of e^2 and $\pi\epsilon_0$:

$$E_n = -\frac{mZ^2 e^4}{8\pi\epsilon_0 \cdot 4\pi\epsilon_0 \hbar^2 n^2} = -\frac{mZ^2 e^4}{2(4\pi\epsilon_0)^2 \hbar^2 n^2}$$

(using $8\pi\epsilon_0 \cdot 4\pi\epsilon_0 = 2 \cdot (4\pi\epsilon_0)^2$). **Evaluating this combination of physical constants numerically** gives the standard Rydberg energy, 13.6 eV (verified explicitly in the Unit 4 Additional Problems):

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

- The full quantum (Schrödinger) treatment reproduces exactly this energy formula — the famous agreement between the old Bohr model and the rigorous quantum result for hydrogen-like energies.

Worked Example 1: Ground-State Energy of Hydrogen

Problem

Find the ground-state energy and the radius r_1 (Bohr radius) of hydrogen ($Z = 1$).

Given: $Z = 1$, $n = 1$ (ground state). **Find:** E_1 and r_1 .

Step 1 — Apply $E_n = -13.6Z^2/n^2$ **eV with** $Z = 1$, $n = 1$.

$$E_1 = -\frac{13.6(1)^2}{1^2} = -\frac{13.6}{1} = -13.6 \text{ eV}$$

Step 2 — Apply the Bohr radius formula $r_1 = 4\pi\epsilon_0\hbar^2/(m_e e^2)$ (here $Z = 1$, $n = 1$).

Step 3 — Substitute the known constants $\epsilon_0 = 8.85 \times 10^{-12}$ F/m, $\hbar = 1.055 \times 10^{-34}$ J s, $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C, and evaluate numerically (a standard constant calculation):

$$r_1 = \frac{4\pi(8.85 \times 10^{-12})(1.055 \times 10^{-34})^2}{(9.11 \times 10^{-31})(1.6 \times 10^{-19})^2}$$

Worked Example 1: Ground-State Energy of Hydrogen

Step 4 — Evaluate:

$$r_1 \approx 0.529 \times 10^{-10} \text{ m} = 0.529 \text{ \AA} \quad (\text{the Bohr radius } a_0)$$

Worked Example 2: Energy Levels of He^+

Problem

Find the first three energy levels of singly-ionized helium, He^+ ($Z = 2$).

Given: $Z = 2$ (one electron remains, nuclear charge $+2e$). **Find:** E_1, E_2, E_3 .

Step 1 — Substitute $Z = 2$ into $E_n = -13.6Z^2/n^2$ eV:

$$E_n = -\frac{13.6(2)^2}{n^2} = -\frac{13.6 \times 4}{n^2} = -\frac{54.4}{n^2} \text{ eV}$$

Step 2 — Evaluate at $n = 1$:

$$E_1 = -\frac{54.4}{1^2} = -54.4 \text{ eV}$$

Step 3 — Evaluate at $n = 2$:

$$E_2 = -\frac{54.4}{2^2} = -\frac{54.4}{4} = -13.6 \text{ eV}$$

Worked Example 2: Energy Levels of He^+

Step 4 — Evaluate at $n = 3$:

$$E_3 = -\frac{54.4}{3^2} = -\frac{54.4}{9} \approx -6.04 \text{ eV}$$

Note $E_2(\text{He}^+) = E_1(\text{H})$ exactly — a useful mnemonic from the Z^2/n^2 scaling.

Worked Example 3: Lyman- α Transition Wavelength

Problem

Find the wavelength of the photon emitted in the hydrogen $n = 2 \rightarrow n = 1$ transition (Lyman- α).

Given: hydrogen, transition $n = 2 \rightarrow n = 1$. **Find:** λ .

Step 1 — Find E_1 and E_2 for hydrogen ($Z = 1$): $E_1 = -13.6 \text{ eV}$, $E_2 = -13.6/4 = -3.4 \text{ eV}$.

Step 2 — Compute the energy released. The photon energy equals the magnitude of the energy difference, $|E_2 - E_1|$:

$$|E_2 - E_1| = |-3.4 - (-13.6)| = |-3.4 + 13.6| = 10.2 \text{ eV}$$

Step 3 — Apply $\lambda = hc/\Delta E$, using the shortcut $hc \approx 1240 \text{ eV nm}$:

$$\lambda = \frac{1240 \text{ eV nm}}{10.2 \text{ eV}}$$

Step 4 — Evaluate:

$$\lambda \approx 121.6 \text{ nm (UV)}$$

Worked Example 4: Ionization Energy and Binding Energy

Problem

A hydrogen atom in its $n = 4$ state is ionized (the electron is removed to $E = 0$, at rest). Find the energy that must be supplied, and the wavelength of a photon that could supply exactly this energy.

Given: hydrogen, $n = 4$. **Find:** ionization energy ΔE and corresponding λ .

Step 1 — Find E_4 , applying $E_n = -13.6/n^2$ eV with $n = 4$:

$$E_4 = -\frac{13.6}{4^2} = -\frac{13.6}{16} = -0.85 \text{ eV}$$

Step 2 — Find the energy needed to reach $E_\infty = 0$ (a free, at-rest electron):

$$\Delta E = E_\infty - E_4 = 0 - (-0.85) = 0.85 \text{ eV}$$

Step 3 — Apply $\lambda = hc/\Delta E$, using the shortcut $hc \approx 1240$ eV nm:

$$\lambda = \frac{1240 \text{ eV nm}}{0.85 \text{ eV}}$$

Worked Example 4: Ionization Energy and Binding Energy

Step 4 — Evaluate:

$$\lambda \approx 1459 \text{ nm (near-infrared)}$$

Note how much less energy is needed to ionize from $n = 4$ compared to the ground state (13.6 eV) — excited states are far more weakly bound, which is why highly excited ("Rydberg") atoms are easily ionized and are of particular interest in atomic physics research.

Worked Example 5: Most Probable Radius for the Hydrogen Ground State

Problem

The hydrogen 1s radial wavefunction is $R_{10}(r) = 2a_0^{-3/2}e^{-r/a_0}$, so the radial probability density is $P(r) = r^2|R_{10}(r)|^2$ (the probability of finding the electron in a thin spherical shell between r and $r + dr$ is $P(r) dr$). Find the radius at which $P(r)$ is maximum.

Given: $P(r) = \frac{4}{a_0^3}r^2e^{-2r/a_0}$. **Find:** $r_{\text{most probable}}$.

Step 1 — Differentiate $P(r)$ with respect to r , using the product rule on r^2e^{-2r/a_0} :

$$\frac{dP}{dr} = \frac{4}{a_0^3} \left[2re^{-2r/a_0} + r^2 \left(-\frac{2}{a_0} \right) e^{-2r/a_0} \right] = \frac{4}{a_0^3} e^{-2r/a_0} \left[2r - \frac{2r^2}{a_0} \right]$$

Step 2 — Set $dP/dr = 0$. Since the exponential factor is never zero, the bracket must vanish:

$$2r - \frac{2r^2}{a_0} = 0 \quad \implies \quad 2r \left(1 - \frac{r}{a_0} \right) = 0$$

Worked Example 5: Most Probable Radius for the Hydrogen Ground State

Step 3 — Solve for r : this gives two roots, $r = 0$ and $r = a_0$.

Step 4 — Identify which root is the maximum. At $r = 0$, $P(0) = 0$ (a minimum, not a maximum, since $P \geq 0$ everywhere and grows initially as r^2); the other critical point, $r = a_0$, must therefore be the maximum (confirmed by $P \rightarrow 0$ as $r \rightarrow \infty$ too, so P must peak somewhere in between).

Step 5 — Conclude:

$$r_{\text{most probable}} = a_0 \approx 0.529 \text{ \AA}$$

Remarkably, the most probable radius in the full quantum treatment **exactly matches** the radius of the electron's orbit in the old Bohr model (Worked Example 1) — even though the physical pictures are completely different: Bohr's electron orbits at a fixed distance a_0 , while the quantum electron is smeared over all $r > 0$, merely *most likely* to be found at $r = a_0$.

Worked Example 6: Comparing Level Spacing — Coulomb Well vs. Infinite Square Well

Problem

Compute the energy gaps $\Delta E_{1 \rightarrow 2} = E_2 - E_1$ and $\Delta E_{2 \rightarrow 3} = E_3 - E_2$ for hydrogen, and compare the trend to how gaps behave in an infinite square well.

Given: $E_n = -13.6/n^2$ eV (hydrogen). **Find:** $\Delta E_{1 \rightarrow 2}$, $\Delta E_{2 \rightarrow 3}$, and the qualitative trend.

Step 1 — Compute the first three hydrogen levels:

$$E_1 = -13.6 \text{ eV}, \quad E_2 = -\frac{13.6}{4} = -3.4 \text{ eV}, \quad E_3 = -\frac{13.6}{9} \approx -1.51 \text{ eV}$$

Step 2 — Compute the two gaps:

$$\Delta E_{1 \rightarrow 2} = -3.4 - (-13.6) = 10.2 \text{ eV}, \quad \Delta E_{2 \rightarrow 3} = -1.51 - (-3.4) = 1.89 \text{ eV}$$

Worked Example 6: Comparing Level Spacing — Coulomb Well vs. Infinite Square Well

Step 3 — Form the ratio of successive gaps:

$$\frac{\Delta E_{2 \rightarrow 3}}{\Delta E_{1 \rightarrow 2}} = \frac{1.89}{10.2} \approx 0.185$$

So hydrogen's level spacing **shrinks** rapidly with increasing n — consistent with $E_n \propto -1/n^2$, whose levels crowd together and accumulate at the ionization threshold $E = 0$ as $n \rightarrow \infty$.

Step 4 — Contrast with the infinite square well, where $E_n \propto n^2$, so the gap between adjacent levels is

$$\Delta E_{n \rightarrow n+1} = E_{n+1} - E_n \propto (n+1)^2 - n^2 = 2n + 1$$

which **grows** linearly with n — the opposite trend.

Step 5 — Physical interpretation. The infinite well's walls are infinitely high, so there is no upper bound on the energy or the level spacing — confinement gets “tighter” relative to the wavelength as n grows. The Coulomb well, in contrast, has a finite total depth (levels are bounded above by $E = 0$), so infinitely many levels must be squeezed into that finite energy range, forcing the spacing to shrink — a

Worked Example 6: Comparing Level Spacing — Coulomb Well vs. Infinite Square Well

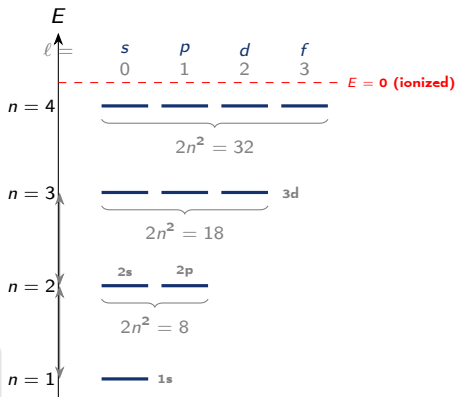
direct consequence of the long range ($\propto 1/r$) of the attractive potential, very different from the sharply confining infinite well.

Quantum Numbers n , ℓ , m_ℓ , m_s — Degeneracy

- **Principal quantum number**
 $n = 1, 2, 3, \dots$ — determines energy.
- **Orbital (azimuthal) quantum number**
 $\ell = 0, 1, \dots, n - 1$ — determines orbital angular momentum magnitude.
- **Magnetic quantum number**
 $m_\ell = -\ell, \dots, +\ell$ — determines z-projection.
- **Spin quantum number** $m_s = \pm \frac{1}{2}$ — intrinsic angular momentum projection.

Degeneracy

For a given n : n^2 spatial states; $2n^2$ including spin.



Derivation: Degeneracy Count n^2

- For fixed n , ℓ ranges over $0, 1, \dots, n-1$ (n values total).
- For each ℓ , m_ℓ ranges over $-\ell, \dots, +\ell$, giving $2\ell + 1$ values.
- **Evaluate** $\sum_{\ell=0}^{n-1} \ell$ **explicitly**. The integers $0, 1, \dots, n-1$ are an arithmetic sequence with n terms, first term $a_1 = 0$ and last term $a_n = n-1$. By Gauss's pairing trick, write the sum forwards and backwards and add:

$$2S = \underbrace{(0) + (1) + \dots + (n-1)}_S + \underbrace{(n-1) + (n-2) + \dots + (0)}_S = \underbrace{(n-1) + (n-1) + \dots + (n-1)}_{n \text{ terms}} =$$

$$\text{Hence } S = \sum_{\ell=0}^{n-1} \ell = \frac{n(n-1)}{2}.$$

- **Total number of (ℓ, m_ℓ) states** for given n , splitting the sum $\sum (2\ell + 1) = 2 \sum \ell + \sum 1$:

$$\sum_{\ell=0}^{n-1} (2\ell + 1) = 2 \cdot \frac{n(n-1)}{2} + n = n(n-1) + n = n^2$$

Derivation: Degeneracy Count n^2

- Including the two spin states $m_s = \pm \frac{1}{2}$ for each spatial state: total degeneracy $= 2n^2$.
- This degeneracy is a special feature of the pure $1/r$ Coulomb potential (an "accidental" degeneracy beyond what spherical symmetry alone requires) — it is lifted by perturbations such as spin-orbit coupling or external fields.

Worked Example 1: Listing States for $n = 3$

Problem

List all allowed (ℓ, m_ℓ) combinations for $n = 3$, and verify the total count is $n^2 = 9$.

Given: $n = 3$. **Find:** all (ℓ, m_ℓ) pairs, and verify the count.

Step 1 — List allowed ℓ values. Since $\ell = 0, 1, \dots, n-1$, for $n = 3$: $\ell = 0, 1, 2$.

Step 2 — For $\ell = 0$, m_ℓ ranges over $-\ell, \dots, \ell$, i.e. just $m_\ell = 0$ (1 state):

$$\ell = 0 : m_\ell = 0 \text{ (1 state)}$$

Step 3 — For $\ell = 1$, $m_\ell = -1, 0, 1$ (3 states):

$$\ell = 1 : m_\ell = -1, 0, 1 \text{ (3 states)}$$

Step 4 — For $\ell = 2$, $m_\ell = -2, -1, 0, 1, 2$ (5 states):

$$\ell = 2 : m_\ell = -2, -1, 0, 1, 2 \text{ (5 states)}$$

Worked Example 1: Listing States for $n = 3$

Step 5 — Sum the counts:

$$1 + 3 + 5 = 9 = 3^2 \checkmark$$

Including spin, the total degeneracy of the $n = 3$ shell is $2(9) = 18$.

From Quantum Numbers to the Periodic Table

- The spectroscopic labels $s, p, d, f, \dots$ (for $\ell = 0, 1, 2, 3, \dots$) are not arbitrary — they originate from 19th-century spectroscopic terms ("sharp", "principal", "diffuse", "fundamental") describing the appearance of atomic spectral lines, long before quantum mechanics explained their cause.
- Each value of ℓ corresponds to a characteristic orbital **shape**: s orbitals are spherically symmetric, p orbitals have a two-lobed "dumbbell" shape, d and f orbitals have progressively more complex angular structure — all encoded in the spherical harmonics $Y_{\ell}^{m_{\ell}}$.
- The $2n^2$ degeneracy formula derived above, combined with the Pauli exclusion principle (one electron per quantum state, covered in later courses), directly explains the lengths of the rows of the periodic table: 2, 8, 18, 32, $\dots$ electrons fill successive shells.
- This is a striking example of how the abstract mathematics of a single exactly-solvable 3D Schrödinger equation underlies the entire structure of chemistry.

Worked Example 2: Spectroscopic Notation

Problem

Identify the spectroscopic term ($n\ell$ notation) for $n = 4, \ell = 2$, and state how many m_ℓ values it has.

Given: $n = 4, \ell = 2$. **Find:** the spectroscopic label and orbital count.

Step 1 — Convert $\ell = 2$ to a letter using the standard correspondence $\ell = 0, 1, 2, 3 \rightarrow s, p, d, f$:
 $\ell = 2 \rightarrow d$.

Step 2 — Combine with n to form the spectroscopic term: $n = 4, \ell = 2 \rightarrow$ the state is labeled $4d$.

Step 3 — Count the m_ℓ values, using $2\ell + 1$:

$$2\ell + 1 = 2(2) + 1 = 5 \quad (m_\ell = -2, -1, 0, 1, 2)$$

Step 4 — Include spin (2 values of m_s for each spatial state):

$$5 \times 2 = 10 \text{ degenerate } 4d \text{ orbitals}$$

Worked Example 3: Maximum ℓ for Given n

Problem

For $n = 5$, what is the maximum possible orbital angular momentum quantum number, and what is the corresponding $|L|$?

Given: $n = 5$. **Find:** $\ell_{\max}$ and $|L|$.

Step 1 — Apply the constraint $\ell = 0, 1, \dots, n - 1$. The maximum value is:

$$\ell_{\max} = n - 1 = 5 - 1 = 4 \quad (\text{an "g" state, following } s, p, d, f, g)$$

Step 2 — Apply $|L| = \hbar\sqrt{\ell(\ell + 1)}$ **with** $\ell = 4$:

$$|L| = \hbar\sqrt{4(4 + 1)} = \hbar\sqrt{4 \times 5} = \hbar\sqrt{20}$$

Step 3 — Evaluate $\sqrt{20}$:

$$\sqrt{20} \approx 4.47$$

Step 4 — So:

$$|L| \approx 4.47 \hbar$$

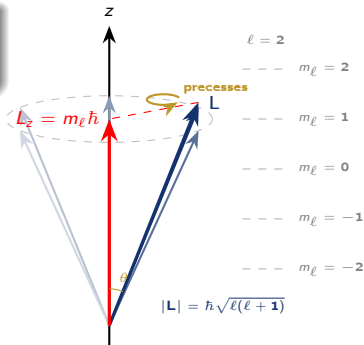
Angular Momentum

Orbital Angular Momentum

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$$

- $\hat{L}^2$ eigenvalues: $\hbar^2 \ell(\ell + 1)$; eigenfunctions are **spherical harmonics** $Y_\ell^{m_\ell}$.
- $\hat{L}_z$ eigenvalues: $m_\ell \hbar$.
- $\hat{L}^2$ and $\hat{L}_z$ commute (simultaneous eigenstates), but $\hat{L}_x, \hat{L}_y, \hat{L}_z$ do **not**.
- $\ell = 0, 1, 2, 3 \rightarrow s, p, d, f$

Problem-Practice Class: Quantum Numbers



Derivation: L_z Eigenvalues from the Azimuthal Equation

- In spherical coordinates, $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$. The azimuthal part of the wavefunction satisfies

$$\hat{L}_z \Phi(\phi) = -i\hbar \frac{d\Phi}{d\phi} = m_\ell \hbar \Phi(\phi)$$

- This first-order ODE has solution $\Phi(\phi) = Ae^{im_\ell \phi}$.
- **Single-valuedness** requires $\Phi(\phi + 2\pi) = \Phi(\phi)$:

$$e^{im_\ell(\phi+2\pi)} = e^{im_\ell \phi} \implies e^{i2\pi m_\ell} = 1 \implies m_\ell \in \mathbb{Z}$$

- This is exactly why m_ℓ must be an **integer** — a direct consequence of requiring the wavefunction to be a well-defined (single-valued) function of position in ordinary 3D space.

Worked Example 1: Angular Momentum Magnitude for a d Electron

Problem

Find $|L|$ and the possible values of L_z for an electron in a d orbital ($\ell = 2$).

Given: $\ell = 2$ (d orbital). **Find:** $|L|$ and all allowed L_z .

Step 1 — Apply $|L| = \hbar\sqrt{\ell(\ell+1)}$ **with** $\ell = 2$:

$$|L| = \hbar\sqrt{2(2+1)} = \hbar\sqrt{2 \times 3} = \hbar\sqrt{6} \approx 2.449 \hbar$$

Step 2 — List the allowed m_ℓ values, $m_\ell = -\ell, \dots, \ell$ **for** $\ell = 2$:

$$m_\ell = -2, -1, 0, 1, 2$$

Step 3 — Apply $L_z = m_\ell \hbar$ **for each value:**

$$L_z = -2\hbar, -\hbar, 0, \hbar, 2\hbar$$

Worked Example 2: Angle Between L and the z -Axis

Problem

For $\ell = 1, m_\ell = 1$, find the angle between L and the z -axis.

Given: $\ell = 1, m_\ell = 1$. **Find:** θ , the angle between L and the z -axis.

Step 1 — Compute $|L|$, applying $\hbar\sqrt{\ell(\ell+1)}$ with $\ell = 1$:

$$|L| = \hbar\sqrt{1(1+1)} = \hbar\sqrt{2}$$

Step 2 — Compute $L_z = m_\ell\hbar$ with $m_\ell = 1$:

$$L_z = (1)\hbar = \hbar$$

Step 3 — Use the geometric relation $\cos\theta = L_z/|L|$ (the z -component is the projection of L onto the z -axis):

$$\cos\theta = \frac{L_z}{|L|} = \frac{\hbar}{\hbar\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Worked Example 2: Angle Between $\mathbf{L}$ and the z -Axis

Step 4 — Take the inverse cosine:

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

Note $\mathbf{L}$ can never point exactly along z (that would require $|L_z| = |\mathbf{L}|$, impossible for $\ell > 0$) — a geometric manifestation of the fact that L_x, L_y cannot simultaneously be zero.

Worked Example 3: Number of Orbitals per Subshell

Problem

How many distinct (m_ℓ) orientations of L are possible for an f electron ($\ell = 3$), and what is $|L|$?

Given: $\ell = 3$ (f orbital). **Find:** number of m_ℓ orientations, and $|L|$.

Step 1 — Apply $2\ell + 1$ to count the orientations:

$$2\ell + 1 = 2(3) + 1 = 7 \quad (m_\ell = -3, -2, -1, 0, 1, 2, 3)$$

Step 2 — Apply $|L| = \hbar\sqrt{\ell(\ell + 1)}$ **with** $\ell = 3$:

$$|L| = \hbar\sqrt{3(3 + 1)} = \hbar\sqrt{3 \times 4} = \hbar\sqrt{12}$$

Step 3 — Evaluate $\sqrt{12}$:

$$\sqrt{12} \approx 3.46$$

Step 4 — So:

$$|L| \approx 3.46 \hbar$$

Worked Example 4: Minimum Angle and the Vector Model

Problem

For $\ell = 2$ (d electron), find the *smallest* possible angle between L and the z -axis.

Given: $\ell = 2$. **Find:** $\theta_{\min}$.

Step 1 — Identify which m_ℓ minimizes θ . Since $\cos \theta = L_z/|L|$ and $|L|$ is fixed for given ℓ , θ is smallest when L_z (hence $|m_\ell|$) is largest: $m_\ell = +2$.

Step 2 — Compute $|L|$, applying $\hbar\sqrt{\ell(\ell+1)}$ with $\ell = 2$:

$$|L| = \hbar\sqrt{2(3)} = \hbar\sqrt{6}$$

Step 3 — Compute $L_z = m_\ell\hbar$ with $m_\ell = 2$:

$$L_z = 2\hbar$$

Worked Example 4: Minimum Angle and the Vector Model

Step 4 — Apply $\cos \theta_{\min} = L_z/|L|$:

$$\cos \theta_{\min} = \frac{2\hbar}{\hbar\sqrt{6}} = \frac{2}{\sqrt{6}}$$

Step 5 — Evaluate $2/\sqrt{6} \approx 0.816$, then take the inverse cosine:

$$\theta_{\min} = \cos^{-1}(0.816) \approx 35.3^\circ$$

Even in the most "aligned" state available, L is never exactly parallel to z — consistent with the general result from Worked Example 2 that perfect alignment ($\theta = 0$) would require $L_x = L_y = 0$ simultaneously, forbidden by the angular-momentum commutation relations developed rigorously in Unit 5. This geometric picture (vectors precessing around a fixed cone of allowed angles) is sometimes called the **vector model** of angular momentum.

Unit 4 Summary

- **Free particle** and **infinite square well**: derived from boundary conditions; continuous vs. quantized spectra.
- **Step potential** and **barrier penetration**: R, T derived from matching conditions; quantum tunneling $T \approx e^{-2\kappa a}$.
- **Dirac delta potential**: jump condition derivation; exactly one bound state, $E = -m\alpha^2/2\hbar^2$.
- **Harmonic oscillator**: ladder-operator derivation of $E_n = (n + \frac{1}{2})\hbar\omega$; zero-point energy from uncertainty principle.
- **Hydrogen-like atom**: separation of variables, Bohr-style energy derivation, quantum numbers n, ℓ, m_ℓ, m_s , degeneracy n^2 , and angular momentum quantization.

Additional Problem 1: Box Energy Scaling

Problem

If the width of an infinite well is tripled, by what factor does E_1 change?

Given: $L \rightarrow 3L$. **Find:** the scaling factor for E_1 .

Step 1 — Recall the scaling law, $E_1 = \pi^2 \hbar^2 / (2mL^2)$, so $E_1 \propto 1/L^2$.

Step 2 — Write the new energy E'_1 **in terms of the new width** $3L$:

$$E'_1 \propto \frac{1}{(3L)^2} = \frac{1}{9L^2}$$

Step 3 — Form the ratio E'_1/E_1 :

$$\frac{E'_1}{E_1} = \frac{1/(9L^2)}{1/L^2} = \frac{1}{9}$$

Step 4 — So:

$$E'_1 = \frac{E_1}{9}$$

Additional Problem 1: Box Energy Scaling

Tripling the width reduces the ground-state energy by a factor of 9.

Additional Problem 2: Quantum Number for a Macroscopic "Box"

Problem

A 1 g marble in a 1D box of width 1 cm has speed 0.1 m/s. Estimate its quantum number n .

Given: $m = 10^{-3}$ kg, $L = 10^{-2}$ m, $v = 0.1$ m/s. **Find:** n .

Step 1 — Compute the classical kinetic energy, $E = \frac{1}{2}mv^2$:

$$E = \frac{1}{2}(10^{-3})(0.1)^2 = \frac{1}{2}(10^{-3})(10^{-2}) = 5 \times 10^{-6} \text{ J}$$

Step 2 — Start from $E_n = n^2\pi^2\hbar^2/(2mL^2)$ and solve for n . Multiply both sides by $2mL^2/(\pi^2\hbar^2)$ and take the square root:

$$n = \sqrt{\frac{2mEL^2}{\pi^2\hbar^2}}$$

Step 3 — Evaluate the numerator, $2mEL^2$:

$$2mEL^2 = 2(10^{-3})(5 \times 10^{-6})(10^{-2})^2 = 2(10^{-3})(5 \times 10^{-6})(10^{-4}) \approx 1.0 \times 10^{-12}$$

Additional Problem 2: Quantum Number for a Macroscopic "Box"

Step 4 — Evaluate the denominator, $\pi^2 \hbar^2$:

$$\pi^2 \hbar^2 \approx 9.87 \times (1.113 \times 10^{-68}) \approx 1.099 \times 10^{-67}$$

Step 5 — Divide and take the square root:

$$n = \sqrt{\frac{1.0 \times 10^{-12}}{1.099 \times 10^{-67}}} = \sqrt{9.1 \times 10^{55}} \approx 2.7 \times 10^{28}$$

An astronomically large quantum number — consistent with the correspondence principle (quantum effects invisible classically).

Additional Problem 3: Transmission Coefficient Comparison

Problem

Compare $T \approx e^{-2\kappa a}$ for two barriers of the same width but $V_0 - E$ differing by a factor of 4 (so κ doubles). If $T_1 = 10^{-6}$, find T_2 .

Given: $T_1 = 10^{-6}$, $\kappa_2 = 2\kappa_1$, same a . **Find:** T_2 .

Step 1 — Find $2\kappa_1 a$ from T_1 . Take the natural log of $T_1 = e^{-2\kappa_1 a}$:

$$\ln T_1 = -2\kappa_1 a \implies 2\kappa_1 a = -\ln T_1 = -\ln(10^{-6}) = 6 \ln 10 \approx 13.8$$

Step 2 — Since $\kappa_2 = 2\kappa_1$, the new exponent is double:

$$2\kappa_2 a = 2(2\kappa_1)a = 2(2\kappa_1 a) = 2(13.8) = 27.6$$

Step 3 — Apply $T_2 = e^{-2\kappa_2 a}$:

$$T_2 = e^{-27.6} \approx 1.0 \times 10^{-12}$$

Doubling κ (by quadrupling the barrier-height excess $V_0 - E$) squares the transmission coefficient, since $T_2 = T_1^2$ when κ exactly doubles.

Additional Problem 4: Reflection Coefficient at a Downward Step

Problem

For a particle of energy $E = 4 \text{ eV}$ crossing a step where V drops from 0 to -2 eV , find R (use $k_2 = \sqrt{2m(E - V_0)}/\hbar$ with $V_0 = -2 \text{ eV}$).

Given: $E = 4 \text{ eV}$, $V_0 = -2 \text{ eV}$ (a downward step). **Find:** R .

Step 1 — Find $E - V_0$:

$$E - V_0 = 4 - (-2) = 6 \text{ eV}$$

Step 2 — Form the ratio k_2/k_1 , using $k_1 \propto \sqrt{E} = \sqrt{4} = 2$ and $k_2 \propto \sqrt{E - V_0} = \sqrt{6}$:

$$\frac{k_2}{k_1} = \frac{\sqrt{6}}{\sqrt{4}} = \sqrt{\frac{6}{4}} = \sqrt{1.5} \approx 1.225$$

Step 3 — Apply the reflection formula, dividing numerator and denominator by k_1 :

$$R = \left(\frac{1 - k_2/k_1}{1 + k_2/k_1} \right)^2 = \left(\frac{1 - 1.225}{1 + 1.225} \right)^2$$

Additional Problem 4: Reflection Coefficient at a Downward Step

Step 4 — Evaluate:

$$R = \left(\frac{-0.225}{2.225} \right)^2 \approx (-0.1011)^2 \approx 0.0102$$

Additional Problem 5: Harmonic Oscillator Energy Gap

Problem

A quantum oscillator has $E_3 - E_2$ corresponding to a photon of wavelength $2\text{ }\mu\text{m}$. Find ω .

Given: $\lambda = 2\text{ }\mu\text{m}$. **Find:** ω .

Step 1 — Recognize that $E_3 - E_2 = \hbar\omega$ (equal level spacing for the harmonic oscillator).

Step 2 — Find the photon energy, $\Delta E = hc/\lambda$:

$$\Delta E = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{2 \times 10^{-6}}$$

Step 3 — Evaluate:

$$\Delta E \approx \frac{1.988 \times 10^{-25}}{2 \times 10^{-6}} \approx 9.94 \times 10^{-20} \text{ J}$$

Additional Problem 5: Harmonic Oscillator Energy Gap

Step 4 — Apply $\Delta E = \hbar\omega$ and solve for ω :

$$\omega = \frac{\Delta E}{\hbar} = \frac{9.94 \times 10^{-20}}{1.055 \times 10^{-34}} \approx 9.42 \times 10^{14} \text{ rad/s}$$

Additional Problem 6: Bound-State Penetration Probability (Delta Well)

Problem

For the delta-potential bound state $\psi(x) = \sqrt{\kappa}e^{-\kappa|x|}$, find the probability of finding the particle at $|x| > 1/\kappa$.

Given: normalized $\psi(x) = \sqrt{\kappa}e^{-\kappa|x|}$. **Find:** $P(|x| > 1/\kappa)$.

Step 1 — Set up the integral, using symmetry to write it as twice the $x > 1/\kappa$ piece:

$$P = 2 \int_{1/\kappa}^{\infty} \kappa e^{-2\kappa x} dx$$

Step 2 — Evaluate the integral using $\int_a^{\infty} e^{-bx} dx = e^{-ab}/b$ with $b = 2\kappa$, $a = 1/\kappa$:

$$\int_{1/\kappa}^{\infty} e^{-2\kappa x} dx = \frac{e^{-2\kappa(1/\kappa)}}{2\kappa} = \frac{e^{-2}}{2\kappa}$$

Additional Problem 6: Bound-State Penetration Probability (Delta Well)

Step 3 — Substitute back:

$$P = 2\kappa \times \frac{e^{-2}}{2\kappa} = e^{-2}$$

Step 4 — Evaluate numerically:

$$P = e^{-2} \approx 0.135$$

About 13.5% of the time, the particle is found beyond one penetration depth from the origin.

Additional Problem 7: Energy Levels of a 3D Cubic Box (Preview)

Problem

For a cubic 3D infinite well of side L , $E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2)$. Find the degeneracy of the first excited level.

Given: the 3D energy formula. **Find:** degeneracy of the first excited state.

Step 1 — Identify the ground state. The minimum of $n_x^2 + n_y^2 + n_z^2$ (each $n_i \geq 1$) occurs at $(1, 1, 1)$, giving sum = 3 — this is the unique, non-degenerate ground state.

Step 2 — Identify the first excited level. The next-smallest sum of three squares (each ≥ 1) is achieved by permutations of $(2, 1, 1)$, giving sum = $4 + 1 + 1 = 6$.

Step 3 — Count the distinct permutations of $(2, 1, 1)$: the "2" can occupy any one of the three positions (n_x, n_y, n_z) :

$$(2, 1, 1), (1, 2, 1), (1, 1, 2) \Rightarrow 3 \text{ distinct states}$$

Step 4 — Conclude: since all three states share the same energy $E = 6\pi^2 \hbar^2 / (2mL^2)$, the degeneracy of the first excited level is 3.

Additional Problem 8: Rydberg Energy Check

Problem

Verify $13.6 \text{ eV} = \frac{m_e e^4}{2(4\pi\epsilon_0)^2 \hbar^2}$ using $m_e = 9.11 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$.

Given: the standard atomic constants. **Find:** verify the Rydberg energy formula.

Step 1 — Compute e^4 :

$$e^4 = (1.6 \times 10^{-19})^4 \approx 6.55 \times 10^{-76}$$

Step 2 — Compute $(4\pi\epsilon_0)^2$: first $4\pi\epsilon_0 \approx 4\pi(8.85 \times 10^{-12}) \approx 1.113 \times 10^{-10}$, then square:

$$(4\pi\epsilon_0)^2 \approx 1.239 \times 10^{-20}$$

Step 3 — Compute the numerator, $m_e e^4$:

$$m_e e^4 \approx (9.11 \times 10^{-31})(6.55 \times 10^{-76}) \approx 5.97 \times 10^{-106}$$

Additional Problem 8: Rydberg Energy Check

Step 4 — Compute the denominator, $2(4\pi\epsilon_0)^2\hbar^2$:

$$2(1.239 \times 10^{-20})(1.113 \times 10^{-68}) \approx 2.76 \times 10^{-88}$$

Step 5 — Divide and convert to eV:

$$\frac{5.97 \times 10^{-106}}{2.76 \times 10^{-88}} \approx 2.16 \times 10^{-18} \text{ J} \approx 13.5 \text{ eV} \checkmark$$

(small rounding differences from truncated constants account for the slight deviation from 13.6).

Additional Problem 9: Ionization Energy of Li^{2+}

Problem

Find the ionization energy of Li^{2+} ($Z = 3$) from its ground state.

Given: $Z = 3$ (lithium nucleus with 2 electrons removed, 1 remaining), $n = 1$. **Find:** ionization energy.

Step 1 — Apply $E_n = -13.6Z^2/n^2$ **eV with** $Z = 3, n = 1$:

$$E_1 = -13.6(3)^2 = -13.6 \times 9$$

Step 2 — Evaluate:

$$E_1 = -122.4 \text{ eV}$$

Step 3 — Find the ionization energy, the energy needed to bring the electron from E_1 to $E_\infty = 0$:

$$\text{Ionization energy} = E_\infty - E_1 = 0 - (-122.4) = 122.4 \text{ eV}$$

Additional Problem 10: Tunneling Current Scaling (STM)

Problem

If raising an STM tip by 0.1 nm reduces the tunneling current by a factor of 10, estimate κ (use $T \propto e^{-2\kappa a}$, current $\propto T$).

Given: $\Delta a = 0.1$ nm, current ratio = 10. **Find:** κ .

Step 1 — Write the ratio of currents (proportional to T) for the two tip heights:

$$\frac{I_{\text{near}}}{I_{\text{far}}} = \frac{e^{-2\kappa a_{\text{near}}}}{e^{-2\kappa a_{\text{far}}}} = e^{-2\kappa(a_{\text{near}} - a_{\text{far}})} = e^{2\kappa \Delta a}$$

where $\Delta a = a_{\text{far}} - a_{\text{near}} = 0.1$ nm is the height increase. Given $I_{\text{near}}/I_{\text{far}} = 10$.

Step 2 — Take the natural log of both sides:

$$\ln 10 = 2\kappa \Delta a$$

Additional Problem 10: Tunneling Current Scaling (STM)

Step 3 — Solve for κ by dividing both sides by $2\Delta a$:

$$\kappa = \frac{\ln 10}{2\Delta a} = \frac{\ln 10}{2(0.1 \times 10^{-9})}$$

Step 4 — Evaluate ($\ln 10 \approx 2.303$):

$$\kappa = \frac{2.303}{2 \times 10^{-10}} \approx 1.15 \times 10^{10} \text{ m}^{-1}$$

(Typical of real STM operating parameters.)

Unit 5: Dirac Notation in Quantum Mechanics

1. Origin of Quantum Mechanics
2. Formulation of Quantum Mechanics
3. Basics of Schrödinger Equation I
4. Basics of Schrödinger Equation II
5. Dirac Notation in Quantum Mechanics

Unit 5: Symbol Glossary

- $|\psi\rangle, |\phi\rangle, |n\rangle$ — ket (state) vectors in Hilbert space $\mathcal{H}$
- $\langle\psi|, \langle\phi|$ — bra vectors (duals of the corresponding kets)
- $\langle\phi|\psi\rangle$ — inner product (overlap) of two states, $\in \mathbb{C}$
- $|\phi\rangle\langle\psi|$ — outer product; this is an operator, not a number
- $\hat{A}, \hat{B}, \hat{C}$ — generic (Hermitian) operators
- $\hat{A}^\dagger$ — adjoint (Hermitian conjugate) of $\hat{A}$
- $\hat{I}$ — identity operator
- $\hat{P}$ — projection operator, $\hat{P} = |n\rangle\langle n|$
- A_{mn} — matrix element of $\hat{A}$, $A_{mn} = \langle m|\hat{A}|n\rangle$

- δ_{mn} — Kronecker delta ($= 1$ if $m = n$, else 0)
- c_n — expansion coefficient of $|\psi\rangle$ in basis $\{|n\rangle\}$
- $[\hat{A}, \hat{B}]$ — commutator, $\hat{A}\hat{B} - \hat{B}\hat{A}$
- $\{\hat{A}, \hat{B}\}$ — anticommutator, $\hat{A}\hat{B} + \hat{B}\hat{A}$
- a_n — eigenvalue of $\hat{A}$ for eigenstate $|\psi_n\rangle$ (real, since $\hat{A}$ is Hermitian)
- $\Delta A, \Delta B$ — standard-deviation uncertainties of observables A, B
- Δt — characteristic evolution timescale of $\langle A \rangle$ (s)
- $\hat{L}^2, \hat{L}_x, \hat{L}_y, \hat{L}_z$ — angular momentum operators (carried over from Unit 4)

Symbols specific to a single derivation are defined where they first appear.

Why a New Notation?

- Throughout Units 2–4, quantum states were described as wavefunctions $\Psi(x, t)$, and operators as differential expressions like $\hat{p} = -i\hbar d/dx$ — the **wave-mechanics** formulation, due to Schrödinger.
- Heisenberg independently developed an equivalent formulation using matrices and abstract vectors — the **matrix-mechanics** approach — before the connection between the two was understood.
- Dirac's bra-ket notation, introduced in this unit, unifies both pictures into a single, representation-independent language: the same symbol $|\psi\rangle$ can represent a wavefunction, a column of numbers, or any other equivalent description, with the specific representation chosen only when needed for calculation.
- This unit also makes rigorous several results used informally earlier: why eigenvalues of observables are real (used since Unit 3), why energy eigenstates are orthogonal (assumed in Unit 4), and why $\Delta x \Delta p \geq \hbar/2$ holds exactly (asserted in Unit 2).

Operators in Quantum Mechanics — Revisited

- An **operator** $\hat{A}$ is a mathematical rule that maps one state (function) to another: $\hat{A}\psi = \phi$.
- Quantum mechanical operators corresponding to observables must be **linear** and **Hermitian**.
- **Linearity**: $\hat{A}(c_1\psi_1 + c_2\psi_2) = c_1\hat{A}\psi_1 + c_2\hat{A}\psi_2$.
- Examples already encountered: $\hat{x}$, $\hat{p} = -i\hbar d/dx$, $\hat{H}$, $\hat{L}^2$, $\hat{L}_z$.
- Dirac notation provides a representation-independent language for all of this.

Worked Example: Verifying Linearity

Problem

Show that $\hat{p} = -i\hbar d/dx$ is linear, and that $\hat{A}\psi = \psi^2$ (squaring) is *not* linear.

Solution: For $\hat{p}$, we distribute the derivative over the linear combination:

$$\hat{p}(c_1\psi_1 + c_2\psi_2) = -i\hbar \frac{d}{dx}(c_1\psi_1 + c_2\psi_2)$$

Using the linearity of the derivative operator ($\frac{d}{dx}(f + g) = \frac{df}{dx} + \frac{dg}{dx}$):

$$= -i\hbar \left(c_1 \frac{d\psi_1}{dx} + c_2 \frac{d\psi_2}{dx} \right)$$

Distributing the constant factor $-i\hbar$:

$$= c_1 \left(-i\hbar \frac{d\psi_1}{dx} \right) + c_2 \left(-i\hbar \frac{d\psi_2}{dx} \right) = c_1 \hat{p}\psi_1 + c_2 \hat{p}\psi_2 \quad \checkmark$$

Worked Example: Verifying Linearity

For squaring, we expand the squared binomial expression:

$$\begin{aligned}\hat{A}(c_1\psi_1 + c_2\psi_2) &= (c_1\psi_1 + c_2\psi_2)^2 \\ &= c_1^2\psi_1^2 + 2c_1c_2\psi_1\psi_2 + c_2^2\psi_2^2\end{aligned}$$

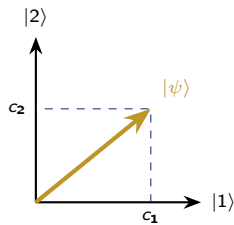
Compare this to a valid linear combination of individual applications:

$$c_1\hat{A}\psi_1 + c_2\hat{A}\psi_2 = c_1\psi_1^2 + c_2\psi_2^2$$

Since $c_1^2\psi_1^2 + 2c_1c_2\psi_1\psi_2 + c_2^2\psi_2^2 \neq c_1\psi_1^2 + c_2\psi_2^2$ in general due to the cross term and squared coefficients, $\hat{A}$ is **not** linear, and cannot represent a quantum observable.

Linear Vector Space & Hilbert Space

- Quantum states live in a **linear vector space**: closed under addition and scalar multiplication (this is precisely the content of the superposition principle, Unit 1).
- A **Hilbert space** $\mathcal{H}$ is a complex vector space equipped with an **inner product**, that is complete (Cauchy sequences converge within the space).
- State vectors are written in **Dirac (bra-ket) notation**:
ket: $|\psi\rangle$ bra: $\langle\psi|$
- $|\psi\rangle$ is representation-independent; $\psi(x) = \langle x|\psi\rangle$ recovers the familiar wavefunction in the position representation.



$|\psi\rangle = c_1|1\rangle + c_2|2\rangle$: a state as a vector in an abstract basis, just like an ordinary 2D vector.

Translating Between Notations

A Dictionary

Wave mechanics (Units 2–4)	Dirac notation (this unit)
$\Psi(x, t)$	$ \Psi(t)\rangle$
$\Psi^*(x, t)$	$\langle\Psi(t) $
$\int \phi^*(x)\psi(x) dx$	$\langle\phi \psi\rangle$
$\int \Psi^* \hat{A} \Psi dx$	$\langle\Psi \hat{A} \Psi\rangle$
$\hat{A}\psi_n = a_n\psi_n$	$\hat{A} n\rangle = a_n n\rangle$

Every expectation-value and normalization formula used in Units 2–4 has an exact Dirac-notation counterpart — the physics is identical, only the bookkeeping changes. From here on, both notations will be used interchangeably, as is standard in the field.

Inner and Outer Products

Inner Product (bra-ket)

$$\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \psi(x) dx \in \mathbb{C}$$

Properties: $\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$, and $\langle \psi | \psi \rangle \geq 0$ (norm-squared).

Outer Product

$|\phi\rangle\langle\psi|$ is an **operator**

- Normalization: $\langle \psi | \psi \rangle = 1$.
- Orthogonality: $\langle \phi | \psi \rangle = 0$.
- **Completeness relation** (resolution of identity) for an orthonormal basis $\{|n\rangle\}$:

$$\sum_n |n\rangle\langle n| = \hat{I}$$

Derivation: The Completeness Relation

- Let $\{|n\rangle\}$ be a complete orthonormal basis of $\mathcal{H}$. Any state $|\psi\rangle$ can be expanded as a linear combination of these basis elements:

$$|\psi\rangle = \sum_n c_n |n\rangle$$

- To isolate a specific coefficient c_m , we act on the left with the bra $\langle m|$:

$$\langle m|\psi\rangle = \langle m| \left(\sum_n c_n |n\rangle \right)$$

- Pulling the scalar constants and the bra inside the summation:

$$\langle m|\psi\rangle = \sum_n c_n \langle m|n\rangle$$

Derivation: The Completeness Relation

- Using the orthonormality relation of the basis vectors, $\langle m|n\rangle = \delta_{mn}$:

$$\langle m|\psi\rangle = \sum_n c_n \delta_{mn} = c_m$$

- So the expansion coefficients are explicitly $c_n = \langle n|\psi\rangle$. Substituting this identity back into our initial expansion formula:

$$|\psi\rangle = \sum_n (\langle n|\psi\rangle) |n\rangle$$

- Rearranging the scalar inner product relative to the ket vector $|n\rangle$:

$$|\psi\rangle = \sum_n |n\rangle \langle n|\psi\rangle = \left(\sum_n |n\rangle \langle n| \right) |\psi\rangle$$

Derivation: The Completeness Relation

- Since this identity holds universally for **every** state vector $|\psi\rangle$ within the space, the operator enclosed in parentheses must behave as the identity operator:

$$\sum_n |n\rangle\langle n| = \hat{I}$$

- This is exactly the resolution of identity / completeness relation — the operator version of "the basis spans the whole space."

Worked Example 1: Inner Product of Two States

Problem

Compute $\langle \phi | \psi \rangle$ for $\phi(x) = e^{-x^2/2a^2}$, $\psi(x) = xe^{-x^2/2a^2}$ on $(-\infty, \infty)$.

Solution: Setting up the inner product integral:

$$\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} \left(e^{-x^2/2a^2} \right)^* \cdot \left(xe^{-x^2/2a^2} \right) dx$$

Since the functions are completely real-valued, the complex conjugate has no effect:

$$= \int_{-\infty}^{\infty} x \cdot e^{-x^2/2a^2} \cdot e^{-x^2/2a^2} dx$$

Combining the exponential terms using standard exponent addition rules:

$$= \int_{-\infty}^{\infty} x \cdot e^{-\left(\frac{x^2}{2a^2} + \frac{x^2}{2a^2}\right)} dx = \int_{-\infty}^{\infty} x e^{-x^2/a^2} dx$$

Worked Example 1: Inner Product of Two States

We evaluate the symmetry of the integrand. Let $f(x) = xe^{-x^2/a^2}$. Testing for parity:

$$f(-x) = (-x)e^{-(-x)^2/a^2} = -xe^{-x^2/a^2} = -f(x)$$

Because the integrand is an odd function integrated over a perfectly symmetric domain about the origin:

$$\int_{-\infty}^{\infty} x e^{-x^2/a^2} dx = 0$$

This reflects the fact that the ground and first-excited harmonic-oscillator states are orthogonal.

Worked Example 2: Expansion Coefficients in a Basis

Problem

A state is $|\psi\rangle = \frac{3}{5}|1\rangle + \frac{4}{5}e^{i\pi/4}|2\rangle$ in an orthonormal basis. Verify normalization, and find $\langle 1|\psi\rangle$ and $\langle 2|\psi\rangle$.

Solution: To check normalization, we construct the corresponding bra vector by taking the conjugate transpose, converting kets to bras and complex-conjugating the scalars:

$$\langle\psi| = \frac{3}{5}\langle 1| + \frac{4}{5}e^{-i\pi/4}\langle 2|$$

Computing the full inner product $\langle\psi|\psi\rangle$:

$$\langle\psi|\psi\rangle = \left(\frac{3}{5}\langle 1| + \frac{4}{5}e^{-i\pi/4}\langle 2|\right) \left(\frac{3}{5}|1\rangle + \frac{4}{5}e^{i\pi/4}|2\rangle\right)$$

Worked Example 2: Expansion Coefficients in a Basis

Expanding the terms by distributing across the expression:

$$= \frac{9}{25} \langle 1|1 \rangle + \frac{12}{25} e^{i\pi/4} \langle 1|2 \rangle + \frac{12}{25} e^{-i\pi/4} \langle 2|1 \rangle + \frac{16}{25} e^{-i\pi/4} e^{i\pi/4} \langle 2|2 \rangle$$

Applying basis orthonormality ($\langle 1|1 \rangle = \langle 2|2 \rangle = 1$ and $\langle 1|2 \rangle = \langle 2|1 \rangle = 0$):

$$= \frac{9}{25}(1) + 0 + 0 + \frac{16}{25} e^0(1) = \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1 \checkmark$$

Now, computing the projection $\langle 1|\psi \rangle$:

$$\langle 1|\psi \rangle = \langle 1| \left(\frac{3}{5}|1 \rangle + \frac{4}{5} e^{i\pi/4} |2 \rangle \right) = \frac{3}{5} \langle 1|1 \rangle + \frac{4}{5} e^{i\pi/4} \langle 1|2 \rangle = \frac{3}{5}(1) + 0 = \frac{3}{5}$$

Similarly, computing the projection $\langle 2|\psi \rangle$:

$$\langle 2|\psi \rangle = \langle 2| \left(\frac{3}{5}|1 \rangle + \frac{4}{5} e^{i\pi/4} |2 \rangle \right) = \frac{3}{5} \langle 2|1 \rangle + \frac{4}{5} e^{i\pi/4} \langle 2|2 \rangle = 0 + \frac{4}{5} e^{i\pi/4}(1) = \frac{4}{5} e^{i\pi/4}$$

Directly reading off the coefficients yields exactly the results guaranteed by our completeness relation.

Worked Example 3: Outer Product as a Projection Operator

Problem

Show that $\hat{P} = |n\rangle\langle n|$ (for normalized $|n\rangle$) satisfies $\hat{P}^2 = \hat{P}$ (a defining property of a projection operator).

Solution: Expanding the expression for $\hat{P}^2$ by multiplying the operator by itself:

$$\hat{P}^2 = \hat{P} \cdot \hat{P} = (|n\rangle\langle n|) \cdot (|n\rangle\langle n|)$$

Using associative properties to group the inner interaction of the inner elements:

$$= |n\rangle \cdot (\langle n|n\rangle) \cdot \langle n|$$

Since the basis state vector is explicitly normalized, we substitute $\langle n|n\rangle = 1$:

$$= |n\rangle \cdot (1) \cdot \langle n| = |n\rangle\langle n| = \hat{P} \quad \checkmark$$

Acting on a general state: $\hat{P}|\psi\rangle = |n\rangle\langle n|\psi\rangle$, which extracts (projects out) the component of $|\psi\rangle$ along $|n\rangle$ — the formal operator version of "the part of ψ in the n -th basis direction."

Worked Example 4: Expressing a Box State in Dirac Notation

Problem

Translate the infinite-well normalization calculation from Unit 3 ($\int_0^L |\psi_1|^2 dx = 1$) into Dirac notation, and write the energy eigenvalue equation for ψ_2 in bra-ket form.

Solution: The spatial integration of the modulus squared is identically equivalent to the inner product of the state with itself:

$$\int_0^L \psi_1^*(x) \psi_1(x) dx = 1 \quad \Longleftrightarrow \quad \langle \psi_1 | \psi_1 \rangle = 1$$

The energy eigenvalue expression in wave mechanics is given by the differential equation $\hat{H}\psi_2(x) = E_2\psi_2(x)$. Converting the state functions directly to state kets gives the abstract relation:

$$\hat{H}|\psi_2\rangle = E_2|\psi_2\rangle$$

Where the specific physical value for the second energy level is given by:

$$E_2 = \frac{2^2 \pi^2 \hbar^2}{2mL^2} = \frac{4\pi^2 \hbar^2}{2mL^2}$$

Worked Example 4: Expressing a Box State in Dirac Notation

Orthogonality between different box states (Unit 5, derived rigorously later in this unit): $\langle \psi_1 | \psi_2 \rangle = 0$. This is purely a notational translation — every calculation from Unit 4 could equally well have been written this way from the start.

- An operator $\hat{A}$ acting on kets: $\hat{A}|\psi\rangle = |\phi\rangle$.
- **Matrix representation**: in a discrete orthonormal basis $\{|n\rangle\}$,

$$A_{mn} = \langle m|\hat{A}|n\rangle$$

so operators become matrices and kets become column vectors — matrix mechanics (Heisenberg's original formulation) and wave mechanics (Schrödinger's) are unified by this language.

- **Adjoint** (Hermitian conjugate) $\hat{A}^\dagger$: defined by $\langle\phi|\hat{A}^\dagger|\psi\rangle = \langle\psi|\hat{A}|\phi\rangle^*$.

Worked Example: Matrix Elements of $\hat{L}_z$

Problem

For the $\ell = 1$ subspace with basis $|-1\rangle, |0\rangle, |1\rangle$ (eigenstates of $\hat{L}_z$ with eigenvalues $-\hbar, 0, \hbar$), write the matrix representation of $\hat{L}_z$.

Solution: The elements of the matrix are calculated using $(L_z)_{mn} = \langle m|\hat{L}_z|n\rangle$, where indices $m, n \in \{-1, 0, 1\}$. Evaluating for every index pair:

$$(L_z)_{-1,-1} = \langle -1|\hat{L}_z|-1\rangle = \langle -1|(-\hbar)|-1\rangle = -\hbar\langle -1|-1\rangle = -\hbar$$

$$(L_z)_{-1,0} = \langle -1|\hat{L}_z|0\rangle = \langle -1|(0)|0\rangle = 0$$

$$(L_z)_{-1,1} = \langle -1|\hat{L}_z|1\rangle = \langle -1|(\hbar)|1\rangle = \hbar\langle -1|1\rangle = 0$$

Repeating this process across the remaining row positions using $\hat{L}_z|n\rangle = n\hbar|n\rangle$:

$$(L_z)_{mn} = n\hbar\langle m|n\rangle = n\hbar\delta_{mn}$$

Worked Example: Matrix Elements of $\hat{L}_z$

Arranging these explicit scalar entries systematically into a standard 3×3 grid:

$$L_z = \begin{pmatrix} \langle -1 | \hat{L}_z | -1 \rangle & \langle -1 | \hat{L}_z | 0 \rangle & \langle -1 | \hat{L}_z | 1 \rangle \\ \langle 0 | \hat{L}_z | -1 \rangle & \langle 0 | \hat{L}_z | 0 \rangle & \langle 0 | \hat{L}_z | 1 \rangle \\ \langle 1 | \hat{L}_z | -1 \rangle & \langle 1 | \hat{L}_z | 0 \rangle & \langle 1 | \hat{L}_z | 1 \rangle \end{pmatrix} = \begin{pmatrix} -\hbar & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \hbar \end{pmatrix} = \hbar \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A Hermitian operator's matrix is always diagonal in its *own* eigenbasis, with real eigenvalues on the diagonal.

Hermitian Operators: Eigenvalue and Eigenfunction Theorems

Hermitian Operator

$$\hat{A}^\dagger = \hat{A} \iff \langle \phi | \hat{A} | \psi \rangle = \langle \hat{A} \phi | \psi \rangle \quad \forall \phi, \psi$$

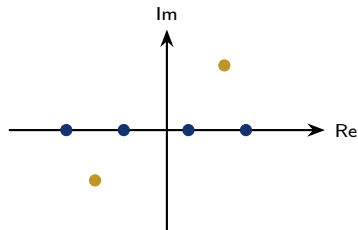
Key Theorems

- 1 Eigenvalues of a Hermitian operator are **real**.
- 2 Eigenfunctions belonging to **different** eigenvalues are **orthogonal**.
- 3 The eigenfunctions of a Hermitian operator form a **complete basis** for the Hilbert space.

Why this matters: since measurement outcomes (eigenvalues) must be real numbers, every quantum observable is represented by a Hermitian operator.

What Hermiticity Really Guarantees

- Hermiticity is the precise mathematical condition that captures the physical requirement: "this operator represents something we can actually measure with a real-numbered instrument reading."
- The three theorems together explain why quantum measurement works the way it does: real eigenvalues (Axiom 3, Unit 2) ensure sensible readouts; orthogonal eigenfunctions ensure that distinct measurement outcomes correspond to genuinely distinguishable states; completeness ensures *every* possible state can be expanded in terms of measurement outcomes, with no "missing" states.
- This is also why arbitrary mathematical combinations of operators are not automatically valid observables: e.g. $\hat{x}\hat{p}$ is *not* Hermitian (its adjoint is $\hat{p}\hat{x} \neq \hat{x}\hat{p}$), so it cannot directly represent a measurable quantity — only specific Hermitian combinations like $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ qualify.
- The next few slides prove the first two theorems explicitly; the completeness theorem requires more advanced functional analysis and is simply assumed (it holds for all operators encountered in this course).



Hermitian operator: eigenvalues confined to the real axis. **General operator:** eigenvalues may be complex.

Derivation: Hermitian Eigenvalues are Real

- Let $\hat{A}\psi_n = a_n\psi_n$, with $\hat{A}$ defined as a Hermitian operator. Let us compute the inner product expression $\langle\psi_n|\hat{A}|\psi_n\rangle$ along two independent pathways.
- **Pathway 1: Acting directly to the right.** We let the operator evaluate directly onto the ket state vector:

$$\langle\psi_n|\hat{A}|\psi_n\rangle = \langle\psi_n|(\hat{A}\psi_n)\rangle = \langle\psi_n|(a_n\psi_n)\rangle$$

Pulling the scalar value a_n entirely out of the bra-ket expression:

$$= a_n\langle\psi_n|\psi_n\rangle$$

- **Pathway 2: Utilizing the definition of Hermiticity.** We shift the operator's action over to the left side onto the bra element:

$$\langle\psi_n|\hat{A}|\psi_n\rangle = \langle\hat{A}\psi_n|\psi_n\rangle$$

Substituting the eigenvalue relation $\hat{A}\psi_n = a_n\psi_n$ into the left-hand position:

$$= \langle a_n\psi_n|\psi_n\rangle$$

Derivation: Hermitian Eigenvalues are Real

When pulling a scalar value out from the bra position of an inner product, it must be complex-conjugated:

$$= a_n^* \langle \psi_n | \psi_n \rangle$$

- Equating our two identical results for $\langle \psi_n | \hat{A} | \psi_n \rangle$:

$$a_n \langle \psi_n | \psi_n \rangle = a_n^* \langle \psi_n | \psi_n \rangle$$

- Bringing all terms to one side of the algebraic relation:

$$(a_n - a_n^*) \langle \psi_n | \psi_n \rangle = 0$$

- Since ψ_n represents a physically meaningful, non-vanishing quantum state vector, its norm-squared cannot be zero ($\langle \psi_n | \psi_n \rangle \neq 0$). This directly forces the scalar term to zero:

$$a_n - a_n^* = 0 \quad \implies \quad a_n = a_n^*$$

Which mathematically asserts that a_n is real.

Derivation: Orthogonality of Hermitian Eigenfunctions

- Let $\hat{A}\psi_m = a_m\psi_m$ and $\hat{A}\psi_n = a_n\psi_n$ where $a_m \neq a_n$ represent distinct eigenvalues, both known to be real numbers. Consider the configuration:

$$\langle \psi_m | \hat{A} | \psi_n \rangle = \langle \psi_m | (a_n \psi_n) \rangle = a_n \langle \psi_m | \psi_n \rangle$$

- Alternatively, utilizing the Hermiticity property to transition the operator action leftward:

$$\langle \psi_m | \hat{A} | \psi_n \rangle = \langle \hat{A}\psi_m | \psi_n \rangle = \langle a_m \psi_m | \psi_n \rangle$$

- Extracting the scalar value out from the first position requires a complex conjugate:

$$= a_m^* \langle \psi_m | \psi_n \rangle$$

- Since we already proved that eigenvalues of Hermitian operators are strictly real, $a_m^* = a_m$:

$$= a_m \langle \psi_m | \psi_n \rangle$$

Derivation: Orthogonality of Hermitian Eigenfunctions

- Setting both evaluated forms equal to one another:

$$a_n \langle \psi_m | \psi_n \rangle = a_m \langle \psi_m | \psi_n \rangle$$

- Collecting the expressions on the left-hand side of the equation:

$$a_n \langle \psi_m | \psi_n \rangle - a_m \langle \psi_m | \psi_n \rangle = 0 \implies (a_n - a_m) \langle \psi_m | \psi_n \rangle = 0$$

- By our original assertion, the two eigenvalues are distinct, meaning $(a_n - a_m) \neq 0$. Therefore, to satisfy the equation, the inner product must vanish:

$$\langle \psi_m | \psi_n \rangle = 0$$

Hence, the corresponding eigenfunctions are proven to be **orthogonal**.

Worked Example 1: Checking Hermiticity of $\hat{p}$

Problem

Show $\hat{p} = -i\hbar d/dx$ is Hermitian for wavefunctions vanishing at $\pm\infty$.

Solution: We evaluate the definition using an integration by parts strategy ($\int u dv = uv - \int v du$):

$$\langle \phi | \hat{p} | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \left(-i\hbar \frac{d\psi}{dx} \right) dx = -i\hbar \int_{-\infty}^{\infty} \phi^*(x) \frac{d\psi}{dx} dx$$

Assigning our parts for integration: $u = \phi^*(x) \implies du = \frac{d\phi^*}{dx} dx$, and $dv = \frac{d\psi}{dx} dx \implies v = \psi(x)$.

$$= -i\hbar \left([\phi^*(x)\psi(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d\phi^*}{dx} \psi(x) dx \right)$$

Because physical boundary conditions dictate that these wavefunctions must vanish at infinity, $\lim_{x \rightarrow \pm\infty} \psi(x) = \lim_{x \rightarrow \pm\infty} \phi(x) = 0$, the boundary term is zero:

$$= -i\hbar \left(0 - \int_{-\infty}^{\infty} \frac{d\phi^*}{dx} \psi(x) dx \right) = i\hbar \int_{-\infty}^{\infty} \frac{d\phi^*}{dx} \psi(x) dx$$

Worked Example 1: Checking Hermiticity of $\hat{p}$

We manipulate the constants to match the structure of the momentum operator format:

$$= \int_{-\infty}^{\infty} \left(-i\hbar \frac{d\phi}{dx} \right)^* \psi(x) dx = \langle \hat{p}\phi | \psi \rangle$$

Since $\langle \phi | \hat{p} | \psi \rangle = \langle \hat{p}\phi | \psi \rangle$ holds under these definitions, we confirm $\hat{p}^\dagger = \hat{p}$. ✓

Worked Example 2: Verifying Orthogonality of Box Eigenstates

Problem

Verify $\langle \psi_1 | \psi_2 \rangle = 0$ for the infinite-well states $\psi_1 = \sqrt{2/L} \sin(\pi x/L)$, $\psi_2 = \sqrt{2/L} \sin(2\pi x/L)$.

Solution: Setting up the integral across the active bounds of the potential well:

$$\langle \psi_1 | \psi_2 \rangle = \int_0^L \left(\sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right) \left(\sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right) \right) dx$$

Pulling out the normalized scalar constants:

$$= \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) dx$$

Worked Example 2: Verifying Orthogonality of Box Eigenstates

Applying the standard trigonometric product-to-sum identification rule,
 $\sin(A)\sin(B) = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$:

$$\begin{aligned} &= \frac{2}{L} \cdot \frac{1}{2} \int_0^L \left[\cos\left(\frac{\pi x}{L} - \frac{2\pi x}{L}\right) - \cos\left(\frac{\pi x}{L} + \frac{2\pi x}{L}\right) \right] dx \\ &= \frac{1}{L} \int_0^L \left[\cos\left(-\frac{\pi x}{L}\right) - \cos\left(\frac{3\pi x}{L}\right) \right] dx \end{aligned}$$

Since cosine is an even function ($\cos(-\theta) = \cos(\theta)$):

$$= \frac{1}{L} \int_0^L \left[\cos\left(\frac{\pi x}{L}\right) - \cos\left(\frac{3\pi x}{L}\right) \right] dx$$

Evaluating the analytical anti-derivatives directly:

$$= \frac{1}{L} \left[\frac{L}{\pi} \sin\left(\frac{\pi x}{L}\right) - \frac{L}{3\pi} \sin\left(\frac{3\pi x}{L}\right) \right]_0^L$$

Worked Example 2: Verifying Orthogonality of Box Eigenstates

Factoring out the shared scale parameters:

$$= \frac{1}{\pi} \left[\left(\sin(\pi) - \frac{1}{3} \sin(3\pi) \right) - \left(\sin(0) - \frac{1}{3} \sin(0) \right) \right]$$

Substituting the values ($\sin(n\pi) = 0$ for all integers n):

$$= \frac{1}{\pi} [(0 - 0) - (0 - 0)] = 0 \checkmark$$

This confirms the general theorem explicitly for this familiar case.

Worked Example 3: Non-Hermitian Operator Example

Problem

Show that $\hat{B} = d/dx$ (without the factor $-i\hbar$) is *not* Hermitian, by identifying its adjoint.

Solution: Setting up the functional inner product integration structure:

$$\langle \phi | \hat{B} | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \left(\frac{d\psi}{dx} \right) dx$$

Using integration by parts with $u = \phi^*(x) \implies du = \frac{d\phi^*}{dx} dx$ and $dv = \frac{d\psi}{dx} dx \implies v = \psi(x)$:

$$= [\phi^*(x)\psi(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d\phi^*}{dx} \psi(x) dx$$

Assuming the wavefunctions drop off to zero at the spatial boundaries:

$$= 0 - \int_{-\infty}^{\infty} \left(\frac{d\phi}{dx} \right)^* \psi(x) dx$$

Worked Example 3: Non-Hermitian Operator Example

Re-incorporating this back into standard bra-ket presentation formatting:

$$= -\langle \hat{B}\phi | \psi \rangle = \langle -\hat{B}\phi | \psi \rangle$$

Therefore, by comparing this against the definition of the adjoint operator, we find:

$$\hat{B}^\dagger = -\hat{B}$$

Since $\hat{B}^\dagger = -\hat{B} \neq \hat{B}$, the operator $\hat{B}$ is **anti-Hermitian**, not Hermitian. This is precisely why the momentum operator needs the factor of $-i\hbar$ (the extra i flips the sign and makes it Hermitian, as shown in Worked Example 1).

Worked Example 4: Hermiticity of $\hat{x}^2$

Problem

Verify directly that $\hat{x}^2$ (multiplication by x^2) is Hermitian.

Solution: Setting up the integration format:

$$\langle \phi | \hat{x}^2 | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \cdot (x^2 \psi(x)) dx$$

Because position values x are strictly real variables, x^2 is purely real, meaning $(x^2)^* = x^2$. This allows us to group it inside the complex conjugation brackets:

$$= \int_{-\infty}^{\infty} \phi^*(x) \cdot (x^2)^* \cdot \psi(x) dx = \int_{-\infty}^{\infty} (x^2 \phi(x))^* \psi(x) dx$$

Converting this integral back into standard inner product notation:

$$= \langle \hat{x}^2 \phi | \psi \rangle$$

Worked Example 4: Hermiticity of $\hat{x}^2$

This confirms $(\hat{x}^2)^\dagger = \hat{x}^2$. More generally, multiplication by *any real-valued function* $f(x)$ — including potentials $V(x)$ in the Hamiltonian — is automatically Hermitian, which is why no special justification was needed when $V(x)$ was added to $\hat{H}$ throughout Unit 4.

Position, Momentum, and Total Energy Operators

Position representation:

$$\hat{x} \rightarrow x, \quad \hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$$

Momentum representation:

$$\hat{x} \rightarrow i\hbar \frac{\partial}{\partial p}, \quad \hat{p} \rightarrow p$$

Total Energy (Hamiltonian) Operator

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad \hat{H}|\psi\rangle = E|\psi\rangle$$

All three are Hermitian, guaranteeing real, measurable eigenvalues for position, momentum, and energy.

Derivation: Why $\hat{x} \rightarrow i\hbar \partial/\partial p$ in Momentum Space

- **Starting point:** the position-space and momentum-space wavefunctions are a Fourier pair:

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi(p) e^{ipx/\hbar} dp, \quad \phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x) e^{-ipx/\hbar} dx$$

- **Step 1 — Represent $\langle x \rangle$ in momentum space.** Start from Born's rule in position space, $\langle x \rangle = \int \psi^*(x) x \psi(x) dx$, and substitute the Fourier representation for $\psi(x)$ and, using a dummy variable p' for the complex-conjugated copy, for $\psi^*(x)$:

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int \phi(p) e^{ipx/\hbar} dp, \quad \psi^*(x) = \frac{1}{\sqrt{2\pi\hbar}} \int \phi^*(p') e^{-ip'x/\hbar} dp'$$

Substituting both into $\langle x \rangle$ and combining the two prefactors $1/\sqrt{2\pi\hbar} \times 1/\sqrt{2\pi\hbar} = 1/(2\pi\hbar)$:

$$\langle x \rangle = \frac{1}{2\pi\hbar} \iiint \phi^*(p') \phi(p) x e^{i(p-p')x/\hbar} dp' dp dx$$

Derivation: Why $\hat{x} \rightarrow i\hbar \partial/\partial p$ in Momentum Space

Rewrite the factor of x as a p -derivative. Differentiating the exponential with respect to p (treating x, p' as constants) gives $\frac{\partial}{\partial p} e^{i(p-p')x/\hbar} = \frac{ix}{\hbar} e^{i(p-p')x/\hbar}$. Multiplying both sides by $-i\hbar$:

$$-i\hbar \frac{\partial}{\partial p} e^{i(p-p')x/\hbar} = -i\hbar \cdot \frac{ix}{\hbar} e^{i(p-p')x/\hbar} = -i^2 x e^{i(p-p')x/\hbar} = x e^{i(p-p')x/\hbar}$$

(using $-i^2 = 1$). So the replacement $x e^{i(p-p')x/\hbar} = -i\hbar \frac{\partial}{\partial p} e^{i(p-p')x/\hbar}$ is exact. Substitute this into $\langle x \rangle$ in place of the factor of x :

$$\langle x \rangle = \frac{1}{2\pi\hbar} \int \int \phi^*(p') \phi(p) \left[-i\hbar \frac{\partial}{\partial p} \int e^{i(p-p')x/\hbar} dx \right] dp' dp$$

Do the x -integral, using the standard Fourier identity $\int e^{i(p-p')x/\hbar} dx = 2\pi\hbar \delta(p - p')$:

$$\langle x \rangle = \frac{1}{2\pi\hbar} \int \int \phi^*(p') \phi(p) \left(-i\hbar \frac{\partial}{\partial p} \right) [2\pi\hbar \delta(p - p')] dp' dp$$

Derivation: Why $\hat{x} \rightarrow i\hbar \partial/\partial p$ in Momentum Space

The factor $2\pi\hbar$ cancels the $1/(2\pi\hbar)$ prefactor:

$$\langle x \rangle = -i\hbar \iint \phi^*(p') \phi(p) \frac{\partial}{\partial p} \delta(p - p') dp' dp$$

Integrate over p first, using the defining property of the derivative of a delta function, $\int f(p) \frac{\partial}{\partial p} \delta(p - p') dp = -f'(p')$ (integration by parts on the delta function, with the boundary terms vanishing), applied here with $f(p) = \phi(p)$:

$$\int \phi(p) \frac{\partial}{\partial p} \delta(p - p') dp = -\left. \frac{\partial \phi}{\partial p} \right|_{p=p'} = -\phi'(p')$$

Substituting this back:

$$\langle x \rangle = -i\hbar \int \phi^*(p') [-\phi'(p')] dp' = i\hbar \int \phi^*(p') \phi'(p') dp'$$

Derivation: Why $\hat{x} \rightarrow i\hbar \partial/\partial p$ in Momentum Space

Finally, relabel the dummy integration variable $p' \rightarrow p$ (its name does not affect the value of the integral) and recognize $\phi'(p) = \partial\phi/\partial p$:

$$\langle x \rangle = \int \phi^*(p) \left(i\hbar \frac{\partial}{\partial p} \right) \phi(p) dp$$

- **Step 2 — Identify the operator.** Comparing with the general Born rule $\langle A \rangle = \int \phi^*(p) \hat{A}_p \phi(p) dp$, the action of $\hat{x}$ in the momentum representation is:

$$\hat{x} \longrightarrow i\hbar \frac{\partial}{\partial p}$$

- **Step 3 — Confirm $\hat{p} \rightarrow p$ by symmetry.** The momentum operator in its own representation is simply multiplication by p , since $\phi(p)$ is already a momentum eigenstate: $\hat{p}\phi(p) = p\phi(p)$.
- **Summary:** the two representations are fully symmetric under the interchange $x \leftrightarrow p$, $\hbar \partial_x \leftrightarrow -\hbar \partial_p$, reflecting the deep Fourier-duality between position and momentum spaces already seen in the uncertainty principle (Unit 2).

Worked Example: Eigenfunction of $\hat{H}$ in Two Representations

Problem

A particle has energy eigenstate $\psi(x) = \sqrt{2/L} \sin(\pi x/L)$ on $[0, L]$ with $V = 0$. Verify $\hat{H}\psi = E\psi$ directly, and state E .

Solution: Since $V(x) = 0$ within the system boundaries, the Hamiltonian simplifies to:

$$\hat{H} = \frac{\hat{p}^2}{2m} = \frac{1}{2m} \left(-i\hbar \frac{d}{dx} \right)^2 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

Applying this explicit derivative definition to the wave function state:

$$\hat{H}\psi = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \left(\sqrt{\frac{2}{L}} \sin \left(\frac{\pi x}{L} \right) \right)$$

Worked Example: Eigenfunction of $\hat{H}$ in Two Representations

Evaluating the first derivative step:

$$\frac{d}{dx} \left(\sqrt{\frac{2}{L}} \sin \left(\frac{\pi x}{L} \right) \right) = \sqrt{\frac{2}{L}} \left(\frac{\pi}{L} \right) \cos \left(\frac{\pi x}{L} \right)$$

Evaluating the second derivative step:

$$\frac{d^2}{dx^2} \left(\sqrt{\frac{2}{L}} \sin \left(\frac{\pi x}{L} \right) \right) = \frac{d}{dx} \left(\sqrt{\frac{2}{L}} \left(\frac{\pi}{L} \right) \cos \left(\frac{\pi x}{L} \right) \right) = -\sqrt{\frac{2}{L}} \left(\frac{\pi^2}{L^2} \right) \sin \left(\frac{\pi x}{L} \right)$$

Substituting this result back into our equation for the full Hamiltonian operation:

$$\hat{H}\psi = -\frac{\hbar^2}{2m} \left(-\frac{\pi^2}{L^2} \sqrt{\frac{2}{L}} \sin \left(\frac{\pi x}{L} \right) \right)$$

Worked Example: Eigenfunction of $\hat{H}$ in Two Representations

Combining negative signs and separating the original state function expression:

$$= \left(\frac{\pi^2 \hbar^2}{2mL^2} \right) \cdot \left(\sqrt{\frac{2}{L}} \sin \left(\frac{\pi x}{L} \right) \right) = \frac{\pi^2 \hbar^2}{2mL^2} \psi$$

So ψ is indeed an eigenfunction, with $E = \frac{\pi^2 \hbar^2}{2mL^2} = E_1$ — consistent with Unit 4.

What a Commutator Really Measures

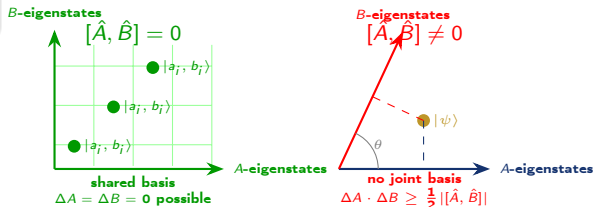
- Classically, the order in which you measure two quantities never matters: measuring a ball's position then its momentum gives the same numbers as measuring momentum then position.
- Quantum mechanically, this is no longer automatic. The commutator $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ precisely quantifies the failure of operators to "commute" — i.e. how much the *order* of measurement (or more precisely, of operator action) matters.
- A nonzero commutator is not merely a mathematical curiosity: it directly encodes a fundamental limit on simultaneous knowledge, culminating in the General Uncertainty Principle derived later in this subsection.
- This subsection completes the conceptual arc of the entire course: Unit 1's wave packets hinted at a position-momentum trade-off; Unit 2 stated Heisenberg's principle as an experimental fact; here, that principle is derived as a mathematical theorem from the single algebraic fact $[\hat{x}, \hat{p}] = i\hbar$.

Commutator Brackets — Simultaneous Eigenfunctions

Commutator

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$$

- $[\hat{A}, \hat{B}] = 0$: operators **commute** $\Rightarrow$ complete set of **simultaneous eigenfunctions**; both observables precisely knowable.
- $[\hat{A}, \hat{B}] \neq 0$: do **not** commute $\Rightarrow$ no joint eigenstates; uncertainty relation follows.
- $[\hat{L}^2, \hat{L}_z] = 0$; but $[\hat{L}_x, \hat{L}_y] \neq 0$.



Derivation: Commuting Operators Share Eigenstates

- Suppose $[\hat{A}, \hat{B}] = 0$ (which implies $\hat{A}\hat{B} = \hat{B}\hat{A}$) and $\hat{A}\psi = a\psi$, assuming a non-degenerate eigenvalue a . Let us consider the action of the operator configuration $\hat{A}(\hat{B}\psi)$:

$$\hat{A}(\hat{B}\psi) = (\hat{A}\hat{B})\psi$$

- Substituting the commuting condition $\hat{A}\hat{B} = \hat{B}\hat{A}$:

$$= (\hat{B}\hat{A})\psi = \hat{B}(\hat{A}\psi)$$

- Substituting the eigenvalue relation $\hat{A}\psi = a\psi$:

$$= \hat{B}(a\psi)$$

- Since a is a scalar constant, it can be factored out to the left front position:

$$= a(\hat{B}\psi)$$

Derivation: Commuting Operators Share Eigenstates

- Therefore, the combined state ($\hat{B}\psi$) is revealed to be an eigenfunction of $\hat{A}$ sharing the exact same eigenvalue a .
- Because we stated that a is non-degenerate, there can only exist a single unique linearly independent eigenfunction for that value. Thus, the state vector ($\hat{B}\psi$) must be directly proportional to ψ by some scalar constant scale factor b :

$$\hat{B}\psi = b\psi$$

- I.e. ψ is proven to be *simultaneously* an eigenfunction of both $\hat{A}$ and $\hat{B}$.

Canonical Commutation Relation

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$$

Derivation sketch: acting on a test function $\psi(x)$,

$$(\hat{x}\hat{p} - \hat{p}\hat{x})\psi = x \left(-i\hbar \frac{d\psi}{dx} \right) - \left(-i\hbar \frac{d}{dx} \right) (x\psi) = i\hbar \psi$$

- Useful commutator identities: $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$; $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$.
- The non-vanishing commutator $[\hat{x}, \hat{p}] = i\hbar$ is the algebraic *origin* of the uncertainty principle.

Derivation: $[\hat{x}, \hat{p}] = i\hbar$ in Full Detail

- Let us evaluate the explicit behavior of the commutator using an arbitrary smooth test function $\psi(x)$.

$$[\hat{x}, \hat{p}]\psi(x) = (\hat{x}\hat{p} - \hat{p}\hat{x})\psi(x) = \hat{x}\hat{p}\psi(x) - \hat{p}\hat{x}\psi(x)$$

- Evaluating the first expression, $\hat{x}\hat{p}\psi(x)$:

$$\hat{x}\hat{p}\psi(x) = x \left(-i\hbar \frac{d\psi}{dx} \right) = -i\hbar x \frac{d\psi}{dx}$$

- Evaluating the second expression, $\hat{p}\hat{x}\psi(x)$, noting that the derivative must act on the combined product ($x\psi$) via the product rule ($\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$):

$$\begin{aligned} \hat{p}\hat{x}\psi(x) &= -i\hbar \frac{d}{dx} (x\psi(x)) = -i\hbar \left(\psi(x) \cdot \frac{dx}{dx} + x \cdot \frac{d\psi}{dx} \right) \\ &= -i\hbar \left(\psi(x) + x \frac{d\psi}{dx} \right) = -i\hbar \psi(x) - i\hbar x \frac{d\psi}{dx} \end{aligned}$$

Derivation: $[\hat{x}, \hat{p}] = i\hbar$ in Full Detail

- Assembling the final difference between our two expressions:

$$(\hat{x}\hat{p} - \hat{p}\hat{x})\psi(x) = \left(-i\hbar x \frac{d\psi}{dx}\right) - \left(-i\hbar\psi(x) - i\hbar x \frac{d\psi}{dx}\right)$$

- Distributing the negative sign:

$$= -i\hbar x \frac{d\psi}{dx} + i\hbar\psi(x) + i\hbar x \frac{d\psi}{dx}$$

- Canceling out the opposing terms:

$$= i\hbar\psi(x)$$

- Since this holds consistently for any choice of test function $\psi(x)$, we drop the function to state the operator identity:

$$[\hat{x}, \hat{p}] = i\hbar \hat{I}$$

- This is the **canonical commutation relation**.

Worked Example 1: Commutator $[\hat{x}^2, \hat{p}]$

Problem

Evaluate $[\hat{x}^2, \hat{p}]$ using the identity $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$.

Solution: We map the identity by setting $\hat{A} = \hat{x}$, $\hat{B} = \hat{x}$, and $\hat{C} = \hat{p}$:

$$[\hat{x}^2, \hat{p}] = [\hat{x} \cdot \hat{x}, \hat{p}] = \hat{x}[\hat{x}, \hat{p}] + [\hat{x}, \hat{p}]\hat{x}$$

Substituting the known canonical commutation value $[\hat{x}, \hat{p}] = i\hbar$:

$$= \hat{x}(i\hbar) + (i\hbar)\hat{x}$$

Since scalars commute with all operators, we consolidate the expression:

$$= i\hbar\hat{x} + i\hbar\hat{x} = 2i\hbar\hat{x}$$

Worked Example 2: Commutator $[\hat{H}, \hat{x}]$ for a Free Particle

Problem

For $\hat{H} = \hat{p}^2/2m$ (free particle), compute $[\hat{H}, \hat{x}]$.

Solution: Setting up the commutator layout:

$$[\hat{H}, \hat{x}] = \left[\frac{\hat{p}^2}{2m}, \hat{x} \right] = \frac{1}{2m} [\hat{p}^2, \hat{x}]$$

Expanding the squared operator into a product representation $[\hat{p} \cdot \hat{p}, \hat{x}]$ and applying our product commutator identity:

$$[\hat{p}^2, \hat{x}] = \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p}$$

Using the anti-symmetric property of commutators, $[\hat{p}, \hat{x}] = -[\hat{x}, \hat{p}] = -i\hbar$:

$$= \hat{p}(-i\hbar) + (-i\hbar)\hat{p} = -2i\hbar\hat{p}$$

Worked Example 2: Commutator $[\hat{H}, \hat{x}]$ for a Free Particle

Plugging this result back into our initial fractional equation:

$$[\hat{H}, \hat{x}] = \frac{1}{2m}(-2i\hbar\hat{p}) = -\frac{i\hbar}{m}\hat{p}$$

This matches (with opposite sign convention) exactly the commutator used in the Ehrenfest's theorem derivation in Unit 3.

Worked Example 3: Do $\hat{x}$ and $\hat{H}$ Commute?

Problem

Using the previous result, determine whether $\hat{x}$ and $\hat{H}$ (free particle) can be simultaneously known precisely.

Solution: We evaluate the commutator value computed directly above:

$$[\hat{H}, \hat{x}] = -\frac{i\hbar}{m}\hat{p}$$

For the operators to commute, this value must equal zero identically across all operations. However, because $\hat{p}$ represents the physical momentum operator, its expectation value depends on the state and is non-zero in general ($\langle p \rangle \neq 0$). Therefore:

$$[\hat{H}, \hat{x}] \neq 0$$

Since they do not commute, position and energy cannot, in general, be known simultaneously and exactly — consistent with the energy–time-like trade-offs and with energy eigenstates being spatially delocalized (e.g. plane waves, infinite-well states spread over the whole well).

Worked Example 4: Do $\hat{L}^2$ and $\hat{L}_z$ Really Commute?

Problem

Unit 4 asserted that $\hat{L}^2$ and $\hat{L}_z$ commute. Verify this is consistent with both operators sharing the simultaneous eigenstates $Y_\ell^{m_\ell}$.

Solution: Let us construct the explicit commutator operation acting directly upon a spherical harmonic basis function:

$$[\hat{L}^2, \hat{L}_z] Y_\ell^{m_\ell} = (\hat{L}^2 \hat{L}_z - \hat{L}_z \hat{L}^2) Y_\ell^{m_\ell} = \hat{L}^2 \hat{L}_z Y_\ell^{m_\ell} - \hat{L}_z \hat{L}^2 Y_\ell^{m_\ell}$$

Evaluating the first path term by letting the innermost operator act first:

$$\hat{L}^2 \hat{L}_z Y_\ell^{m_\ell} = \hat{L}^2 (m_\ell \hbar Y_\ell^{m_\ell}) = m_\ell \hbar \cdot (\hat{L}^2 Y_\ell^{m_\ell}) = m_\ell \hbar \cdot \hbar^2 \ell(\ell + 1) Y_\ell^{m_\ell}$$

Consolidating the factors:

$$= m_\ell \ell(\ell + 1) \hbar^3 Y_\ell^{m_\ell}$$

Worked Example 4: Do $\hat{L}^2$ and $\hat{L}_z$ Really Commute?

Evaluating the second path term similarly:

$$\hat{L}_z \hat{L}^2 Y_\ell^{m_\ell} = \hat{L}_z (\hbar^2 \ell(\ell+1) Y_\ell^{m_\ell}) = \hbar^2 \ell(\ell+1) \cdot (\hat{L}_z Y_\ell^{m_\ell}) = \hbar^2 \ell(\ell+1) \cdot m_\ell \hbar Y_\ell^{m_\ell}$$

Consolidating the factors:

$$= m_\ell \ell(\ell+1) \hbar^3 Y_\ell^{m_\ell}$$

Subtracting the two evaluated paths from one another:

$$[\hat{L}^2, \hat{L}_z] Y_\ell^{m_\ell} = m_\ell \ell(\ell+1) \hbar^3 Y_\ell^{m_\ell} - m_\ell \ell(\ell+1) \hbar^3 Y_\ell^{m_\ell} = 0$$

The two results are identical, so $[\hat{L}^2, \hat{L}_z] Y_\ell^{m_\ell} = 0$ for every spherical harmonic — and since these form a complete basis, $[\hat{L}^2, \hat{L}_z] = 0$ as an operator identity. This confirms, via the "commuting operators share eigenstates" derivation earlier in this subsection, exactly why ℓ and m_ℓ can be assigned simultaneously as good quantum numbers in Unit 4.

General Form of the Uncertainty Principle

Robertson Uncertainty Relation

For any two Hermitian operators $\hat{A}, \hat{B}$:

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

Applying to $\hat{A} = \hat{x}$, $\hat{B} = \hat{p}$:

$$\Delta x \Delta p \geq \frac{1}{2} |\langle i\hbar \rangle| = \frac{\hbar}{2}$$

- This **rigorously derives** the Heisenberg uncertainty principle introduced qualitatively in Unit 2.
- The general relation applies to *any* pair of non-commuting observables, e.g. angular momentum components: $\Delta L_x \Delta L_y \geq \frac{1}{2} |\langle L_z \rangle| \hbar$.

Problem-Practice Class: Uncertainty Principle

Derivation: The Robertson Uncertainty Relation — Setup

- Define shifted (zero-mean) operators $\Delta\hat{A} = \hat{A} - \langle A \rangle$, $\Delta\hat{B} = \hat{B} - \langle B \rangle$, and let $|f\rangle = \Delta\hat{A}|\psi\rangle$, $|g\rangle = \Delta\hat{B}|\psi\rangle$.
- By the **Cauchy–Schwarz inequality**: $\langle f|f\rangle\langle g|g\rangle \geq |\langle f|g\rangle|^2$. Let us compute each piece explicitly.

$$\langle f|f\rangle = \langle\psi|\Delta\hat{A}^\dagger\Delta\hat{A}|\psi\rangle$$

Since $\hat{A}$ is Hermitian, its shifted variant is also Hermitian ($\Delta\hat{A}^\dagger = \Delta\hat{A}$):

$$\langle f|f\rangle = \langle\psi|(\Delta\hat{A})^2|\psi\rangle = (\Delta A)^2$$

Similarly, $\langle g|g\rangle = (\Delta B)^2$. Now evaluating the overlap expression:

$$\langle f|g\rangle = \langle\psi|\Delta\hat{A}^\dagger\Delta\hat{B}|\psi\rangle = \langle\psi|\Delta\hat{A}\Delta\hat{B}|\psi\rangle = \langle\Delta\hat{A}\Delta\hat{B}\rangle$$

Substituting these explicit components into the Cauchy-Schwarz inequality statement:

$$(\Delta A)^2(\Delta B)^2 \geq \left| \langle\Delta\hat{A}\Delta\hat{B}\rangle \right|^2$$

Derivation: The Robertson Uncertainty Relation — Setup

- Write the product operator $\Delta\hat{A}\Delta\hat{B}$ explicitly as a sum of its symmetric and anti-symmetric operator components:

$$\begin{aligned}\Delta\hat{A}\Delta\hat{B} &= \frac{1}{2} \left(\Delta\hat{A}\Delta\hat{B} + \Delta\hat{B}\Delta\hat{A} \right) + \frac{1}{2} \left(\Delta\hat{A}\Delta\hat{B} - \Delta\hat{B}\Delta\hat{A} \right) \\ &= \frac{1}{2} \{ \Delta\hat{A}, \Delta\hat{B} \} + \frac{1}{2} [\Delta\hat{A}, \Delta\hat{B}]\end{aligned}$$

where $\{\cdot, \cdot\}$ is the anticommutator. The first term is real (Hermitian), the second purely imaginary (since $[\hat{A}, \hat{B}]^\dagger = -[\hat{A}, \hat{B}]$ for Hermitian $\hat{A}, \hat{B}$).

Derivation: The Robertson Uncertainty Relation — Result

- Taking the expectation value of our split operator expression:

$$\langle \Delta \hat{A} \Delta \hat{B} \rangle = \frac{1}{2} \langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle + \frac{1}{2} \langle [\Delta \hat{A}, \Delta \hat{B}] \rangle$$

Since the commutator expands directly as $[\Delta \hat{A}, \Delta \hat{B}] = [\hat{A} - \langle \hat{A} \rangle, \hat{B} - \langle \hat{B} \rangle] = [\hat{A}, \hat{B}]$, we substitute this simplification:

$$= \frac{1}{2} \langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle + \frac{1}{2} \langle [\hat{A}, \hat{B}] \rangle$$

- Let $x = \frac{1}{2} \langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle$ (which is purely real) and $iy = \frac{1}{2} \langle [\hat{A}, \hat{B}] \rangle$ (which is purely imaginary). The magnitude squared of any complex value $z = x + iy$ evaluates as $|z|^2 = x^2 + y^2$:

$$\begin{aligned} |\langle \Delta \hat{A} \Delta \hat{B} \rangle|^2 &= \left(\frac{1}{2} \left| \langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle \right| \right)^2 + \left(\frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right| \right)^2 \\ &= \frac{1}{4} \left| \langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle \right|^2 + \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \end{aligned}$$

Derivation: The Robertson Uncertainty Relation — Result

- Dropping the non-negative anticommutator term only weakens the inequality:

$$\frac{1}{4} \left| \langle \{\Delta \hat{A}, \Delta \hat{B}\} \rangle \right|^2 + \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2$$

- Combining this tracking chain with our initial Cauchy–Schwarz configuration step:

$$\begin{aligned} (\Delta A)^2 (\Delta B)^2 &\geq \left| \langle \Delta \hat{A} \Delta \hat{B} \rangle \right|^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \\ \implies (\Delta A)^2 (\Delta B)^2 &\geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \end{aligned}$$

- Taking the square root of both sides yields the final product inequality:

$$\Delta A \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|$$

- Setting $\hat{A} = \hat{x}$, $\hat{B} = \hat{p}$ and using $[\hat{x}, \hat{p}] = i\hbar$ recovers exactly $\Delta x \Delta p \geq \hbar/2$ — the Heisenberg relation is now a **theorem**, not just a heuristic (as it was in Unit 2).

Worked Example 1: Energy–Time from a Generalized Relation

Problem

Using $d\langle A \rangle / dt = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle$ (Ehrenfest-type relation) together with the Robertson bound applied to $\hat{A}, \hat{H}$, motivate $\Delta E \Delta t \gtrsim \hbar/2$, where $\Delta t \equiv \Delta A / |d\langle A \rangle / dt|$.

Solution: Applying the Robertson uncertainty equation specifically to our chosen operators $\hat{A}$ and $\hat{H}$:

$$\Delta A \Delta H \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{H}] \rangle \right|$$

We identify ΔH as the uncertainty in energy, ΔE . From the stated Ehrenfest relationship, we isolate the commutator term algebraically:

$$\frac{d\langle A \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle \quad \Longrightarrow \quad \langle [\hat{A}, \hat{H}] \rangle = i\hbar \frac{d\langle A \rangle}{dt}$$

Worked Example 1: Energy–Time from a Generalized Relation

Taking the absolute magnitude of this value:

$$\left| \langle [\hat{A}, \hat{H}] \rangle \right| = \left| i\hbar \frac{d\langle A \rangle}{dt} \right| = \hbar \left| \frac{d\langle A \rangle}{dt} \right|$$

Substituting this directly back into our Robertson expression:

$$\Delta A \Delta E \geq \frac{1}{2} \left(\hbar \left| \frac{d\langle A \rangle}{dt} \right| \right) = \frac{\hbar}{2} \left| \frac{d\langle A \rangle}{dt} \right|$$

Dividing both sides of the inequality by the absolute rate of change factor $\left| \frac{d\langle A \rangle}{dt} \right|$:

$$\frac{\Delta A}{\left| \frac{d\langle A \rangle}{dt} \right|} \cdot \Delta E \geq \frac{\hbar}{2}$$

Worked Example 1: Energy–Time from a Generalized Relation

Substituting our definition for the characteristic evolution timescale, $\Delta t = \frac{\Delta A}{|d\langle A \rangle/dt|}$:

$$\Delta t \Delta E \geq \frac{\hbar}{2}$$

Here Δt is interpreted as the time for $\langle A \rangle$ to change by one standard deviation — a natural "characteristic evolution time," giving the energy–time uncertainty relation used qualitatively in Unit 2.

Worked Example 2: Angular Momentum Uncertainty Product

Problem

Given $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$, find the minimum value of $\Delta L_x \Delta L_y$ for a state with $\langle L_z \rangle = 2\hbar$.

Solution: Applying the Robertson bounding equation directly to the tracking components:

$$\Delta L_x \Delta L_y \geq \frac{1}{2} \left| \langle [\hat{L}_x, \hat{L}_y] \rangle \right|$$

Substituting our given commutator expression into the equation:

$$= \frac{1}{2} \left| \langle i\hbar\hat{L}_z \rangle \right|$$

Pulling the scalar multipliers outside of the expectation value brackets and magnitude constraints:

$$= \frac{1}{2} \cdot |-i| \cdot \hbar \cdot \left| \langle \hat{L}_z \rangle \right| = \frac{\hbar}{2} |\langle L_z \rangle|$$

Worked Example 2: Angular Momentum Uncertainty Product

Plugging in our explicit expectation target value $\langle L_z \rangle = 2\hbar$:

$$= \frac{\hbar}{2} \cdot (2\hbar) = \hbar^2$$

Therefore, the absolute minimum product threshold is $\hbar^2$.

Worked Example 3: Numeric Check of $\Delta x \Delta p$ for a Gaussian

Problem

For the Gaussian ground state of the harmonic oscillator, $\Delta x = \sqrt{\hbar/2m\omega}$ and $\Delta p = \sqrt{\hbar m\omega/2}$. Verify the Robertson/Heisenberg bound is saturated.

Solution: We evaluate the direct product multiplication of both uncertainty values:

$$\Delta x \Delta p = \left(\sqrt{\frac{\hbar}{2m\omega}} \right) \cdot \left(\sqrt{\frac{\hbar m\omega}{2}} \right)$$

Consolidating both terms under a single unified radical sign:

$$= \sqrt{\frac{\hbar}{2m\omega} \cdot \frac{\hbar m\omega}{2}}$$

Worked Example 3: Numeric Check of $\Delta x \Delta p$ for a Gaussian

Canceling matching mass terms m and frequency terms ω within the fraction:

$$= \sqrt{\frac{\hbar^2}{4}}$$

Extracting the final value from the square root:

$$= \frac{\hbar}{2}$$

Exactly equals the lower bound — the harmonic oscillator ground state, like the general Gaussian wave packet of Unit 2, is a **minimum-uncertainty state**.

Additional Problems 1–2

Problem 1: Matrix Elements of $\hat{x}$ for a Box

For the infinite well, compute $\langle \psi_1 | \hat{x} | \psi_1 \rangle$ (you may quote the Unit 3 result).

Solution: Setting up the expectation value integral explicitly using the ground state wavefunction

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right):$$

$$\langle \psi_1 | \hat{x} | \psi_1 \rangle = \int_0^L \psi_1^*(x) \cdot x \cdot \psi_1(x) dx$$

Substituting the real-valued function expressions directly into the integral:

$$= \int_0^L \left(\sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right) \cdot x \cdot \left(\sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right) dx$$

Pulling out the constant normalization factors and combining the sine terms:

$$= \frac{2}{L} \int_0^L x \sin^2\left(\frac{\pi x}{L}\right) dx$$

Additional Problems 1–2

Using the trigonometric identity $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$:

$$= \frac{2}{L} \int_0^L x \cdot \left(\frac{1 - \cos\left(\frac{2\pi x}{L}\right)}{2} \right) dx = \frac{1}{L} \int_0^L \left[x - x \cos\left(\frac{2\pi x}{L}\right) \right] dx$$

Splitting this expression into two separate integration components:

$$= \frac{1}{L} \int_0^L x dx - \frac{1}{L} \int_0^L x \cos\left(\frac{2\pi x}{L}\right) dx$$

Evaluating the first power-integral directly:

$$\frac{1}{L} \left[\frac{x^2}{2} \right]_0^L = \frac{1}{L} \left(\frac{L^2}{2} - 0 \right) = \frac{L}{2}$$

Additional Problems 1–2

Evaluating the second integral via integration by parts ($\int u dv = uv - \int v du$) where $u = x \implies du = dx$, and $dv = \cos\left(\frac{2\pi x}{L}\right) dx \implies v = \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right)$:

$$\int_0^L x \cos\left(\frac{2\pi x}{L}\right) dx = \left[x \cdot \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_0^L - \int_0^L \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) dx$$

Since $\sin(2\pi) = \sin(0) = 0$, the boundary term drops out completely:

$$= 0 - \frac{L}{2\pi} \left[-\frac{L}{2\pi} \cos\left(\frac{2\pi x}{L}\right) \right]_0^L = \frac{L^2}{4\pi^2} (\cos(2\pi) - \cos(0)) = \frac{L^2}{4\pi^2} (1 - 1) = 0$$

Combining our solved integration pieces together yields:

$$\langle \psi_1 | \hat{x} | \psi_1 \rangle = \frac{L}{2} - 0 = \frac{L}{2}$$

Additional Problems 1–2

Problem 2: Commutator $[\hat{x}, \hat{p}^2]$

Evaluate $[\hat{x}, \hat{p}^2]$.

Solution: Expanding the squared momentum operator into a sequential product expression $[\hat{x}, \hat{p} \cdot \hat{p}]$ and applying the commutator identity $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$:

$$[\hat{x}, \hat{p}^2] = [\hat{x}, \hat{p} \cdot \hat{p}] = [\hat{x}, \hat{p}]\hat{p} + \hat{p}[\hat{x}, \hat{p}]$$

Substituting the core canonical commutation value $[\hat{x}, \hat{p}] = i\hbar$:

$$= (i\hbar)\hat{p} + \hat{p}(i\hbar)$$

Because the scalar values i and $\hbar$ naturally commute with all physical operators, they can be pulled freely to the front:

$$= i\hbar\hat{p} + i\hbar\hat{p} = 2i\hbar\hat{p}$$

Additional Problems 3–4

Problem 3: Normalization Check via Bra-Ket

If $|\psi\rangle = N(|1\rangle + 2i|2\rangle - |3\rangle)$ with $\{|n\rangle\}$ orthonormal, find N (real, positive).

Solution: To build the left-hand bra element $\langle\psi|$, we take the complete conjugate transpose of the expression, changing kets to bras and complex-conjugating the scalar terms ($i \rightarrow -i$):

$$\langle\psi| = N^* \left(\langle 1| + (2i)^* \langle 2| - \langle 3| \right) = N \left(\langle 1| - 2i \langle 2| - \langle 3| \right)$$

Now, constructing the full inner product equation $\langle\psi|\psi\rangle = 1$ to check normalization bounds:

$$\langle\psi|\psi\rangle = \left[N(\langle 1| - 2i \langle 2| - \langle 3|) \right] \cdot \left[N(|1\rangle + 2i|2\rangle - |3\rangle) \right]$$

Factoring out the matching real scaling coefficients to the front:

$$= N^2 \left(\langle 1| - 2i \langle 2| - \langle 3| \right) \left(|1\rangle + 2i|2\rangle - |3\rangle \right)$$

Additional Problems 3–4

Expanding this multiplication completely across all inner basis interactions:

$$\begin{aligned} &= N^2 \left(\langle 1|1 \rangle + 2i\langle 1|2 \rangle - \langle 1|3 \rangle \right. \\ &\quad \left. - 2i\langle 2|1 \rangle - 4i^2\langle 2|2 \rangle + 2i\langle 2|3 \rangle \right. \\ &\quad \left. - \langle 3|1 \rangle - 2i\langle 3|2 \rangle + \langle 3|3 \rangle \right) \end{aligned}$$

Applying the orthonormality conditions ($\langle m|n \rangle = 1$ if $m = n$, and 0 if $m \neq n$):

$$= N^2 (1 + 0 - 0 - 0 - 4(-1)(1) + 0 - 0 - 0 + 1)$$

Consolidating the numerical terms inside the parenthesis:

$$= N^2(1 + 4 + 1) = 6N^2$$

Additional Problems 3–4

Setting this expression equal to 1 for proper state normalization:

$$6N^2 = 1 \implies N^2 = \frac{1}{6} \implies N = \frac{1}{\sqrt{6}}$$

Problem 4: Hermiticity of a Sum

If $\hat{A}$, $\hat{B}$ are both Hermitian, show $\hat{A} + \hat{B}$ is Hermitian.

Solution: We evaluate the definition of the adjoint on a sum using the linear distributive rule for Hermitian conjugation, $(\hat{X} + \hat{Y})^\dagger = \hat{X}^\dagger + \hat{Y}^\dagger$:

$$(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$$

Because both individual operators are already established as inherently Hermitian, we can substitute $\hat{A}^\dagger = \hat{A}$ and $\hat{B}^\dagger = \hat{B}$:

$$= \hat{A} + \hat{B}$$

Since applying the full adjoint returns the exact unchanged starting sum expression, $(\hat{A} + \hat{B})^\dagger = \hat{A} + \hat{B}$, the compound operator sum is proven to be Hermitian. ✓

Additional Problems 5–6

Problem 5: Is $i\hat{A}$ Hermitian?

If $\hat{A}$ is Hermitian, is $i\hat{A}$ Hermitian?

Solution: We apply the adjoint operator operation to the product expression. Using the scalar multiplication rule for adjoints, $(c\hat{X})^\dagger = c^*\hat{X}^\dagger$:

$$(i\hat{A})^\dagger = (i)^* \cdot \hat{A}^\dagger$$

Evaluating the complex conjugate of the imaginary unit factor ($i^* = -i$):

$$= -i \cdot \hat{A}^\dagger$$

Since our condition states that $\hat{A}$ is fundamentally a Hermitian operator, we swap $\hat{A}^\dagger \rightarrow \hat{A}$:

$$= -i\hat{A}$$

Additional Problems 5–6

Comparing this result directly against the starting configuration:

$$-i\hat{A} \neq i\hat{A} \quad (\text{unless } \hat{A} = 0 \text{ identically})$$

So $i\hat{A}$ is **anti-Hermitian**, not Hermitian — this is exactly why $\hat{p} = -i\hbar d/dx$ needs that specific factor of i , as seen earlier.

Problem 6: Eigenvalue of $\hat{a}_+\hat{a}_-$ for $n = 5$

For the harmonic oscillator number operator $\hat{N} = \hat{a}_+\hat{a}_-$, what is $\hat{N}|\psi_5\rangle$?

Solution: Let us break down the action step-by-step using the standard raising ($\hat{a}_+$) and lowering ($\hat{a}_-$) ladder definitions, where $\hat{a}_-|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}_+|n\rangle = \sqrt{n+1}|n+1\rangle$:

$$\hat{N}|\psi_5\rangle = \hat{a}_+\hat{a}_-|\psi_5\rangle = \hat{a}_+(\hat{a}_-|\psi_5\rangle)$$

Additional Problems 5–6

Evaluating the internal lowering operator step first on index level 5:

$$\hat{a}_-|\psi_5\rangle = \sqrt{5}|\psi_4\rangle$$

Substituting this resultant back into our processing track:

$$\hat{a}_+\left(\sqrt{5}|\psi_4\rangle\right) = \sqrt{5}\left(\hat{a}_+|\psi_4\rangle\right)$$

Now, evaluating the remaining raising operator step upon index level 4:

$$\hat{a}_+|\psi_4\rangle = \sqrt{4+1}|\psi_{4+1}\rangle = \sqrt{5}|\psi_5\rangle$$

Multiplying the radical constant values together to reach the final eigenvalue outcome:

$$= \sqrt{5} \cdot \sqrt{5}|\psi_5\rangle = 5|\psi_5\rangle$$

This explicitly confirms the definition of the number eigenstate.

Additional Problems 7–8

Problem 7: Probability of Measuring a Particular Eigenvalue

For $|\psi\rangle = \frac{1}{\sqrt{3}}|1\rangle + \sqrt{\frac{2}{3}}|2\rangle$, find the probability of measuring the eigenvalue associated with $|2\rangle$.

Solution: According to Born's statistical interpretation rule, the probability is found by taking the absolute modulus squared of the inner product projection overlap, $P_2 = |\langle 2|\psi\rangle|^2$. First, let us expand the internal projection term $\langle 2|\psi\rangle$:

$$\langle 2|\psi\rangle = \langle 2|\left(\frac{1}{\sqrt{3}}|1\rangle + \sqrt{\frac{2}{3}}|2\rangle\right)$$

Distributing the bra vector across the expansion parts:

$$= \frac{1}{\sqrt{3}}\langle 2|1\rangle + \sqrt{\frac{2}{3}}\langle 2|2\rangle$$

Additional Problems 7–8

Applying basis state orthonormality ($\langle 2|1\rangle = 0$ and $\langle 2|2\rangle = 1$):

$$= \frac{1}{\sqrt{3}}(0) + \sqrt{\frac{2}{3}}(1) = \sqrt{\frac{2}{3}}$$

Now, computing the absolute value squared of this outcome amplitude:

$$P_2 = \left| \sqrt{\frac{2}{3}} \right|^2 = \frac{2}{3}$$

Problem 8: Commutator $[\hat{L}_y, \hat{L}_z]$

Given the cyclic relations $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$, $[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$, $[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$, find $\Delta L_y \Delta L_z$ in terms of $\langle L_x \rangle$.

Additional Problems 7–8

Solution: We evaluate this relationship by passing the operators into the general Robertson Uncertainty Relation statement:

$$\Delta L_y \Delta L_z \geq \frac{1}{2} \left| \langle [\hat{L}_y, \hat{L}_z] \rangle \right|$$

Substituting the specific given cyclic commutator identity $[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$ directly into the expectation brackets:

$$= \frac{1}{2} \left| \langle i\hbar\hat{L}_x \rangle \right|$$

Pulling the deterministic scalar constants completely out from the absolute expectation value calculation:

$$= \frac{1}{2} \cdot |i| \cdot \hbar \cdot \left| \langle \hat{L}_x \rangle \right|$$

Since $|i| = 1$, simplifying this expression yields the clear lower bound:

$$\Delta L_y \Delta L_z \geq \frac{\hbar}{2} |\langle L_x \rangle|$$

Additional Problems 9–10

Problem 9: Verifying a Resolution of Identity (2-state system)

For an orthonormal basis $\{|0\rangle, |1\rangle\}$, verify $|0\rangle\langle 0| + |1\rangle\langle 1| = \hat{I}$ by acting on a general state $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$.

Solution: Let us apply the combined outer-product operator directly to the expanded definition of our general state vector:

$$\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right)|\psi\rangle = \left(|0\rangle\langle 0| + |1\rangle\langle 1|\right)(c_0|0\rangle + c_1|1\rangle)$$

Expanding this expression by distributing the operator terms across the linear combination:

$$= |0\rangle\langle 0|(c_0|0\rangle + c_1|1\rangle) + |1\rangle\langle 1|(c_0|0\rangle + c_1|1\rangle)$$

Distributing the inner bra projections onto the inner state vectors and pulling scalars to the front:

$$= c_0|0\rangle\langle 0|0\rangle + c_1|0\rangle\langle 0|1\rangle + c_0|1\rangle\langle 1|0\rangle + c_1|1\rangle\langle 1|1\rangle$$

Additional Problems 9–10

Applying our core basis orthonormality rules ($\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$):

$$= c_0|0\rangle \cdot (1) + c_1|0\rangle \cdot (0) + c_0|1\rangle \cdot (0) + c_1|1\rangle \cdot (1)$$

Eliminating the zeroed terms and combining what remains:

$$= c_0|0\rangle + c_1|1\rangle = |\psi\rangle \checkmark$$

Since acting with this operator combination returns the exact state completely unchanged, it is equivalent to the identity operator $\hat{I}$.

Problem 10: Minimum Uncertainty Product for Spin- $\frac{1}{2}$ -like System

Two Hermitian operators satisfy $[\hat{A}, \hat{B}] = 2i\hbar\hat{C}$, with $\langle\hat{C}\rangle = 0.5\hbar$. Find the minimum value of $\Delta A \Delta B$.

Solution: We begin by introducing our unique operator parameters directly into the standard baseline Robertson bounding equation:

$$\Delta A \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|$$

Additional Problems 9–10

Substituting our explicit given commutator identity expression, $[\hat{A}, \hat{B}] = 2i\hbar\hat{C}$:

$$= \frac{1}{2} \left| \langle 2i\hbar\hat{C} \rangle \right|$$

Factoring out the clear constant numbers outside of the absolute modulus constraints:

$$= \frac{1}{2} \cdot |2i| \cdot \hbar \cdot \left| \langle \hat{C} \rangle \right|$$

Evaluating the magnitude of the imaginary scalar coefficient ($|2i| = 2$):

$$= \frac{1}{2} \cdot (2) \cdot \hbar \cdot \left| \langle \hat{C} \rangle \right| = \hbar \left| \langle \hat{C} \rangle \right|$$

Plugging in our given value for the expectation level, $\langle \hat{C} \rangle = 0.5\hbar$:

$$= \hbar \cdot (0.5\hbar) = 0.5\hbar^2$$

Thus, the minimum allowed product threshold is exactly $0.5\hbar^2$.

Course Recap: Five Units

- ➊ **Origin of QM:** blackbody radiation, photoelectric & Compton effects, de Broglie waves, wave packets.
- ➋ **Formulation of QM:** axioms, uncertainty principle, the wavefunction, double-slit experiment.
- ➌ **Schrödinger Equation I:** probability density, expectation values, TDSE & TISE, probability current.
- ➍ **Schrödinger Equation II:** infinite well, tunneling, harmonic oscillator, hydrogen atom, angular momentum.
- ➎ **Dirac Notation:** Hilbert space, Hermitian operators, commutators, the general uncertainty principle.

From Planck's quanta to the algebra of observables — the conceptual arc of quantum mechanics.

Recommended Resources

Textbooks

- Eisberg, R., & Resnick, R. (2006). *Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles*, 2nd ed. Wiley.
- Griffiths, D. (2004). *Introduction to Quantum Mechanics*, 2nd ed. Pearson.
- Aruldas, G. (2013). *Quantum Mechanics*, 2nd ed. PHI.

Online Resources

- MIT 8.04 Quantum Physics I (YouTube playlist)
- Introduction to Quantum Mechanics (YouTube playlist)

THANK YOU!!!